



The Currency Book

September 29, 2026

Introduction

With trillions of dollars, euros and yen changing hands every day, foreign exchange is indisputably the largest and most liquid financial market in the world. Yet despite its size, the currency market is also one of the least understood. Unlike bonds or equities, there is no widely accepted framework for explaining currency moves, which reflect a complex interplay of macroeconomics, capital flows, market structure, investor behaviour and (geo)politics.

Over the past two decades, advances in technology and market infrastructure have transformed how currencies are traded, placing FX at the forefront of broader changes in global finance. Yet because currencies sit at the intersection of national economies, the FX market is also fundamentally connected to international trade and cross-border capital flows. It is therefore increasingly exposed to the forces reshaping the global economic landscape—from geopolitical fragmentation and changing trade relationships to shifts in regulation, payments infrastructure and the architecture of the international monetary system.

This Deutsche Bank Currency Book brings together academic, industry and Deutsche Bank expertise to explain how FX markets function, how exchange rates are determined, how investors seek to generate returns from currencies, and how FX risk can be managed by investors and corporates alike. Our aim throughout the *Book* is to set out the enduring principles driving markets alongside the developments defining FX today. Ultimately, the structure of the FX market is not simply a question of how currencies trade, but of who controls the financial infrastructure through which money moves across borders. The evolution of FX is increasingly intertwined with questions of monetary sovereignty, financial resilience and geopolitical power.

Chapter 1 reviews the structure of the FX market, examining its participants, ecosystem and evolution over time. Chapter 2 discusses exchange rate determination, moving from traditional academic theories to modern macro-financial approaches and practical decision frameworks. Chapter 3 reviews FX as an asset class and the evidence on currency investing, while Chapter 4 considers currency hedging for both investors and corporates. Finally, Chapter 5 discusses the future of FX, including the impact of technology, payment systems and digital assets.

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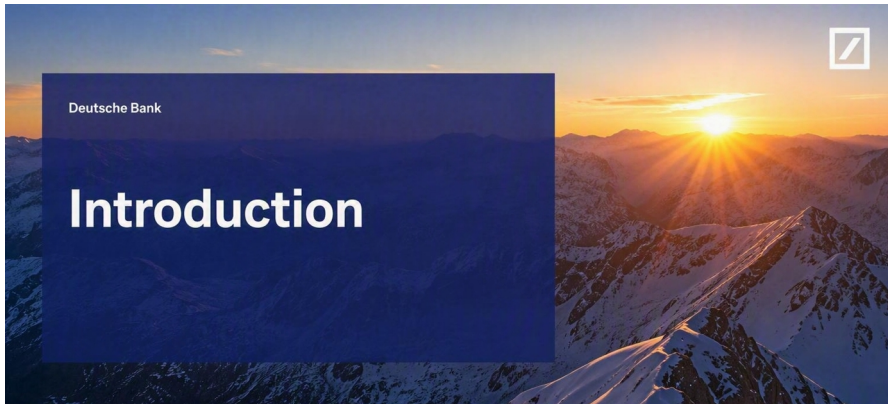
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Introduction

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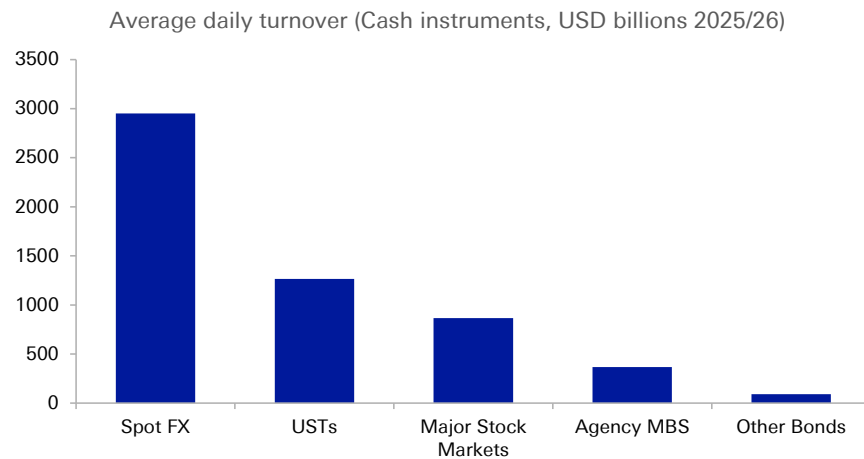
Over the past two decades, advances in technology and market infrastructure have transformed how currencies are traded, placing FX at the forefront of broader changes in global finance. Yet because currencies sit at the intersection of national economies, the FX market is also fundamentally connected to international trade and cross-border capital flows. It is therefore increasingly exposed to the forces reshaping the global economic landscape—from geopolitical fragmentation and changing trade relationships to shifts in regulation, payments infrastructure and the architecture of the international monetary system.

This **Deutsche Bank Currency Book** brings together academic, industry and Deutsche Bank expertise to explain how FX markets function, how exchange rates are determined, how investors seek to generate returns from currencies, and how FX risk can be managed by investors and corporates alike. Our aim throughout the *Book* is to set out the enduring principles driving markets alongside the developments defining FX today. Ultimately, the structure of the FX market is not simply a question of how currencies trade, but of who controls the financial infrastructure through which money moves across borders. The evolution of FX is increasingly intertwined with questions of monetary sovereignty, financial resilience and geopolitical power.



[Chapter 1](#) reviews the structure of the FX market, examining its participants, ecosystem and evolution over time. [Chapter 2](#) discusses exchange rate determination, moving from traditional academic theories to modern macro-financial approaches and practical decision frameworks. [Chapter 3](#) reviews FX as an asset class and the evidence on currency investing, while [Chapter 4](#) considers currency hedging for both investors and corporates. Finally, [Chapter 5](#) discusses the future of FX, including the impact of technology, payment systems and digital assets.

Figure 1: Turnover in spot FX dwarfs other asset classes



Source : Federal Reserve Bank of New York, Bank of England, BIS.
Note: Major Stock Markets covers all US stock Markets, LSE and Nikkei



Introduction

The foreign exchange market is the world's largest and most liquid financial market, but it differs fundamentally from a centralised exchange. Trading takes place across a fragmented, largely over-the-counter network of banks, non-bank liquidity providers, electronic venues and end-users, using instruments ranging from spot and forwards to swaps, futures and options. How do transactions move through this network? Who provides liquidity, who consumes it and what risks arise between execution and settlement? This chapter maps the FX ecosystem from instruments and trading infrastructure to the participants ultimately responsible for currency demand.

1.1 Instruments

The FX market is not one single product. Instead, it is made up of a group of instruments that allow market participants to exchange currencies today, lock in an exchange rate for the future and/or hedge against adverse exchange-rate moves. Below, we outline the main instruments in FX markets and their users.

Spot: This is the simplest and most widely known FX instrument. When market participants refer to "the exchange-rate" or the "spot exchange-rate," they generally mean the price agreed today for exchanging one currency for another on the standard spot settlement date. For most major currency pairs spot settlement is two business days later (T+2).¹ For example, a spot purchase of EUR 1 million against the USD at a EUR/USD exchange-rate of 1.20 means that the buyer receives EUR 1 million in his or her bank account on the settlement date and the seller receives USD 1.2 million from the buyer on the same date.

¹ There are exceptions, for example USD/CAD settles T+1.



By convention, the first currency in an exchange-rate quotation is known as the base currency, while the second is the quote currency. The exchange-rate therefore represents the number of units of the quote currency required to purchase one unit of the base currency. In USD/JPY, for example, the USD is the base currency and the exchange-rate indicates the number of Japanese yen required to purchase one US dollar. A USD/JPY exchange-rate of 160 means 1 USD can be exchanged for 160 JPY.

Spot transactions are the most commonly used foreign exchange payments for cross-border payments for goods and services, the purchase of foreign financial assets, as well as portfolio rebalancing and proprietary trading. In 2025, the Bank for International Settlements (BIS) estimates spot turnover at \$3trn per day, accounting for [31%](#) of global over-the-counter (OTC) FX turnover.

Forwards and futures: An (outright) forward is similar to a spot transaction, but is instead an agreement to exchange two currencies at a rate (price) agreed today, but with settlement taking place at a date more than T+2 days in the future. They are also traded OTC. Forwards are most commonly used by those who have a known future cash flow and want to remove uncertainty around the future exchange rate. In 2025, the BIS estimates that they account for [19%](#) of global turnover, or \$1.8trn per day.

Futures are economically similar to forwards but are standardised contracts traded on exchanges and centrally cleared. This provides transparent pricing and reduces bilateral counterparty risk, but offers less flexibility over contract size, maturity and settlement terms than an OTC forward. Exchange-traded FX futures remain considerably smaller than the OTC FX market.

Non-deliverable forwards (NDFs): An NDF is a cash-settled forward commonly used for currencies where convertibility or offshore deliverability is restricted. Two parties agree today on an exchange rate for a future date, but unlike a standard forward, the underlying currencies are not exchanged. At maturity, the agreed NDF rate is compared with a pre-specified fixing or reference rate. The difference is usually settled in USD. NDFs are commonly used by corporates, banks and investors to hedge or take exposure to currencies subject to capital controls.

Swaps: An FX swap combines two transactions: an exchange of currencies on an initial date and a reverse exchange at a later date, with both exchange rates agreed when the swap is initiated. They are used by institutions that hold cash in one currency (e.g. EUR) but need funding in another (e.g. USD) for investment, hedging or other needs. They are by far the largest FX instrument by turnover, with the BIS estimating turnover of \$4trn per day in April 2025, [42%](#) of global FX turnover. A swap is, in effect, a form of collateralised borrowing, with the collateral being cash in a different currency to that of the currency borrowed.

Cross-currency (XCCY) swaps: A cross-currency swap is similar to an FX swap in that principal amounts denominated in two different currencies are typically exchanged at inception and reversed at maturity. The key difference is that a cross-currency swap also involves periodic exchanges of interest payments denominated in the two currencies over the life of the contract. These payments may be fixed or floating. Cross-currency swaps therefore allow market participants to transform both the currency denomination and, where relevant, the interest-rate characteristics of an asset or liability.



For example:

- Suppose a European corporate issues a USD-denominated bond to US-based investors. It receives the bond proceeds in USD, but must periodically pay interest in USD and repay the USD principal when the bond matures. However, because the corporate primarily earns revenue in EUR, it needs to convert these USD liabilities into EUR liabilities to avoid currency and potentially interest-rate risk.
- The corporate removes FX and interest-rate risk by entering into a xccy swap, where, in essence, it converts this USD-liability into a EUR-liability.²
- They are commonly used by banks, corporates and investors that need to raise money in one currency but need funds in another. The basis is the extra spread added to one side of the swap to balance unequal demand for funding in the two currencies. See [Section 2.11](#) for more on cross-currency basis and [here](#).

Options: An option gives the holder the right, but not the obligation, to buy or sell one currency against another at a specified exchange rate during a specified period. Options are used to hedge currency risk or speculate on currency movements and are particularly useful when the end-user wants an asymmetric payoff profile. Unlike a forward, an option can limit losses from an adverse currency movement while allowing the holder to retain some or all of the benefit from a favourable movement. In 2025, they represented around 7% of global FX turnover. Options are widely used by corporates and investors for hedging, and by investors and dealers to express directional or volatility views.

The simplest and most widely used instruments are *vanilla options*, which give the holder the right to buy (a call) or sell (a put) a currency at a predetermined strike price. A European-style option can be exercised only at expiry, while an American-style option can be exercised at any point up to expiry. More structured products include barrier options, whose payoff is activated or extinguished if the exchange rate reaches a specified level—for example, knock-in and knock-out options. One-touch and no-touch options pay a fixed amount depending on whether the exchange-rate does, or does not, reach a predetermined level during the life of the option. These instruments are commonly used to express views not just on the direction of a currency pair, but also on the probability, timing and magnitude of exchange-rate moves.

1.2 FX spot trading life cycle and risks

Having discussed the main FX instruments, the next step is to examine how these transactions are executed and settled in practice. The complete trade life cycle, meaning the sequence of events and processes involved in executing a trade, is crucial to understanding how FX transactions work. We therefore describe below the broad steps involved in trading FX, focusing on spot transactions. They are:

1. **Pre-trade:** This stage encompasses all the necessary preparations before a trade is executed, including client onboarding, platform selection, credit checks, and regulatory compliance.

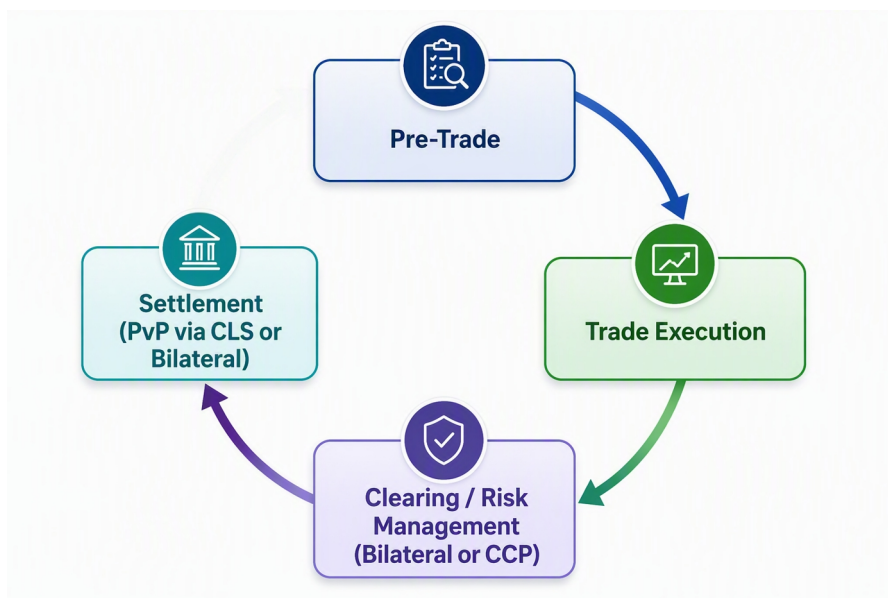
2 Specifically, it exchanges the USD bond proceeds for EUR at inception, receives USD interest payments (from the swap counterparty to pay U.S. bond investors) and pays EUR interest (to the swap counterparty) over the life of the swap, and receives USD principal (from the swap counterparty to pay U.S. investors) in exchange for EUR principal at maturity. The USD receipts from the swap are used to meet the bond's USD interest and principal payments, effectively converting the USD-denominated bond into a EUR-denominated liability.



2. **Trade execution:** This is the point where the trade is agreed upon and executed at a specific exchange rate, creating a legally binding contract.
3. **Trade clearing:** This stage involves validating trade details and managing counterparty credit risk before settlement, either bilaterally or through a central clearing counterparty (CCP). In centrally cleared FX markets, the CCP becomes the buyer to every seller and the seller to every buyer, reducing direct counterparty exposures between market participants. Specialist FX CCPs, such as LCH's [ForexClear](#), clear both non-deliverable and deliverable FX products. While FX spot transactions can be cleared, they are not subject to mandatory clearing requirements. ForexClear supports 25 non-deliverable forward (NDF) currency pairs, 9 non-deliverable option (NDO) currency pairs, and 8 deliverable (Spot, Forwards and Options) currency pairs, at the time of writing.
4. **Trade settlement:** This final stage represents the completion of the transaction, with the actual exchange of currencies taking place. Settlement for spot trades typically occurs two business days (T+2) after the trade date, although exceptions exist. For example, USD/CAD settles on a T+1 basis.

A Central Clearing Counterparty (CCP), also known as a Central Counterparty, stands between buyers and sellers in financial markets, becoming the buyer to every seller and the seller to every buyer. This mitigates counterparty risk.

Figure 2: FX Spot Trade Life Cycle



Source : Deutsche Bank

Risks surrounding FX spot trading

The over-the-counter (OTC) nature of the FX market, while offering flexibility and adaptability to participant needs, does present certain risks that are often more pronounced compared to exchange-traded markets. Below we discuss some of the key risks in FX.

Credit risk: Credit risk in FX trading, stemming from a counterparty's potential inability to fulfill its obligations, is present throughout the trade lifecycle:

- **Pre-trade:** Counterparty creditworthiness is a critical consideration in the



pre-trade stage of FX transactions. A thorough assessment should involve evaluating the potential counterparty's financial strength, credit rating, and reputation. Choosing a creditworthy counterparty significantly reduces the risk of default throughout the trade lifecycle.

- **Post-trade** (Clearing and Settlement): While credit risk exists in both the clearing (post-execution/pre-settlement) and settlement stages of a spot FX trade, it primarily materializes during settlement. This stage, involving the actual exchange of currencies, carries the highest risk due to the potential for settlement failure. This risk, often referred to as settlement risk, arises when one party delivers their currency while the counterparty defaults. Cross-border trades are particularly vulnerable due to time zone differences, which can create a window where one party has performed but the other has not. Additionally, mark-to-market risk from FX rate fluctuations between trade and settlement dates can exacerbate settlement risk by increasing potential losses from a default. This is particularly pronounced with forward contracts, where the longer pre-settlement period leads to greater mark-to-market risk and, consequently, higher settlement risk.

Market Risk: This risk in FX trading arises from potential losses due to adverse exchange-rate movements. This is influenced by factors like economic data, geopolitical events, and interest rates. For customers, asymmetric information, where one party has more or better information than another, can lead to inefficient pricing and increased volatility, thereby exacerbating market risk.

- For dealers, a key component of market risk is inventory management risk – the potential for losses from unfavourable price movements in the currencies they hold to facilitate client trading. Effective inventory management, including position limits and hedging, mitigates this risk.

Liquidity Risk: This refers to the overall market's ability to facilitate the buying and selling of a currency pair quickly and efficiently at a desired price. It's influenced by factors such as market depth, trading volume, volatility, and fragmentation.

Operational Risk: This encompasses risks associated with trade processing, settlement, and technology. Errors in trade execution, system failures, and fraud can all lead to significant losses.

Regulatory Risk: This refers to the risk arising from changes in regulations or their interpretation. The evolving regulatory landscape in FX, particularly concerning non-bank dealers and systemic risk, can impact market participants and create uncertainty.

1.3 Market participants

The FX market comprises a variety of participants with different objectives. FX market participants fall into two broad categories: financial and non-financial. Financial participants can act as both liquidity providers and consumers, while non-financial participants act largely as liquidity consumers.

Financial participants: In the FX market, financial participants are primarily banks acting as dealers.

- **Dealers:** These are mainly large commercial banks, investment banks and securities houses that participate in the inter-dealer market for proprietary



trading.³ and/or for meeting customer demand through market-making.⁴

- **Institutional investors:** These are mutual funds, pension funds, insurance companies and endowment funds, collectively referred to as “real-money investors”, which need to buy currencies to satisfy end-user demand for an underlying asset. For instance, a UK pension fund needs to buy USD to purchase US equities. These investors also engage in hedging, proprietary trading and risk management activities.
- **Hedge Funds & Principal Trading Firms (PTFs):** Hedge funds are investment funds that follow a broad range of investment strategies and are generally not subject to the same borrowing and leverage restrictions as traditional investment funds. They are users of FX and derivatives for speculative purposes. PTFs are principal trading firms, such as XTX Markets and Citadel Securities, that hedge or speculate for their own account and also participate in market-making. This category may include high-frequency trading firms that employ high-speed algorithmic trading strategies.

Non financial institutions: Non-financial participants, such as corporations involved in international trade, retail and the official sector, also primarily consume liquidity.

- **Corporates** are active in currency markets primarily to manage FX exposures⁵ arising from international trade, foreign-currency financing, overseas operations and cross-border mergers and acquisitions (M&A).⁶
- **Official sector:** This category includes central banks, sovereign wealth funds, international financial institutions, such as the IMF and BIS, development banks and government agencies. Monetary-policy, financial-stability and mercantilist considerations have all, at various times, motivated central-bank intervention in FX.⁷
- **Other:** This category includes all other market participants, including retail participants. Retail participants engage in FX both to speculate and to meet transactional demand arising from travel, trade or investment.

1.4 Profit versus non-profit seekers

In 2025, daily FX turnover reached \$9.5 trillion according to the BIS triennial survey (BIS, 2025) making currency trading the world's largest financial market. Yet, size and liquidity do not necessarily imply efficiency. Unlike most equity or bond transactions, a large share of FX activity serves an underlying transactional purpose. For instance, corporates must convert revenues and meet foreign-currency invoices; institutional investors hedge overseas assets; and dealers intermediate these flows. For these participants, expected FX returns may be secondary, or even irrelevant, to the decision to trade. In this section, we investigate

Institutional investors are subject to a variety of (1) regulatory, (2) mandate and (3) process constraints.

(1) Regulatory constraints have had an increasingly [important](#) influence on cross-border investment since 2008.

(2) Mandate constraints relate to limits placed on investment allocations by a fund's internal rules. These constraints may arise from, among other things, fund design (i.e., fund characteristics, regions, target benchmarks, target returns), founding charters or guidelines (i.e. many endowments have externally [imposed](#) constraints on investment allocations and government policy (i.e. Norway's pension fund mandate is determined by the [government](#)). These restrictions can materially limit a fund's ability to take currency risk. The simplest mandate distinction is between passive and active funds, with the former following an index and therefore being explicitly required to track its returns.

(3) Process constraints relate to limits placed on the decision-making process. Most investment funds abide by internal risk rules designed to limit exposure to losses during periods of market volatility. These rules can generate value-at-risk (VaR) shocks.

³ Proprietary trading is a profit-seeking activity that dealers once engaged in more extensively, but which has been severely constrained by regulatory change. Most notably, the Volcker Rule, implemented in 2014-15 under the Dodd-Frank regulations, bans banks from engaging in proprietary trading activities, including many foreign exchange instruments.

⁴ Market making: This is an activity that seeks to capture returns by matching buyers and sellers and charging a price for the service through the bid-offer spread. This spread represents the price that liquidity seekers must pay to access the market and is a “frictional” cost of executing transactions. As such, market-making is a liquidity-provision service rather than a source of the ‘excess returns’ associated with activities such as proprietary trading

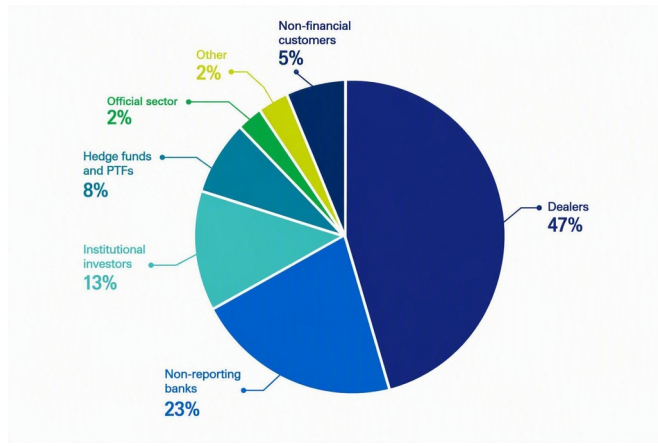
⁵ For more see [here](#)

⁶ Corporates using foreign exchange to hedge future exposures may also seek hedge accounting to



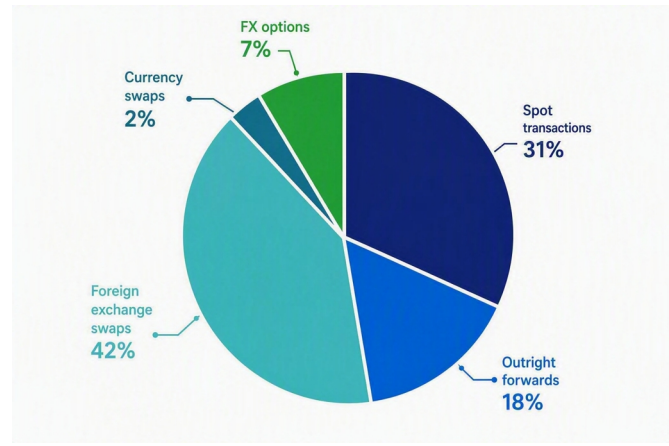
the division of the market between profit-seeking and non-profit-seeking participants. This is the first step in understanding exchange-rate determination.

Figure 3: Daily FX turnover by counterparty



Source : BIS, Deutsche bank

Figure 4: Daily FX turnover by instrument



Source : BIS, Deutsche Bank

Approach 1: The Triennial BIS Survey of FX market volumes

We use the 2025 BIS Triennial Survey to update Deutsche Bank's previous work ([Saravolos et al, 2018](#)) and estimate the share of FX turnover initiated with a profit-seeking currency objective. We construct a "realistic scenario", which applies what we deem to be "realistic" assumptions about participants' ability and mandate to take FX risk.⁸ We find that profit-seeking activity accounts for approximately 40% of turnover under the realistic scenario.⁹

Financial participants

- **Dealers:** Dealer-to-dealer activity accounts for approximately 47% of total daily FX turnover (\$4.44trn) and we classify 10% of dealer flow as profit-seeking (\$444bn/day). Given that much of the daily FX turnover reflects market-making, we assume that inter-dealer transactions conducted by voice are not profit-seeking and that only electronic transactions conducted through order-matching systems such as EBS are profit-seeking.
- **Hedge Funds and PTFs:** Hedge Funds and PTFs account for approximately \$760bn (8%) of headline daily FX turnover and we classify 90% of this as profit-seeking (\$684bn/day). The \$684bn/day figure represents the profit-seeking nature of hedge-funds where the remaining 10% reflects the

ensure that the mark-to-market value of their hedge does not introduce unwanted volatility in the income statement. IFRS 9 provided an updated set of standards for hedge accounting, designed to increase flexibility and better link financial reporting to firms' risk management practices. However, hedge accounting still places limitations on the types of instruments and kinds of hedges that can be implemented. For example, to qualify for hedge accounting, corporates must adhere to hedge effectiveness and demonstrate that an economic relationship between the underlying cash flow or asset and the hedge exists, as well as quantitative limits to hedge ratios.

7 Exchange rates are explicitly targeted by some central banks as part of their overall monetary policy framework (for instance the Monetary Authority of Singapore or the Hong Kong Monetary Authority); other central banks have intervened to defend their inflation target or financial stability reasons (most recently, the Swiss National Bank); and others to prevent competitiveness losses in global trade.



expansion of PTFs to act not only as profit-seekers but also as liquidity providers.

Figure 5: Profit-seekers make up 40% of the FX market

 Counterparty	 Profit-seeking assumptions
 Dealers	Mainly market-makers, small profit-seeking 10%
 Hedge Funds and PTFs	All profit-seeking, except PTF liquidity provision 90%
 Non-Reporting Banks	Mainly market-makers, small profit-seeking incentive 10%
 Institutional Investors	Some fixed-income and equity managers take FX risk 30%
 Official Sector	Reserve managers profit seeking 50%
 Corporates	All corporate activity liquidity-seeking 0%
 Other	Retail participation 50%
 Undistributed	Unknown 0%
Overall	40% of daily FX turnover is profit-seeking

Source : BIS, TIC, Federal Reserve Bank of New York, Deutsche Bank

- **Non-Reporting Banks:** Non-reporting banks account for approximately \$2.19trn, or 23% of headline daily turnover and we classify 10% of this as profit-seeking (\$220bn/day). These largely intermediate for institutional and non-financial customers and apply the same profit-seeking assumptions as we do for dealers.
- **Institutional investors:** Institutional investors account for \$2.56tn of daily BIS turnover¹⁰ and we classify 30% of this as profit-seeking. We assume 80% of active fixed-income managers have discretion to take FX risk and 50% of active equity managers have discretion to take FX risk, implying a profit-seeking share of approximately 30%.
- **Official Sector:** The official sector accounts for approximately 1-2% of headline FX turnover and we classify 50% of the turnover as profit-seeking. The motivations within this category vary. Central-bank intervention may pursue monetary-policy, financial-stability or exchange-rate objectives, while reserve managers may also seek to preserve capital and earn returns.

Non-Financial Sector:

- **Corporates and governments:** Corporates and governments account for 10% of daily FX turnover and we classify 0% of this flow as profit-seeking.¹¹

8 An upper-bound scenario, which relaxes those "realistic" assumptions and produces a higher estimate of profit-seeking activity by classifying ambiguous transactions as profit-seeking is outlined in Appendix .

9 See Appendix for full methodology

10 After reallocation flow from intermediated by non-reporting banks, see Appendix for full methodology.

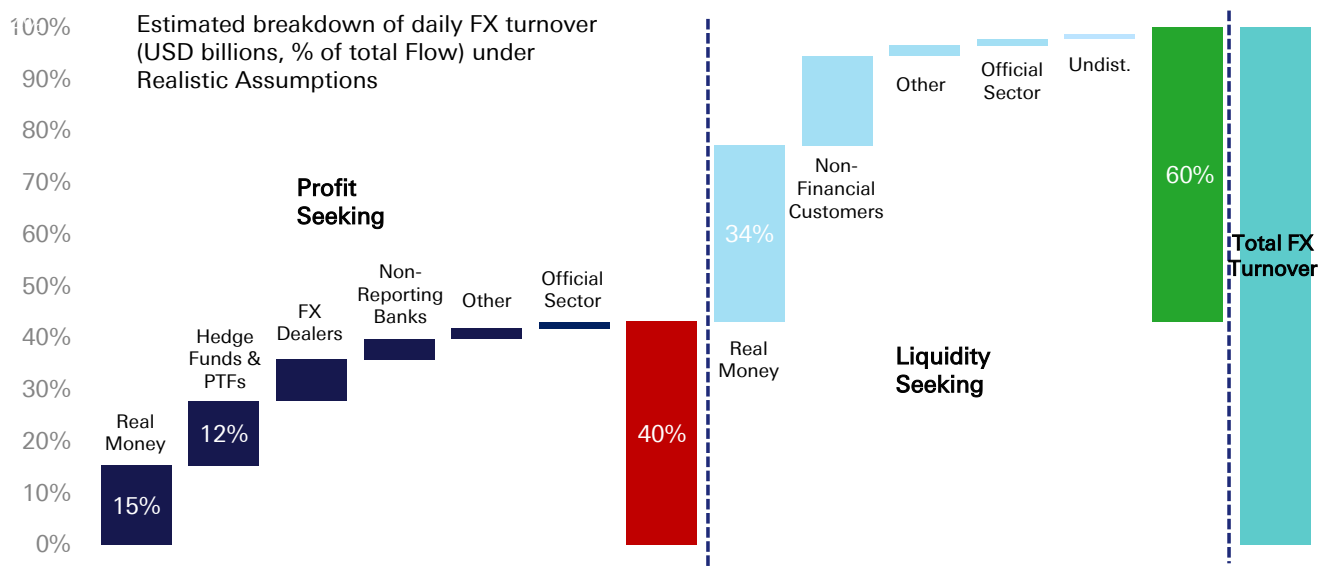


- **Other and Undistributed:** The BIS residual “other” category accounts for approximately 2% of headline turnover. We classify 50% as profit-seeking.

¹² The undistributed category accounts for less than 1% of headline daily turnover and we classify this as 0% profit-seeking.

Bringing it all together: Taken together, we estimate that approximately 40% of adjusted FX turnover is profit-seeking, while approximately 60% is liquidity-seeking or transactional.

Figure 6: There are more liquidity seekers than profit seekers in our baseline estimates of FX market size



Source : BIS, TIC, Federal Reserve Bank of New York, Deutsche Bank

Approach 2: Commitment of Traders CFTC data

While the BIS data provide the most comprehensive view of the global FX market, the survey is only updated every three years. For a higher-frequency perspective, we therefore complement it with the US Commodity Futures Trading Commission's Commitments of Traders (CoT) report, which is published weekly and provides a breakdown of positions in exchange-traded currency futures and options, principally those traded on the Chicago Mercantile Exchange (CME). The report shows the number of outstanding long and short contracts held by different categories of market participant, based on positions as of each Tuesday. Although the futures market represents only a small subset of total global FX activity—most of which takes place over the counter—the CoT data provide a timely and consistent measure of how speculative and hedging positions are evolving.

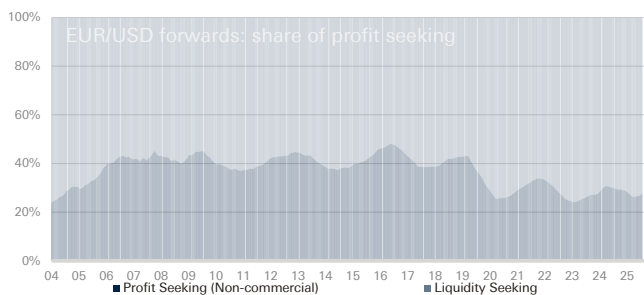
11 Corporates use FX to settle foreign-currency revenues and expenses, finance cross-border activity, hedge forecast cash flows and manage earnings volatility. Governments similarly transact to meet an underlying funding or expenditure requirement.

12 Some investors trade currencies to earn returns, while other transactions relate to travel, remittances or purchases of foreign assets



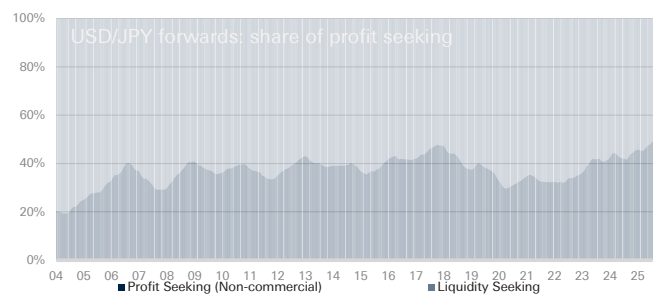
In the traditional or “legacy” CoT report, reportable positions are divided into commercial and non-commercial traders, alongside a residual category of smaller, non-reportable positions. Commercial traders are generally entities that use futures primarily in connection with an underlying business activity and are therefore commonly interpreted as hedgers—for example, corporates or financial institutions managing currency exposures. We classify these positions as liquidity seeking. Non-commercial traders, by contrast, are participants whose futures activity is not classified as commercial hedging and typically includes hedge funds, commodity trading advisers and other leveraged investors. We therefore treat non-commercial positions as profit-seeking. [Figure 7](#) and [Figure 8](#) show how the relative size of the profit-seeking FX market has developed over time. Using one-year rolling averages to reduce the noise of the data, we see that roughly 30-40% of the futures market (*EUR/USD*, *GBP/USD*, *USD/JPY*) is profit-seeking.

Figure 7: Profit-seeking activity in EUR futures



Source : CFTC, Bloomberg Finance LP

Figure 8: Profit-seeking activity in JPY futures



Source : CFTC, Bloomberg Finance LP

Approach 3: Options market DTCC data

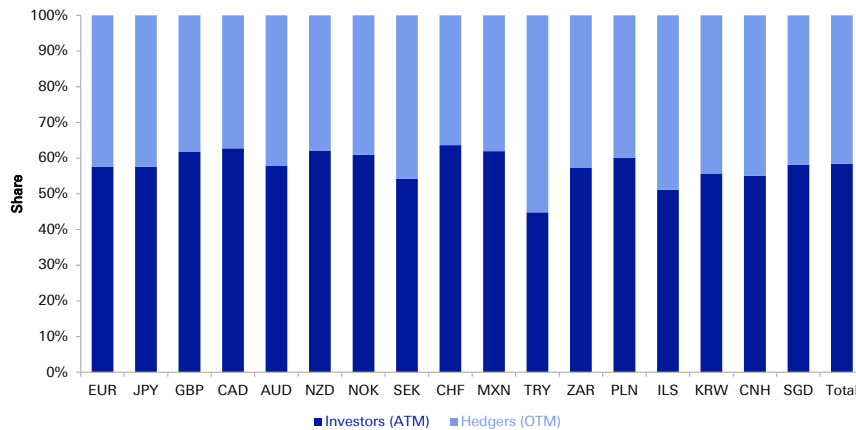
Following the implementation of the Dodd-Frank Act, OTC FX options falling within the US regulatory perimeter are required to be reported to a CFTC-registered swap data repository (SDR). This includes transactions involving US market participants, among others. DTCC operates one of the largest data repositories and therefore captures a substantial share of global institutional FX option activity. We estimate that the DTCC data capture more than 20% of global FX derivative volumes, providing another way of estimating the relative importance of profit- and non-profit-seeking activity in the FX market.

While the data does not provide client granularity, we can back out the likely motivation of the transaction by distinguishing between out-of-the-money and in-the-money option trades. We argue that strikes preferred by the different client groups will likely differ, with hedgers favouring out of the money options (0-25 delta). While speculators focus on directional leverage, the latter often translates to options with strikes nearer to the money (25-75 delta).

[Figure 9](#) shows the split in options volumes for 2025/26 across the two investor categories. We find that option volumes are roughly split equally between investors and hedgers. FX options are instruments that are favoured by hedge funds over corporates and institutional investors, so the results here likely materially overstate the relative importance of profit-maximizing participants in other markets. Again, this shows the FX market is split roughly 50/50 by those who are profit-seeking and those who are not.



Figure 9: Size of profit-seekers in the FX options market



Source : Deutsche Bank , Bloomberg Finance LP, DTCC

*Note: Option volumes captured only for currencies against USD.

1.5 The FX ecosystem

Following [Grover and Saravelos \(2025\)](#), this section examines the significant changes in FX market structure in recent decades, with a particular focus on the FX spot market, and highlights the key challenges that have emerged as a result of this transformation. To understand how these challenges emerged, it is useful to begin by examining the evolution of the FX spot market.

1.5.1 The Evolution of FX Trading: From Voice to Electronic

We analyse the evolution of the FX spot market, focusing on the shift from a pre-electronification era of voice-based trading and bank-centric risk management to the phased electronification of interdealer and dealer-client segments. This transformation has led to the emergence of diverse execution styles and trading venues, which will be explored to provide a comprehensive overview of the changing FX landscape and its implications for risk mitigation. The discussion begins with an examination of the pre-electronification era.

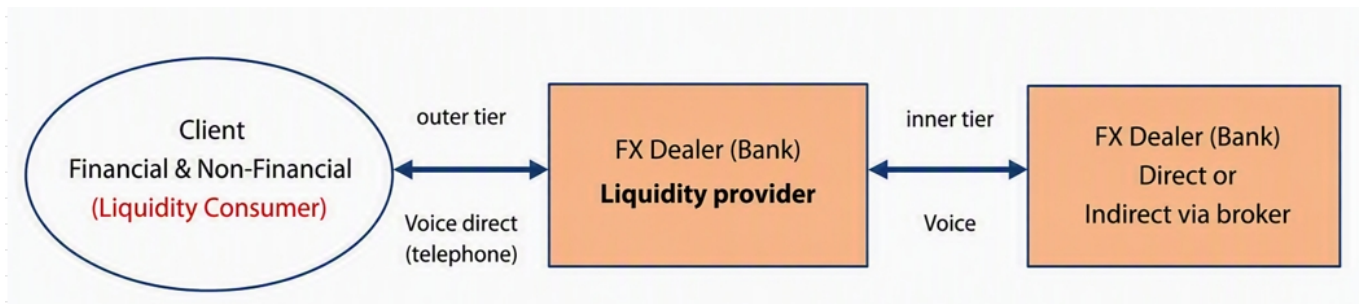
1. Voice trading: The age of the telephone

Early foreign exchange (FX) trading was a two-tiered system conducted over the phone ([Figure 10](#)). Clients contacted dealers (outer tier) directly to inquire about bid and ask prices. Dealers profited from the difference between these prices (the bid-ask spread). Dealers further traded amongst themselves (the "inter-dealer market" - inner tier) to manage inventory imbalances arising from trading with clients. They did this either directly or anonymously through voice brokers.

In this two-tier system, *banks* acted as the sole liquidity providers and risk bearers in the FX market. Inter-dealer trading, used for hedging and inventory management, formed a significant portion of overall trading volume.



Figure 10: The age of the telephone



Source : Deutsche Bank

2. Rise of the machines

The first wave of electronic trading in FX markets primarily affected the inter-dealer market, replacing phone-based trading with computer systems while leaving the dealer-client relationship unchanged. Here, we first discuss the main innovations in the inter-dealer market and then move on to the dealer-client segment.

(a) Electronification of inter-dealer market

Two major innovations in the trading process facilitated by technological advances led to major transformations in the inter-dealer market. They are:

- The **introduction of electronic brokers**, Reuters/Refinitiv (now LSEG) and EBS, in the early 1990s brought a centralized, anonymous limit order book to FX trading, similar to equity market exchanges. This increased the efficiency of price discovery and inventory risk management. These two platforms are still considered primary trading venues.
- Large banks increasingly **internalized trades** (settling within their own books) in the early to mid-2000s. This reduced their need to manage inventory risk through inter-dealer platforms. [Figure 11](#) demonstrates this shift in FX market dynamics, showing a decline in dealer-reported spot trading from roughly 70% in the 1990s to 40% recently. For the largest dealers, internalization now accounts for an estimated 80% or more of major currency pair trades (Moore et al., 2016; Schrimpf and Sushko, 2019).

The two innovations (electronic brokers and trade internalisation) primarily impacted the inter-dealer market, improving efficiency and narrowing bid-ask spreads (King et. al, 2012). Thus, resulting in a more efficient inventory risk management ([Figure 12](#)).

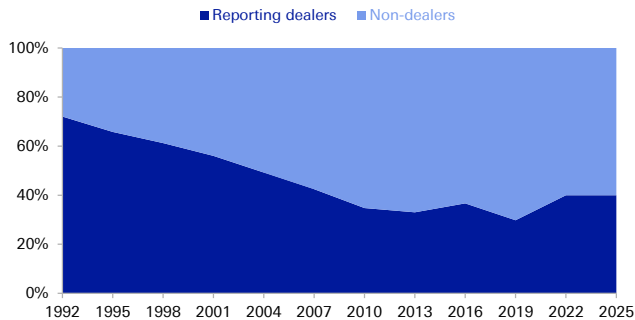
The benefits of electronification were not limited to the trading process. It also helped FX markets address **settlement risk** in the late 1990s. **Settlement risk (a form of credit risk)**, also known as "Herstatt risk," arose due to cross-border currency transfers being limited by central bank operating hours. This created a window where one party to a transaction could default after the other had already settled.

Settlement risk posed a significant challenge. The potential for a default after one leg of an FX transaction had settled created a substantial credit exposure for banks. This exposure could be particularly large for major dealers, where transactions with



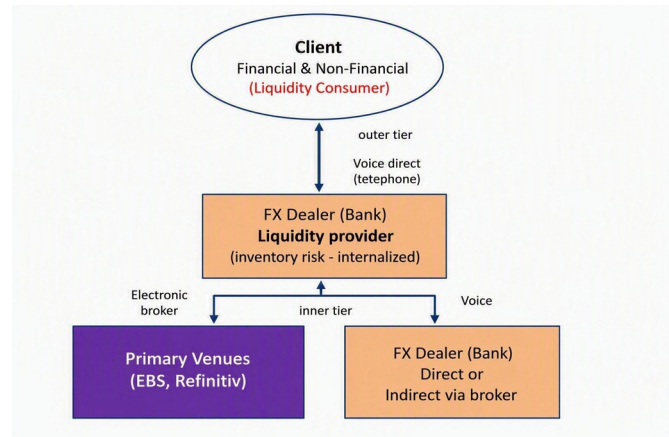
even a single counterparty could exceed the bank's capital.

Figure 11: Drop in dealer share of spot trading as a result of internalization



Source : Deutsche Bank, BIS Triennial Survey

Figure 12: Rise of the machines Part I – electrification of the inter-dealer



Source : Deutsche Bank

To address this, a group of banks developed a new electronic settlement system – the **Continuous Linked Settlement (CLS) Bank**, which began operating in 2002. CLS mitigates settlement risk by simultaneously settling both sides of a trade and by netting payments between member banks.

This netting process drastically reduces the actual funds transferred for settlement to just 4% of the total traded value. *More than \$14 trillion in gross financial obligations were settled on an average day in April 2025, with just over one-third (36%) settled through PvP systems, predominantly CLS, eliminating FX settlement risk for those transactions.*

(b) Electrification of the dealer-client segment

Electronic trading in the inter-dealer market increased efficiency and narrowed bid-ask spreads within that market. However, the dealer-client segment remained largely unchanged, and did not experience the same efficiency gains and reduced bid-ask spreads as the inter-dealer market. This resulted in a surge of new electronic trading platforms specifically designed for clients.

With the electrification of the FX market, new liquidity providers emerged, aiming to reduce market-related liquidity risk and reshape the trading landscape. These new players, enabled by technological advancements, include:

1. Multi-dealer platforms (MDP), emerging in the early 2000s, allowed clients to request for quotes (RFQs) from multiple dealers at once. This streamlined the price discovery process, enabling clients to compare pricing from various dealers through a single platform rather than individual phone calls. The increased competition and transparency benefited clients. Example of MDPs include but are



not limited to FXconnect, FXall, 360T.

2. Single dealer platforms (SDP): Many banks developed their own single-dealer platforms (SDPs) to facilitate direct electronic trading with clients. These SDPs largely replaced older communication methods such as telephone and telex for bilateral trading. Examples include J.P. Morgan eXecute, UBS Neo, and Citi Velocity.

Financial and non-financial consumers of FX (liquidity consumers) can access both multi-dealer platforms (MDPs) and single-dealer platforms (SDPs) for liquidity. Both platform types operate using a request for quote (RFQ) execution protocol and full pre-trade identity disclosure.

A key distinction, however, is that MDPs primarily facilitate connections between liquidity consumers (e.g., asset managers, hedge funds, corporations) and liquidity providers (mainly banks). They do not bear the risk of inventory imbalances stemming from client trades. Conversely, SDPs are integrated within individual bank infrastructures and, consequently, are part of the bank's broader risk management function.

3. Prime-brokerage: The emergence of prime brokerage (PB) arrangements in the late 1990s marked a significant shift in FX market structure, allowing clients such as hedge funds to access the inter-dealer market directly or through electronic brokers. Under a prime-brokerage arrangement, a large bank effectively lends its credit standing and trading relationships to a client—typically a hedge fund, asset manager or other active non-bank institution. Rather than having to establish separate bilateral credit lines and trading documentation with every dealer it wishes to face, the client establishes a primary credit relationship with a single prime broker. The prime broker sets trading limits, monitors the client's exposures, requires collateral or margin where appropriate and charges a fee, typically linked to trading volume and the amount of credit consumed.

Request for Quote (RFQ): It's a request from a liquidity consumer to a liquidity provider (like a bank) asking for the price at which they are willing to buy or sell a specific currency pair. The RFQ will typically specify the currency pair, the amount to be exchanged, and potentially the desired settlement date. The FX provider then responds with a quote, which includes the exchange rate they are offering.

This structure allows hedge funds to access liquidity that was historically available mainly to banks in the interdealer market. A hedge fund can execute an FX trade with one of many approved dealing banks, or through an electronic venue linking multiple dealers, using the credit of its prime broker. The executing dealer therefore does not need to extend a separate bilateral credit line to the hedge fund itself: subject to pre-agreed limits, it accepts the prime broker as the ultimate credit counterparty. Once the client executes the trade, it is "given up" to the prime broker. In legal and settlement terms, the transaction is generally transformed into two matching trades: one between the executing dealer and the prime broker, and another between the prime broker and the hedge fund client. The prime broker consequently stands between the client and the wider dealer network from a credit perspective, even though it need not be involved in choosing the price or executing the original trade.

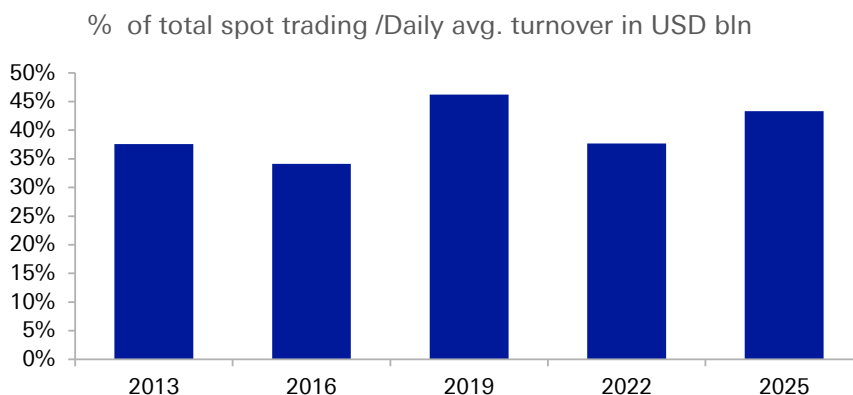
Prime brokerage therefore separates execution from credit provision. A hedge fund can seek the best price across many banks and electronic liquidity pools while concentrating its credit, collateral and settlement relationship with one or a small number of prime brokers. Before PB became widespread, a hedge fund's ability to trade with a dealer depended much more directly on whether that dealer was willing to establish a bilateral credit relationship with it, limiting the number of counterparties it could access and often preventing it from trading on the same electronic venues or at the same prices as large banks. PB substantially lowered this



barrier and helped non-bank institutions become much more active participants in what had previously been a predominantly bank-to-bank market.

Prime brokerage also generates important operational and balance-sheet efficiencies. Because trades with multiple executing dealers are ultimately consolidated at the prime broker, clients can net exposures across counterparties, reducing the amount of collateral and settlement infrastructure required. PB arrangements also facilitate access to a wider range of liquidity providers—including smaller and specialist banks—and support automated trade matching, confirmation and straight-through processing. At the same time, the prime broker assumes significant counterparty and operational risk: it is responsible to executing dealers for trades conducted under its name and must therefore impose credit limits and monitor the client's positions, leverage and trading activity in real time. Prime brokerage thus does not eliminate credit risk; rather, it centralises it at the prime broker while allowing execution to be dispersed across the wider market. Today, a substantial share of spot FX trading is conducted through prime brokerage arrangements ([Figure 13](#)).

Figure 13: Around 40% of total spot trading is done via prime brokerage



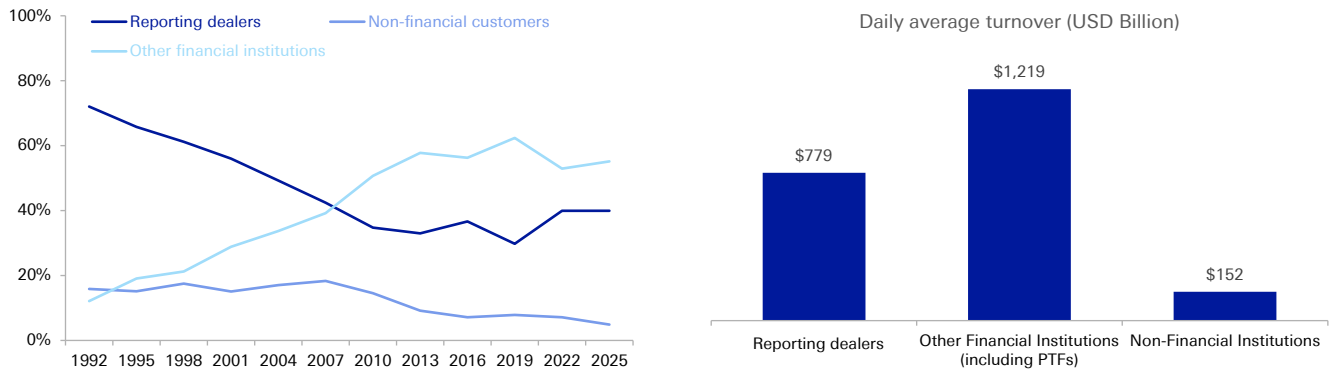
Source : Deutsche Bank, BIS Triennial Survey

4. Principal trading firms (non-bank dealers): Traditionally, banks acted as the primary dealers, or liquidity providers, in the decentralised OTC FX market. However, prime brokerage facilitated client access to primary FX trading venues (EBS/Reuters). This allowed principal trading firms (PTFs), such as XTX Markets and Citadel Securities, to leverage their technological capabilities for electronic market-making, becoming key liquidity providers and serving a wide range of clients, from regional banks to institutional investors.

While Principal Trading Firms (PTFs) are key FX market liquidity providers, they also engage in proprietary trading (investing, hedging, speculating) including high-frequency trading (HFT). HFT firms utilise sophisticated algorithms to execute a large number of rapid trades, often holding positions for extremely short durations. In this capacity, some PTF firms also act as liquidity consumers. This dual role has contributed to the growth of algorithmic traders in FX, impacting inter-dealer trading and the rise of other financial institutions' trading, as noted by BIS Triennial Surveys ([Figure 14](#)).



Figure 14: Increase in financial institution spot trading due to the emergence of principal trading firms (algorithmic firms)



Source : Deutsche Bank, BIS Triennial Survey

5. Electronic communication networks (ECNs): While prime brokerage has facilitated client access to primary FX trading venues (EBS/Reuters), numerous secondary venues, such as Currenex and LMAX, have also emerged as a result of competition driven by technological advances. This has enabled clients (non-banks) to both provide and consume liquidity through limit orders or RFS/RFO model.

Beyond these venues, futures exchanges and dark pools are other alternative liquidity sources. Exchanges utilise centralized limit order books with standardised products, while dark pools operate with hidden order books, matching buy and sell interest at a venue-calculated midpoint. BGC MidFX and FXSpotStream(FSS) MidMatch are prominent examples of FX dark pools.

The rise of secondary and alternative trading venues has also changed how banks manage inventory risk. Banks also utilise multiple venues for hedging client trades. Crucially, the inter-dealer market is no longer exclusive to banks; some clients now also provide liquidity through primary, secondary, and alternative venues.

1.5.2 Execution styles: voices versus electronic trading

This section explores the various FX execution styles, adhering to the Bank for International Settlements (BIS) definitions for the dealer-client segment and the inter-dealer segment.

Outer tier: Dealer-client segment

The two main styles of execution in the dealer-client (outer tier) segment are: Voice and Electronic (BIS Triennial Survey). Each of these can further be classified as direct or indirect. As per BIS, they are defined as follows:

1. Voice-direct (telephone): Trades originated in person, by phone, by telefax or through general messaging systems (eg Outlook, Gmail or Yahoo mail) regardless of how trades are subsequently matched, and not intermediated by a third party.

A Request for Stream (RFS) allows a liquidity consumer to receive a continuous flow of quotes for a specified size and duration, such as a trading day. This functions as a continuous Request for Quote (RFQ), providing real-time price updates for the desired transaction.



2. Voice-indirect (prime-broker): Trade agreed by a voice method and intermediated by a third party (eg a voice broker).
3. Electronic-direct (SDP): Trades executed via an electronic trading system, not intermediated by a third party. These include transactions originated through specific messaging systems that are part of trading platforms. This covers single dealer platforms. It also includes direct API streams: Bloomberg Chat.
4. Electronic-indirect (MDP/ECN): Trades executed via an electronic medium, a third party electronic platform (eg via a matching system). Here, we have multiple platforms that stream prices (RFS/RFO), run a central limit order book, or matching buy and sell interest at a venue-calculated midpoint. [Figure 15](#) highlights the different platforms available where multi dealer platforms are primary RFS/RFO focussed while ECNs do both live streaming and matching.

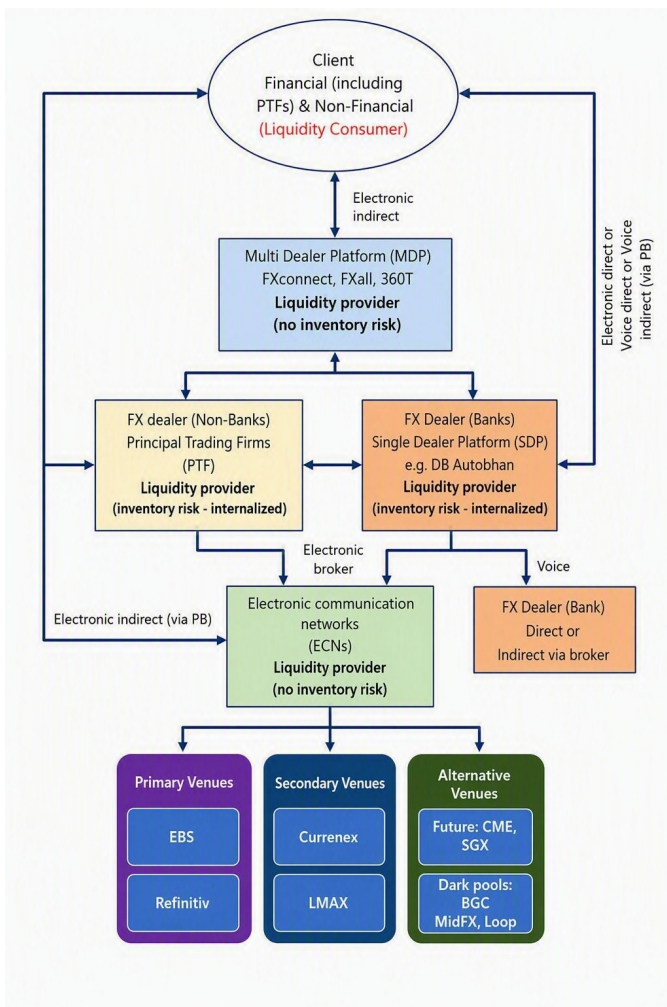
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Single dealer platform (SDP): Bank developed single-dealer platforms (SDPs) to facilitate direct electronic trading with clients.

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include but not limited to FXconnect, FXall, 360T.

Electronic Communication Network (ECN): Primary, secondary and alternative trading venues are collectively referred to as ECN.

Figure 15: Rise of the machines Part II – electronification of dealer-client segment – current market structure



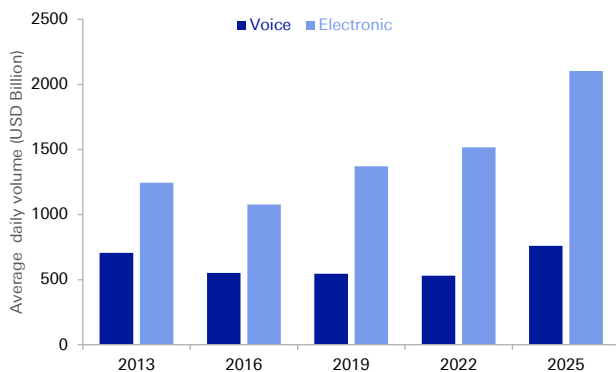
Source : Deutsche Bank

[Figure 16](#) highlights the impact of electronification on FX markets, revealing a



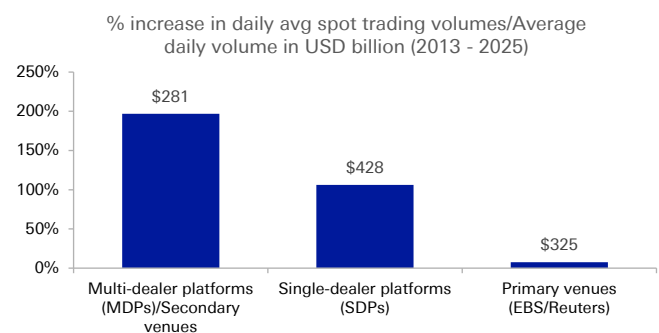
substantial increase in electronic trading in recent years. Notably, the growth of third-party venues has significantly outpaced that of single-bank dealer platforms (Figure 17). These third-party venues provide price streaming aggregation technology and typically charge a pre-trade brokerage fee to liquidity providers.

Figure 16: Share of FX spot trading using electronic methods has been on the rise



Source :Deutsche Bank, BIS Triennial Survey

Figure 17: More than 100% growth in e-spot trading via third party intermediaries (MDPs/Secondary Venues)



Source :Deutsche Bank, BIS Triennial Survey

Inner tier: Inter-dealer segment

For trades that are not internalized (settled across own books), dealers can choose between voice or electronic trading to manage remaining inventory. While traditionally relying on primary venues like EBS/Reuters, they now also have the option of hedging through secondary or alternative venues. This expanded choice, along with increased internalisation, contributes to the observed decline in inter-dealer platform activity (Figure 17).

1.5.3 Trading Venues: Anonymity and Liquidity

Different trading venues can be distinguished based on their offerings in terms of two key characteristics: pre-trade anonymity and liquidity firmness (Chaboud et al., 2023). Pre-trade anonymity refers to the degree to which counterparties remain unknown to each other before a trade is executed. Liquidity firmness, on the other hand, describes the certainty of execution at a quoted price.

Pre-trade anonymity: The level of anonymity can range from fully disclosed to fully anonymous. Multi-dealer platforms (MDPs), acting as aggregators of dealer liquidity, typically operate on a fully disclosed basis. Clients, both financial institutions and corporations, submit requests for quote (RFQs) via the platform to multiple dealers, revealing their identity before trade execution. Similarly, bank-owned single-dealer platforms also function with full disclosure, as the bank dealer knows the identity of the client submitting the RFQ.

At the other end, primary venues running a centralized limit order book (such as EBS/Reuters) generally operate on a fully anonymous basis. The displayed orders do not show a source and all participants can be matched based on the typical price and time priority rules of a CLOB. However, it is important to note that unlike exchanges, this still preserves bilateral relationship of an OTC market structure



since only those that have established bilateral credit relationships, either directly or via their PBs can participate on such venues. Exchange venues, on the other hand, provide the highest possible level of pre-trade (and post-trade) anonymity, basically by definition, as the exchange is the counterparty to each trade.

This anonymity stems from the exchange acting as the central counterparty to every trade, effectively shielding the identities of the ultimate buyers and sellers. This characteristic distinguishes exchange-based trading from other FX venues and contributes to a distinct trading environment.

Secondary FX trading venues occupy a middle ground between the full disclosure of MDPs/SDPs and the complete anonymity of primary venues/exchanges. They offer partial pre-trade disclosure, often employing "tags" to identify counterparties based on specific characteristics or trading styles. This allows participants to select counterparties based on factors like execution quality, creating a tiered system of interaction.

Liquidity firmness: The firmness of liquidity in FX trading, meaning the certainty of execution at a quoted price, is significantly influenced by the practice of "last look." This practice allows liquidity providers (LPs) a brief window, typically milliseconds, to review a trade request before accepting or rejecting it (Oomen, 2016; Cartea, Jaimungal, and Walton, 2019).

Last look was initially developed to address challenges posed by network latencies, platform multiplicity, and varying technological sophistication among market participants. Its acceptable use includes near-instantaneous checks for credit availability and price validity. However, more controversial applications involve imposing additional hold times before trade confirmation.

Liquidity on primary venues is generally considered firm, as last look is prohibited. Conversely, last look is widely employed on secondary venues and universally available on MDPs and SDPs, where LPs retain the option to use it. Consequently, liquidity on these platforms is viewed as less firm, introducing uncertainty regarding trade execution upon hitting a quoted price.

1.5.4 Current market structure challenges

The adoption of electronic trading has generally enhanced FX market quality. Benefits include increased pre-trade transparency, improved operational efficiency through electronic clearing and settlement, and tighter bid-ask spreads and lower intraday volatility in the spot market.

[Figure 18](#) and [Figure 19](#), which track G10 FX high-frequency realized volatility and top-of-the-book spreads respectively, both calculated using one-minute mid-price changes, indicate a downward trend in these metrics. This suggests an improvement in FX market quality over time. The sustained decrease in spreads over time reflects intensified competition for client business in the FX market. The proliferation of trading venues has fostered a competitive environment among liquidity providers, driving them towards consistently narrower, approaching zero, spreads.

A Request for Stream (RFS) allows a liquidity consumer to receive a continuous flow of quotes for a specified size and duration, such as a trading day. This functions as a continuous Request for Quote (RFQ), providing real-time price updates for the desired transaction.

Single dealer platform (SDP): Bank developed single-dealer platforms (SDPs) to facilitate direct electronic trading with clients.

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include but not limited to FXconnect, FXall, 360T.

Centralised limit order book (CLOB): It is a centralised electronic trading platform used to match buy and sell orders.

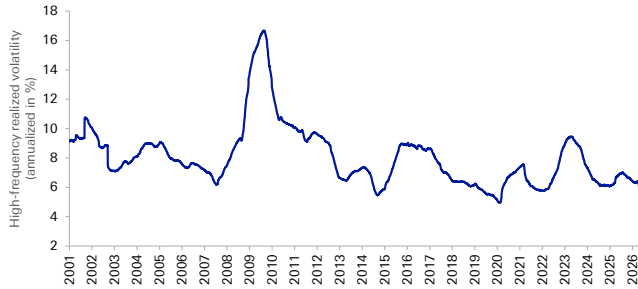
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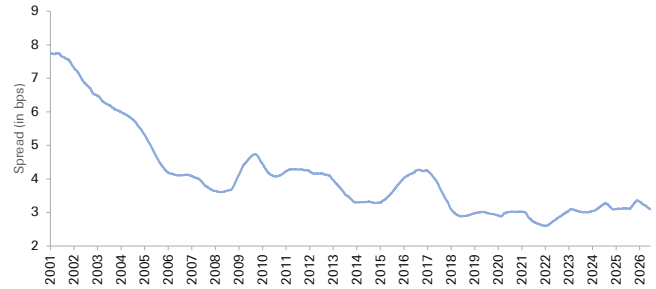


Figure 18: High-frequency G10 FX vol has fallen over time



Source : Deutsche Bank, Reuters. Daily HF realised vol= square root of sum of squared intraday price changes. Annual vol= Daily vol * sqrt(252)

Figure 19: Top of the book spread has also fallen over time



Source : Deutsche Bank, Reuters
Spread=(Ask-Bid)/(Ask+Mid)/2

However, the current structure is not without its flaws. We highlight a few key areas of research for our future publications.

The FX Liquidity Mirage: Myth or Reality?

The fragmentation of the FX market, driven by the decline of inter-dealer electronic brokerage and the rise of diverse trading venues, particularly in the dealer-to-customer segment, presents significant challenges for liquidity assessment. The proliferation of platforms can create a deceptive appearance of increased liquidity, a "liquidity mirage," as the same liquidity might be reflected across multiple interconnected venues. Executing a trade on one platform can trigger a simultaneous reduction in liquidity on others, leading to an overestimation of available liquidity when simply aggregating displayed market depth.

Further, accessing liquidity across this fragmented landscape can be technologically challenging and costly due to latency issues and access fees. While aggregated depth might suggest the feasibility of executing a large order, attempting to do so could significantly impact prices as liquidity is tapped across various platforms.

Accurately measuring FX liquidity requires a more sophisticated approach than simply aggregating depth. Metrics like effective spreads (inclusive of slippage and fees) and price impact offer a more realistic representation of transaction costs and market depth. Analysing volume distribution across venues can help identify concentrations of genuine liquidity, but practical constraints such as technological limitations and access fees must also be factored into the assessment.

Electronification of markets and flash crashes

The increasing attention given to flash crashes in recent years has coincided with the electronification of financial markets. Computer algorithms are now used to automatically make trading decisions, submit orders, and manage those orders after submission. Algorithmic trading (AT) now contributes to more than 70% of trading volumes in equity and FX markets (Hendershott, 2012 and BIS, 2018). Recent flash crashes have triggered an important policy debate over the merits of algorithmic trading in financial markets. As argued in Deutsche Bank research (Glover et al., 2019), establishing a direct link between flash crashes and electronification is far from straightforward. First, there is no universal agreement

Algorithmic trading: Automated transactions where a computer algorithm decides the order-submission strategy.



on what constitutes a flash crash event, though recently Christensen et al. (2017) have made some important contributions on this topic. Second, limited high-frequency data makes it difficult to identify flash crash events going back in time, in turn making it difficult to compare the frequency of such events with changes in market structure. Third, there is a risk of confounding the presence of flash crashes and market electrification. Market automation allows trading to take place during highly illiquid hours with limited human presence, raising the question of whether flash crashes as a phenomenon are in any case only possible due to the electrified nature of the market - the alternative in previous eras being a complete market breakdown where no trades took place.

New intermediaries and Bank dealers: Disruption or Evolution?

Changes in the composition of market intermediaries in the FX spot market also raise interesting research questions.

1. Third party multi-dealer platforms: Multi-dealer platforms (MDPs) offer price aggregation technology and charge liquidity providers a pre-trade brokerage fee, enabling clients to access liquidity from multiple dealers. However, MDPs do not assume inventory risk. Increased competition among dealers, driven in part by MDPs, has compressed spreads, reducing bank dealer profit margins. This presents a challenge for banks, who face declining profits while still incurring MDP brokerage fees and bearing inventory risk from client trades executed through these platforms. Thus, potentially reducing the bank's risk-taking capacity.

2. Non-bank firms: Bank dealers now face competition from principal trading firms (PTFs) in providing FX liquidity. These lightly capitalised, non-bank PTFs leverage speed and cross-asset correlations to manage their risk. The rise of multiple trading venues, in contrast to the pre-electrification era where banks primarily relied on platforms like EBS/Reuters for inventory management, adds another layer of complexity. This fragmentation makes it challenging for banks to discern true liquidity from the "liquidity mirage" of aggregated, but potentially inaccessible, depth across various platforms.

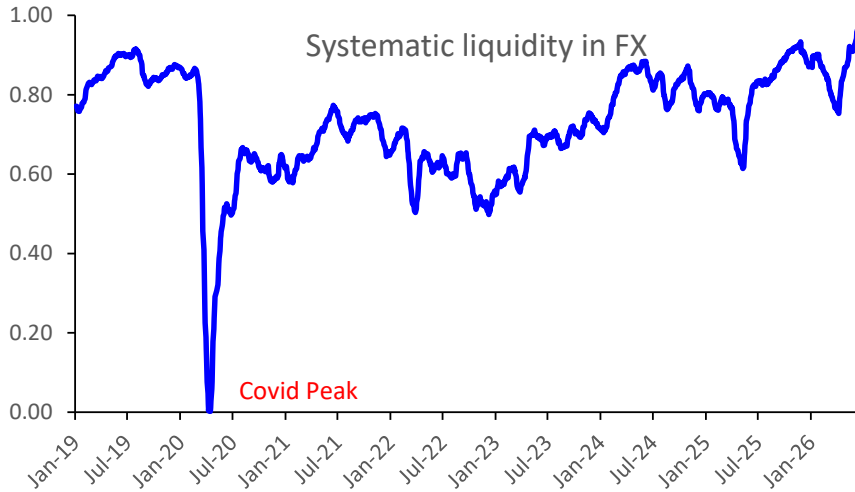
Has the risk-bearing capacity of FX market intermediaries, in total, changed as a result of (1) and (2)? Has this affected market volatility or the frequency of flash events?

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include but not limited to FXconnect, FXall, 360T.

For example, during the initial stages of the Covid crisis in March 2020 ([Figure 20](#)), top of the book bid-ask spreads in the FX market widened significantly. For instance, the spread for USD/JPY doubled from 1 basis point (bp) to 2 bps, while the spread for GBP/USD increased from 1.5 bps to 3 bps over a 10-day period see [Grover, \(2020\)](#).



Figure 20: FX liquidity dropped significantly during Covid Peak (March 2020)



Source : Deutsche Bank

Systematic FX Liquidity is computed as scaled first principal component of Z-scores across measures and currencies. Here, liquidity measures include but not limited to top of book spreads, order book depth, price impact etc. A higher score implies higher liquidity. For more details, please see [FX Liquidity Snapshot: Introducing the DB FX liquidity monitor](#)

New intermediaries: Is FX Price Discovery Broken?

FX market fragmentation raises key questions about price discovery. Have primary CLOBs, despite lower trading volumes, retained their central role? Or has it shifted, potentially to futures markets, given the smaller CLOB size and the increased influence of PTFs using futures-based spot pricing? Analysing latency, order flow, market impact, and correlations across venues is crucial to understanding price discovery in this evolving landscape.

Clearing in FX: Balancing Benefits and Risks in the Modern Market

Regulations like the Standardised Approach for Counterparty Credit Risk (SA-CCR) and the Uncleared Margin Rules (UMR) are increasing capital and margin requirements for non-cleared FX derivatives, making bilateral trading more costly and complex. This is driving some market participants to consider clearing a portion of their FX exposure through CCPs.

Besides exchange-based clearing for standardised FX products using CLOBs, initiatives are emerging to facilitate clearing for bilateral trades like block trades and Exchange For Physical (EFPs). Block trades allow private negotiation of FX futures/options with later clearing house reporting, subject to size/time thresholds. EFPs enable migration of OTC FX exposures to cleared positions, enhancing risk management, operational efficiency, and access to diverse liquidity pools. These initiatives broaden the scope of FX clearing. For more details, please see [Grover \(2025\)](#).

The potential advantages of increased FX clearing include enhanced operational efficiency, potential margin benefits compared to some bilateral arrangements under SA-CCR, expanded counterparty access, and regulatory support. However, drawbacks remain, including increased clearing fees, reduced flexibility in trade

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include, but not limited to FXconnect, FXall, 360T.

Centralized limit order book (CLOB): It is a centralised electronic trading platform used to match buy and sell orders.

Central Counterparty (CCP): It is a financial market infrastructure that reduces bilateral counterparty risk by acting as the intermediary to both sides of a trade and managing the resulting credit exposure through collateral and default-management mechanisms.

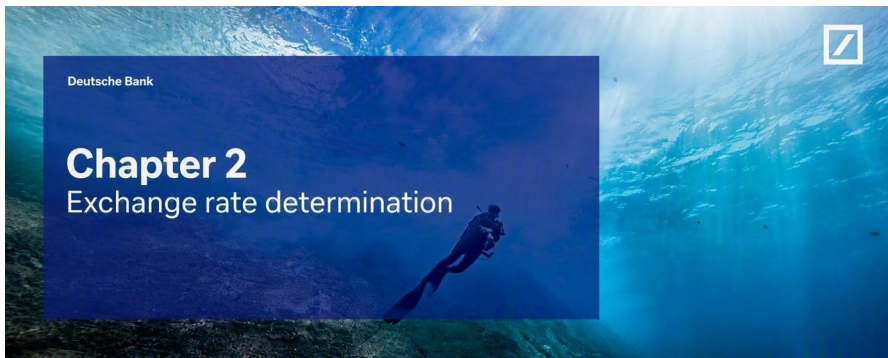
CLS is an independent multi-currency settlement system for FX transactions.



execution and settlement, potential market fragmentation from diverse clearing venues, and the operational challenges of adapting to new technologies and processes. The future of FX clearing depends on the evolving balance between these advantages and disadvantages.

Is FX Settlement Still a Concern?

Settlement risk (Herstatt risk) is the risk of paying out a currency in an FX transaction without receiving the purchased currency. This arises from separate payments in different currencies, often settling across time zones. PvP (Payment versus Payment) mitigates this risk by ensuring simultaneous currency exchange, eliminating the possibility of one party paying while the other defaults. PvP is crucial for FX transactions, especially those settling across time zones, where settlement failures are more likely. CLS operates a PvP settlement service for eligible FX transactions. However, not all FX trades settle through CLS. CLS settlement is not universal due to currency eligibility (not all currencies are included), cost (membership and transaction fees), operational requirements, the availability of alternative methods (like bilateral netting), and internal settlement of intra-group transactions.



Introduction

Exchange rates are famously difficult to forecast. Why can currencies move sharply when macroeconomic fundamentals appear unchanged? Why do apparently compelling exchange-rate models so often fail out of sample? This chapter provides an overview of FX drivers over the short, medium, and long-run horizon. We rely on academic evidence, Deutsche Bank research and practitioner experience to show that while there is no single, universally reliable model of exchange-rate determination, there is a compelling set of frameworks and drivers that can help explain and forecast exchange rate movements over time.

2.1 Why the efficient market hypothesis breaks down in FX

As the world's largest and most liquid financial market, foreign exchange is often regarded as a particularly strong candidate for satisfying the efficient market hypothesis. This states that the current exchange rate fully reflects all publicly available information and it should not be possible to use that information systematically to earn abnormal risk-adjusted profits. In his seminal paper Fama (1970) identified three conditions that would be sufficient to ensure market efficiency:

(a) there are no transaction costs;

(b) all available information is freely available to all market participants; and

(c) all participants agree on the implications of that information for current prices and the underlying model of the economy.

Omitted from these conditions is, however, a fourth and critically important one that is only left implicit in Fama's paper:



(d) a sufficient number of market participants are profit-seeking and are able to move the market rapidly to equilibrium valuations.

FX market structure itself provides perhaps the most compelling reason why the mechanism supporting market efficiency in FX may be significantly weaker than is commonly assumed. Earlier in this book we presented evidence from several sources suggesting that FX turnover is divided roughly evenly between transactions motivated by profit and transactions undertaken for commercial, hedging, institutional or policy purposes. Moreover, with the expansion of passive investment and the increasing importance of regulatory, mandate and balance sheet constraints, the relative importance of unconstrained, return-seeking risk capital may be declining.

A high share of non-profit-motivated or price-insensitive demand has important implications. If transactional flows are large relative to the capital available to arbitrageurs, prices may then reflect not only expectations of future fundamentals, but also hedging demand, corporate cash flows, reserve-management decisions, benchmark rebalancing, regulation and the balance sheet capacity of intermediaries. These forces will generate persistent deviations from conventional measures of fundamental value and predictable returns as compensation for providing liquidity and bearing inventory or balance sheet risk.

Note that the market inefficiency argument does not require non-speculative participants to be irrational or poorly informed. It rests on the simpler observation that, unlike equities and bonds, currencies are required to complete a broad range of underlying economic transactions unrelated to FX profit-seeking. Other participants may seek to maximise returns but be constrained by leverage limits, risk budgets, benchmarks, regulation or investment mandates. The presence of such participants weakens the practical arbitrage mechanism through which information is presumed to be incorporated into prices. Where non-profit-motivated flows are large and the capital of informed speculators is limited, exchange-rate determination is likely to depend on a broader and more distinctive set of forces than in traditional asset markets.

2.2 Underlying principles of exchange rate determination

First, like all asset prices, the exchange rate is forward-looking: today's level embeds expectations about the entire future path of fundamentals and of risk premia. Engel and West (2005) formalised this, showing that in a rational-expectations present-value model when fundamentals are shifting over time the exchange rate will behave almost exactly like a random walk under certain conditions. This is not because fundamentals are irrelevant, but because the exchange rate optimally aggregates expected future fundamentals. This reconciles the empirical failure of fundamentals to *forecast* currencies with the belief that currencies are ultimately driven by fundamentals.

Second, exchange rates are not purely driven by macroeconomic but also financial drivers. In the traditional academic paradigm, the exchange rate equilibrated goods and money markets. In the modern paradigm, it is set at the margin by financial intermediaries and investors operating in imperfect, segmented capital markets subject to balance sheet constraints and time-varying risk-bearing capacity. This shift, associated with Gabaix and Maggiori (2015), Itskhoki and Mukhin (2021), Jiang, Krishnamurthy and Lustig (2021) and the broader intermediary-asset-pricing literature explains not only long-standing "puzzles" of exchange-rate determination but also begins to explain phenomena like the cross-currency basis.



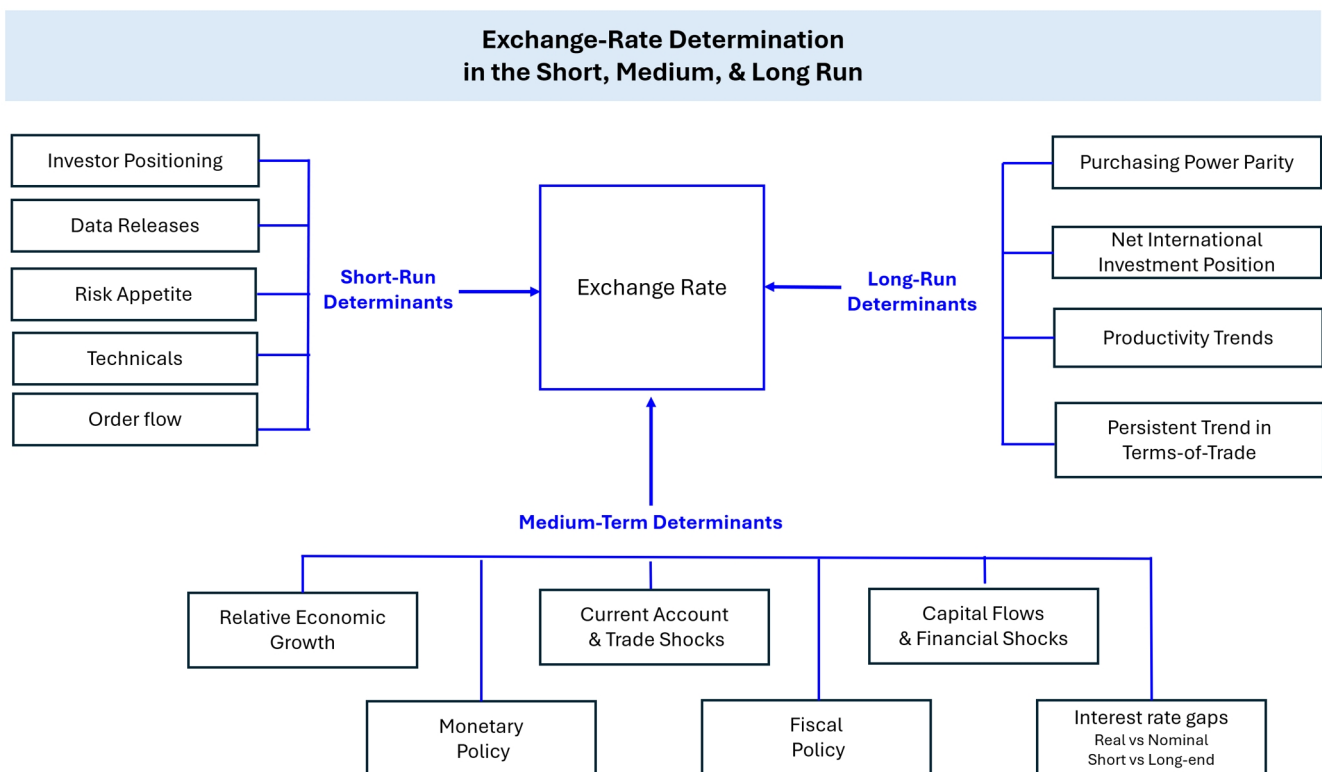
2.3 Time horizons of exchange rate determination

Predicting exchange rates is famously difficult. A helpful organising principle, and the one that structures the remainder of this chapter, is to think about exchange rate determination across three horizons:

- In the **short run** (intraday to a few weeks), exchange rates are dominated by order flow, positioning, sentiment, risk appetite, data releases and technical/trend-following behaviour.
- In the **medium run** (months to a couple of years), cyclical macro forces — interest rate differentials, the monetary policy cycle, relative growth, the current account and capital flows pull currencies around their long-run anchors.
- In the **long run** (several years and beyond), fundamentals reassert themselves: relative prices (PPP), productivity (Balassa-Samuelson), terms of trade, and net international investment positions shape the equilibrium real exchange rate.

In the next few sections we will proceed by horizon and by mechanism. We begin with the long-run anchors (PPP, productivity, terms of trade, external balances and equilibrium-exchange-rate concepts), then turn to the medium-run parity conditions and macro models, then to the short-run microstructure of the market (order flow, positioning, options and technical analysis). We then review forecasting performance tests, before moving on to modern financial theories of the exchange rate, central bank intervention, currency regimes and crises, and finally a set of frontier topics — machine learning and digital currencies.

Figure 21: Overview of Exchange Rate Determination



Source : Deutsche Bank



2.4 Purchasing Power Parity and the long-run real exchange rate

The concept

Purchasing power parity is the oldest and most intuitive theory of the exchange rate.

- *Absolute* PPP says that once converted into a common currency, a representative basket of goods should cost the same everywhere; the nominal exchange rate should equal the ratio of national price levels. This absolute PPP form has little empirical support for reasons discussed later.
- *Relative* PPP says that the change in the exchange rate should offset the difference in inflation rates, so that the real exchange rate (the nominal rate adjusted for relative prices) is stationary and mean-reverting around a constant.

The real exchange rate can be written simply as:

$$q = s + p^* - p$$

Where s is the nominal exchange rate (domestic price of foreign currency, in logs), p is the domestic price level and p^* the foreign price level. PPP holds in the long run if q reverts to a constant mean.

The PPP puzzle and the half-life debate

The central empirical problem, crystallised by Rogoff (1996) as the "PPP puzzle", is FX deviations from PPP are highly persistent and volatile. In Rogoff's (1996, p. 648) own words, "consensus estimates for the rate at which PPP deviations damp... suggest a half-life of three to five years, seemingly far too long to be explained by nominal rigidities." Such slow reversion is very hard to reconcile with nominal price stickiness that lasts one to two years; something other than sticky goods prices must be driving the persistence.

The survey by Taylor and Taylor (2004, Journal of Economic Perspectives) documents how the profession's view on PPP has evolved, concluding that by the early 2000s "the idea of long-run PPP now enjoys perhaps its strongest support in more than thirty years." Two methodological advances drove this:

1. Panel and long-horizon data greatly increased the power to reject the random-walk hypothesis and detect mean reversion.
2. More importantly, nonlinear mean reversion modelling is used: because arbitrage in goods markets is subject to transaction costs, trade barriers and large non-traded sectors, the real exchange rate reverts faster the further it is from equilibrium, and can wander almost freely within a "band of inaction". Once nonlinearity is modelled, estimated half-lives fall and PPP reversion looks far more consistent with economic theory. The more recent survey by Vo and Vo (2023, Journal of Economic Surveys) confirms the long-run validity of PPP and a 3-5 year half-life.

Practical implications and DB house research

For investors, the operational lesson is that PPP is a poor timing tool but a powerful valuation anchor. Deviations can persist for years, an overvalued currency can become more overvalued, but they do eventually correct, and the correction is exploitable if the holding period is long enough. This is precisely what Deutsche

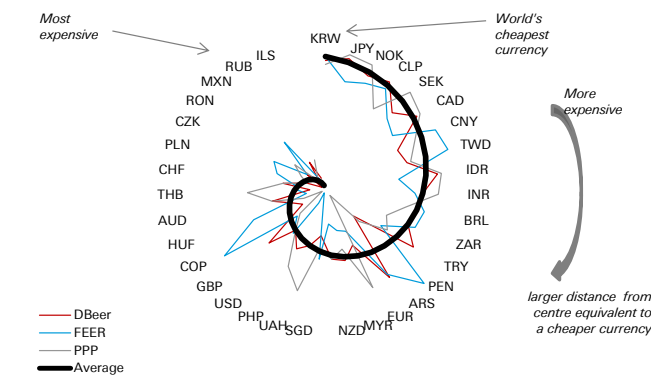


Bank's own back-testing confirms.

[Grover \(2025\)](#), in a study of long-term FX value investing, finds that PPP-based valuation signals generate economically significant mean-reversion signals whose performance improves as the holding period lengthens, peaking at around 24 months - consistent with a PPP misalignment half-life in excess of two years, echoing Lothian and Taylor (2000), Ricci et al. (2013) and [Winkler \(2017\)](#).

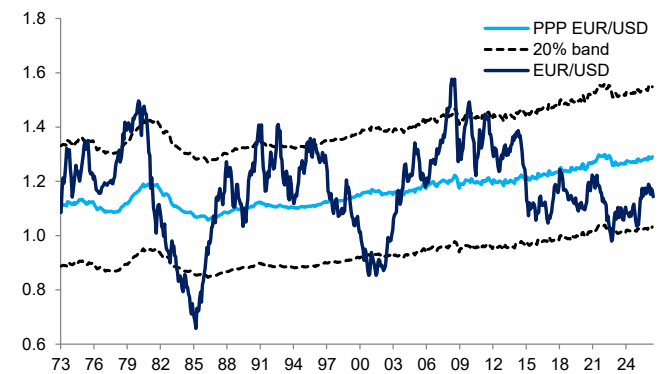
A cross-sectional portfolio that goes long the most undervalued and short the most overvalued currencies on a 12-month hold delivers a Sharpe ratio of 0.60 on the PPP signal after costs, with PPP working best in G10. Volatility-adjusted valuation signals outperform simple sign-based signals, particularly in G10. Notably, [Grover \(2025\)](#) also finds that valuation models underperformed over the two years to early 2025, especially in G10, plausibly because cyclical drivers and extreme macro divergence (aggressive and desynchronised hiking cycles) dominated valuation over that window, a reminder that valuation is a long-horizon strategy that can suffer extended drawdowns.

Figure 22: Valuations ranked



Source : Haver Analytics, Deutsche Bank

Figure 23: EURUSD versus PPP



Source : Refinitiv Workspace Datastream, Deutsche Bank Research

Relative PPP does not imply that real exchange rates must remain constant over all horizons. Even when nominal exchange rates broadly adjust to differences in national inflation rates, equilibrium real exchange rates can drift as the relative price of domestic and foreign consumption baskets changes. Such movements may reflect persistent differences in productivity growth between tradable and non-tradable sectors, shifts in the terms of trade, changes in trade barriers and consumption preferences, pricing-to-market, or movements in international capital flows and risk premia.

Real exchange rate appreciation or depreciation should therefore not automatically be interpreted as a deviation from equilibrium or evidence against PPP; it may instead represent an adjustment in the equilibrium real exchange rate itself. The best-known explanation for such structural movements is the Balassa–Samuelson effect, to which we turn in the following section, before considering other forces that can generate persistent cross-country differences in prices and real exchange rates.



2.5 Productivity, the Balassa-Samuelson effect and terms of trade

Balassa-Samuelson

Absolute PPP based on broad consumer-price baskets fails partly because those baskets include large non-traded components (services, housing). These components cannot be arbitrated over borders. The Balassa-Samuelson (BS) hypothesis provides the classic explanation for systematic trend deviations from absolute PPP.

The logic is that countries that experience faster productivity growth in their traded-goods sector can pay higher wages there; wage equalisation across sectors pushes up wages, and hence prices, in the non-traded sector; this raises the country's overall price level and appreciates its real exchange rate. Rich, high productivity countries therefore have higher price levels and currencies that look "overvalued" on a naive PPP metric but are in fact in equilibrium. Productivity convergence should thus produce trend real appreciation in catching-up economies, or even in developed economies with different productivity rates.

The empirical record on Balassa-Samuelson (BS) is mixed. Egert, Halpern and MacDonald (2006) and the survey by Tica and Druzic (2006) document that the hypothesis has been tested dozens of times across scores of countries. They find that the evidence is strongest for the intermediate links in the mechanism: richer or more productive economies have higher relative prices, and faster tradable-sector productivity growth is often associated with rising non-tradable prices. Cross-sectional tests are overwhelmingly supportive, while panel and time-series results are more heterogeneous and sensitive to the countries, periods and econometric specifications used.

Evidence for the full proposition, that productivity differentials reliably explain aggregate real exchange-rate appreciation, is considerably weaker, in part because purchasing-power parity often fails even within the tradable sector. Productivity catch-up tends to raise relative service prices and can generate real appreciation, but its magnitude is highly uncertain and it is rarely sufficient on its own to explain observed exchange-rate movements.

Terms of trade and commodity currencies

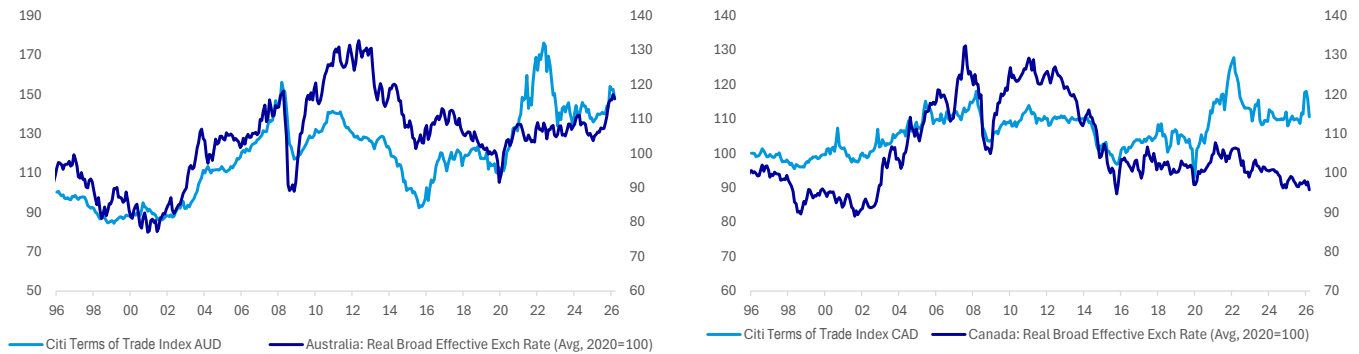
For commodity-exporting economies, the terms of trade, the price of exports relative to imports, can be a more important driver of the real exchange rate than productivity. A rise in the world price of a country's main export commodity increases the purchasing power of its exports and raises national income and wealth. Higher income increases domestic demand, including for non-tradable goods and services whose prices are determined locally. Because the prices of internationally traded goods are more tightly anchored by world markets, the resulting rise in the relative price of non-tradables produces a real exchange-rate appreciation.

This mechanism is sometimes associated with what is known as "Dutch disease". In the classic framework, a commodity boom generates a spending effect, as higher national income raises demand and prices in the non-traded sector, and potentially a resource-movement effect, as labour and capital shift toward the booming commodity sector and away from other tradable industries. Both



channels tend to raise the relative price of non-tradables and appreciate the real exchange rate. The resulting currency appreciation can in turn reduce the competitiveness of other non-commodity exporters in the country.

Figure 24: Commodity terms of trade vs. commodity-currency real exchange rates for AUD and CAD



Source : Deutsche Bank

Terms of trade commodity shocks should be distinguished from the Balassa–Samuelson effect. Balassa–Samuelson attributes real appreciation to relatively rapid productivity growth in the tradable sector: higher tradable-sector productivity raises wages, which spill over into the non-tradable sector, increasing non-tradable prices and appreciating the real exchange-rate. A commodity terms-of-trade shock can produce a similar observable outcome but originates from an improvement in export prices and national income rather than faster productivity.

Chen and Rogoff (2003) formalised the empirical importance of this mechanism for “commodity currencies,” showing that the real exchange-rates of major commodity exporters such as Australia, New Zealand and Canada have strong long-run relationships with world commodity prices.

A striking and much-cited result is the forecasting reversal in Chen, Rogoff and Rossi (2010, Quarterly Journal of Economics): because exchange rates are forward-looking asset prices that embed expectations about future fundamentals, commodity-exporter exchange rates actually help predict world commodity prices, more reliably than the reverse. The follow-up by Ferraro, Rogoff and Rossi (2015, Journal of International Money and Finance) sharpens the caveat for practitioners: the relationship between oil prices and the Canadian dollar is robust and strong at the daily (contemporaneous) frequency but is fragile and largely disappears at the monthly and quarterly horizons and out of sample. Commodity terms of trade are a real-time co-movement story more than a forecasting tool. Terms of trade and the BS effect are incorporated into our [DBeer](#) model, below.



2.6 External balances, net foreign assets and equilibrium exchange rate concepts

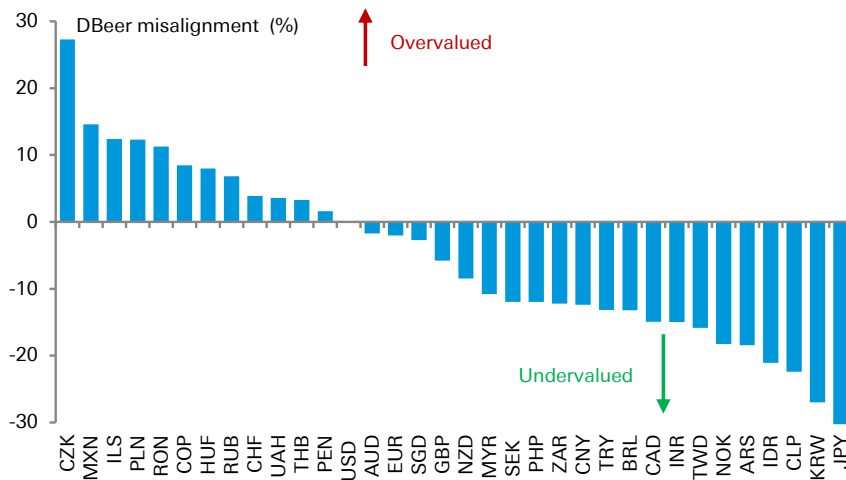
The flow and stock approaches

Two related traditions link the exchange rate to a country's external position. The flow (current account) approach holds that a currency must adjust to bring the current account toward a sustainable level: persistent deficits require a weaker real exchange rate to restore competitiveness. The stock (portfolio/net-foreign-asset) approach emphasises that a country running deficits accumulates external liabilities, and the real exchange rate must eventually be weak enough to generate the trade surpluses that service them or strong enough, for creditor nations, to permit the running down of surpluses. The policy and sell-side community operationalises these ideas through equilibrium exchange-rate models, of which there are three main families:

- The **FEER (Fundamental Equilibrium Exchange Rate)**, (Williamson 1994) and the closely related DEER (Desired EER) are normative, macroeconomic-balance concepts: they identify the real effective exchange rate consistent with the economy achieving internal balance (full employment, target inflation) and external balance (a "sustainable" current account) over the medium term. FEER requires an assumption about a current account equilibrium and trade elasticities but is largely model-free about the adjustment path. Such an approach is used as part of our [FX Valuation Snapshot](#).
- The **BEER (Behavioural Equilibrium Exchange Rate)**, Clark and MacDonald 1998) is essentially an econometric model. It regresses the observed real exchange rate on a set of fundamentals, net foreign assets, productivity/relative prices (Balassa-Samuelson), the terms of trade, and real interest-rate differentials, and defines the equilibrium as the fitted value, with the residual measuring misalignment. One of our proprietary valuation frameworks use a BEER model, called [DBeer](#). It uses terms of trade, a productivity ratio and trade openness. [Grover \(2025\)](#) shows it produces mean-reversion signals and that, in contrast to PPP, DBeer performs better in emerging markets, where idiosyncratic and terms-of-trade dynamics dominate, while PPP dominates in G10.
- The **IMF's External Balance Assessment (EBA)**, introduced in 2012, is not a separate exchange rate theory so much as a framework that combines three approaches. First, its principal current account model estimates an underlying current account balance and compares it with a "norm" consistent with economic fundamentals and policies set at their desired levels. The resulting current account gap is then converted, using estimated trade elasticities, into the real-exchange-rate adjustment that would be required to close it. In this respect, the operational procedure is similar to a FEER calculation, although the current-account norm is itself estimated econometrically rather than imposed entirely outside the model. Second, EBA also contains direct REER panel regressions that are closer in spirit to the BEER approach, estimating the exchange rate implied by fundamentals and desired policy settings. A third, external-sustainability approach calculates the current-account balance needed to stabilise or move the net foreign asset position towards an appropriate benchmark. IMF staff use the three approaches as cross-checks and combine them with country-specific evidence and judgement to arrive at an overall assessment of the current-account and REER gaps.



Figure 25: Dbeer estimated over/undervaluation (%) versus USD across G10 and major EM currencies



Source : Haver Analytics, Deutsche Bank

2.7 Medium-run macro models

The monetary model of exchange rates

The monetary approach treats the exchange rate as the relative price of two monies, determined by money supplies, output (money demand) and interest rates. In Dornbusch's (1976) sticky-price version this is how exchange rate overshooting is generated. Because goods prices are sticky, a monetary expansion causes the exchange rate to depreciate more than proportionately on impact before reverting, explaining why exchange rates are more volatile than the underlying fundamentals.

The original monetary model dominated academic thinking in the 1970s but was challenged by the forecasting failure documented by Meese and Rogoff (1983). It nonetheless remains a useful framework for thinking about the exchange rate consequences of relative monetary policy and the interest rate differential is a key input used by FX market practitioners to this day.

Covered and uncovered interest parity, carry trades in FX

The relationship between interest rate differentials and exchange rates is one of the most widely studied in academic research and has given birth to what are known as the two "parity conditions".

- Covered interest parity (CIP) is an arbitrage relationship linking spot and forward exchange rates to interest rates across currencies. It states that an investor should earn the same return from investing in domestic cash as from converting funds into a foreign currency, investing at the foreign interest rate, and simultaneously using a forward contract to lock in the exchange rate at which the proceeds will be converted back. If the two hedged returns differed, investors could borrow in the lower-returning market and invest in the higher-returning one, using the forward market to eliminate exchange-rate risk, until the arbitrage opportunity disappeared.



CIP held almost exactly in the data until 2008 (see later section).

- Uncovered interest parity (UIP) is a looser condition: it says the expected change in the exchange rate should equal the interest rate differential, so that a high-interest-rate currency should be expected to depreciate by exactly enough to offset its yield advantage. UIP is tested through the Fama (1984) regression of realised exchange-rate changes on the interest differential (or forward premium):

$$\Delta s = \alpha + \beta(i - i^*) + \varepsilon$$

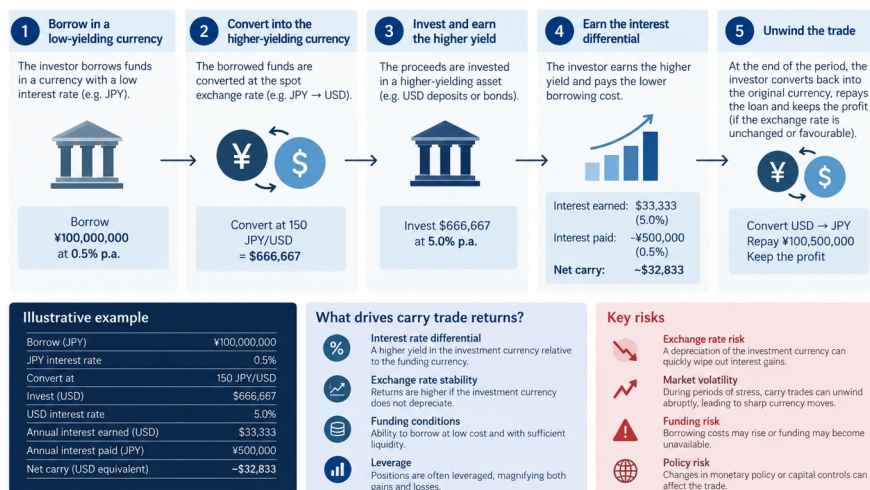
Where Δs is the change in the base currency and i is the base-currency interest rate.

UIP predicts $\beta = 1$. The overwhelming empirical finding for decades was that the estimated β was not only below one but frequently negative. This is what is known as the forward premium puzzle in academia or as the "carry trade" in daily market practice: high-yield currencies tend, on average, to appreciate rather than depreciate, which is exactly what makes the carry trade profitable. This is one of the most robust anomalies in international finance.

Figure 26: Illustration of FX carry trade

How the FX Carry Trade Works

The FX carry trade borrows in a low-yielding currency, converts into a higher-yielding currency, invests the proceeds, and profits from the interest rate differential – as long as the exchange rate does not move against the position.



Source :Deutsche Bank

The literature offers several competing explanations for the existence of carry trades. One interpretation is that carry returns represent compensation for bearing state-dependence or crash risk rather than a free arbitrage. Brunnermeier, Nagel and Pedersen (2009) show that carry returns are negatively skewed: high-yielding currencies can fall sharply and funding currencies appreciate when risk appetite and market liquidity deteriorate. Such episodes can trigger simultaneous unwinding of leveraged positions, amplifying exchange-rate movements. Menkhoff, Sarno, Schmeling and Schrimpf (2012) similarly find that high interest rate currencies tend to perform poorly following unexpected increases in global FX volatility, while low interest rate funding currencies provide insurance in precisely those adverse states.



The broad conclusion is therefore more nuanced than a simple rejection of UIP. Carry trades have historically earned positive average returns, but those returns are negatively skewed, perform poorly during volatility and liquidity shocks, and appear partly related to global risk exposure. At the same time, conventional models have struggled to explain fully the magnitude and time variation of the returns.

Taylor-rule fundamentals and interest rate differentials

In its most recent form, monetary modelling of exchange rates replaces money supplies with the monetary policy reaction function and interest rates. If central banks set interest rates according to a Taylor rule (responding to inflation and the output gap), then the interest differential and hence, via UIP, the expected exchange rate change can be expressed in terms of relative inflation and output gaps. Engel and West (2005), Molodtsova and Papell (2009) and Rossi (2013) showed that Taylor-rule fundamentals can help model exchange rates. With Rossi finding that Taylor-rule and net-foreign-asset fundamentals as the most promising predictors. The strong academic support of monetary exchange-rate modelling is reflected in the strong emphasis on interest rate differentials in market practitioner analysis.

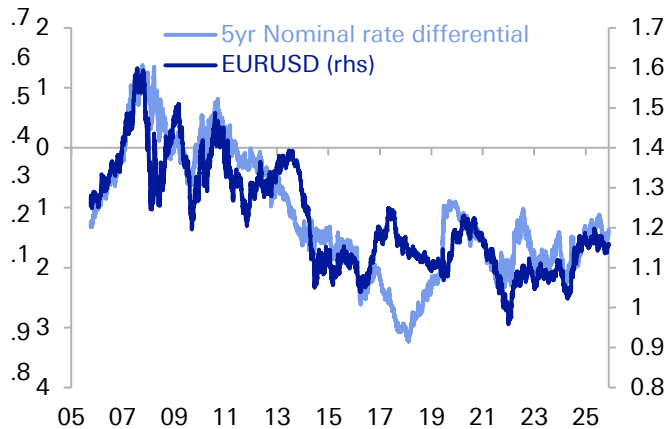
Relative interest rates remain among the most closely watched drivers of exchange rates among market participants, but the distinction between nominal and real differentials is critical:

- Nominal interest-rate differentials determine the carry earned on an unhedged currency position and therefore matter directly for short-run portfolio allocation; in practice, currencies often respond most strongly to revisions in the expected path of policy rates rather than to the current policy rate itself.
- For medium-run valuation, however, the more relevant concept is the expected ex ante real interest-rate differential, the nominal rate adjusted for expected inflation, because it measures the relative return on purchasing power. A high nominal yield generated by tighter monetary policy and a higher real rate should, other things equal, support a currency, whereas the same nominal yield caused by high expected inflation may provide little support and can instead be associated with depreciation.

This distinction is central to Frankel's (1979) real-interest-differential model, although the empirical relationship is not stable. Meese and Rogoff (1988) find little robust connection between real exchange rates and real-rate differentials, while Baxter (1994) found a stronger relationship at business-cycle and lower frequencies. The practical conclusion is that nominal differentials matter most for carry and near-term policy pricing, while expected real rate differentials are the more meaningful, though still imperfect, guide to medium-run exchange-rate valuation.

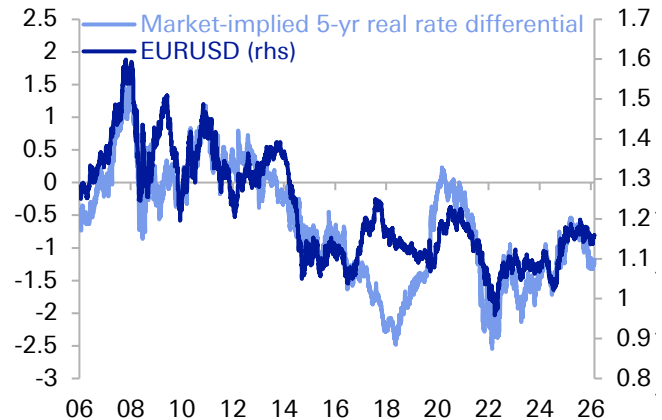


Figure 27: EUR/USD and the nominal interest rate differential



Source : Deutsche Bank, Bloomberg Finance LP

Figure 28: EUR/USD and the real interest rate differential



Source : Deutsche Bank, Bloomberg Finance LP

Current accounts, capital flows and the basic balance

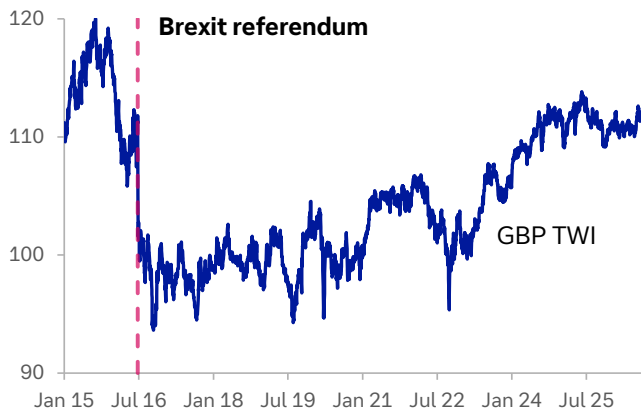
Current accounts provide one of the clearest medium-run anchors for exchange rates because they summarise the gap between national saving and domestic investment. Persistent current-account deficits imply that domestic consumption exceeds output and must be financed by net capital inflows, while persistent surpluses imply the opposite. In the medium run, those external positions cannot drift without consequence: a weaker net foreign asset position must eventually be stabilised through stronger future net exports, favourable valuation effects on the external portfolio, or both, and the exchange rate is one of the key relative prices through which that adjustment is delivered (Chinn and Prasad, 2003; Gourinchas and Rey, 2007).

The composition of capital flows matters as much as the headline balance. Portfolio equity and bond flows are typically more sensitive to relative returns, hedging demand and global risk appetite, so marginal reallocations in these categories can produce sizeable currency moves. FDI is usually the stickier component of external financing, reflecting the partial illiquidity of direct ownership. For FX, a deficit financed by long-horizon FDI is therefore very different from one financed by return-seeking portfolio debt or equity inflows that can reverse quickly when relative yields or risk sentiment change (Hau and Rey, 2006; Albuquerque, 2003; Gabaix and Maggiori, 2015; Greenwood et al., 2023).

Large changes in the trading regime can also shift the expected path of the external balance abruptly. Events such as Brexit, the imposition of tariffs or a major trade agreement can alter expected export demand, import costs, investment and the location of production, with potentially significant consequences for the equilibrium exchange rate. Importantly, currencies are forward-looking asset prices: they need not wait for a trade shock to appear in realised current account data. Once markets assign a sufficiently high probability to a future change in trade arrangements, the exchange rate can adjust immediately in anticipation of its expected effects on future trade flows, investment and national income. The sizeable adjustment in sterling around the 2016 Brexit referendum is a prominent example of the exchange rate moving well before the full effects of the new trading relationship became visible in the balance of payments data.



Figure 29: GBP moved well before the full effects of Brexit took place



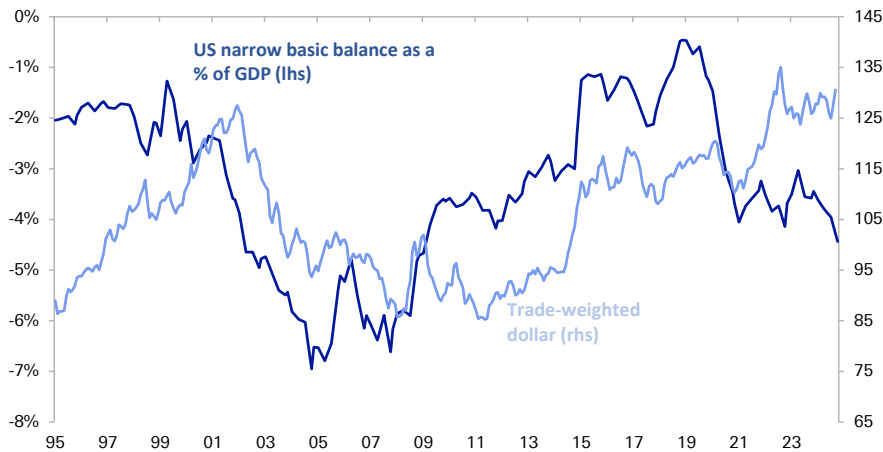
Source : BIS, Bloomberg Finance LP, Deutsche Bank

For medium-run FX analysis, the key issue is whether saving-investment trends and the composition of external financing are moving in a currency supportive or negative direction. Economies with chronically low saving relative to investment are more likely, over time, to require a weaker real exchange rate to generate the external adjustment needed to service a larger stock of foreign liabilities, whereas economies with excess saving and stronger external balance sheets can sustain firmer currencies for longer (Chinn and Prasad, 2003; Gourinchas and Rey, 2007; Lane and Milesi-Ferretti, 2004).

Practitioners often construct measures of the basic balance to assess shifts in external financing and the underlying demand for a currency. A narrow measure typically combines the current account balance with net foreign direct investment, on the grounds that these are among the slowest-moving components of the balance of payments. Broader measures may also include longer-term portfolio flows, particularly bond and equity investment. These indicators are useful but incomplete, because other financial flows can at times dominate exchange rate dynamics. This is especially true for currencies used extensively as funding currencies: during the 2000s, for example, large-scale borrowing in the low-yielding yen generated substantial capital outflows, while the unwinding of those positions periodically produced sharp yen appreciation.



Figure 30: Basic balance metrics are loosely correlated to currencies



Source : Deutsche Bank, Bloomberg Finance LP

The importance of hedge ratios as a medium-run FX driver

Alongside the current account and portfolio flow channels discussed above, the hedging decision on existing foreign asset stocks may become a first-order driver of exchange rates. The mechanism is a stock adjustment one: when a non-US pension fund or insurer holds unhedged US securities, a decision to raise its hedge ratio does not require selling the underlying bond or equity, but it does generate a flow of dollar sales to enact the forward hedge that is then rolled in perpetuity. Because the pool of cross-border foreign ownership of assets is very large, even modest shifts in the desired aggregate hedge ratios of foreign investors can dominate other flows in the balance of payments. The medium-run character of the channel comes from its persistence: hedge ratios are sticky, adjusted slowly and in cycles rather than day-to-day.

The incentive to hedge is governed largely by the cost of hedging, which for a rolling short-dated hedge would tend to be in the 3m to 1-year tenor and approximated by the front-end interest rate differential plus the cross-currency basis. A wide positive rate differential between the US and the rest of the world makes selling the dollar forward expensive, it subtracts directly from the hedged return and, when the hedged yield on a US bond turns negative, removes the entire rationale for holding the asset in hedged form. This is precisely why hedge ratios fell across the developed world as US front-end rates rose after 2022, leaving investors structurally under-hedged and dollar exposure near historical highs ([Gopal, Baker and Dynan, 2025](#)).

The relationship is not mechanical, however and after the US administration's trade actions in 2025 investors raised dollar hedges sharply in response to surprise shifts in US economic and foreign policy (Shin, Wooldridge and Xia, 2025 and [Saravolos, 2025](#)). The practical conclusion is that hedge ratio dynamics are best read as a joint function of the short-end rate differential (the cost) and broader shifts in underlying currency views ([Grover and Harvey, 2024](#)).

The yield curve and FX

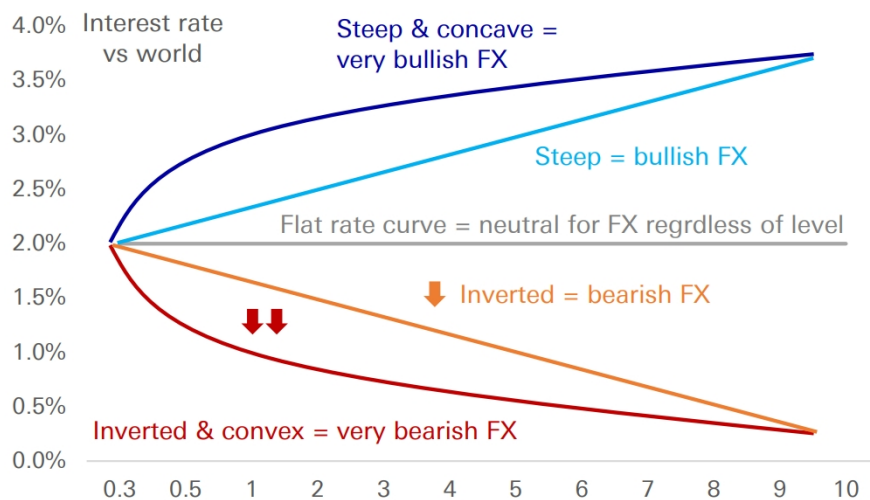
The shape of the yield curve carries independent information for the medium-run exchange rate, and does so through two complementary channels. The first is



predictive. Because the term structure embeds the market's expectations of future short rates, growth, inflation and policy risk (term premium), cross-country differences in the slope of the curve can help forecast currency returns at longer horizons. Ang and Chen (2010) show that term spreads and changes in interest rates significantly predict excess FX returns for up to twelve months, and that portfolios built on these signals carry low skewness and little correlation with the carry trade.

[Winkler \(2022\)](#) breaks down term structures into three curve factors: level, slope and curvature. He shows that the dollar depends more on the slope and curvature of the US rate curve than its level. A steep and hump-shaped rate curve relative to the rest of the world, implying a young business cycle and rising real rates, predicts a stronger dollar, and vice versa. The practical reading is that a relatively steep domestic curve is a signal of expected future easing or of a term premium demanded for macro risk, both of which are currency-negative over the medium run.

Figure 31: FX implications of the shape of the rate curve relative to the world



Source : Deutsche Bank

The second channel via which the yield curve affects FX is a flow mechanism that connects directly to the hedging discussion above. Nenova, Schrimpf and Shin (2025) show that the relative slope of yield curves governs where hedged bond capital is allocated: a steeper curve draws foreign bond inflows and, with them, a rise in FX swap hedging volumes because hedging is cheap. Flatter curves reduce the attractiveness of FX hedging encouraging more un-hedged inflow.



2.8 Short-run determination I: Order flow and market microstructure

Order flow and market microstructure

Because macro fundamentals are near-useless at high frequency, the short-run behaviour of exchange rates is the domain of market microstructure, the study of how the trading process itself conveys information and moves prices. The initial insight, due to Evans and Lyons (2002), is that order flow (signed transaction volume — the net of buyer-initiated and seller-initiated trades) is the proximate driver of exchange rates and the channel through which dispersed information about fundamentals, risk appetite and hedging demand is aggregated into price. Order flow explains a large share of daily and intraday exchange-rate variation, more than macro news alone.

The most important modern contribution is Menkhoff, Sarno, Schmeling and Schrimpf (2016), where using a large dealer's disaggregated customer order flow, they show that order flow is highly informative about future exchange rates. Crucially, different customer segments (asset managers, hedge funds, corporates, private clients) differ markedly in their trading styles, information content and risk exposures. Financial and non-financial customers tend to trade in opposite directions, so the dealers intermediate risk sharing between them, and the information in financial-customer flow has a persistent, longer-term impact on returns consistent with it carrying fundamental information.

DB house research: CORAX and the anatomy of FX flows

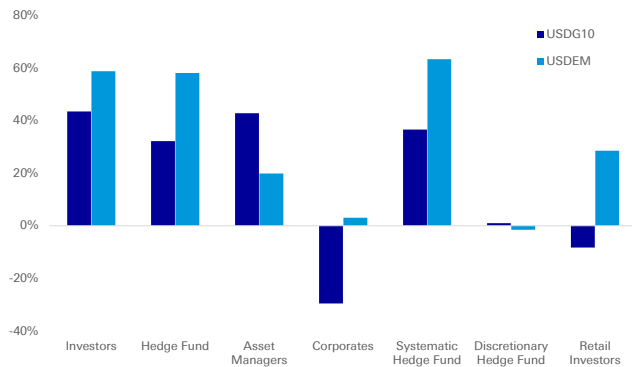
Deutsche Bank's [CORAX](#) indicators (Categorised ORderflow from Autobahn FX), derived from the bank's globally aggregated electronic spot flow, allow this literature to be applied directly. [Grover \(2021\)](#), building explicitly on Menkhoff et al. (2016), studies discretionary versus systematic hedge-fund flows across G10 and liquid EM from January 2016. The findings largely mirror the results of the academic literature:

- Both discretionary and systematic hedge-fund flows are positively correlated with contemporaneous spot, the relationship being stronger for systematic funds.
- Looking forward, discretionary investor flows in G10 positively predict spot for up to about ten days, turning negative at longer horizons — while systematic flows provide more of a mean reversion signal. In EM, discretionary flows show strong mean reversion.
- Across the broader client taxonomy, real-money flows are positively correlated with contemporaneous and one-day-ahead spot; corporate flows are negatively correlated contemporaneously but positively related to future spot in the near-to-medium term in G10; and retail flows are negatively correlated (contrarian in the near term, becoming trend-following at the medium term).

Flows are powerful contemporaneous and short-horizon indicators, and the sign of their predictive content depends on which client segment is trading and over what horizon.

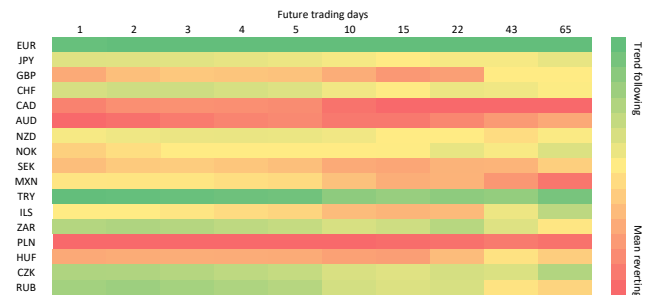


Figure 32: Systematic HF flows show a higher degree of contemporaneous correlation than other client categories



Source : Deutsche Bank

Figure 33: Correlation between CORAX systematic HF flows and future spot move



Source : Deutsche Bank

2.9 Short-run determination II: Positioning, options flow, sentiment, data releases and technical analysis

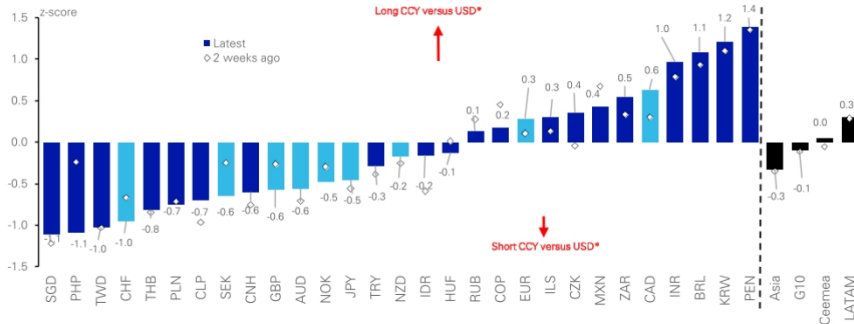
Positioning

Because currencies can overshoot and trend, measures of market positioning are valuable, particularly as contrarian signals at extremes. Positioning refers to the stock of existing long and short currency exposures held by investors: a market is described as “long” a currency when investors have, in aggregate, built positions that benefit from its appreciation, and “short” when they are positioned for depreciation. This differs from flows, which measure new buying and selling over a period; positioning instead captures the cumulative result of past flows and therefore indicates how much directional risk is already embedded in investor portfolios.

Positioning matters because the impact of new information depends partly on how investors are already positioned. When a currency has been rising for some time, investors may progressively add to long positions, reinforcing the trend. But once positioning becomes crowded, fewer investors remain to initiate additional longs, while a negative surprise can force existing holders to unwind positions simultaneously. Stop-losses, margin constraints and systematic trading strategies can amplify this process, turning an initially modest reversal into a sharp move. Conversely, an extremely short currency can rally strongly even on modestly positive news as investors rush to cover their positions. Positioning is therefore most useful not as a mechanical timing indicator, but as a measure of market vulnerability and asymmetry: stretched positioning raises the probability that an adverse surprise generates a disproportionately large price response.



Figure 34: DB EPIC+ positioning (as of 01-Sep-2026)



Source : Deutsche Bank, Bloomberg Finance LP, DTCC, EBS, Reuters, IMM, CORAX Note: 1) * Long (or short) versus EUR for CZK, HUF, PLN, CHF, NOK and SEK. 2) The black bars on the right-hand side are average positioning levels for G10, Asia, CEEMEA and LatAm. 3) The y-axis shows readings of positioning index in a z-score format, which is an average of rolling one year z-scores for individual sub-indicators. Thus, the long or short of each currency is relative to its one year's average instead of absolute level.

The classic public gauges are the CFTC/IMM speculative futures positions, which provide weekly data on long and short positions held by different investor categories, and dealer or investor sentiment surveys. These measures capture only parts of the global FX market, however, and practitioners therefore supplement them with other metrics.

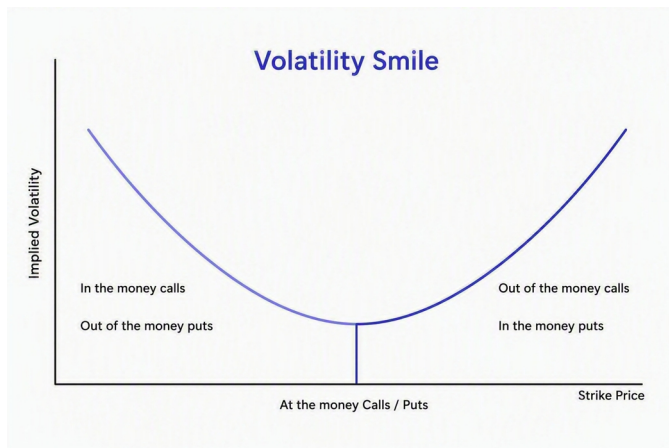
Deutsche Bank's [EPIC+ framework](#) aggregates positioning across both direct measures (CFTC/IMM) and indirect ones (CORAX & spot flows, inter-dealer order-book data, risk reversals, and the DTCC-derived Skew Volume/COFFEE options indicators). [Grover, Nandy, Charles and Xu \(2024\)](#) evaluate the live, out-of-sample performance of position-fading (contrarian) strategies since the EPIC+ launch in January 2021 and find them still profitable for most pairs over the medium term: an equal-weighted position-fading portfolio delivered a Sharpe ratio of 0.54 after costs (0.75 excluding EUR, CNH and THB, which are better traded as position-following), while a cross-sectional long-short portfolio that fades excessive positioning delivered a Sharpe ratio of 1.2 after costs (against 1.7 in backtest). Importantly, performance dips when FX volatility is high and the dollar is trending strongly — contrarian signals work best in less-trending markets. Risk reversals, CORAX and IMM are the most effective inputs for spotting fading opportunities in DM; the COFFEE options-skew indicator and spot imbalance work better in EM.

Options market information

Given the size and depth of the currency option market, option prices can contain information about market expectations and risk sentiment that is not fully visible in spot or forward rates. Implied volatility measures the market price of protection against exchange rate movements, while the volatility smile provides information about the perceived balance of tail risks.



Figure 35: Simplistic depiction of a volatility smile



Source : Investopedia, Deutsche Bank

The most widely followed measure of this asymmetry is the risk reversal, usually calculated as the implied volatility of an out-of-the-money currency call minus that of an equivalent-delta currency put. A positive risk reversal means that calls on the base currency are more expensive than comparable puts, indicating greater demand for protection against, or exposure to, base-currency appreciation. Risk reversals should not, however, be interpreted as pure forecasts: they reflect the market's risk-neutral distribution and therefore combine expected outcomes with risk premia, hedging demand and position imbalances. More generally, option prices can be used to recover information about the market-implied distribution of future exchange rates, including its volatility, skewness and tail probabilities (Campa, Chang and Reider, 1998).

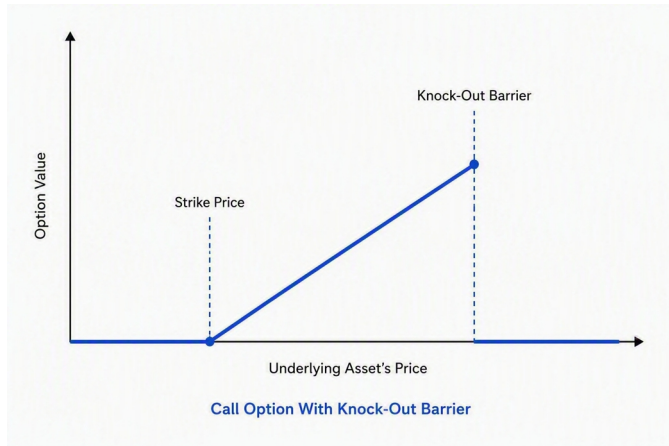
Options can also affect the exchange rate mechanically through the hedging activity they generate. Dealers typically manage the delta of their option books (the change to the value of their positions driven by spot exchange-rate movements) by buying or selling the underlying currency as spot moves. The direction of these flows depends importantly on the market's aggregate gamma, which measures how rapidly an option portfolio's delta changes with the exchange rate. Dealers who are long gamma generally sell the currency after it rises and buy it after it falls, creating countercyclical flows that can dampen volatility and sometimes pull spot towards large strikes around expiry. Dealers who are short gamma must instead buy as the currency rises and sell as it falls, producing procyclical flows that can amplify an existing move. Empirical evidence using trade-repository data confirms that negative market-maker gamma can increase spot-market volatility, while positive gamma has the opposite effect (Anderegg, Ulmann and Sornette, 2022).

These effects can become especially pronounced around barrier options, which are activated or extinguished when the exchange rate reaches a predetermined level. As spot approaches a barrier, the option's delta and gamma may change sharply, requiring increasingly large hedge adjustments. Investors with an interest in preventing the barrier from being triggered may also trade defensively around the level. If the barrier is ultimately breached, the associated option exposure and hedges can disappear or change abruptly, generating a sudden reversal of previous hedging flows or accelerating the move through the level. The precise direction depends on whether the option is a knock-in or knock-out, whether it is a call or put,



and which side of the transaction dealers hold. Consequently, large strikes, expiries and barriers do not provide deterministic support or resistance, but they can materially alter the short-run elasticity of market flows and help explain why exchange-rates occasionally become unusually stable near particular levels.

Figure 36: Simplistic depiction of a knock-out call



Source : Financialtrading.com Deutsche Bank

Weng and Grover (2017) introduce Deutsche Bank's [Skew Volume Indicators](#), built from DTCC data (the DTCC registers almost all US currency-option trades, over a fifth of global derivative volumes). The key innovation is separating option transactions into investors (at-the-money options, 25-75 delta, the SVA indicator), versus hedgers (out-of-the-money options, 0-25 delta, the SVO indicator). The findings are tradeable: investor flows positively lead currency moves (it pays to follow investors), whereas hedging flows negatively lead returns (it pays to fade hedgers). A momentum strategy on SVA earned a Sharpe ratio of 0.83, a contrarian strategy on SVO a Sharpe of 0.67, and a blended strategy 0.96 in their sample.

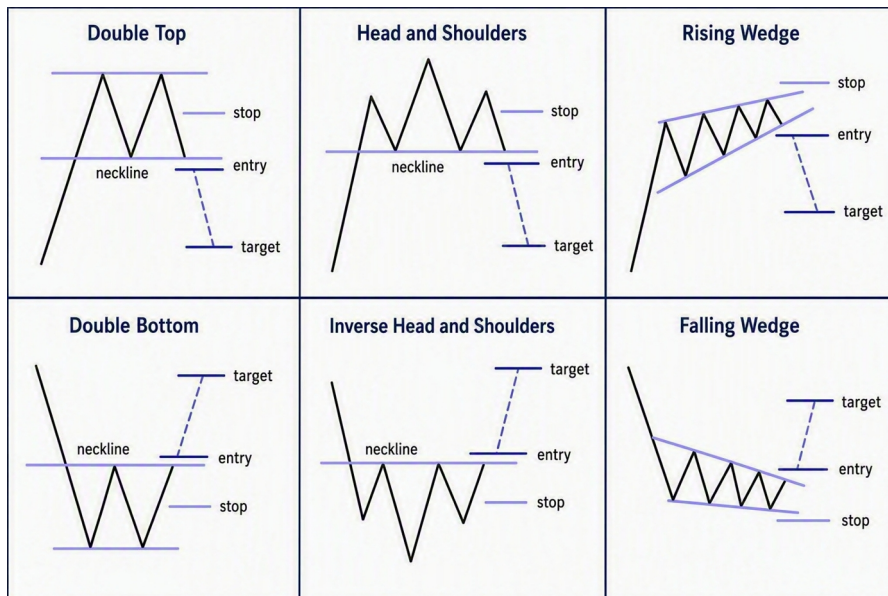
Technical analysis and trend-following

Technical analysis refers to forecasting future price movements from patterns in past market data rather than from an explicit model of economic fundamentals. In FX, this typically means analysing the history of exchange rates and sometimes volume, volatility or positioning to identify trends, momentum, turning points and important price levels.

Common techniques include moving averages, which generate signals when a short-term price trend crosses a longer-term one; momentum and oscillator indicators, which attempt to identify overbought or oversold conditions; and support and resistance levels, where previous highs, lows or other widely watched prices are expected to influence future trading behaviour. More sophisticated systematic versions of technical analysis translate these ideas into mechanical trend-following or breakout rules.



Figure 37: Examples of technical analysis



Source : Aimarow.com, Deutsche Bank

The underlying premise is not necessarily that past prices contain fundamental information, but that investor behaviour, herding, stop-loss activity and institutional trading practices can generate persistence or recurring patterns in prices. Technical analysis is pervasive in FX markets, with surveys finding widespread use among FX dealers and professional fund managers, associated with the belief that psychological and non-fundamental factors can influence prices (Gehrig and Menkhoff, 2006; Menkhoff, 2010).

There is also a substantial literature showing that simple technical trading rules have historically generated statistically and economically significant returns in FX. Levich and Thomas (1993), using bootstrap tests on currency futures, find that moving-average and filter rules produced profits that were difficult to reconcile with simple random exchange-rate processes. Neely, Weller and Dittmar (1997) similarly find significant out-of-sample excess returns across several major exchange rates using rules selected by genetic-programming techniques. Menkhoff and Taylor (2007) survey this literature and evaluate four broad explanations for the use of technical analysis: irrational behaviour, exploitation of central bank intervention, efficient processing of dispersed information, and the ability of technical signals to capture non-fundamental influences on exchange rates. They regard the final explanation as the most persuasive. In this interpretation, technical analysis is useful because FX prices can be driven for extended periods by order flow, positioning, herding and other market dynamics that are not immediately captured by standard macroeconomic models.

News, Announcements and High-Frequency Dynamics

A distinct microstructure literature studies how exchange rates react to scheduled macroeconomic news at high frequency. The work of Andersen, Bollerslev, Diebold and Vega (2003) showed, using real-time data, that exchange rates respond within seconds to the surprise component of macro announcements (relative to survey



expectations) and that announcement surprises explain a meaningful share of high-frequency volatility. This body of work reconciles the "fundamentals don't matter" result with intuition: fundamentals do move exchange rates, but they do so in sharp, news-driven jumps around data releases and central-bank communications, which are hard to see in monthly data. It also underpins the modern reality that the largest intraday FX moves cluster around US payrolls, CPI and FOMC decisions.

Risk sentiment and impact on FX

Risk sentiment is an important cross-cutting driver of exchange rates because it changes investors' willingness and capacity to hold leveraged or risky currency positions. During periods of strong risk appetite and low volatility, investors tend to borrow in low-yielding funding currencies and invest in higher-yielding currencies, supporting carry-trade returns. When volatility rises or market liquidity deteriorates, however, losses, tighter funding constraints and margin calls can trigger a simultaneous unwinding of these positions: high-yielding and relatively illiquid currencies are sold, while funding currencies are repurchased, amplifying the initial exchange-rate move.

Brunnermeier, Nagel and Pedersen (2009) document the resulting negative skewness and crash risk of carry trades, while Menkhoff, Sarno, Schmeling and Schrimpf (2012) show that high-yield currencies perform poorly following increases in global FX volatility. Risk appetite is generally perceived to be negatively correlated with the broad US dollar: the dollar often strengthens when global risk sentiment deteriorates, reflecting its central role in international funding, liquidity and safe-asset demand. The relationship between the dollar and risk appetite is however not stable over time, with shocks that disproportionately impact the US such as the bursting of the 2000 dot-com bubble often associated with a weaker dollar.

2.10 Forecasting performance and the Meese-Rogoff Puzzle

Meese and Rogoff (1983), showed that the structural exchange-rate models of the day, monetary, portfolio-balance, could not out-forecast a naive random walk out of sample, even when fed the realised (future) values of the fundamentals. Four decades on, the Meese-Rogoff puzzle remains the benchmark against which every forecasting claim is measured, and the random walk without drift has proven extraordinarily hard to beat at horizons under a year.

One of the most useful modern surveys on exchange-rate forecasting is Rossi (2013, *Journal of Economic Literature*). Her review of the post-1983 literature reaches a mixed verdict: predictability does exist, but it is fragile and conditional. Parameter instability is one of the biggest challenges to systematic exchange-rate modelling: it is not only that fundamentals change over time, but the relationship between these and exchange rates changes too.



2.11 Modern theories of the exchange rate

Exchange rates dominated by financial markets

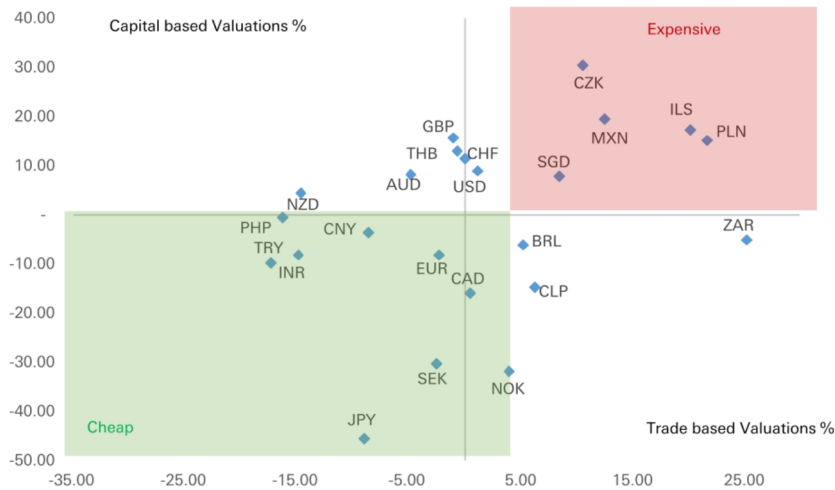
With financial markets outpacing the real economy in size and growth, Gabaix and Maggiori (2015) provide a theory in which exchange rates are determined by capital flows intermediated through imperfect financial markets. Gabaix and Maggiori introduce financial intermediaries (primarily banks) with limited risk-bearing capacity into their model. When, for example, investors in one country increase their demand for foreign assets, intermediaries must take the opposite currency exposure to clear the market. As their balance sheets become more exposed, they require greater compensation for bearing that risk, and the exchange rate moves sufficiently to induce them to absorb the imbalance. The larger the required flow relative to intermediaries' risk-bearing capacity, the larger the resulting exchange-rate adjustment. This provides an important explanation for why exchange rates can move substantially even when trade flows or conventional macroeconomic fundamentals change little. Shifts in international portfolio preferences, financial regulation, intermediary balance sheet capacity or global risk appetite can all generate exchange-rate movements by changing the willingness of financial institutions to intermediate the resulting flows.

The framework is therefore particularly useful for understanding episodes such as the 2008 global financial crisis, when large exchange-rate movements were associated with deleveraging, balance-sheet constraints cross-border portfolio adjustment rather than simply changes in goods trade. A closely related perspective is provided by Rey (2015), who argues that the US dollar sits at the core of the global financial cycle. The large pool of dollars sitting outside the US banking system means easier US financial conditions are associated with expanding cross-border leverage and capital flows and, typically, a weaker dollar. Conversely, tighter financial conditions and rising global risk aversion trigger deleveraging, repatriation and demand for dollar liquidity, tending to strengthen the dollar. The dollar therefore acts not simply as the relative price of US and foreign goods, but as a key price through which changes in global financial conditions and risk-bearing capacity are transmitted internationally.

Conventional PPP and real effective exchange rate measures weight currencies according to bilateral trade, implicitly reflecting the importance of different countries in goods and services markets. This financial-market perspective is the motivation behind the Deutsche Bank's [Capital-weighted PPP model](#). A capital-weighted approach instead gives greater weight to currencies against which a country has large cross-border financial exposures, recognising that the relevant counterparties for exchange rate determination may increasingly be determined by the geography of portfolios rather than trade. Our capital-based valuation measures therefore construct a real effective financial exchange rate using capital weights and compare its level with a historical equilibrium in much the same way that conventional PPP compares a trade-weighted real exchange rate with its long-run norm. For countries with very large external asset and liability positions, this can yield materially different measures of currency valuation from traditional trade-weighted PPP and provides a useful complement in a world where gross capital flows substantially exceed the flows associated with international trade.



Figure 38: Scatter plot of current capital and trade-based valuations



Source : Haver Analytics, Deutsche Bank

Convenience yields and safe-asset demand for dollars

A complementary financial mechanism, specific to the dollar, is the convenience yield on safe dollar assets. Jiang, Krishnamurthy and Lustig (2021) develop a theory in which the dollar's value reflects the convenience (safety and liquidity) services that global investors derive from holding US Treasuries. This convenience yield is measurable via the Treasury basis (the yield gap between US government bonds and currency-hedged foreign government bonds); when it widens, the dollar appreciates immediately and then depreciates subsequently. In their words, "the variation in the Treasury basis accounts for 25 percent of the quarterly variation in the dollar between 1988 and 2017," and "on average, foreign investors earn an extra convenience yield of 2% on Treasury holdings," 90% of which is directly attributable to the dollar denomination.

Covered Interest Parity Deviations and the Cross-Currency Basis

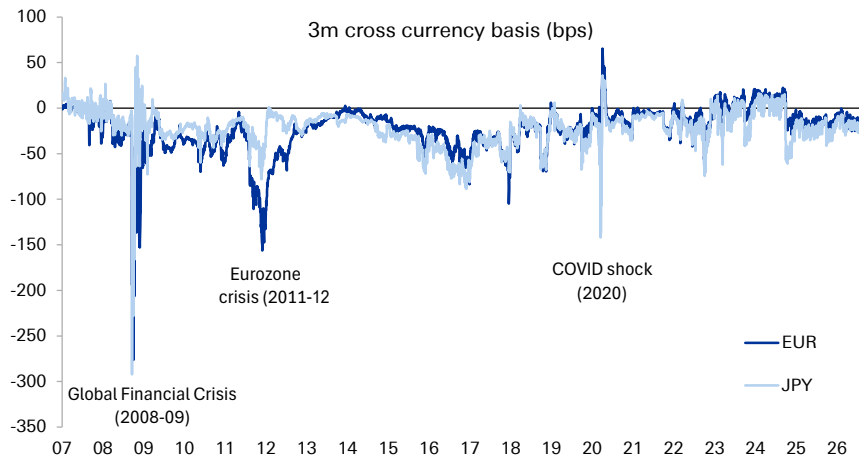
Among the most consequential post-crisis developments is the breakdown of covered interest parity. CIP is a pure no-arbitrage identity and held almost perfectly before 2008. Since the crisis, persistent, systematic deviations have opened up, measured by the cross-currency basis: the extra cost, over and above the interest differential, of borrowing dollars synthetically through the FX swap market. Du, Tepper and Verdelhan (2018) document that these deviations are large, persistent and systematic, and are not explained by credit risk or transaction costs alone. They are strongest for contracts that sit on bank balance sheets at quarter-ends, pointing to post-crisis banking regulation (leverage ratios, risk-weighted capital) as the cause: the balance-sheet cost of arbitrage means dealers no longer close the gap. The basis is typically negative for currencies whose investors demand dollar hedges (JPY, EUR), reflecting a structural shortage of dollar funding. See more [here](#).

The macro-financial significance was crystallised by Avdjiev, Du, Koch and Shin (2019), who document a "triangular" relationship: a stronger broad dollar goes hand-in-hand with a higher cost of dollar funding and a contraction in cross-border dollar bank lending. This links the cross-currency basis (a microstructure/funding



variable) directly to the broad-dollar cycle and global credit conditions.

Figure 39: Cross currency basis for EUR and JPY



Source : Deutsche Bank calculations, Bloomberg Finance LP

The Dominant Currency Paradigm and Invoicing

The Dominant Currency Paradigm (DCP) of Gopinath, Boz, Casas, Díez, Gourinchas and Plagborg-Møller (2020) and alters the textbook framework in which prices are set either in the producer's currency or the destination's. Using a dataset of more than 2,500 country pairs covering 91% of world trade, they show that the vast majority of world trade is invoiced in a small number of dominant currencies, overwhelmingly the US dollar, and that those invoice prices are sticky in dollars ([Sachdeva, 2026](#)).

This has important implications: exchange rate movements against the dollar, rather than bilateral rates, drive trade prices and quantities and expenditure switching operates weakly and asymmetrically. The dollar therefore exerts a systemic influence on global trade and inflation that no other currency matches, reinforcing the demand for dollar invoicing, dollar funding and dollar reserves in a self-sustaining equilibrium. Combined with the safe-asset/convenience-yield channel and the funding/basis channel, this literature completes the picture of dollar dominance as a structural feature of the international monetary system.

2.12 Central bank intervention

Official intervention in the FX market, the purchase or sale of foreign currency by the central bank, is a long-standing but contested policy tool. The theoretical channels are the portfolio-balance channel (intervention changes the relative supply of assets), the signalling channel (intervention reveals information about future policy) and, more recently, an order-flow/microstructure channel (intervention



affects price like any large order). Sterilised intervention¹⁴ by advanced economies was long viewed sceptically. The best modern evidence is more encouraging. Fratzscher, Gloede, Menkhoff, Sarno and Stöhr (2019) using a novel daily dataset covering 33 countries from 1995–2011, find that "intervention is widely used and an effective policy tool, with a success rate in excess of 80 percent under some criteria." It works best at smoothing the exchange-rate path and defending narrow bands; moving the level of a floating currency requires that interventions be large, public and supported by communication.

Menkhoff (2013) surveys the specific and generally more favourable evidence for emerging markets, where intervention and reserve management are pervasive. Recent episodes illustrate the spectrum: the Swiss National Bank's EUR/CHF floor (2011–15) and its abrupt abandonment in January 2015; large-scale Japanese intervention to slow yen depreciation in 2022 and 2024; and routine EM intervention. Intervention is most effective when it works with fundamentals and market positioning rather than against them.

2.13 Currency regimes and emerging market crises

The distinction between fixed and floating exchange rates is in practice a continuum. The IMF's de facto classification distinguishes ten regimes according to the strength of the authorities' commitment to an exchange-rate level or path, the observed behaviour of the currency and the extent of official intervention. At one extreme, dollarisation and currency boards largely eliminate monetary autonomy; at the other, freely floating currencies are almost entirely market determined. Between them sit a range of pegs, bands, crawling regimes and managed arrangements that trade greater exchange-rate stability for progressively less monetary-policy independence.

¹⁴ Intervention by a central bank to buy or sell a currency changes the currency's money supply. A sterilization involves an offsetting transaction by the central bank in the domestic money market to remove or add back the liquidity.



Figure 40: Exchange rate regimes according to the IMF

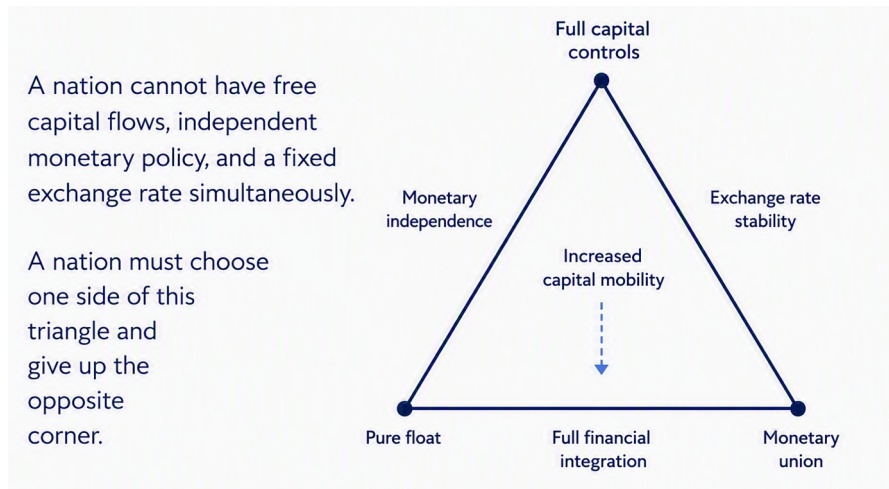
Broad FX regime	IMF category	What it means	Main IMF classification criteria
Hard peg	No separate legal tender	The country uses another country's currency as its legal tender—for example, through formal dollarisation—so there is no independent domestic exchange rate.	Authorities' de jure arrangement is confirmed; another country's currency is the sole legal tender. This effectively eliminates independent national monetary policy.
Hard peg	Currency board	Domestic currency exists, but its value is rigidly fixed to an anchor currency and its issuance is constrained by the availability of foreign reserves.	There is an explicit legal commitment to exchange domestic currency for a specified foreign currency at a fixed rate. Domestic currency is normally backed by foreign assets and the issuing authority faces tight restrictions on discretionary monetary policy.
Soft peg	Conventional pegged arrangement	The currency is formally pegged to another currency or a basket, but unlike a currency board the peg can ultimately be changed and the central bank retains more policy flexibility.	Authorities formally announce or notify the IMF of the peg and stand ready to defend it through direct or indirect intervention. The rate must generally stay within $\pm 1\%$ of a central parity, or within a total 2% range, for at least six months.
Soft peg	Stabilized arrangement	A de facto peg: the authorities may not formally announce a peg, but the exchange rate behaves as though one exists.	The spot rate remains within a 2% margin for six months or more relative to an identifiable currency or basket, and the stability is judged to result from official action or structural market rigidities. Unlike a conventional peg, there need not be an announced commitment.
Soft / intermediate	Crawling peg	Instead of fixing the currency permanently at one level, the authorities periodically adjust the central rate in small steps.	A formally announced rule determines the crawl—for example a fixed depreciation rate or adjustments based on inflation differentials or other indicators. The rule and its parameters are public or notified to the IMF.
Soft / intermediate	Crawl-like arrangement	A de facto crawling peg: the currency follows a relatively smooth trend even though the authorities have not formally committed to a crawling rule.	The rate remains within a 2% margin around a statistically identified trend for at least six months and cannot qualify as floating. Generally the trend must be sufficiently persistent; an annualised movement of at least 1% can qualify if appreciation or depreciation is sufficiently monotonic and continuous.
Soft / intermediate	Pegged within horizontal bands	The authorities commit to keeping the exchange rate inside a band around a central parity rather than at one precise level. The band gives monetary policy somewhat more room than a conventional peg.	The central rate and band are formally announced or notified to the IMF. The permitted fluctuation is at least $\pm 1\%$ around the central rate, or the distance between the maximum and minimum permitted exchange rates exceeds 2%.
Floating	Floating	The exchange rate is primarily determined by market supply and demand. The central bank can still intervene to dampen volatility, but does not defend a particular exchange-rate level.	The rate is largely market determined and has no ascertainable or predictable path. Intervention may moderate excessive movements, but targeting a particular exchange-rate level is inconsistent with the classification.
Floating	Free floating	The strongest form of floating: the market determines the exchange rate and official intervention is very rare.	Must first meet the criteria for floating. Intervention should occur only exceptionally to address disorderly markets; under the IMF methodology it is limited to no more than three episodes in the previous six months, each lasting no more than three business days.
Residual / managed	Other managed arrangement	The authorities materially influence the exchange rate, but its behaviour does not fit any of the IMF's peg, crawl or floating definitions. This can include regimes where the intervention strategy or exchange-rate objective changes over time.	Residual category used when none of the other definitions is satisfied; regimes characterised by frequent changes in exchange-rate policy can fall here.

Source : Deutsche Bank, IMF



The exchange-rate regime shapes how shocks are absorbed and how vulnerable a currency is to crisis. The "Impossible Trinity" (fixed rate, free capital mobility, monetary autonomy: choose two) remains the organising principle.

Figure 41: The Impossible Trinity



Source : Feridun, M. (2014), Deutsche Bank

The literature on early warning systems for emerging-market currency crises identifies recurrent symptoms: real overvaluation, weak growth, adverse terms-of-trade shocks, deteriorating current accounts, excessive domestic credit expansion, banking-system fragility, large fiscal deficits, high ratios of short-term external liabilities (and of M2) to reserves, and reserve losses (Frankel and Saravelos, 2012). Successive generations of crisis models – first-generation (fiscal/monetary fundamentals), second-generation (self-fulfilling expectations and multiple equilibria), and third-generation (balance-sheet and banking/currency-mismatch channels, motivated by the 1997–98 Asian crisis) — map onto these symptoms.

2.14 Frontier topics: machine learning and digital currencies

The application of machine learning (ML), random forests, gradient boosting, neural networks, and now large language models for news processing, to exchange-rate forecasting has grown significantly in recent years. Several studies report ML models beating the random walk in directional (sign) forecasting and when many predictors and time effects are combined, but the improvements are typically small and concentrated in specific regimes.

Rigorous recent work continues to find that ML models frequently fail to beat a random walk on error-magnitude (RMSE) criteria under strict out-of-sample, expanding-window evaluation — the Meese-Rogoff result proves robust even to modern methods (Pfahler, 2022 and Jackson & Magkonis (2024). ML is most promising for (i) directional and volatility forecasting, (ii) exploiting high-frequency and alternative data (order flow, news, positioning), and (iii) nonlinear combination of the fundamentals and factors described in this section rather than as a magic



solution to level forecasting.

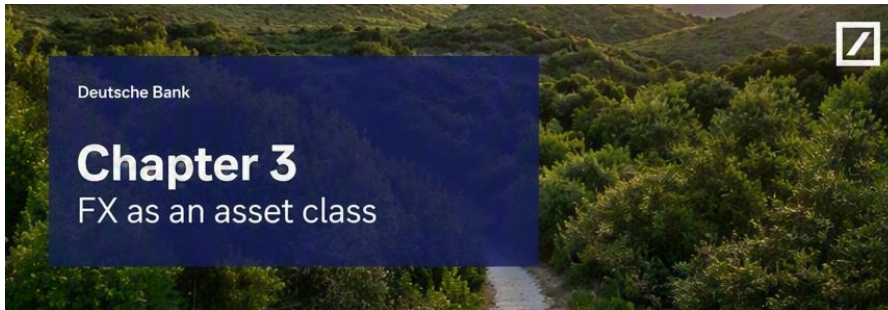
2.15 Cryptocurrencies, stablecoins and CBDCs

Digital assets are increasingly relevant to FX. Dollar-pegged stablecoins are emerging as a potentially meaningful conduit for dollar demand, particularly in emerging markets with weak currencies or capital controls, effectively extending dollar dominance. A recent Deutsche Bank [study](#) has expanded into this topic in more detail (Sachdeva and Saravelos, 2025). Central-bank digital currencies (CBDCs) and cross-border CBDC projects could, over time, alter FX-market plumbing, settlement and the mechanics of intervention, though their macro impact on exchange-rate determination remains speculative as of 2026. Cryptocurrencies themselves remain too volatile and too small to function as genuine exchange rate anchors, but the infrastructure they have spawned — 24/7 settlement, tokenised deposits — is beginning to matter for market microstructure.

2.16 Bringing it all together

No single model of exchange rate determination is right all the time; the challenge lies in matching the tool to the horizon and the regime. The evidence assembled here supports a staged approach to exchange rate determination.

- First, *anchor exchange rates with valuation (strategic, 3 years+)*. Use PPP, BEER and FEER to establish a fair-value range and the direction of long-run misalignment. Prefer PPP for G10 and DBBeer for EM.
- Second, *align FX forecasting with the economic cycle (cyclical, 1–2 years)*. Use interest rate differentials, both level (carry) and change, the monetary policy cycle and Taylor rule fundamentals for medium-run direction. Take account of shifts in the balance of payments, both on the current account and the capital flow side of the ledger.
- Third, *flow and positioning matter (tactical overlay)*. Use order flow, options flow and positioning as high-frequency short-horizon mean-reversion signals.
- Finally, be aware of regime changes. This can take the form of a change in exchange rate regime from a peg to a float, an underlying abrupt shift in macroeconomic fundamentals (Brexit) or a large financial shock that shifts the market to a new equilibrium (2008 Financial Crisis or Eurozone crisis).



Introduction

Currency investing starts from a different premise to equities or bonds: a currency has no stand-alone expected return and every exposure is inherently relative, involving a long position in one currency and a short position in another, typically implemented through spot, forwards or options. Do investors make money from trading FX? Can FX be treated as an investable asset class? Our first chapter demonstrated the importance of non-profit-seeking participants in the FX market. This weakens the application of the efficient-market hypothesis and creates scope for excess returns. Our second chapter introduced the exchange-rate frameworks that can inform an investment approach to currencies. This chapter turns from theory to real-world evidence. We examine whether systematic currency factors such as carry, momentum and value have generated persistent returns, whether quantitative investors can capture them after costs and whether discretionary currency managers generate alpha beyond these systematic exposures.

3.1 FX "beta" and factor investing

The poor out-of-sample performance of models designed to forecast exchange rate levels, together with the persistent empirical rejection of uncovered interest parity (UIP) has generated a large investor ecosystem focused on harvesting currency risk premia via systematic strategies and indices across baskets of currencies rather than individual exchange rates. Three investment-style factors have become canonical:

- *Carry* involves buying high-interest-rate currencies and funding the position by selling low-interest-rate currencies. Its profitability is the investment counterpart of the forward-premium puzzle: high-yielding currencies have not, on average, depreciated sufficiently to eliminate their interest-rate advantage. Carry returns are often interpreted as compensation for exposure to global risk, since high-yielding currencies



tend to underperform during episodes of rising volatility, deteriorating liquidity and widespread deleveraging (Lustig, Roussanov and Verdelhan, 2011; Menkhoff et al., 2012a).

- *Momentum* involves buying currencies with strong recent excess returns—the “winners”—and selling recent “losers.” Menkhoff, Sarno, Schmeling and Schrimpf (2012b) document substantial cross-sectional momentum in currencies. Proposed explanations include the gradual incorporation of information into exchange rates, investor underreaction and overreaction, and limits to arbitrage, although transaction costs reduce the returns available to implementable strategies.
- *Value*, sometimes described as a valuation or mean-reversion strategy, involves buying currencies judged to be undervalued and selling those judged to be overvalued relative to an equilibrium benchmark such as purchasing-power parity or a behavioural equilibrium exchange rate model. Value signals generally operate over longer horizons than carry or momentum, reflecting the slow tendency of real exchange rates to revert towards equilibrium. Evidence for currency value strategies forms part of the broader finding that value and momentum premia appear across several asset classes (Asness, Moskowitz and Pedersen, 2013). Practitioner implementations include Deutsche Bank’s PPP and DBeer frameworks ([Grover, 2025](#)).

One of the earliest practical attempts to package these systematic returns into a currency “beta” equivalent to the S&P 500 for equities was the Deutsche Bank Currency Returns (DBCR) Index, introduced in 2007. The index equally weights transparent, rules-based carry, momentum and valuation strategies across liquid developed-market currencies and was designed to be investable after transaction costs. In its original backtest over 1980–2006, DBCR generated an annualised excess return of approximately 4 per cent, volatility of 5.2 per cent and a Sharpe ratio of 0.77, with a maximum drawdown of 11 per cent. Its relatively low correlations with global bonds and equities illustrated the potential diversification benefits of combining systematic currency exposure with traditional asset classes (Hafeez, 2007).

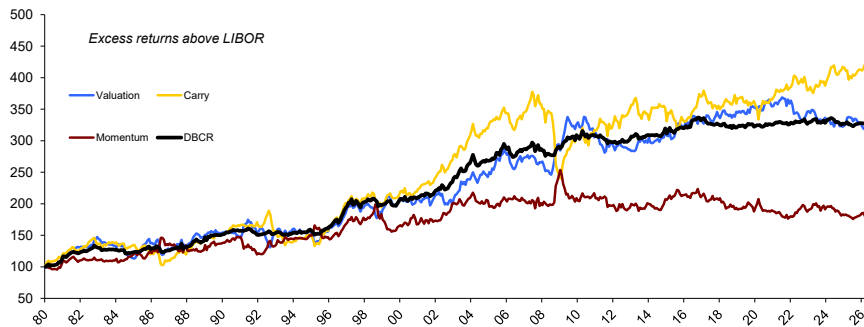
A subsequent reassessment by [Saravolos et al. \(2018\)](#) found that the risk-adjusted returns of the combined carry, momentum and value approach had remained resilient after the global financial crisis. Over the longer sample beginning in the 1980s, systematic currency returns were found to be comparable on a risk-adjusted basis with those of global bond and equity portfolios, while adding FX exposure to a conventional portfolio reduced historical volatility and drawdowns.

These benefits arise partly because the component factors perform well in different environments: momentum can offset some of the losses suffered by carry and value during sharp regime changes, while valuation may benefit from the eventual reversal of prolonged trends.

Factor returns nevertheless remain cyclical and can be exposed to crash risk, crowding, changing monetary regimes and implementation costs; they should therefore be understood as compensation for bearing a collection of risks rather than as mechanical arbitrage profits.



Figure 42: DBCR returns vs risk-free-rate



Source : Deutsche Bank

3.2 Systematic and quant-based investors

Systematic currency and macro managers construct portfolios using predefined, rules-based models that are more advanced than the drivers highlighted above. These may combine price-based signals, such as trend, momentum and volatility, with macroeconomic and valuation variables, including interest-rate differentials, purchasing power parity, external balances and policy expectations. Traditional factor strategies, econometric models and more recent machine-learning approaches all fall within this category. Their defining feature is the consistent translation of observable data into positions through a repeatable portfolio-construction and risk-management process. Commodity Trading Advisors, or CTAs, are an important subset of systematic macro managers. Despite their name, CTAs are not confined to commodities: they typically trade diversified portfolios of equity-index, government-bond, interest-rate, commodity and currency markets. Their currency positions are therefore usually part of a broader cross-asset portfolio rather than a dedicated FX mandate.

CTAs commonly implement currency exposure through exchange-traded futures. Futures are standardised, centrally cleared and highly liquid in the major currency contracts, while allowing investors to take long or short positions with relatively little initial capital through margining.

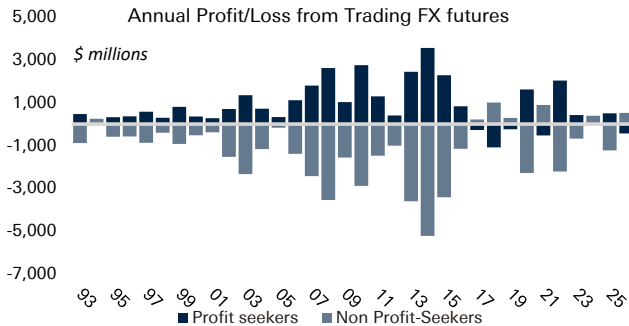
We thus return to the CFTC weekly commitment of traders report to produce an estimate of returns from profit-seeking activity from quant-based models using the methodology of Kearns and Manners (2004). As noted above, the IMM report gives the FX futures position of commercial and non-commercial every Tuesday. Given this information, it is a relatively straightforward exercise to calculate the profit or loss made every week by the respective parties.

We find that profit seekers make billions of dollars trading FX futures at the expense of hedgers almost every year since 1993.¹⁶ The breakdown for the individual currency pairs is also as expected, with annualised losses for non-profit seekers highest in EUR and JPY.

¹⁶ (We use dollar rates for CAD, CHF, EUR (since 1999), GBP, AUD (since 2000) and JPY)

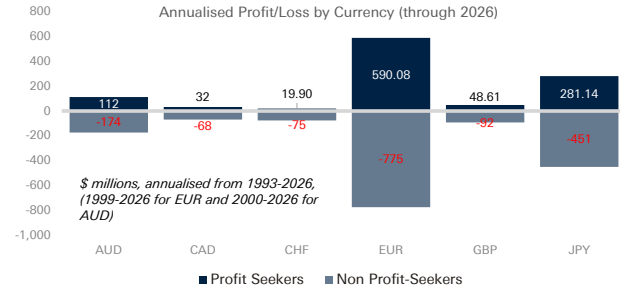


Figure 43: CFTC data shows that speculators consistently make money



Source : CFTC, Bloomberg Finance LP, Deutsche Bank

Figure 44: Profit-seekers consistently make money across all currencies



Source : CFTC, Bloomberg Finance LP, Deutsche Bank

3.3 Discretionary currency manager "alpha"

FX has been one of the asset classes of choice for macro investors since the 1980s so in contrast to other asset classes, we have a long history of "active" manager returns too. Discretionary macro investors take currency positions as part of a broader assessment of the global economic and policy environment. Their decisions may reflect views on monetary-policy divergence, fiscal policy, political risk, capital flows, commodity prices or shifts in the international monetary system.

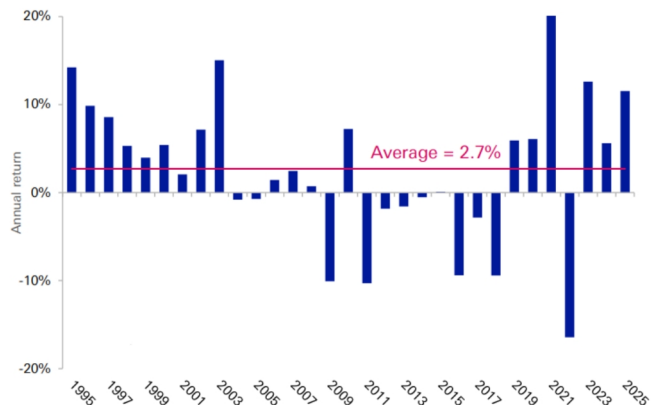
Unlike purely systematic strategies, discretionary investors can place greater weight on events that are difficult to incorporate into historical models, including elections, policy regime changes and geopolitical shocks. This flexibility can be valuable around turning points, but performance depends heavily on the quality of the investor's judgement, timing and risk management.

How have these returns behaved? We take the average of the Barclays, Parker, Hedge Fund Research and Bloomberg macro indices - where returns are available - to come up with an estimate of active manager returns. We show that since 1995, the annual return has been 2.7% on average.

Finally, the Deutsche Bank research team has been producing a tri-annual publication of top trade "[FX Blueprint](#)" recommendations since 2008, intended to replicate the trade recommendations that would appeal to a "traditional" macro hedge fund. The returns of this discretionary macro portfolio are shown below and have also been positive over time.

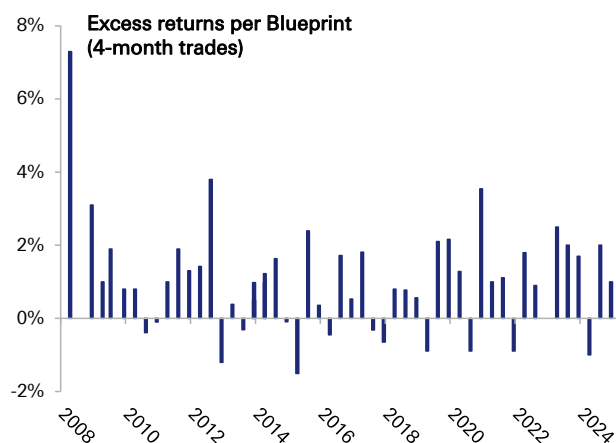


Figure 45: Active FX manager currency returns

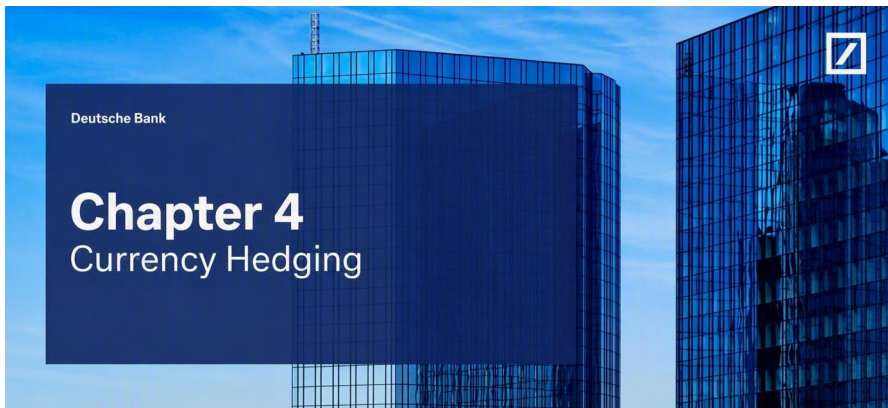


Source : Deutsche Bank, Bloomberg Finance LP [Average returns of Barclays, Parker, HFR and Bloomberg Finance LP FX macro indices]

Figure 46: DBFX Research trade recommendations



Source : Deutsche Bank, Bloomberg Finance LP



Introduction

Currency exposure is unavoidable whenever assets, liabilities or cash flows are denominated in a foreign currency. Yet the decision to hedge is not as simple as choosing between being hedged and unhedged. What is the objective of the hedge? How much risk should be removed, over what horizon, with what instrument and at what cost? Every hedge involves a trade-off. It can reduce uncertainty around portfolio returns, revenues or costs, but it also incurs direct costs through forward points, option premia and transaction expenses, as well as an opportunity cost if the currency subsequently moves in the hedger's favour. This chapter develops a framework for making those decisions. Rather than treating FX hedging as a standalone transaction, it considers currency risk within the investor's total portfolio or the corporate's full set of exposures. It examines passive and dynamic hedge ratios, currency overlay programmes, cash-flow and balance-sheet hedging, and the additional complications created by restricted emerging market currencies. The central argument is that there is no universally optimal hedge ratio: the appropriate strategy depends on the hedger's objective, constraints, underlying exposures and tolerance for cost and risk.

4.1 The problem of defining the hedging "objective"

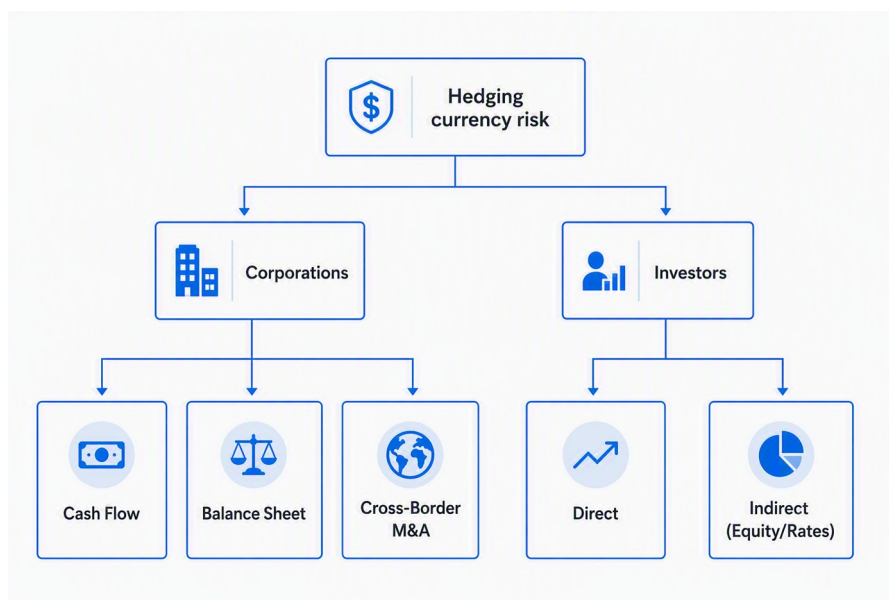
So far we have looked at currency returns from the perspective of an active investor. The objective here is clear: maximise returns. But what about market participants where FX risk is an unwanted by-product of underlying economic exposure. While in theory simple to understand, a hedging objective is far more complicated to implement in practice. The first challenge is that a hedging objective is not well-defined. An investor aims to maximise returns; a "hedger" aims to offset underlying



currency exposure, but to what end? The most frequent objective of hedging is to minimise earnings volatility for a corporate or returns volatility for a passive investor versus a benchmark. In practice, this objective faces multiple difficulties.

First, particularly for corporates, there is no widely accepted and well-defined "benchmark" against which to measure the success or cost of a hedging decision. Second, the objective of earnings volatility minimisation may not be achievable because of imperfect foresight: cashflows and balance sheet exposure are not infinitely predictable and hedgeable into the future. Third, hedging involves a cost. There is the direct cost that needs to be paid to purchase the underlying hedging instrument, usually in the form of forward points. There is also the indirect cost of the "counter-factual" exchange-rate that would have prevailed had the underlying exposure not been hedged. The simple point is that there is an opportunity cost to hedging; and that the volatility minimisation or other objective needs to be set in the context of this cost.

Figure 47: The broad landscape of hedging currency risk



Source : Deutsche Bank

4.2 Currency hedging for investors

Institutions with global investments face unavoidable currency exposure and therefore currency risk. A EUR-based investor holding US equities, for example, is exposed not only to US equity market risk but also to movements in EUR/USD. Because exchange rates can be volatile, economically meaningful and correlated with the underlying asset exposure, FX can materially affect overall portfolio risk and returns.

From static hedging to portfolio-aware hedging

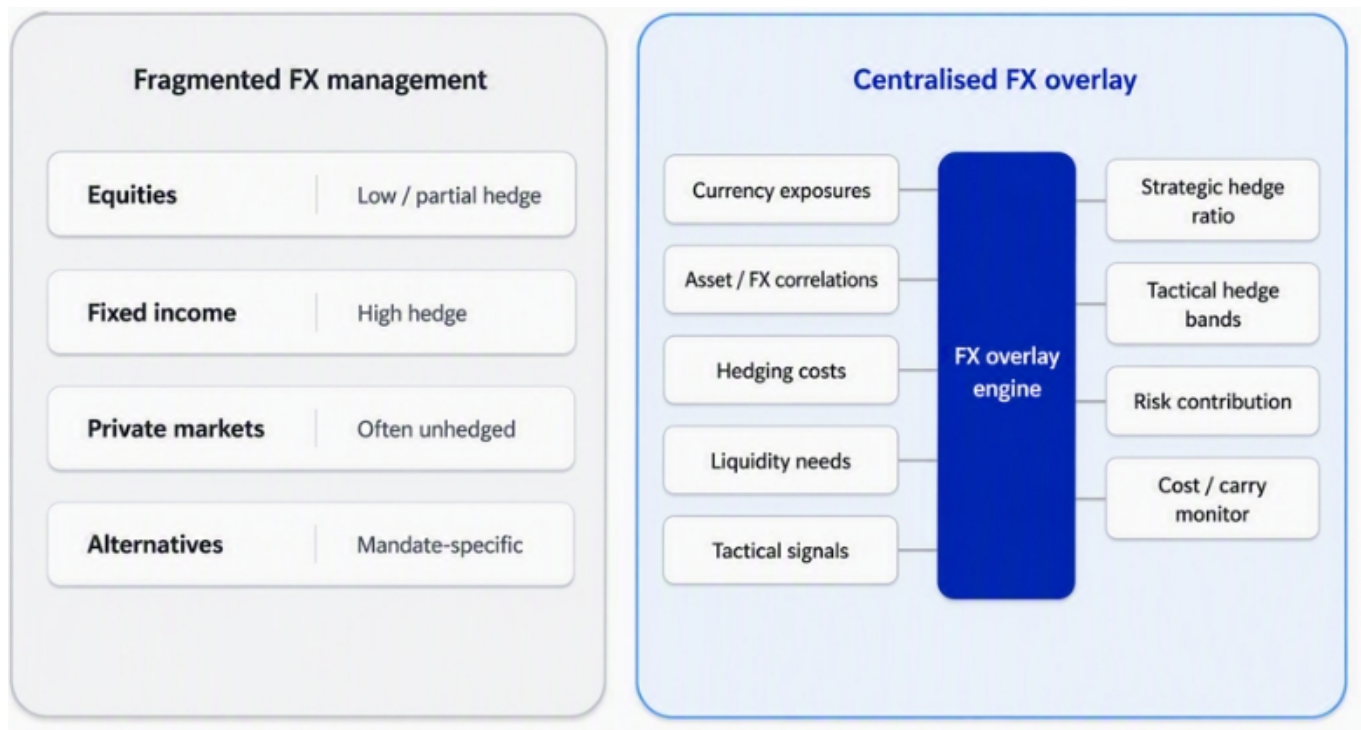
Traditional FX hedging has often been managed in a siloed way: investors may leave exposures unhedged, fully hedge them, or apply a fixed hedge ratio such as 50%



across the portfolio. While simple to implement, this approach can overlook important interactions between currencies, asset classes and portfolio objectives.

A more integrated approach treats currency exposure as part of the total portfolio rather than as a standalone transaction cost. This allows investors to assess whether FX should be hedged to reduce risk, retained as a potential source of return, or managed dynamically through an overlay strategy.

Figure 48: Currency risk – the underrated opportunity under the total portfolio approach



Source : Deutsche Bank, Grover and Anand (2026)

Dynamic FX hedging and total-portfolio risk

Currency risk management is increasingly being viewed through a holistic portfolio lens. [Grover and Anand \(2026\)](#) argue that currency exposure should be assessed within the full portfolio rather than treated as a standalone overlay.

The hedging decision can be viewed through three key considerations:

- How FX contributes to total portfolio risk.
- The cost or benefit of hedging, particularly through forward carry.
- The expected direction of spot, informed by tactical signals such as carry, momentum and valuation.

Cost of carry is only one part of the decision. The realised cost or benefit of hedging ultimately depends on how spot evolves over the hedging period. [Grover and Anand](#)



(2026) find that tactical hedging can help reduce the opportunity cost of hedging by identifying periods when currency returns are more likely to support portfolio performance. The main conclusions of their paper include:

- Dynamic hedging strategies can deliver stronger risk-adjusted outcomes than static hedge ratios.
- Investors generally benefit more from prioritising portfolio volatility reduction than maximising FX carry.
- Incorporating a view on future spot movements through currency signals can improve risk-adjusted returns.
- Tactical signals can help reduce the opportunity cost of hedging by selectively participating in favourable currency moves.

Why currency risk is moving up the agenda

Currency risk has often been overlooked or managed in silos, but investor interest in more active FX management is increasing as institutions adopt a Total Portfolio Approach (TPA) to investing. Choosing not to hedge is itself a risk decision and, if implemented passively, can materially influence portfolio outcomes. While bond portfolios have traditionally been hedged more actively than equity portfolios, many institutional investors are increasingly exploring centralised management of FX exposures across the full portfolio.

Grover and Anand (2026) identify key themes in institutional FX risk management:

1. **Defining the objective:** A fundamental question is whether currency risk should be managed primarily to enhance returns or to improve portfolio risk management. While FX can generate returns in certain market environments, a hedging approach should aim to design programmes that improve portfolio stability while preserving returns where possible.
2. **Benchmark-aware hedging:** An important theme in currency risk management is how to manage currency risk without straying too far from benchmark performance. Many funds are measured against market indices that are unhedged—for example, equity portfolios often track the MSCI World. Another common reference point is peer performance. These practical constraints can be incorporated into a total portfolio approach framework and may still allow for efficiency gains over standard approaches.
3. **Hedging instruments:** While forwards remain the main tool for managing currency risk, options are becoming more widely used. Their embedded cost efficiencies have made them increasingly attractive as a secondary instrument. Funds that must maintain a fully hedged currency exposure are also using options to introduce more flexibility and improve efficiency.
4. **Looking beyond direct currency holdings:** A holistic view of currency risk extends beyond the currencies directly held within a portfolio. For example, if a fund owns a US stock whose business is materially exposed to the Japanese yen, the fund may also have indirect yen exposure. While there is no simple way to measure this, one possible approach is to review company annual and quarterly reports to estimate those exposures.
5. **Private markets and uncertain cash flows:** Currency management is more difficult in private markets because the timing and size of cash flows are uncertain. One emerging approach is to model expected private-asset



distributions under different scenarios, using long-term currency volatilities and correlations, and then design hedging strategies that balance risk reduction against the carry cost of forwards and options.

6. **Cash-flow and liquidity management:** A key practical issue in hedging is managing margin calls, especially when the hedge moves against the hedger. For example, if a currency someone is hedging against starts to strengthen, the hedged position can experience large swings. Solutions are emerging to help clients manage the resulting liquidity needs.

Currency overlay management

Many of these hedging approaches are implemented through currency overlay programmes managed by specialist managers. These firms specialise in managing the foreign exchange exposures that arise from international investment portfolios. Their traditional role has been to implement hedging programmes for institutional investors such as pension funds, insurance companies and asset managers, helping to reduce the impact of currency fluctuations on portfolio returns. These programmes are typically managed separately from the underlying investment portfolio. Systematic currency overlay programmes can range from simple passive hedging to more active approaches. Passive programmes maintain a predefined hedge ratio across foreign currency exposures, while dynamic programmes adjust hedge ratios according to predefined signals such as valuation, momentum or carry. Some overlay managers also run currency alpha mandates, taking active currency positions with the objective of generating returns.

4.3 Currency hedging for corporates

While many of the principles of dynamic FX hedging also apply to corporates, the underlying hedging problem is different. Institutional investors typically hedge currency exposure embedded within assets whose value changes over time. Because the asset value itself fluctuates, a 100% hedge does not necessarily eliminate risk and may even increase volatility if currency and asset returns naturally offset one another. The objective is therefore to manage FX risk as part of the total portfolio.

Corporates, by contrast, are usually hedging known or forecast foreign-currency cash flows, revenues, costs, assets or liabilities. Because the size of the underlying exposure is often known in advance, a corporate can in principle hedge a much larger proportion of its FX risk. The challenge is therefore less about portfolio optimisation and more about balancing risk reduction against hedging costs and opportunity costs. While investors ask how much currency risk should be retained within a portfolio, corporates are typically focused on how much uncertainty around future cash flows should be eliminated and at what cost.

For corporates, the most common source of FX risk arises when purchases, sales, assets or liabilities are denominated in a currency other than the entity's functional currency. The main corporate hedging strategies are cash flow hedging, balance sheet hedging and, less frequently, deal-contingent hedging linked to transactions such as cross-border M&A.

- **Cash flow hedging:** This is one of the most common hedging strategies that require hedging currency risk when the underlying item is not in the entity's functional currency. For example, a sale that has been forecast but has not yet happened does not get recorded on the income statement of the company. These types of transactions that are forecast but not recorded



require a cash flow hedge. In this case, hedge accounting is important as the hedge's mark to market does not affect net income until the underlying transaction is recorded in earnings. It does not create profit/loss volatility during the hedging period. Hedging strategies that are commonly used to hedge future cash flow are the purchase or sale of FX forwards or net purchased options. Let's say a US-based company is contracted to manufacture tyres for a German automaker. This results in a predictable stream of euro cashflows. However, given the company's cost base is in US dollars, it may not be able to protect its revenue margin if euro depreciates considerably against the dollar. The company here then opts for a cash flow hedge to protect its margins.

- **Balance sheet hedging:** In the example above, the company is hedging a transaction which has not yet been recognised as a sale or expense on the income statement. Now say the manufacturer decides to shift its cost base to Europe by building a factory in Europe. Now, the parent company has a EUR asset on its USD balance sheet. Once they start billing out of their EUR subsidiary, for accounting purposes a regular balance sheet hedge may be required as the parent company's functional currency is still USD. In summary, this type of hedging hedges assets and liabilities on corporate balance sheets that are not in the entity's functional currency. A typical balance sheet hedging strategy may be rolling short dated forwards.
- **Deal-contingent and M&A hedging:** Continuing with the example, say the US company instead chooses to merge or acquire a European company. Thus, the cross-border merger and acquisition (M&A) now creates a need for hedging the foreign currency. In certain respects, this type of hedge is conceptually similar to cash flow hedges, as it is based on a potentially unpredictable transaction, but with additional complexities, including whether the underlying transaction is in stock or cash.

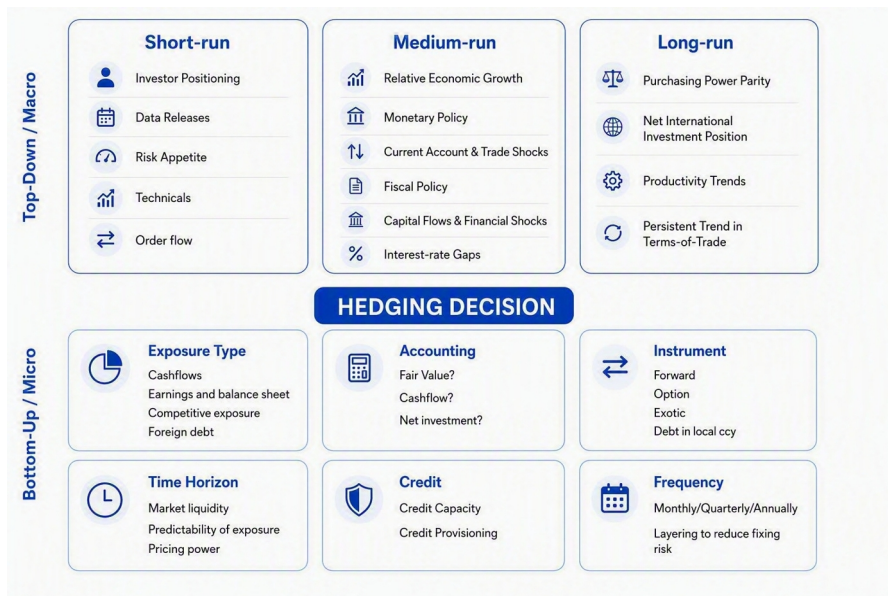
The corporate hedging decision

The "bottom-up" hedging decision involves a large number of parameters the optimal mix of which is unique for each firm. While it is true that many firms share the same basic exposure profile, each firm's strategy can vary dramatically depending on its objectives and constraints. Empirical investigations into the hedging practices of multinational firms show that hedging decisions are influenced by a large number of case-specific factors. These include accounting treatment, competitive price concerns and recent hedging outcomes. Overall empirical studies of corporate risk management suggest that a single, clear and value-maximising policy for risk management is not obvious (Brown, GW 2001). The micro hedging decision will also vary depending on market conditions. In all, the evidence cited above argues against recommending a universally applicable hedging programme for all firms.

The "top-down" elements of the hedging decision relate to the market drivers of FX that we identified in the earlier sections of the report. Unlike the multiple parameters involved in the micro hedging decision, the macro decision boils down to market variables: where does the Treasurer or CFO expect the exchange rate(s) to be over the relevant hedging period and how does this relate to the underlying exposure of the firm? One approach is for the hedger to make a discretionary decision based on his or her underlying assessment of market drivers. Another is to adopt a rules-based approach based on the systematic excess returns we have identified in the FX market. Such an approach would be no different to the dynamic hedging strategy employed by a passive "profit-constrained" investor discussed in the section above.



Figure 49: The Hedging Decision



Source : Deutsche Bank

Dynamic FX hedging for corporates

Dynamic FX hedging differs for corporates and institutional investors. Corporates typically hedge known exposures tied to contracted or forecast cash flows, rather than currency risk embedded in assets whose value can grow over time. In principle, they can remove much of this risk by hedging 100% of the identified exposure. However, a full hedge can carry opportunity costs. For multi-currency exposures, it may overlook diversification benefits when currency moves partly offset one another. It can also lock in hedging costs before the full-period outcome is known, potentially causing corporates, like institutional investors, to forgo gains from favourable currency moves. For corporates, the hedging framework should aim to reduce these opportunity costs without materially increasing risk.

For companies with a single-currency exposure, signal-based hedging frameworks can be useful. Rather than relying solely on forward points, hedge decisions can incorporate carry, momentum and valuation signals to assess whether hedging is likely to be more or less beneficial. [Grover \(2024\)](#) suggests this can reduce the opportunity cost of fully hedging, or of hedging based only on current forward points.

For companies with multiple currency exposures, hedging becomes more portfolio-like. Managing the joint risk of the currency basket through Value-at-Risk or cross-currency correlation analysis can help capture diversification benefits that a full-hedge approach may miss. [Grover and Nandy \(2025\)](#) show that, over recent decades, a holistic FX framework has delivered lower-cost and higher-return hedging outcomes than fully hedging each exposure. This result is particularly clear for euro-based exporters, where carry costs have been higher and correlations between euro crosses have been lower, increasing the value of a portfolio-based approach. Overall, corporate hedging is best approached as a two-step process:



- Manage risk across the full currency exposure, particularly where diversification between currencies exists.
- Use currency signals such as carry, momentum and valuation to reduce the opportunity cost of hedging and retain some participation in favourable currency moves.

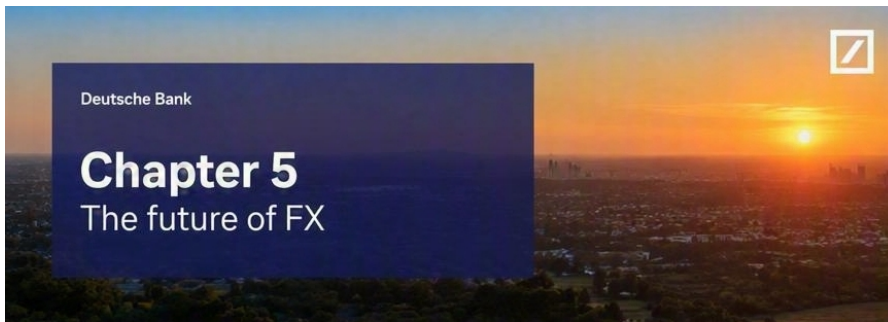
This approach recognises that, while corporates and investors start from different objectives, both can benefit from combining strategic risk management with dynamic currency signals. The difference is that institutional investors seek to optimise FX exposure within a broader investment portfolio, whereas corporates seek to minimise uncertainty around future cash flows while controlling the cost of doing so.

4.4 Hedging emerging market currencies

While the framework above is applicable to both G10 and EM currencies, some EM currencies require additional consideration when hedging. In particular, currencies that are subject to current account restrictions, capital controls, or both operate under different market constraints. We refer to these as restricted markets. Many of these currencies trade offshore through non-deliverable forwards (NDFs), and foreign institutional investors and corporations have historically relied on NDFs to hedge their currency exposures.

However, offshore markets are not always the most cost-effective venue for hedging. On average, onshore forward curves have traded around 40 basis points below offshore curves over the past decade. That creates a potential opportunity for investors. By accessing onshore markets, they may be able to achieve lower hedging costs. For currencies, where operational challenges exist, tactical hedging via cheaper onshore forward markets can further improve hedging efficiency ([Grover, 2026](#) and [Harvey et al., 2025](#)).

Another important consideration when hedging EM currencies is the cross-currency basis, which represents a hidden component of hedging costs for corporate and institutional investors ([Harvey and Friedemann, 2023](#)). Although it has attracted significant attention since the 2008-09 financial crisis, its role in emerging markets is often overlooked despite being larger and more volatile than in many developed markets. The cross-currency basis can create either costs or opportunities for hedgers and issuers. While instruments such as cross-currency basis swaps can be used to manage these exposures, liquidity and market access vary considerably across EM currencies, and usage is typically concentrated at longer tenors. As most shorter-dated EM hedging remains centred on FX forwards and swaps, investors should remain aware of the cross-currency basis when assessing the true cost of hedging.



Introduction

The foreign exchange market has always evolved alongside the technologies, institutions and regulations through which money moves across borders. Payment systems determine which currencies are easiest to use, where liquidity accumulates and who controls the infrastructure supporting cross-border transactions. This final chapter considers how technology, regulation and geopolitics may reshape FX market structure, payments and currency demand.

5.1 Technology and market structure

While electrification aimed to improve the FX market by reducing risk and increasing efficiency, it has inadvertently introduced greater complexity and fragmentation. This has resulted in a more decentralised and dispersed market structure, increasing operational risks such as vulnerability to cyberattacks, even in a traditionally liquid market like FX. Despite these challenges, technological innovation continues to drive change. Distributed ledger technology (DLT), with its decentralised and shared database, offers potential benefits like improved efficiency, transparency, and reduced settlement risk. Furthermore, automation promises cost reductions and efficiency gains in trading processes, while AI and machine learning are being increasingly applied to price prediction, risk management, and fraud detection. The interplay between these technological advancements and the evolving market structure will continue to shape the future of the FX market.

5.2 Stablecoins and payment rails as the new geopolitical battle lines

The dollar is hugely dominant in FX markets today. According to BIS, 88% of FX transactions have a USD leg. The USD acts as a global vehicle currency. It offers the deepest pools of liquidity which makes it cheaper to trade third currencies via their USD crosses. The USD accounts for 50% of messages exchanged on SWIFT for cross-border payment, and its share in trade invoicing far exceeds the US' economic share of global trade. Cross-border USD payments today are mostly



intermediated through the USD correspondent banking system. Global banks are able to transact in USD with each other because they hold accounts with large correspondent banks that settle with each other via the Fed balance sheet in wholesale central bank USD.

But [Sachdeva and Saravelos \(2025\)](#) show that the dollar correspondent banking has been at growing risk of disruption from faster distributed ledger technology, geopolitical competition, alternative payment systems, and underserved corridors in the Global South. Correspondent banking still features slow settlement times, high fees, and fractured access. Many parts of the Global South have declining access to correspondent bank relationships due to growing AML/CFT. Combined with the use of sanctions that cut certain countries from SWIFT, there has been interest in developing alternative payment rails, particularly by BRICS. China has developed CIPS as an equivalent RMB correspondent banking system. China has also been a core member of Project mBridge that explores cross-border CBDC payments on a distributed ledger.

Keeping the dollar dominant in payments and transactions will be of significant importance to the US in a world of heightened geopolitical competition. Remaining the preferred currency of payment is closely linked to being the currency in which external surpluses are built. The world's willingness to save in USD in turn lowers the cost of USD financing which is a big advantage for the US government and corporates. To support the dollar's role in payments in the future, the US appears to be turning to the power of distributed ledger technology and a private form of money: stablecoins.

Stablecoins are new forms of private digital money tokens hosted on blockchains that maintain a peg to a fiat currency. Under GENIUS Act regulation, USD stablecoins will be fully backed by liquid, short-term USD reserves like T-bills to help maintain their 1:1 peg with the USD. 99% of stablecoins outstanding today are linked to the USD. They are issued and traded on distributed ledgers, making them accessible anywhere through the internet. Their on-chain attributes mean stablecoins offer the possibility of near-instant settlement - where all aspects of monetary value transfer from messaging to reconciliation happen together in the form of a ledger update. Stablecoins do however disintermediate the Fed from the settlement process for international payments in USD. Europe and China have taken a very different approach thus far to digital money, concentrating more efforts on central bank digital currencies as their response to the future of cross-border payments.

5.3 Tokenization and programmability

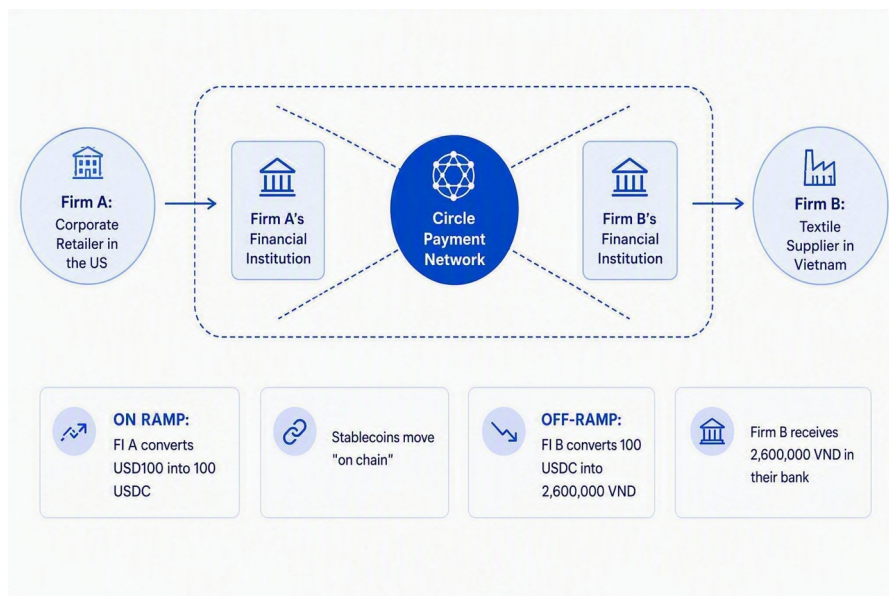
Stablecoins are arguably just a subset of a broader trend towards tokenisation that could open many new possibilities for how FX markets operate in the future. Tokenisation is the process of creating claims on underlying money or assets, that can then be represented as digital tokens and made transferrable on a shared infrastructure. Stablecoins are a new type of tokenized private money, backed by cash, bank deposits or bills. But tokenisation also extends into existing commercial bank deposits, central bank money, and traditional assets like equities, money markets and bonds. In the earlier section we covered how stablecoins might help with cross-border payments, by offering 24/7 infrastructure to move USD that does not rely on traditional banking relationships and hours. But tokenized money can also help with other aspects of the FX settlement process - the exchange of currencies for each other, or for assets. Because FX trading takes place primarily in the over-the-counter market, once a trade is executed and enters the post-trade



process there is settlement risk that is created: payment-versus-payment mechanisms such as CLS or pre-settlement netting reduce the risk to some extent, but tokenized money could offer another feature that addresses it more directly: programmability.

Programmability allows certain automated rules and conditions to be embedded into money processes. This is often implemented through smart contracts that execute transactions when predefined conditions are met, helping improve timing frictions, settlement risk, or the need to pre fund and trap liquidity to meet payment obligations. In a programmable world, a payment can be released only if a corresponding asset is transferred, or an FX transaction can settle only if both currency legs are ready for delivery. This technology can thus support more synchronised settlement processes, linking cash, securities and collateral movements through predefined rules and automated execution. While tokenised money can improve payments and settlement on a standalone basis, its full potential may emerge when combined with increasingly tokenised securities.

Figure 50: How stablecoins might change a cross-border payment



Source : Deutsche Bank, Circle Payment Network White Paper

5.4 Concluding thoughts

The foreign exchange market is one of the most innovative and rapidly evolving financial markets in the world. Over the past two decades, advances in technology, electronic trading, data and market infrastructure have transformed how currencies are traded, priced and managed, placing FX at the forefront of broader changes in global finance. Yet because currencies sit at the intersection of national economies, the FX market remains fundamentally connected to international trade and cross-border investment. It is therefore increasingly exposed to the forces reshaping the global economic landscape—from geopolitical fragmentation and changing trade relationships to shifts in regulation, payments infrastructure and the architecture of the international monetary system. Ultimately, the structure of the FX market is not simply a question of how currencies trade, but of who provides liquidity, intermediates capital and controls the financial infrastructure through



which money moves across borders. In that sense, the evolution of FX is increasingly intertwined with questions of monetary sovereignty, financial resilience and geopolitical power. One thing is therefore clear: understanding what moves and shapes exchange rates, and the market through which they are traded, is more important than ever.



Appendix

Profit vs non-profit seekers in the FX market

We use the 2025 BIS Triennial Survey to update [previous work](#) and estimate the share of FX turnover initiated with a profit-seeking currency objective. We construct two estimates:

- A realistic scenario, applying what we deem realistic assumptions about participants' ability and mandate to take FX risk;
- And an upper-bound scenario, which relaxes those assumptions by classifying ambiguous transactions as profit-seeking.

We find profit-seeking activity accounts for approximately 40% of turnover under the realistic scenario and 60% under the upper-bound scenario.

Financial participants

Dealers:

Dealer-to-dealer activity accounts for approximately 47% of [total daily FX turnover](#) (~\$4.44tn). However, treating all inter-dealer activity as speculative would materially overstate profit-seeking demand. Much of this turnover reflects market-making: dealers absorb client flow, manage the resulting inventory and redistribute risk across the market. In other words, market-making activity provides liquidity to the FX market and should not be interpreted as reflecting speculative objectives, especially given regulatory change since 2008.¹⁷

In order to find the percentage of dealers who engage in profit-seeking behaviour, we assume that inter-dealer transactions conducted by voice are not profit-seeking and that only electronic transactions conducted through order-matching systems such as EBS are profit-seeking.¹⁸

By doing this we find that:

- 20% of dealer flow, approximately \$889bn per day, goes through electronic broking systems, as such, we class all of this as profit-seeking under our "upper-bound" scenario. Given the impact of Dodd-Frank and the Volcker rule on banks' proprietary activities we believe this to be a relaxed assumption.
- Therefore, under a more realistic set of assumptions we classify half of this

¹⁷ At the same time, post-global-financial-crisis regulation has increased the balance-sheet cost of some forms of dealer intermediation and arbitrage. Basel III introduced tighter risk-based capital requirements alongside new liquidity and leverage constraints, including a leverage ratio that captures both on- and off-balance-sheet exposures and derivative positions. These constraints can affect banks' willingness and capacity to deploy balance sheets to market-making and arbitrage activities. One important manifestation has been the persistent, although time-varying, deviations from covered interest parity (CIP) observed since the global financial crisis, reflected in the cross-currency basis. See Borio et al., "Covered Interest Parity Lost: Understanding the Cross-Currency Basis," BIS Quarterly Review, September 2016

¹⁸ More specifically, we use data from the April 2017 Federal Reserve Bank of New York FX Survey and data from the 2025 FXJSC FX turnover survey. In the FRBNY paper, trade execution method is split



flow, 10% of dealer flow as profit-seeking (\$444bn/day). To account for the decline in bank proprietary trading and the importance of inventory management within inter-dealer activity

Hedge Funds and PTFs

Hedge Funds and PTFs account for approximately \$760bn (8%) of headline daily FX turnover and are reported together in the BIS data. Traditionally, banks acted as the primary dealers, or liquidity providers, in the decentralised OTC FX market. However, the expansion of prime brokerage, which has facilitated client access to primary FX trading venues such as EBS and Reuters, has allowed principal trading firms to act not only as profit-seekers but also as liquidity providers.

We assume that:

- All hedge-fund activity is profit-seeking. Hedge funds generally have a mandate to take FX risk, meaning that even transactions associated with purchases and sales of foreign assets may incorporate an active currency view.
- Because the BIS data does not disaggregate the dual role of PTFs (liquidity providing and profit-seeking) we assume liquidity provision is 10% of the BIS' headline "hedge fund & PTFs" \$760bn figure. PTFs, we assume that 90% of headline activity is profit-seeking and 10% reflects liquidity provision.¹⁹

Therefore, under both the realistic and upper-bound scenario we assume that \$684bn (90% of their daily turnover) of hedge fund and PTF activity is profit-seeking.

Non-Reporting Banks

Non-reporting banks account for approximately \$2.19tn, or 23% of headline daily turnover. However, these institutions frequently intermediate transactions for institutional investors and non-financial customers rather than representing a separate source of final currency demand.

We assume that:

- Under our realistic (upper-bound) scenario, we assume that 90% (80%) of this daily flow represents the fact that non-reporting banks act as an intermediary for institutional and other non-financial customers. Therefore, we distribute the 90%, \$1.97tn of daily turnover proportionately, across institutional investors and other non-financial customers.²⁰
- We therefore assume that under the realistic (upper-bound) scenario 10% (20%) of non-reporting bank flow is profit-seeking, in line with our dealer

across two dimensions - voice/electronic and direct/indirect. We assume all voice transactions are liquidity seeking, as well as direct electronic transactions that are conducted through single-bank electronic trading systems. Transactions executed via voice are highly unlikely to be speculative in nature since they typically involve large-ticket customer flow. Foreign exchange desks at banks would be unlikely to be executing large proprietary transactions of such size which would typically take place on electronic matching systems such as EBS. More recent data from the Bank of England finds that 20% of reporting dealer flow goes through these electronic broking systems. We apply the final 20% figure to the worldwide BIS survey from 2025 (10% under our more realistic set of assumptions) to come to an estimate.

¹⁹ Our first assumption is that liquidity provision by PTFs is largely dominated by the spot market. We therefore focus on Hedge fund and PTF volumes from spot trading. According to [BIS](#) 45% of Hedge



assumptions.²¹

Institutional investors:

Institutional investors account for \$1.26tn of daily BIS turnover and present the most difficult classification challenge. After reallocating the relevant flow intermediated by non-reporting banks, their estimated turnover increases to \$2.56tn under the realistic scenario.

We assume that:

- First, given that BIS data does not divide institutional-investor turnover according to the underlying asset class we instead use US Treasury International Capital (TIC) data on gross cross-border purchases and sales of securities to estimate the relative contribution of fixed-income and equity portfolios.²² This adjustment produces an estimated asset-allocation split of 63% equities and 37% fixed income.
- Second, considering this asset allocation split, we use [Morning Star](#) and [ICI](#) data on the share of passive versus active investment funds. We estimate that approximately 45% of equity assets and 60% of fixed-income assets are actively managed. Passive investors consist of index or benchmark-tracking funds and we assume they are active in FX on a transactional non-profit-seeking basis. Active investors aim to outperform a stated benchmark or offer unconstrained positive returns. In practice, this type of investor would still face limits on the amount of exchange-rate risk that can be taken but for the purposes of this exercise we assume all "active" investor flow is profit-seeking in FX.

Putting this together we find that:

Using the US TIC data, it implies an asset allocation split of 66% equity and 34% fixed income split. As per data from Morningstar and ICI, 45% of those equities will be actively managed, compared with 60% of those fixed income holdings.

- **Under our upper-bound scenario**, we assume that all of these actively managed holdings are with firms that are mandated to take FX risk. This implies that approximately 50% of institutional-investor turnover, or around \$1.28tn per day, is profit-seeking.
- Under the **realistic assumption**, active management of the underlying portfolio does not necessarily imply an active currency mandate. An investor may actively select foreign bonds or equities while remaining required to hedge the associated FX exposure mechanically. So, instead of assuming all actively managed equities and fixed income assets are with funds that are mandated to take FX risk, we assume 80% of active fixed-income managers have discretion to take FX risk and 50% of active equity

Fund and PTF volume comes from spot trading. We assume that this is split 50/50 between Hedge Funds, an upper-bound estimate. Further, we also find that [30% of PTF trading can be attributed to market making](#). Together, these assumptions leave us with an estimate of approximately 10% liquidity provision.

20 We do not allocate a share to hedge funds or the official sector, which are more likely to transact directly with large reporting dealers.

21 Recent evidence shows that a proportion of non-reporting bank activity is growing <https://www.greenwich.com/market-structure-technology/behind-market-structure-future-nbfp-landscape>, as is their share of profit-seeking activity. As a result of their growing importance, we apply the same split as we do for dealers.

22 TIC data reports (1) monthly purchases and sales of domestic and foreign securities by security type



managers have discretion to take FX risk. This implies a profit-seeking share of approximately 30% under the realistic scenario. Applied to \$2.56tn of institutional-investor turnover, this would equal approximately \$800bn per day.

Official Sector

The official sector includes central banks, sovereign wealth funds and international financial institutions. Its activity accounts for approximately 1%–2% of headline FX turnover. The motivations within this category vary. Central-bank intervention may pursue monetary-policy, financial-stability or exchange-rate objectives, while reserve managers may also seek to preserve capital and earn returns.

We therefore divide official-sector activity evenly between the two categories under both realistic and upper-bound scenarios.

- We classify 50% of the turnover as profit-seeking. Because official-sector activity represents a small share of aggregate turnover, the final result is relatively insensitive to this assumption.

Non-Financial Sector:

Corporates and governments

After allocating non-reporting-bank turnover, estimated non-financial-customer activity is equivalent to around \$952 bn per day in the realistic scenario and \$896 bn in the upper-bound scenario.

We assume that:

- Corporates use FX to settle foreign-currency revenues and expenses, finance cross-border activity, hedge forecast cash flows and manage earnings volatility. Governments similarly transact to meet an underlying funding or expenditure requirement. We classify all non-financial-customer turnover as liquidity-seeking under both scenarios.

Other

The BIS residual “other” category accounts for approximately 2% of headline turnover. It includes activity for which the underlying participant or motivation cannot be identified precisely, including some retail-driven flow. Retail and other residual transactions can serve both speculative and transactional purposes.

Some investors trade currencies to earn returns, while other transactions relate to travel, remittances or purchases of foreign assets. We assume that:

- We classify 100% as profit-seeking under the upper-bound scenario and 50% as profit-seeking under the realistic scenario.

Undistributed

(Table 4). and; (2) foreign holdings of US securities, split between official and private investors and by asset class (Table B). Because we assume official-sector investors have different profit-seeking incentives from private institutional investors, we exclude their activity from the asset-class split. When doing so we find the proportion of institutional investors transacting in fixed income vs equity markets.



Final residual category accounts for less than 1% of headline turnover. Because there is insufficient information to identify its underlying motivation, we classify this turnover as

- 100% liquidity-seeking rather than assume that it represents directional risk-taking. The treatment has little effect on the aggregate result because the category is very small.

Bringing it all together:

Taken together, **under our realistic assumptions** the current model estimates that approximately 40% of adjusted FX turnover is profit-seeking, while approximately 60% is liquidity-seeking or transactional.

Under the upper-bound assumptions, the split is approximately 60% profit-seeking and 40% liquidity-seeking.



Glossary of terms

A

- Algorithmic trading (AT) — Automated trading in which a computer algorithm decides the order-submission strategy.
- Alpha — Return generated by skill or active judgement in excess of a benchmark or of systematic factor exposure, as pursued by discretionary currency managers.
- American-style option — An option that can be exercised at any point up to expiry.
- At-the-money (ATM) — An option whose strike is at or near the prevailing spot/forward rate.

B

- Balance sheet hedge — A corporate hedge of assets or liabilities on the balance sheet that are denominated in a currency other than the entity's functional currency.
- Balassa–Samuelson effect — Theory that faster productivity growth in the tradable sector raises wages and non-tradable prices, generating real exchange-rate appreciation.
- Barrier option — An option whose payoff is activated (knock-in) or extinguished (knock-out) when spot reaches a specified level.
- Basel III — Post-crisis bank regulation introducing tighter capital, liquidity and leverage requirements that raise the balance-sheet cost of dealer intermediation and arbitrage.
- Basic balance — A measure combining the current account with the slowest-moving financing flows (net FDI, and sometimes long-term portfolio flows) to gauge underlying currency demand.
- Basis point (bp) — One hundredth of a percentage point (0.01%).
- Behavioural Equilibrium exchange-rate (BEER) — An equilibrium model estimating the exchange-rate implied by economic fundamentals through panel regression.
- Beta (FX beta) — The systematic, rules-based return available from generic currency factor exposures such as carry, momentum and value.
- Bid-ask (bid-offer) spread — The difference between the price at which a dealer buys (bid) and sells (offer); a frictional cost of trading and the market maker's compensation.

C

- Call option — The right, but not the obligation, to buy one currency against another at a predetermined strike price.
- Capital controls — Official restrictions on cross-border capital movements, common in restricted and emerging-market currencies.
- Capital-weighted PPP — A valuation approach that weights currencies by cross-border financial exposures rather than by trade, reflecting the



geography of portfolios.

- Carry / carry trade — Borrowing a low-yielding currency to invest in a higher-yielding one to earn the interest differential; historically profitable but with negative skew and crash risk.
- Central Bank Digital Currency (CBDC) — A digital form of central-bank money.
- Central Counterparty / Central Clearing Counterparty (CCP) — Financial market infrastructure that interposes itself between buyer and seller, becoming the buyer to every seller and the seller to every buyer, reducing counterparty risk.
- Commodity currencies — Currencies of major commodity exporters (e.g. AUD, NZD, CAD) whose real exchange-rates have strong long-run relationships with world commodity prices.
- Commodity Futures Trading Commission (CFTC) — The US derivatives regulator; source of the IMM speculative-positioning data used to gauge FX positioning.
- Commodity Trading Advisor (CTA) — A systematic macro manager trading diversified futures across asset classes, including currencies, typically via exchange-traded futures.
- Continuous Linked Settlement (CLS) — An independent multi-currency payment-versus-payment settlement system that mitigates FX settlement risk by settling both legs simultaneously and netting payments.
- Convenience yield — The safety and liquidity premium global investors derive from holding US Treasuries, which supports the dollar; measurable via the Treasury basis.
- CORAX — Categorised ORderflow from Autobahn FX — Deutsche Bank's categorised indicators built from globally aggregated electronic spot order flow.
- Covered Interest Parity (CIP) — The no-arbitrage condition that the forward premium/discount equals the interest-rate differential; held almost exactly until 2008.
- Cross-currency basis — The extra spread, over and above the interest differential, of borrowing dollars synthetically through the FX swap market; reflects structural dollar-funding demand.
- Cross-currency swap (XCCY) — A swap that exchanges principal in two currencies at inception and maturity and also involves periodic interest payments in each currency over the life of the contract.
- Cryptocurrency — A digital asset secured through cryptographic and distributed-ledger technologies, such as Bitcoin.
- Currencyboard — A hard-peg regime in which domestic currency issuance is rigidly backed by foreign reserves at a fixed rate, tightly constraining monetary policy.
- Currency overlay — A programme, often run by specialist managers, to manage the FX exposures arising from an international investment portfolio, separately from the underlying assets.
- Current account — The balance of a country's trade and income flows; a key medium-run anchor summarising the gap between national saving and investment.



D

- Dark pool — A venue with hidden order books that matches buy and sell interest at a venue-calculated midpoint (e.g. BGC MidFX, FXSpotStream MidMatch).
- Deal-contingent / M&A hedge — A hedge linked to a potentially unpredictable transaction such as a cross-border merger or acquisition.
- Dealer — A large commercial or investment bank (or securities house) that makes markets and trades in the inter-dealer market for proprietary purposes and to meet customer demand.
- Delta — The change in an option's value for a change in spot; also used to describe moneyness (e.g. 0–25 delta is out-of-the-money).
- Depository Trust & Clearing Corporation (DTCC) — Operator of a major swap data repository that captures a substantial share of global institutional FX option activity.
- Deutsche Bank Currency Returns Index (DBCR) — A 2007 index packaging transparent, rules-based carry, momentum and valuation strategies into an investable currency 'beta'.
- Deutsche Bank Equilibrium exchange-rate (DBer) — DB's proprietary behavioural-equilibrium model using the terms of trade, a productivity ratio and trade openness; performs better in emerging markets.
- Discretionary manager — An investor who takes active currency positions based on judgement rather than a purely systematic, rules-based process.
- Dodd-Frank — The US post-crisis regulatory framework; includes the Volcker Rule and OTC-derivative reporting requirements.
- Dollarisation — Use of another country's currency as legal tender, which largely eliminates independent national monetary policy.
- Dornbusch overshooting — Model in which sticky goods prices cause the exchange-rate to overshoot its long-run level on impact before reverting, explaining excess FX volatility.
- Drawdown — The peak-to-trough decline in the value of a strategy or portfolio.
- Dutch disease — A situation in which a commodity boom raises the real exchange-rate and erodes the competitiveness of a country's other tradable sectors.

E

- EBS — A primary inter-dealer electronic broking venue, historically central to banks' inventory management.
- Effective spread — A transaction-cost measure inclusive of slippage and fees, giving a more realistic picture of liquidity than quoted depth alone.
- Efficient Market Hypothesis (EMH) — The theory that asset prices fully reflect available information.
- Electronic Communication Network (ECN) — A third-party electronic venue that both streams prices and matches buy and sell orders.
- Electronification — The shift of FX trading from voice to electronic execution across both the inter-dealer and dealer-client segments.
- Emerging markets (EM) — Developing-economy currencies, often subject



to capital controls, higher basis volatility and offshore NDF trading.

- EPIC+ — Deutsche Bank's positioning framework aggregating direct measures (CFTC/IMM) and indirect measures (CORAX, inter-dealer orderbook, risk reversals, DTCC options).
- Equilibrium exchange-rate — The exchange-rate consistent with economic fundamentals and internal/external balance; estimated via frameworks such as FEER, BEER and DBeer.
- European-style option — An option that can be exercised only at expiry.
- External Balance Assessment (EBA) — The IMF methodology that estimates current-account and REER 'gaps' using several complementary approaches.

F

- Fama regression — A test of UIP that regresses realised exchange-rate changes on the interest differential (or forward premium); typically yields a slope below one — the forward-premium puzzle.
- Flash crash — A sudden, severe and short-lived price dislocation, increasingly studied alongside the electronification of markets.
- Foreign Direct Investment (FDI) — The stickier, long-horizon component of external financing, reflecting the partial illiquidity of direct ownership.
- Forward (FX forward) — A contract to exchange currencies at a future date at an exchange-rate agreed today.
- Forward points — The forward premium or discount added to spot; the direct carry cost or benefit of hedging with a forward.
- Forward premium puzzle — The robust empirical finding that high-yield currencies tend on average to appreciate rather than depreciate, making the carry trade profitable and contradicting UIP.
- Functional currency — The primary currency in which an entity operates and reports; FX risk arises from items denominated in other currencies.
- Fundamental Equilibrium exchange-rate (FEER) — The real exchange-rate consistent with the economy operating at internal and external balance.
- FX swap — A contract combining a spot exchange of currencies with a reverse exchange at a later date; the largest FX instrument by turnover and effectively collateralised borrowing.

G

- G10 — The group of major, most liquid developed-market currencies.
- Gamma — The rate at which an option's delta changes with spot; long-gamma dealers dampen volatility while short-gamma dealers amplify moves through their hedging.
- GENIUS Act — US legislation requiring USD stablecoins to be fully backed by liquid, short-term USD reserves (such as T-bills) to maintain a 1:1 peg with the dollar.
- Give-up — The process by which a trade executed by a client is transferred ('given up') to its prime broker, becoming two matching trades.
- Global financial cycle — Rey's concept in which the US dollar and global risk appetite drive worldwide leverage, capital flows and financial



conditions.

H

- Hedge accounting — Accounting treatment that defers a hedge's mark-to-market so it does not affect net income until the underlying transaction is recorded in earnings.
- Hedge ratio — The proportion of an exposure that is hedged; typically sticky and adjusted in cycles, and an increasingly first-order driver of FX.
- Herstatt risk — See Settlement risk.
- High-frequency trading (HFT) — Very high-speed algorithmic trading, often conducted by principal trading firms.

I

- Implied volatility — The market's expected volatility embedded in option prices; the price of protection against exchange-rate movements.
- In-the-money (ITM) — An option that has intrinsic value because its strike is favourable relative to spot.
- Inter-dealer market — The 'inner tier' in which dealers trade among themselves to manage inventory imbalances; no longer exclusive to banks.
- Intermediary asset pricing — A framework in which exchange-rates are set at the margin by constrained financial intermediaries with limited, time-varying risk-bearing capacity.
- Internalisation — Dealers matching offsetting client flow on their own books rather than hedging externally, which has reduced their share of inter-dealer trading.
- International Monetary Market (IMM) — The CME division whose weekly CFTC futures positioning data are used to proxy speculative positioning in FX.
- Intervention — Central-bank buying or selling of a currency to influence the exchange-rate; most effective when large, public, communicated and aligned with fundamentals.

K

- Knock-in / knock-out option — Barrier options that are activated (knock-in) or cancelled (knock-out) when spot reaches a predetermined level.

L

- Last look — A brief window (typically milliseconds) allowing a liquidity provider to accept or reject a trade request after receiving it; prohibited on primary venues.
- Leverage ratio — A non-risk-weighted Basel III capital constraint capturing both on- and off-balance-sheet exposures and derivative positions.
- Liquidity consumer (seeker) — A participant that takes liquidity to meet an underlying need, such as corporates, real-money investors and retail.
- Liquidity firmness — The certainty of execution at a quoted price; reduced by the use of last look on secondary venues, MDPs and SDPs.



- Liquidity mirage — The appearance of depth created by aggregated but potentially inaccessible liquidity spread across fragmented venues.
- Liquidity provider (LP) — A participant that supplies two-way prices to the market, principally banks and some PTFs.

M

- Market maker / market making — Providing two-way prices and capturing the bid-offer spread; a liquidity-providing service rather than a source of the 'excess returns' from proprietary trading.
- Market microstructure — The study of how the trading process itself conveys information and moves prices.
- Meese-Rogoff puzzle — The 1983 finding that structural exchange-rate models cannot out-forecast a naive random walk out of sample, even given realised fundamentals.
- Momentum — A factor and technical signal based on the tendency of price trends to persist.
- Monetary model — An exchange-rate model based on relative money supplies and interest rates; in modern form it uses the monetary policy reaction function and Taylor-rule fundamentals.
- Multi-dealer platform (MDP) — A venue that lets clients request quotes (RFQs) from multiple dealers at once (e.g. FX Connect, FXall, 360T); it bears no inventory risk.

N

- Net Foreign Assets / Net International Investment Position (NFA / NIIP) — A country's external assets minus its liabilities; a long-run anchor for the real exchange-rate.
- Nominal exchange-rate — The quoted price of one currency in terms of another.
- Non-deliverable forward (NDF) — A cash-settled forward for a restricted currency, settled (usually in USD) against a fixing rather than by physical delivery.
- Non-deliverable option (NDO) — A cash-settled option on a restricted, non-deliverable currency pair.
- Non-reporting banks — Smaller banks that do not report turnover directly to the BIS and largely intermediate for institutional and non-financial customers.

O

- Official sector — Central banks, sovereign wealth funds, international financial institutions (IMF, BIS), development banks and government agencies.
- One-touch / no-touch option — Options that pay a fixed amount depending on whether spot does (one-touch) or does not (no-touch) reach a predetermined level during the option's life.
- Order flow — Signed transaction volume (buyer-initiated minus seller-initiated trades); the proximate driver that aggregates dispersed



information into price.

- Out-of-the-money (OTM) — An option with no intrinsic value (0–25 delta); the band favoured by hedgers.
- Over-the-counter (OTC) — Bilateral trading conducted away from an exchange; the dominant structure of the FX market.

P

- Payment-versus-Payment (PvP) — The simultaneous exchange of both currency legs of a trade, which eliminates settlement risk (e.g. through CLS).
- Positioning — The stock of existing long and short currency exposures held by investors; most useful as a contrarian signal at extremes and a measure of market vulnerability.
- Pre-trade anonymity — The degree to which counterparties remain unknown to each other before execution; highest on exchanges and lowest on fully disclosed MDPs/SDPs.
- Prime brokerage (PB) — An arrangement in which a bank lends its credit standing and trading relationships to a client, separating execution from credit provision.
- Principal Trading Firm (PTF) — A lightly capitalised non-bank firm (e.g. XTX Markets, Citadel Securities) that trades for its own account and also makes markets, often via HFT.
- Programmability — The embedding of automated rules and conditions into money, usually via smart contracts, enabling conditional and synchronised settlement.
- Proprietary trading — Profit-seeking trading with a firm's own capital; heavily constrained for banks by the Volcker Rule.
- Purchasing Power Parity (PPP) — The theory that exchange-rates adjust so that identical baskets of goods cost the same across countries; a powerful long-run anchor but a poor short-run timing tool.
- Put option — The right, but not the obligation, to sell one currency against another at a predetermined strike price.

R

- Random walk — A price process with no predictable drift; the benchmark that FX forecasting models struggle to beat at short horizons.
- Real effective exchange-rate (REER) — A trade-weighted measure of a currency's real value against a basket of trading partners.
- Real exchange-rate — The nominal exchange-rate adjusted for relative price levels; it mean-reverts under PPP in the long run.
- Real interest-rate differential — The nominal interest differential adjusted for expected or realized inflation; the more relevant concept for medium-run valuation (Frankel's real-interest-differential model).
- Real money investors — Mutual funds, pension funds, insurance companies and endowments that buy currencies to satisfy end-user demand for underlying assets.
- Reporting dealers — The major banks that report turnover to the BIS; their



share of spot trading has fallen as internalisation has grown.

- Request for Quote (RFQ) — A request from a liquidity consumer to a liquidity provider for the price to buy or sell a specified currency pair, amount and settlement date.
- Request for Stream (RFS) — A continuous flow of quotes for a specified size and duration (such as a trading day) — in effect a continuous RFQ providing real-time price updates.
- Restricted markets — Currencies subject to capital controls and/or current-account restrictions, often traded offshore through NDFs.
- Risk reversal — The implied-volatility difference between an out-of-the-money currency call and an equivalent-delta put; a gauge of skew and directional demand.

S

- Settlement risk (Herstatt risk) — The risk of paying out one currency in an FX trade without receiving the other, arising from settlement across time zones; mitigated by PvP/CLS.
- Sharpe ratio — A measure of risk-adjusted return, defined as excess return per unit of volatility.
- Single-dealer platform (SDP) — A bank's own electronic platform for direct client trading (e.g. Deutsche Bank Autobahn), integrated within the bank's risk-management function.
- Skew Volume Indicators (SVA/SVO) — Deutsche Bank indicators built from DTCC data that separate investor flows (at-the-money, SVA) from hedger flows (out-of-the-money, SVO).
- Slippage — The difference between the expected price of a trade and the price at which it is actually executed.
- Smart contract — Self-executing code that performs a transaction automatically when predefined conditions are met.
- Spot / FX spot — The purchase or sale of one currency against another for near-immediate delivery, typically settling two business days later (T+2).
- Stablecoin — A digital token that maintains a peg to a fiat currency (about 99% are USD-linked), backed by reserves; a potential conduit for dollar demand.
- Standardised Approach for Counterparty Credit Risk (SA-CCR) — A regulatory approach for measuring counterparty credit risk on derivatives, relevant when comparing cleared and bilateral margining.
- Sterilised intervention — Central-bank intervention offset by a domestic money-market operation so that the money supply is left unchanged.
- Straight-through processing (STP) — The automated, end-to-end matching, confirmation and settlement of trades.
- Strike price — The predetermined exchange-rate at which an option can be exercised.
- Swap Data Repository (SDR) — A CFTC-registered repository to which in-scope OTC FX derivatives must be reported under Dodd-Frank.
- Systematic investor — A manager who translates observable data into positions through a repeatable, rules-based portfolio-construction and risk-management process.



T

- T+1 / T+2 — Settlement one or two business days after the trade date; spot normally settles T+2, while USD/CAD settles T+1.
- Taylor rule — A monetary-policy reaction function relating the interest rate to inflation and the output gap; central to modern monetary models of the exchange-rate.
- Technical analysis — Forecasting future price moves from patterns in past market data rather than fundamentals (e.g. moving averages, momentum, support and resistance).
- Terms of trade — The price of a country's exports relative to its imports; a key real-exchange-rate driver for commodity-exporting economies.
- Tokenization — The process of creating transferable digital tokens that represent claims on underlying money or assets on a shared infrastructure.
- Top-of-book spread — The best (tightest) bid-offer spread available on a venue; used to track FX market quality over time.
- Total Portfolio Approach (TPA) — Managing currency exposure as an integral part of the whole portfolio rather than as a standalone overlay or transaction cost.
- Treasury basis — The yield gap between US government bonds and currency-hedged foreign government bonds; a measure of the dollar's convenience yield.
- Treasury International Capital data (TIC) — US data on gross cross-border purchases and sales of securities, used to split investor turnover between fixed income and equities.

U

- Uncovered Interest Parity (UIP) — The condition that the expected change in the exchange-rate equals the interest-rate differential; strongly rejected in the data, which underpins the carry trade.

V

- Value (valuation factor) — A strategy that goes long undervalued and short overvalued currencies relative to an equilibrium anchor such as PPP or DBBeer.
- Value-at-Risk (VaR) — A statistical estimate of the potential loss on a position or portfolio over a horizon at a given confidence level; used in portfolio hedging.
- Vanilla option — A standard call or put option with no exotic features.
- Voice trading — Trade negotiation by telephone or messaging rather than electronic systems; classified as voice-direct or voice-indirect (via a broker).
- Volatility smile — The pattern of implied volatility across option strikes, conveying the market's perceived balance of tail risks.
- Volcker Rule — A Dodd-Frank provision (implemented 2014–15) banning banks from most proprietary trading, including many foreign-exchange instruments.



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Other Sources

Artificial Intelligence has been used to generate and/or enhance certain diagrams and parts of the text in this handbook, including the cover image.



Appendix 1

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The background features a stylized illustration of a globe in shades of blue. Several chess pieces, including a king, queen, and pawns, are positioned around the globe. Two hands, one in a blue sleeve and one in a red sleeve, are shown moving the pieces. The overall theme is geopolitical strategy.

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The Next 90: Your Geopolitical Risk Radar

Oct–Dec 2026

September 2026

Miha Hribernik | Dr. Helen Belopolsky

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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The Next 90: Your Geopolitical Risk Radar | Oct–Dec 2026



A geopolitical calendar mapping the next 90 days

OCTOBER			NOVEMBER			DECEMBER		
3	Latvia	General election	3	US	Midterm elections (House, Senate, gubernatorial and state legislatures)	3-4	OSCE	Ministerial Council in Switzerland
4	Brazil	General election	4	South Africa	Municipal elections	4	US	Section 232 tariffs (15% ad valorem) and minimum import prices on polysilicon, derivatives and solar products set to enter into force
4	Bosnia and Herzegovina	General election	4-5	Spain	Ibero-American Summit	5	The Gambia	Presidential election
5-28	Canada	General election in Quebec, municipal elections in British Columbia, Ontario and Manitoba	7	New Zealand	General election	5-6	Doha Forum	Qatar's flagship dialogue themed "Redefining Global Trust"
9-17	Czech Republic	Senate and municipal elections	9	EU	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings	10-11	EU	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings
8-9	EU/China	Trade talks in Beijing; EU deadline for "tangible results"	9-20	COP31	COP31 in Türkiye	13	Haiti	General election
12-18	IMF/World Bank	Annual meetings of the IMF and WB in Bangkok	10-12	ASEAN	49 th ASEAN Summit in Metro Manila	14	EU	Foreign Affairs Council meeting
15-16	EU	European Council meeting	18-19	APEC	APEC Economic Leaders' Meeting in Shenzhen	14-15	G20	G20 Summit in Miami
19	Canada	Alberta independence referendum	28	Taiwan	Local elections	17-18	EU	European Council Summit
20-22	Germany	Berlin Global Dialogue	28	Palestinian National Authority	Legislative election	31	OPEC+	Expiry of the current group-wide production framework
24	Slovakia	Municipal and regional elections	30-1 Dec	EU/Argentina	EU-Argentina Investment and Business Forum in Buenos Aires	TBD	US	US Congress is set to adopt annual-must pass defence legislation
24	EU/GCC	Second EU-GCC Summit	TBC	US/North Korea	Potential meeting between Donald Trump and Kim Jong-un	TBD	China	Annual Central Economic Work Conference
25	Bulgaria	Presidential election						
26-29	China	Fifth Plenum of China's Central Committee						
27	Israel	Legislative election						
29-30	EU/Canada	EU-Canada Summit						
30-31	G20	G20 Foreign Ministers' Meeting in Atlanta						

Note: The calendar contains selected key events and is not intended to provide an exhaustive list.

Appendix 1



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Deutsche Bank
Research

Rules of Engagement

Mapping the Next Phase of China-US Ties

September 2026

Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Tactical détente: The near-term outlook



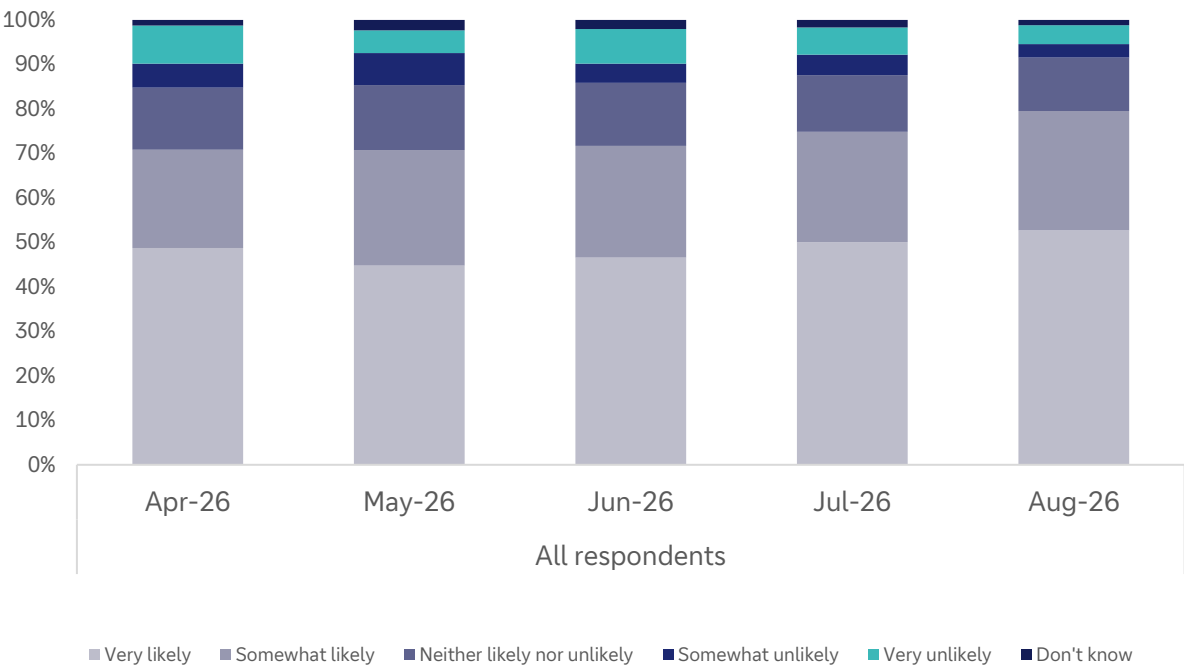
Political commitment and shifting domestic priorities create space for stability into 2027

The Trump-Xi meeting in Washington met our key expectation: It delivered a subdued-yet-constructive summit resulting in a modest number of [concrete deliverables](#) that will minimise the near-term risks of abrupt decoupling, a security crisis, or a disruptive trade war. **This was the best plausible outcome for markets, given the structural nature of China-US competition.** The brief two-month extension of the Busan agreement until 10 Jan 2027 will further delay new tariffs and export controls on both sides. It will also increase pressure to deliver a broader trade and investment package, and further trade-truce extensions, by the end of the year. We anticipate more talks between US Treasury Secretary Scott Bessent and Chinese Vice Premier He Lifeng in Q4 as both sides seek [a larger deal](#).

For China, stability with Washington will likely remain a priority as it manages **rising trade tensions with the EU** and prepares for top leadership reshuffles at its 2027 Party Congress. Combined with shifting US public perceptions of China, these factors **could create near-term space for more dealmaking** on the sidelines of the upcoming APEC (18-19 Nov) and G20 (14-15 Dec) summits. **As our surveys below show, US voters are now more open to increased trade and investment with China, even at the cost of geopolitical concessions**, while **concerns over the cost of living** have intensified. Both trends are now at their highest levels since April.

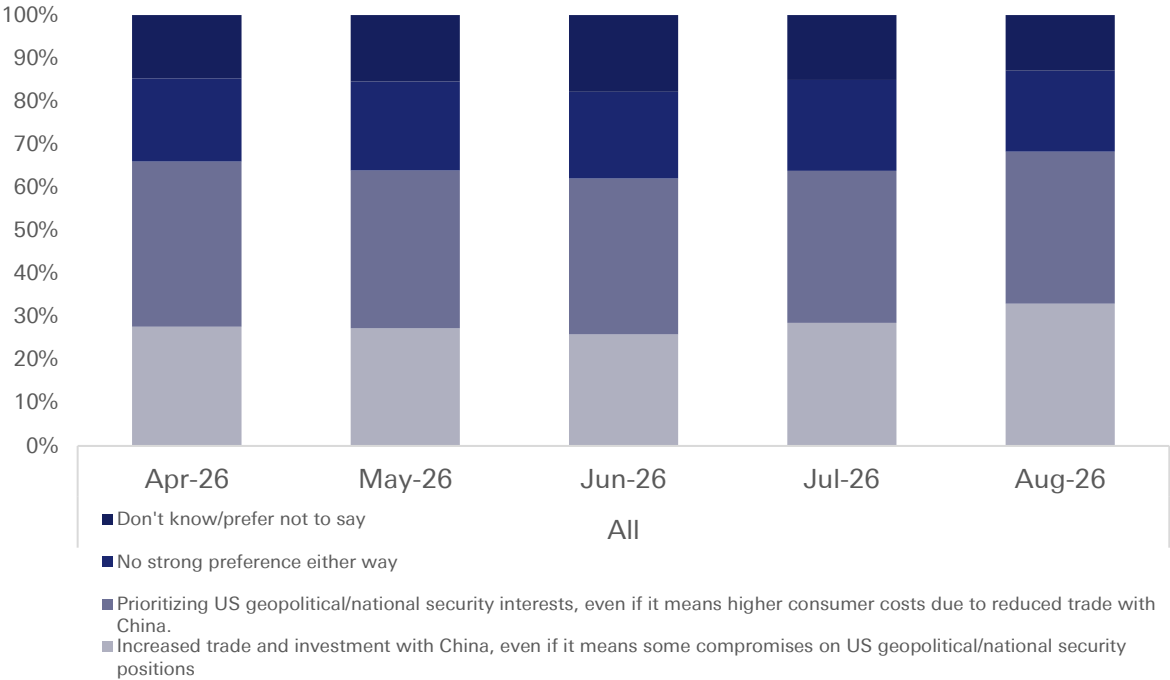
Nevertheless, political guardrails are not a panacea for a relationship that remains characterized by mistrust and competing national security priorities. **As the next slide will show, the outlook for China-US relations becomes more complex beyond this near-term horizon....**

US participants were asked: How likely or unlikely is it that current cost-of-living concern will influence your vote in the midterms?



Source: dbDataInsights. Responses collected April-August 2026 from an Online Nationally Representative sample of ~600 US respondents.

Which of the following outcomes for US relations with China do you most prefer?



Source: dbDataInsights. Responses collected April-August 2026 from an Online Nationally Representative sample of ~600 US respondents.

The policy horizon: Key events to watch (2026–2029)

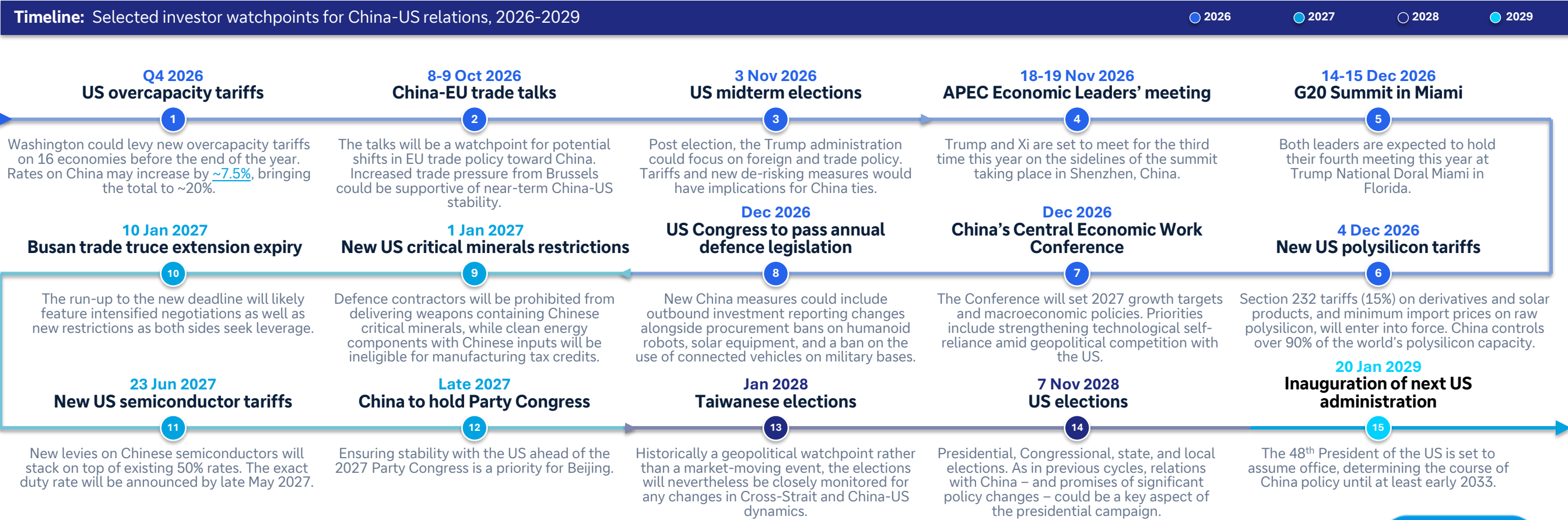
The next several years will see more summits, tariffs, major political events, and key elections



An analysis of the geopolitical calendar reveals a complex slate of tariffs, import curbs, export controls, and sensitive political events. Even as leader-level diplomacy has reduced the risk of a near-term breakdown, it has done little to stem the tide of new bilateral economic restrictions. These have proliferated during 2026 to include new export controls, sanctions, and tariffs (see [list](#)), with significant implications for high-growth industries including drones, AI, electric vehicles, and robotics.

Competition is unlikely to slow, and events could repeatedly test the fragile trade truce. Q4 2026 will see further China-US dialogue and China-EU trade talks. Should talks with Brussels fall short of averting increased European trade pressure, Beijing may further prioritise near-term US stability. But the coming months will also bring challenges in the form of new **US overcapacity tariffs and polysilicon levies**, while next year will usher in **curbs on critical minerals and additional semiconductor tariffs**. The run-up to the **10 Jan Busan truce** expiry could also bring new economic restrictions as both sides seek negotiating leverage. Both countries will also hold major political events (**China's Party Congress in 2027** and **US elections in 2028**), with potential implications for bilateral ties, while **Taiwan's Jan 2028 election** will be another key watchpoint.

But these are just the events whose outcomes and impacts we can estimate in advance. **Understanding probable China-US developments that are not on the calendar is equally important for investors.** We should assume that competitive dynamics across myriad policy issues will drive other developments and policy changes, often announced with little or no advance notice. **We examine some of the key issues in the next slide.**



Structural strains: Testing the guardrails

Navigating the trade, technology, and regional pressure points

Unless consistently managed, structural competition could move markets and test stability through the end of the decade and beyond. We highlight five watchpoints for markets:

- 1. The AI race:** Dialogue is set to strengthen AI safety guardrails, but national and economic security imperatives mean it is unlikely to significantly slow the China-US AI race, much less meaningfully reverse existing restrictions. Any attempts to pace AI development would touch on questions of tech restrictions, including Beijing's perceptions of US technological containment, which could increase amid potential restrictions on [remote access to AI chips](#) and Chinese [data centre components](#).
- 2. Tariff pressures:** Trade tensions have subsided given a lower US trade deficit and the new Board of Trade, which is set to reduce tariffs on US\$60 billion of non-sensitive goods. But significant [differences](#) persist: Washington plans to unveil new **polysilicon and semiconductor tariffs**, as well as **overcapacity levies** on 16 economies (including China). The US is also seeking to restrict Chinese inputs in North American supply chains and target alleged **transshipment**, while Beijing shores up its supply chain dominance in areas of US import dependence, such as **advanced drone components** and processed **critical minerals** – which continue to be a source of [friction](#).
- 3. Regulatory toolkits:** The wider tech sector remains central to expanding export controls and restrictions. China is strengthening its toolkit. Its 2026 codification of **anti-derisking regulations** strengthens Beijing's ability to take action over foreign sanctions or export controls. Meanwhile, its **outbound investment regulations**, effective 1 Jul, cover technology, data, talent and capital. In the US, Congress is drafting legislation to change outbound investment reporting requirements for **biotech and biopharma**, restrict government procurement of **humanoid robots and solar equipment**, and ban the use of **China-linked connected vehicles** on military installations.
- 4. Geopolitical stress-tests:** The **war in Iran** is unlikely to derail relations in the near term, but the conflict has [strained](#) bilateral ties. Any potential broadening of the conflict in scale and intensity, or a further expansion of US [sanctions](#) that impact Chinese interests, could test guardrails. Additionally, while the probability of conflict in the Taiwan Strait remains low, differences over **Taiwan** could remain unresolved across successive administrations in both Washington and Beijing into the 2030s and beyond.
- 5. Political anchors: Two domestic events will shape relations into the 2030s.** Beijing's priority is stability ahead of the late **2027 Party Congress**. For China, dialogue can help reduce the risk of costly escalation, potentially slow the pace of new US economic restrictions, and buy time for efforts to strengthen self-reliance in strategic sectors such as semiconductors (where China is aiming for **80% self-sufficiency by 2030**).

In the US, a durable bipartisan consensus on China could make visible concessions to Beijing increasingly difficult ahead of the **2028 presidential election**. Recent bipartisan criticism of the Trump administration's decision to allow limited Nvidia H200 chip sales to China reflects the political constraints facing a transactional approach to China, raising the prospect that guardrails will strain closer to election day.



Source: istockphoto

Appendix 1



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The end of the safety net: Life without the macroeconomic stabilisers

28 September 2026

For several decades, financial markets and the wider economy benefited from an implicit safety net. If a slowdown or a downturn arrived, there was a monetary put from central banks, and a fiscal put from governments. The monetary put would see interest rates slashed and liquidity injected via QE. The fiscal put would see governments run larger deficits to cushion any downturn. These policy stabilisers smoothed out the business cycle, and supported asset valuations too.

These policy interventions were increasingly normalised from the late-1980s, and became more aggressive with each passing cycle. This reached its peak during the Covid-19 pandemic, when governments ran fiscal deficits unprecedented in peacetime, whilst many central banks expanded their balance sheets with huge asset purchases.

But after an era of ever-bigger interventions, these old stabilisers now face major obstacles. On the fiscal side, policy is constrained today by sovereign yields at multi-year highs, alongside the highest debt-to-GDP ratios in several decades. Then on the monetary side, policy is constrained by persistent inflation that's increasingly supply-driven.

This represents a big shift from the last few decades, but it does echo previous regimes where the macroeconomic stabilisers were either blunted or left unused. A good example is the 1970s experience, when stagflation and supply shocks posed a dilemma to traditional policy tools. Even further back, the pre-WWI environment of the classical gold standard and balanced budget orthodoxy saw governments play a very different and more limited role in macroeconomic stabilisation.

Looking forward, this environment means policymakers will have a much more difficult time stabilising the business cycle in the coming years. So absent a sharp fall in yields or a collapse in inflation, higher macro volatility will be here to stay.

Authors

Henry Allen

Macro Strategist



1. Background: How the stabilisers used to work.

Economic policymakers have long used interventions to stabilise output. This became increasingly fashionable after WWII, when Keynesian economics established a more active use for fiscal policy to stimulate the economy in a downturn. This meant budget deficits became a more regular feature of life, rather than used solely for wartime emergencies.

It was from the late 1980s, however, that the extent of policy interventions became increasingly sizeable. Indeed, it was here that the origins of a central bank “put” for financial markets began, in the crisis of Black Monday in 1987. That was a huge stock market crash, where the S&P 500 fell over 20% in a single day, marking the biggest daily decline in its entire history. The size of the falls even went beyond the Great Depression, raising fears of wider contagion, particularly if it were to spark a liquidity crisis that caused financial institutions to fail. However, the Federal Reserve responded by cutting rates and providing liquidity, offering a macroeconomic stabiliser against the prospect of wider economic stress. This offered support against a wider downturn, and the US economic expansion continued until 1990.

Whilst the genesis of a central bank “put” came in 1987, the idea gained further traction in the late-1990s, with the Asian financial crisis in 1997, followed by the Russian financial crisis in 1998. Fears of an economic contraction mounted, particularly after issues at Long-Term Capital Management (LTCM) raised concerns that the US wasn’t immune from these global pressures. So once again, the Fed delivered three 25bp rate cuts pre-emptively, seeking to support the economy.

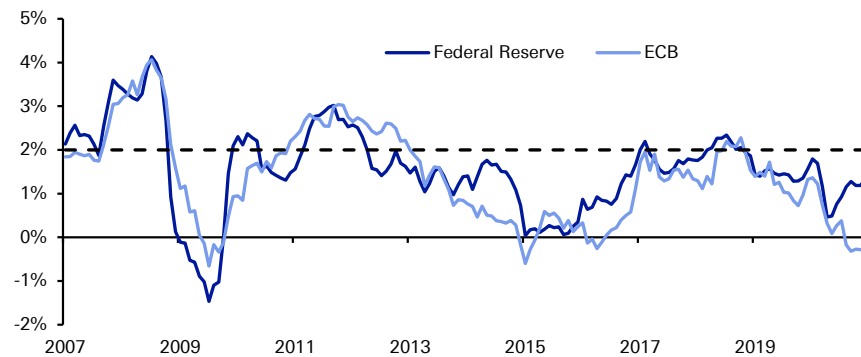
A few years later, the bursting of the dot com bubble in early 2000 set the stage for yet further stimulus. As equity markets declined, the Fed began to cut rates from early 2001, ultimately delivering 475bps of cuts that year that took rates from 6.5% to just 1.75%. That was followed up by further rate cuts in 2002 and 2003, leaving rates at 1% by mid-2003 into 2004. Moreover, the Bush tax cuts were also delivered in 2001 and 2003, acting as a fiscal stimulus as the wider US economy entered recession.

In hindsight, these moves seem modest compared to what then happened in 2008, when the Global Financial Crisis led to even bigger policy interventions. Suddenly, the world faced the biggest economic contraction since the Great Depression, with genuine fears that a repeat might happen. To avoid this, policymakers launched into the biggest economic stimulus ever seen in a peacetime environment. That included massive fiscal stimulus, with the US federal deficit at 9.8% of GDP in 2008, the highest since 1945. Moreover, central banks slashed rates to the zero lower bound, and many like the Federal Reserve launched into Quantitative Easing programmes, purchasing assets to ease financial conditions even more. Over the coming years, many began pledging to keep rates low well into the future, pioneering the concept of forward guidance.

These levels of policy intervention weren’t confined to the immediate crisis however. In the subsequent years, active attempts were made to prolong and extend the business cycle, particularly as the 2010s became an era of low inflation and low interest rates. In this environment, borrowing was historically cheap, and low inflation meant central banks could cut rates without seeing a big overshoot of their targets. Indeed, both the Fed and the ECB undershot their 2% inflation target on average in the 2010s, and the ECB took their own deposit rate into negative territory in 2014.



Figure 1: The Federal Reserve and the ECB frequently missed their 2% inflation targets on the downside in the 2010s (PCE inflation for the US, HICP inflation for the Euro Area)

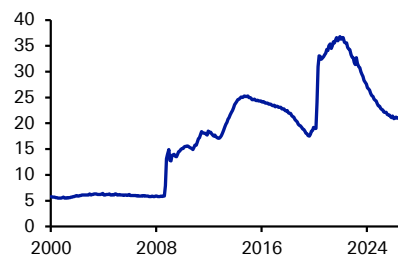


Source : Haver Analytics, Deutsche Bank

By 2019, the year before the pandemic, the intellectual climate had continued to move in the direction of more policy intervention. Government bond yields hit new lows, with Germany's 10-year bund yield at -0.7% in summer 2019. This environment of negative yields meant borrowing was basically a "free lunch", and that summer also saw the Fed begin a 75bp easing cycle, as inflation was below their target and they wanted to sustain the expansion.

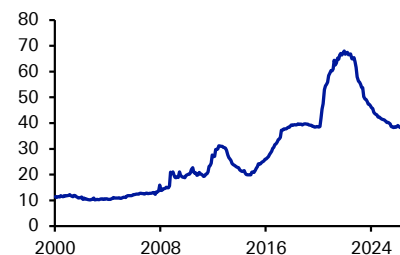
The peak of this policy interventionism was seen with the Covid-19 pandemic in 2020. Just as in 2008, the world was facing a huge and sudden economic contraction. Moreover, the consensus among policymakers was that they could have done more to speed up the recovery after 2008, potentially avoiding years of very high unemployment. As such, 2020 saw the most aggressive economic stimulus packages ever seen in a peacetime environment, well above those in 2008. Central banks immediately slashed rates back to the zero lower bound, pledged to keep rates low for years to come, and launched into major QE programmes again. Then on the fiscal side, record peacetime deficits were run. Indeed, the US federal deficit of 14.5% of GDP in 2020 was the highest since 1945.

Figure 2: Federal Reserve Balance Sheet (% of GDP)



Source : Bloomberg Finance LP, Deutsche Bank

Figure 3: ECB Balance Sheet (% of GDP)



Source : Bloomberg Finance LP, Deutsche Bank



During the pandemic period, many initially expected these interventions to remain for years to come. After all, in mid-2020, financial markets continued to price a continuation of the low inflation and low interest rates of the 2010s. This belief was clear on several fronts, and the 10-year Treasury yield closed at just 0.51% in August 2020. Meanwhile, the Federal Reserve even adjusted its own framework to allow for average inflation targeting, i.e. permitting an overshoot of the inflation target after periods of an undershoot. This reflected the consensus view that inflation was unlikely to be a serious threat.

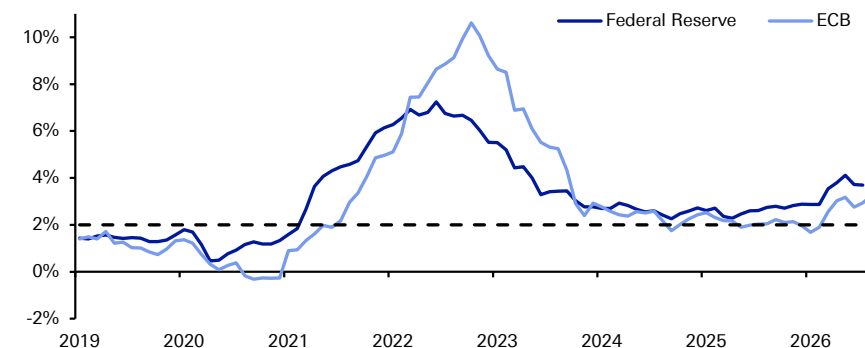
But as we now know, inflation returned in a serious way after the pandemic, forcing central banks into rate hikes. Meanwhile, public debt-to-GDP ratios rose significantly thanks to the pandemic stimulus, helping to push up sovereign bond yields too. In other words, many of the usual policy tools of the previous few decades were either blunted or no longer available.

2. The Monetary Constraint: High inflation means the central bank put isn't available

In the pre-2020 world described above, central banks regularly grappled with how to generate enough inflation. In part, this was thanks to several structural forces that put downward pressure on global prices. For instance, deepening globalisation acted as a positive supply shock that kept costs down. Meanwhile, the expansion of the global labour force in these years also boosted the supply side. In other words, inflation was low and central banks generally had the space to cut interest rates when they needed to.

But since the pandemic, we have entered a very different environment of above-target inflation across many of the world's biggest economies. Initially, that was caused by a huge wave of pent-up demand after the pandemic, alongside disruption to supply chains and labour markets. But the subsequent years also brought further shocks. Those included Russia's invasion of Ukraine in early 2022, driving a big spike in European energy prices, and pushing Euro Area inflation above 10% briefly. In 2026, the closure of the Strait of Hormuz then acted as a fresh energy shock, pushing inflation further away from target and contributing to a fresh tightening cycle from the major central banks.

Figure 4: In the 2020s, inflation has generally been above central bank targets (PCE inflation for the US, HICP inflation for the Euro Area)



Source : Haver Analytics, Deutsche Bank



Alongside those shocks, the structural forces keeping inflation down have gone into reverse. For instance, demographic trends have seen working-age populations shrink in many places. Meanwhile, the trend towards deeper globalisation has stalled, and the focus has turned to near-shoring and supply-chain resilience, rather than locating production in the cheapest place. Simultaneously, other trends like the push for higher defence spending are also adding to demand-side pressures.

Because of all this, the 2020s so far have brought persistently above-target inflation, and central bankers have found it much more difficult to ease policy in response to growth wobbles or financial stability fears.

This initially became apparent in 2022, amidst the biggest inflation wave for several decades. Central banks had to keep hiking rates, even in the face of slowing growth and financial market stress. Indeed, despite the S&P 500 being in bear market territory by mid-2022, the Federal Reserve still launched into a series of 75bp hikes, the biggest hikes since the 1990s. Then in 2023, despite the collapse of Silicon Valley Bank and fears of wider financial contagion, the Fed and ECB still continued with their rate hikes to combat inflation.

That tightening cycle in 2022-23 was a significant moment, because high inflation meant monetary policy could not simply act as a stabiliser against growth or financial stress. After all, policymakers realised that if they didn't act to tackle high inflation, they risked a repeat of the 1970s and the potential for even harsher monetary tightening to get inflation under control again.

Today in 2026, this is a familiar problem again. We are in a globally synchronised hiking cycle, with the Fed, ECB and Bank of Japan all having delivered rate hikes. So far at least, global growth has been robust and financial conditions remain accommodative by historical standards. But because inflation remains clearly above target, central banks don't have the space they did in the pre-pandemic world to simply cut rates at the first sign of difficulty. There is still an inflation mandate to be achieved, and given we have now seen multiple years of elevated inflation, there is a much higher risk of inflation expectations becoming unanchored in that context, particularly if another shock occurred.

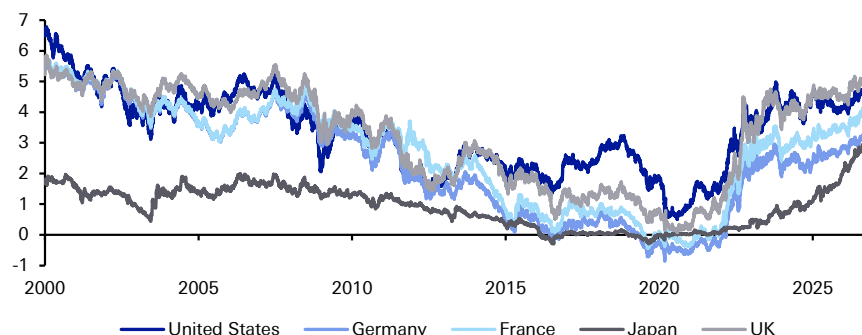
3. The Fiscal Constraint: High yields and high public debt mean fiscal stimulus isn't easily available either

Just as monetary policy is constrained by high inflation, fiscal policy today is also constrained by higher yields and higher public debt. This is a big transition from the world of 1980-2020, when yields were generally falling throughout. Indeed, despite all the fiscal stimulus around the GFC, the continuous decline in yields over the 2010s meant that debt dynamics remained manageable, because lower yields helped to offset the impact of higher debt.

Since the pandemic however, we've been in a very different world, as the dual forces of higher debt and higher yields have made debt dynamics far more unfavourable around the world. And this isn't just a nominal story, as real yields have also risen to multi-year highs as well.



Figure 5: 10yr Government Bond Yields (%) have risen to multi-year highs across several countries

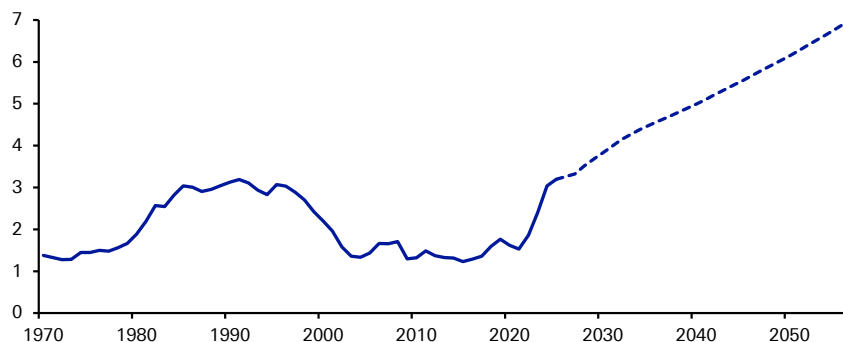


Source : Bloomberg Finance LP, Deutsche Bank

In essence, there are 4 new constraints around fiscal stimulus.

First, is the significant rise in global yields. By definition, the cost of incurring debt has grown, in both nominal and real terms. Relative to the pre-pandemic baseline in early 2020, the 10-year US Treasury yield has risen from 1.9% to 5.2%. Similarly in Germany, the 10-year bund yield has risen from -0.2% to 3.6%. Moreover, with yields already elevated, any further borrowing risks crowding out the private sector, and further dampening growth. And for governments, interest spending has ramped up significantly. For instance, US net interest spending as a share of GDP exceeded its recent peak from the early 1990s in the most recent fiscal year, and the CBO forecast this is set to continue.

Figure 6: US Net Interest Costs (% of GDP) with CBO Forecasts



Source : CBO, Deutsche Bank

The second constraint is that public debt itself is now at a higher level. This interacts with the higher yields mentioned above, because even though yields have only returned to their levels from the mid-2000s, that is much more impactful because they are being paid on a much larger debt pile. Using IMF numbers for consistency, net public debt across the advanced economies has risen from 52% of GDP in 2000, to 80% in 2025, with further increases projected to 2030. This means the fiscal space available to governments is more limited than it used to be.



The third constraint on fiscal stimulus is the high starting level of budget deficits. So to generate a meaningful fiscal stimulus, it would require deficits to be pushed up to even higher levels than they are today. But that in turn would add to the risks of bond market disruption that cascades through the economy, leading to higher borrowing costs for corporates and consumers as well.

The fourth constraint is higher inflation. When inflation is high, any fiscal stimulus that boosts demand risks pushing up prices rather than boosting output, and in turn makes monetary tightening more likely that could offset the impact of the stimulus. This trade-off was apparent in 2022, when central banks were hiking rates aggressively, and even further back in the inflationary environment of the 1970s. In a world of high inflation and supply constraints, fiscal stimulus is more likely to overheat the economy.

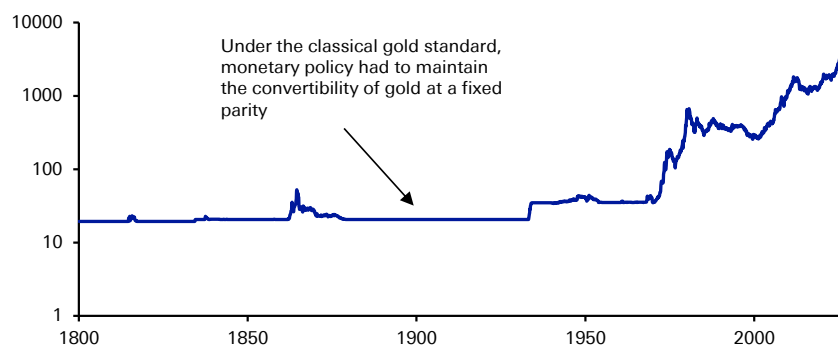
4. Are there any other eras of history where the stabilisers weren't available?

Whilst history never quite repeats exactly, there have been other eras in which the traditional policy stabilisers were either unavailable or severely blunted, which hold some lessons for our own time.

The pre-WWI classical gold standard

Before WWI, the international monetary system was known as the gold standard, in which currencies were convertible into a fixed quantity of gold. In practice however, this acted as a huge constraint on macroeconomic stabilisation. After all, monetary policy had to maintain the convertibility of gold at a fixed parity. This meant contemporary options like QE weren't available in a crisis. Meanwhile on the fiscal side, governments generally adhered to balanced budget orthodoxy. This was before Keynes wrote the General Theory, so the modern understanding of countercyclical deficit spending was not in place. In the US, there wasn't even a central bank until the Federal Reserve's establishment in 1913.

Figure 7: Gold Prices (US Dollars per ounce, log scale)



Source: Finaeon, Deutsche Bank

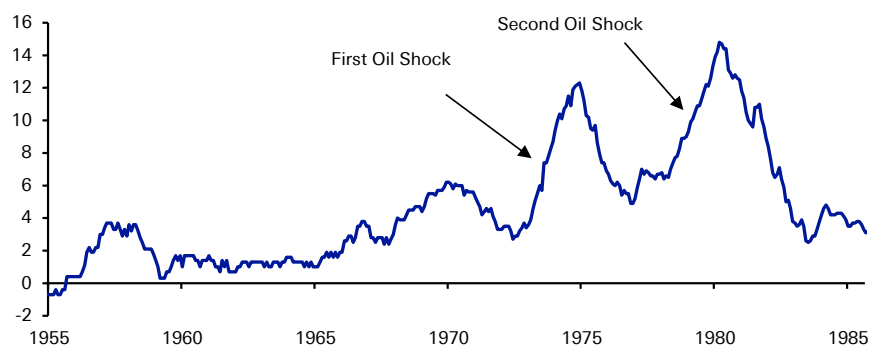


Without modern policy stabilisers however, this was also an era of huge macroeconomic volatility. There was a major financial panic across Europe and the US in 1873, which marked the start of the so-called “Long Depression”. In the US, that contraction lasted until 1879, which in length was even longer than the Great Depression contraction of 1929-33. And this was followed by other major panics, including the 1893 panic that led to another US contraction, along with the 1907 panic that saw the Dow Jones almost halve relative to its 1906 peak.

The stagflationary 1970s

The more recent example comes from the 1970s, another era of multiple supply shocks. The biggest were the two oil shocks in 1973 and 1979, which both represented negative supply shocks that pushed up inflation and hurt growth in the US and Europe. However, this meant policymakers faced a dilemma. If central banks cut rates, that would add to the inflationary pressures and risk worsening the wage-price spirals already underway. Meanwhile, if fiscal stimulus were pursued, that would also push up inflation.

Figure 8: US CPI (year-on-year % change)



Source : Haver Analytics, Deutsche Bank

This context meant that policy couldn’t smooth out the business cycle as successfully as it had in other decades. Indeed, in the decade from the first oil shock in 1973, the US experienced 3 separate recessions. That is significantly more turbulent than the last 40 years, when there have been just 4 recessions in total, or once a decade on average. For financial markets too, the 1970s were a terrible period for asset performance, with both equities and bonds losing ground in real terms across multiple countries.

5. Implications

If economic policymakers can’t use traditional monetary and fiscal policy tools to stabilise output and smooth the business cycle, then this will result in higher macro volatility.

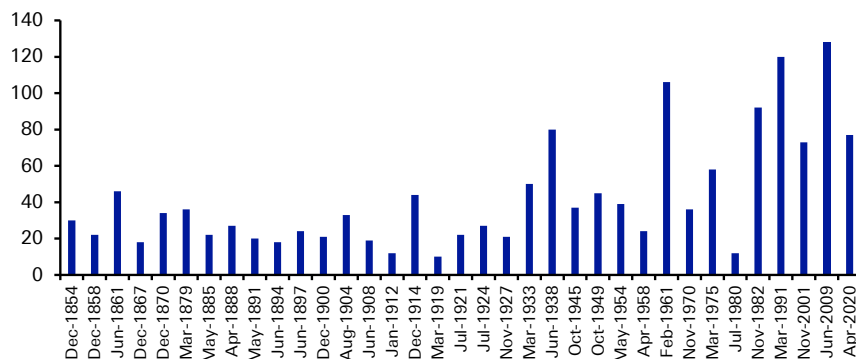
If we look back at the last 40 years, it is notable that the 5 most recent US expansions all rank in the top 7 longest expansions since business-cycle dating began. This has several causes, and some of it reflects a move away from an agriculture-based economy where output was dependent on factors like the weather. But a key role has also been played by pre-emptive policy interventions, such as cutting rates to stop a slowdown turning into a recession. Indeed, recent examples include the dovish shifts in monetary policy after the 1987 stock market crash, the 1997-98



financial crises, and the mid-cycle slowdown of 2015-16.

Beyond the actual policy interventions, these moves also had a confidence effect that supported financial markets. The fact that policymakers stood ready to take action helped create a virtuous circle, as optimism on the economic outlook boosted financial asset values, supported wealth effects, and avoided a tightening in financial conditions that would itself cause growth to slow.

Figure 9: Length of US economic expansions (in months) from the starting point



Source : NBER, Deutsche Bank

Clearly, policy is not as constrained today as in some periods of history. There is no longer a gold standard where the convertibility needs to be maintained. And unlike the 1970s, inflation is not in double-digits. Nevertheless, this is still a more constrained world than we have been used to in the last 4 decades. Higher inflation and more frequent supply shocks are constraining monetary policy. Meanwhile, higher yields and higher public debt levels are constraining fiscal policy. We have already seen the impacts since the pandemic, most notably in 2022, when central banks hiked rates to combat the inflation surge, and there was a simultaneous selloff across bonds and equities of a kind unseen in modern times.

Looking forward, this environment will offer new challenges of the sort we haven't been used to in recent decades. The policy trade-offs are now more acute, and if pre-emptive stabilisation is no longer available, then higher macro volatility lies ahead.



Appendix 1

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German Defense: An Ecosystem in Transformation

September 25, 2026

This is the first section of a report available to clients via our Deutsche Bank Research site [here](#).

Launching Private Company Analysis with the German Defense Sector

Traditional analysis that focused on public companies overlooked a critical market segment: the many important competitors that are privately owned, typically by families or private equity firms. To address this, we are launching dedicated analysis of Europe's private-market giants, starting with Germany, where a traditionally dominant private sector means large firms outnumber public ones almost two to one. We begin with the German defense sector, where understanding the private ecosystem is essential: it is the source of critical innovation, the solution to supply-chain issues, and the venue for a future wave of M&A, consolidation, and IPOs.

From Incremental Modernization to Industrial Scaling

Germany's defense landscape is undergoing its most significant transformation in decades, shifting from a paradigm of 'incremental modernization' to 'industrial scaling'. Triggered by the Zeitenwende in 2022, the new era is underpinned by substantial financial commitments, including a €100bn special fund and constitutional reforms enabling sustained long-term spending. The objective is no longer merely to fill discrete capability gaps but to execute a full 'ecosystem build-out' that addresses structural choke points and enhances industrial capacity. This transformation is creating a more resilient and scalable German Defense Industrial Base (DIB), prepared to meet surging domestic and European demand.

Technology, Integration, and the Rise of New Players

This strategic pivot is also reshaping the industrial and technological environment. EU initiatives (EDF, EDIRPA, ASAP) are fostering joint funding and integrated supply chains, while the market is experiencing a 'next wave' of technological change, with a clear shift in value towards drones, software-defined systems, electronic warfare, and AI-driven capabilities. These dynamics force industry consolidation but also create openings for agile start-ups, which are vital for innovation, as speed and software become paramount. This report provides an in-depth primer on this evolving sector. We identify Germany's industrial strengths and technology gaps, analyze the key private players poised to benefit, and examine the structural themes shaping the market - from funding dynamics and procurement reforms to technology roadmaps.

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Half the Industry is Invisible

We are Changing that.

Equity research has always looked at the listed half of the economy. That half is shrinking. Companies stay private for longer, the largest capital pools are now private, and in Germany the listed universe was never the industry to begin with. **There are roughly 120 listed German companies with sales above €1bn - and roughly 200 unlisted ones.** In our home market, large private companies outnumber public ones almost two to one.

That gap is not a curiosity. It is an analytical problem.

If a sector's innovation, its supply-chain frictions, its pricing behavior and its eventual consolidation sit largely outside the listed universe, then covering only the listed names means covering only part of the argument. With this report we start closing that gap. We are launching dedicated, ongoing analysis of Europe's private-market giants, beginning in Germany.

Why we are starting with German defense

Defense is the sharpest version of the problem we are trying to solve. Germany is in the middle of its largest rearmament cycle since reunification, but much of the capability being bought is not primarily produced by the listed defense groups. It is produced by an ecosystem: family-owned system houses that have supplied the Bundeswehr for decades, venture-capital-backed companies that did not exist three years ago, and the tier-2 and tier-3 suppliers who sit on the actual constraints - energetics, seekers, qualified electronics, test capacity, cleared labor. That ecosystem is where the innovation happens, where the supply chain either breaks or scales.

Understanding it is not a side-project to covering Rheinmetall, Hensoldt or Renk. It is a prerequisite.

Executive Summary - Eight Points Carry this Report

1. **The spending shift is structural, not cyclical.** The €100bn special fund was the headline; the 2025 constitutional reform exempting defense spending above 1% of GDP from the debt brake was the substance. Government budgets now foresee more than €150bn of annual defense spending by 2029, against roughly €52bn of core budget in 2024. NATO's 2025 pledge of 3.5% of GDP on core defense by 2035 puts a floor under the back-end.
2. **Budgets are not revenue.** Between an approved budget and a delivered capability sit parliamentary approval for anything above €25m, tender design, contracting, qualification, test capacity and acceptance. The February 2026 procurement acceleration reform helps, but the conversion chain - not demand - sets the delivery slope.
3. **Value is moving to software, autonomy and effects.** Drones, software-defined systems, electronic warfare and AI-enabled decision layers are taking share from platform-centric spend. That favors companies whose iteration cycle is measured in weeks, and it is the clearest reason private and venture-capital-backed firms now sit inside mainstream force design.
4. **Production is deterrence - and production is where Germany is short.** Scaling breaks at specific points: energetics permits, seekers and guidance electronics, qualification burden, specialized labor, and limited



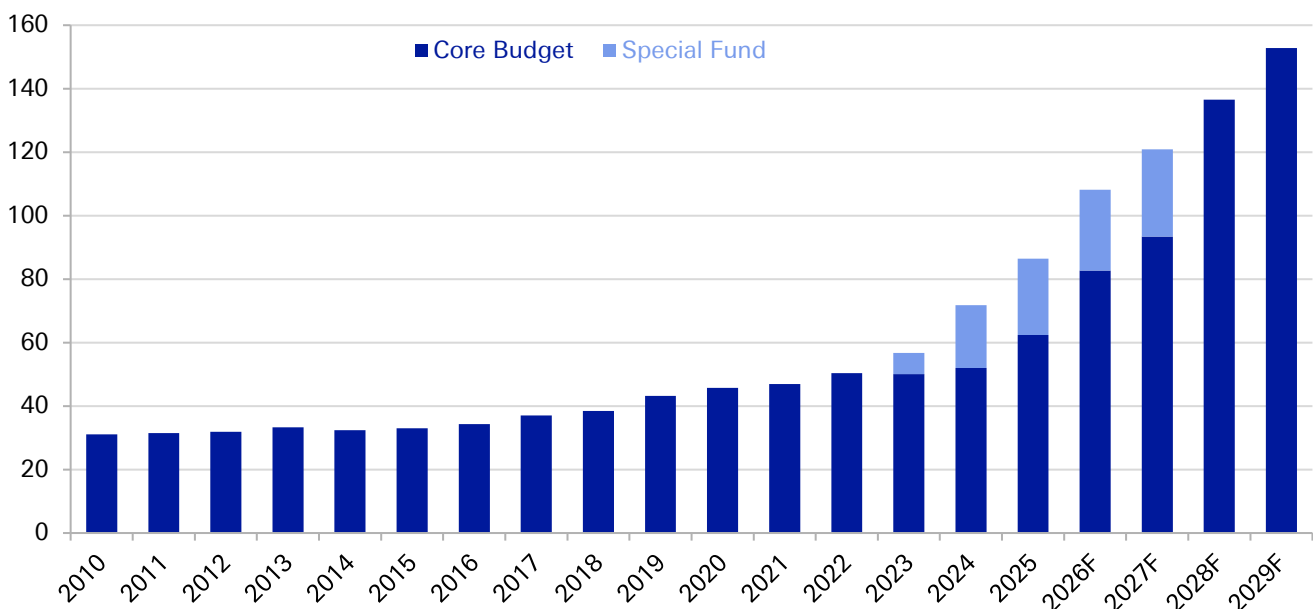
test windows. Almost every one of those nodes is occupied, at least in part, by a private company.

5. **The private ecosystem is now procurement-relevant, not just innovation-relevant.** The February 2026 Bundeswehr loitering-munitions framework, awarded to Helsing and STARK from an initial package of roughly €540m, is the clearest proof point. Sovereign buyers are contracting with companies two to five years old.
6. **Scale still sits with the incumbents.** Diehl generated €5.4bn of sales and €703m of EBIT in FY2025, with Defence at roughly €2.3bn; Rohde & Schwarz passed €3bn of revenue in FY2024/25 with no bank debt. These are not start-ups with a defense angle - they are among Europe's more significant defense industrial assets, and they happen to be family-owned.
7. **Regulation is the binding constraint, in both directions.** Export licensing, foreign investment screening, NATO interoperability and compliance intensity limit scaling and exits. But policy also accelerates - fast-track procurement, EU joint programs, state-backed vehicles for defense tech.

Why defense, in numbers

The money. Germany's core defense budget has moved from roughly €47bn in 2021 to a projected path above €150bn by 2029, with the €100bn special fund bridging the middle of that ramp. The 2025 debt-brake reform matters more than the fund itself: by exempting defense spending above 1% of GDP, it converts a one-off program into a sustained baseline. Procurement approvals have already responded: 97 major defense projects approved in 2024 against 55 in 2023, with over €58bn allocated.

Figure 1: German defense budget (€bn) - government budgets foresee >€150bn annual spending by the end of the decade

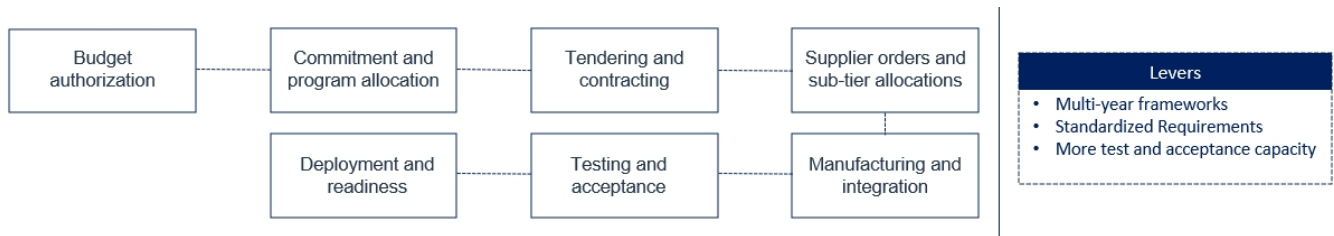


Source : German government, Deutsche Bank research



The conversion chain. What is in doubt is throughput. Contracts above €25m require binding Bundestag budget-committee approval, and tendering, supplier orders, integration, testing and acceptance each add time. Europe's fragmentation compounds it: European states operate around 179 different major weapon systems against 33 in the US, and 19 different main battle tanks against one. Between 2020 and 2024, European arms imports rose 155%, and European NATO members sourced 64% of those imports from the US, up from 52% in 2015-19. Whether 'buy European' becomes policy rather than rhetoric is the single largest swing factor for the domestic industrial base.

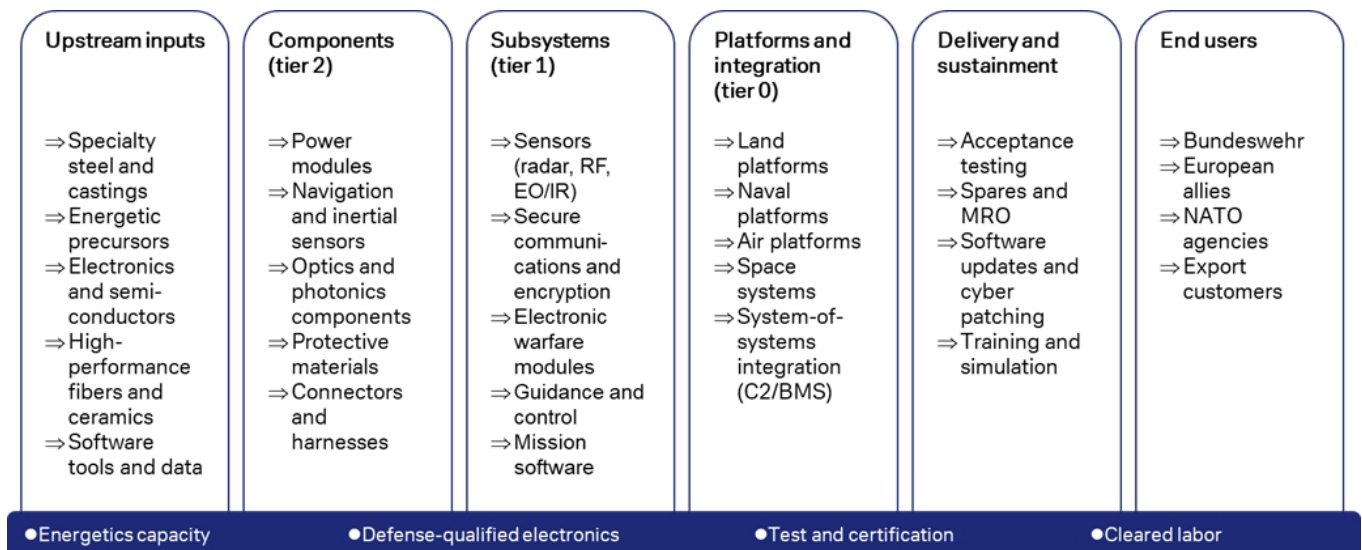
Figure 2: Budget to Output: The Conversion Chain



Source : Deutsche Bank research

Where the private sector sits. The constrained nodes are consistent across almost every program: energetics capacity, defense-qualified electronics, test and certification, and cleared labor. Private companies are disproportionately present in exactly those nodes. Diehl's acquisition of Dynamit Nobel was, in substance, the acquisition of explosives-related operating authorizations it describes as unique in Germany. STARK's dependence on TDW for German-manufactured warheads is the same story one tier down. Rohde & Schwarz occupies the secure-communications and SIGINT/EW layer no platform program can route around.

Figure 3: German Defense Industrial Value Chain

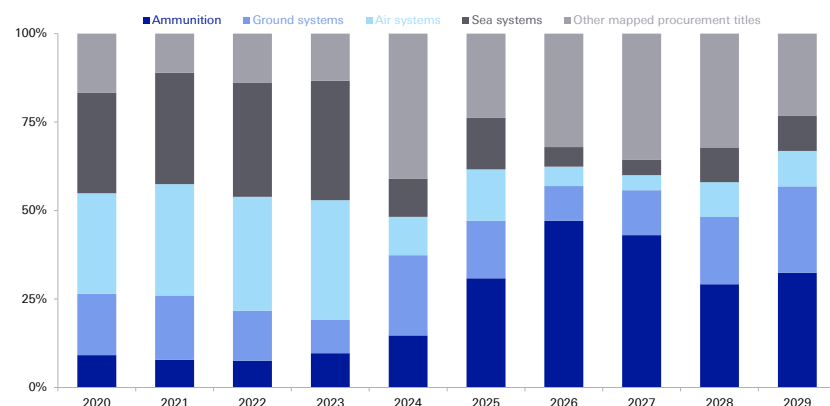


Source : Deutsche Bank research



Where the new money is going. Germany attracted close to €1bn of defense and security venture-capital funding in 2025, second only to the US. The deal mix since 2021: unmanned systems (34 deals), surveillance and space (24), other defense tech (24), electronic warfare (10) and cyber (8). Helsing raised \$1.8bn at an \$18bn valuation in July 2026; STARK Defence raised €500m in June 2026 above €3.5bn. Capital of that scale changes the industrial map.

Figure 4: Spending Percentage of Germany on Equipment by Category



Source : BMVG – Budget

The five debates - and where we go next

1. **Can Germany convert budget into capability fast enough?** Demand is settled; throughput is not. Watch acceptance capacity, qualification timelines and framework design rather than headline budgets.
2. **Do the newcomers industrialize, or stall at the factory door?** Helsing, STARK Defence, ARX Robotics and Quantum Systems are all making the same transition from software and prototypes into serial hardware. That is where defense start-ups historically fail. The next 18 months are the test.
3. **Do incumbents get disrupted, or absorb?** The likelier outcome is a denser partner network - Diehl with Hensoldt and Airbus, Helsing with Saab and Kongsberg, ARX Robotics with RENK and Daimler Truck. Partnership maps are becoming a better guide to competitive position than product catalogs.
4. **Does 'buy European' become real?** A conscious substitution of even part of the 64% US import share would be a structural repricing of the European industrial base. Fragmentation and interoperability standards cut against it.
5. **Who is financed, and how?** Rising capital intensity, thin private disclosure, constrained exit routes under foreign-investment screening, and uneven ESG treatment will all influence the financing and consolidation wave.



The Argument in Five Parts

1. The Strategic Landscape: Russia's 2022 invasion of Ukraine, combined with the distancing of the US as Europe's protector, represents a turning point for the safety of Europe and its defense industry. The shift in policy is creating immense demand across Europe for arms in the decades to come, leading to significant M&A and investing opportunities. German companies are uniquely suited to serve the rising demand due to their immense technical know-how and their already-existing industrial sector. Combined with targeted funding programs, these developments both stimulate existing companies and incentivize start-ups that push the technical frontier forward.

2. New Wars Change the Old Battlefield: Recent wars in Ukraine and the Middle East have reshaped modern warfare, shifting capital from traditional contractors to start-ups that can keep pace with rapid innovation cycles. Key trends defining this new era include: unmanned systems creating a transparent battlefield that demands adaptability; huge munitions consumption proving that production itself is deterrence; the battlespace expanding into commercial space and cyberspace; and societal resilience against disinformation becoming a vital security element. This all occurs amid a new great power competition, heightening the risk of multi-theater wars and the use of nuclear threats to back conventional aggression.

3. How to Gain Exposure – Investment Opportunities in Private Gems: Rapid innovation in modern warfare is creating significant investment opportunities in Germany's defense sector, as start-ups outperform slow, traditional procurement. This operational shift is fueled by a long-term surge in European defense spending targeted at R&D and equipment. Investors can gain exposure through key 'private gems', from established firms like Diehl Group and Rohde & Schwarz to high-growth start-ups like Helsing (AI) and Quantum Systems (drones). These start-ups leverage a mix of venture capital and government grants with innovative business models, making the German private defense market a compelling investment area.

4. Trend – Europe Buys European & Europe Buys between Middle Powers: Recent energy and security shocks have shattered Europe's post-Cold War assumptions, creating an urgent push for strategic self-reliance amid uncertainty over the long-term US security guarantee. This pivot is evident in new security partnerships, such as Germany deepening defense ties in the Indo-Pacific (Japan) and North Atlantic (Canada). Paradoxically, Europe's rearmament has increased its reliance on US hardware (64% of imports '20-'24) and its defense sector remains highly fragmented. The core challenge is whether Europe's quest for autonomy will lead to a 'buy European' policy; doing so would unlock a monumental internal market, driving consolidation and creating a truly independent defense capability.

5. The Regulatory Landscape & Investment Constraints: The defense industry is uniquely constrained and regulated, and investors must understand its landscape. Growth is not driven simply by demand but is filtered through government budgets, procurement laws, and export controls. This creates risks: budgets don't translate to immediate revenue, market access can be denied politically, and production cannot be scaled freely. High compliance costs and national security reviews that limit exit strategies add further complexity. Successful investment, therefore, requires mastering these political and regulatory realities, not just identifying demand.



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Research

The House View: Snapshot

A broadening tightening cycle

Deutsche Bank

Macro views

World

- Global growth of 3.3% in 2026 and 3.4% in 2027.
- **Rationale:** AI investment supports activity despite the energy shock. Brent is forecast to average \$95/bbl in Q4 2026, with supply normalisation expected from Q1 2027.
- **Key risks:** Prolonged Middle East conflict, higher energy costs and tighter financial conditions. Upside risk from stronger AI investment and productivity.

Europe

- Euro-area GDP upgraded to 0.8% (prev. 0.5%) in 2026; 1.2% (prev. 1.1%) in 2027. Growth has been surprisingly resilient in the face of the energy shock.
- **Rationale:** Consumption, defence and AI spending, modest fiscal easing and bank lending have cushioned the energy shock.
- **Key risks:** A worsening energy shock, higher borrowing costs and vulnerable public finances around elections could lift risk premia.

China

- GDP growth lowered to 4.6% in 2026; 2027 remains 4.5%.
- **Rationale:** Exports and AI support growth, but consumption, investment and property remain weak. Fiscal spending is picking up; PBoC rates remain unchanged.
- **Key risks:** Delayed fiscal support, further property weakness and weaker external or AI-related demand.

United States

- 2026 & 2027 GDP growth forecast at 2.2%.
- **Rationale:** Easy financial conditions, fiscal policy and AI-driven capex support growth. Unemployment is expected near 4.1% into year-end; PCE inflation stays above 3% into 2027.
- **Key risks:** A larger oil shock weakens real income; financial conditions tighten; or tighter labour market and AI productivity surprise on the upside.

Germany

- GDP growth forecast at 1.0% in 2026 and 1.3% in 2027.
- **Rationale:** Strong H1 offset by loss of purchasing power caused by the energy shock, although expansionary fiscal policy continues to provide tailwinds.
- **Key risks:** Energy exposure, higher financing costs and political uncertainty could delay the improvement.

India

- 7.3% y/y GDP for FY27 (prev. 6.7%). FY28 growth outlook unchanged at 7.2%.
- **Rationale:** Growth remains strong even as higher oil prices raise inflation and external-balance sensitivity. Current policy looks too accommodative, with strong growth and rising headline inflation (above 6% through Oct-Dec 2026); 100bps of rate hikes starting in Oct 2026.
- **Key risks:** Persistently high oil prices, weaker global demand and tighter domestic financial conditions.

Key downside risks

- H** **Middle East conflict.** Possibility that the war can extend into 2028 pushes Brent forecast to \$130-150/bbl, leading to higher stagflation concerns.
- H** **Inflation:** Higher energy costs, AI investment, fiscal expansion, and higher defence spending could require more monetary tightening.
- M** **Political and fiscal risks.** European elections and fragile public finances could increase risk premia. US cost-of-living concerns remain important ahead of the midterms.

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September 2026

Deutsche Bank AG/London

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Market views

	Market Sentiment	– Risk assets remain resilient despite rate hikes and higher yields. – The global economy resilient despite geopolitical conflict and energy risks, supported partly by the AI investment boom. – Key question is how long activity can withstand high energy costs, rising rates and limited fiscal space.
	Equities	– S&P 500 YE-2026 target: 8000; STOXX 600 YE-2026 target: 670. – Earnings recovery has broadened beyond technology. – Overweight mega-cap growth & tech, financials, industrials, and metals & mining.
%	Rates	– YE-2026: 10Y UST at 4.80%; 10Y Bund at 3.20%. – Neutral duration after a long-standing bearish bias. Equilibrium rates and term premia remain structurally higher.
	Credit	– YE 2026: \$IG 85bp; \$HY 315bp; €IG 95bp; €HY 345bp. – Spreads expected to widen modestly. – Prefer US over European credit amid stronger US growth and greater ECB risk.
	FX	– YE: EUR/USD: 1.17; USD/JPY: 150. – Neutral EUR/USD with a mild upside bias.
	Oil & Gold	– Q4-2026: Brent: \$95/bbl; WTI: \$90/bbl. – Supply normalisation expected from Q1 2027, leading Brent to average \$74/bbl in 2027. – Gold: YE-2026 target \$4,600/oz. Higher real yields and a more hawkish Fed weigh near term; 2027 average forecast is \$5,100/oz.
	Monetary Policy	– Fed: Two 25bp hikes in Dec and Mar, taking policy rate to 4.375%. – ECB: 25bp hikes in Dec, taking the deposit rate to 2.75%. – BoJ: Two 25bp hikes in Jan and Apr, bringing policy rate to 1.75%. – BoE: Two 25bp hikes in Nov and Feb, taking Bank Rate to 4.25%. – PBoC: Unchanged at 1.40% through 2026.

Key macro & markets forecasts

GDP growth (%)			Central bank policy rate (%)				Key market forecasts			
	2026F	2027F		Current	Dec 26F	Jun 27F		Current	Dec 26F	Jun 27F
Global	3.3	3.4	US: Federal Funds Rate	3.875	4.125	4.375	US 10Y yield (%)	5.11	4.80	4.75
US	2.2	2.2	Eurozone: Deposit Facility Rate	2.50	2.75	2.75	EUR 10Y yield (%)	3.55	3.20	3.20
Eurozone	0.8	1.2	UK: Bank Rate	3.75	4.00	4.25	S&P 500	7,706	8,000	-
Germany	1.0	1.3	Japan: Policy Balance Rate	1.25	1.25	1.75	Gold	4,288	4,600	5,000
Japan	0.7	0.9	China: 7d OMO Rate	1.40	1.40	1.40	EUR/USD	1.14	1.17	-
UK	1.3	1.4					USD/JPY	158	150	-
China	4.6	4.5					Oil WTI (USD/bbl)	91	90	67
							Oil Brent (USD/bbl)	102	95	70

2026 Macro events calendar

October 2026			November 2026			December 2026		
28	CA	Bank of Canada Decision	3	US	Midterm Elections	9	CA	Bank of Canada Decision
28	US	Fed Decision	5	UK	Bank of England Decision	9	US	Fed Decision
29	EZ	ECB Decision				17	UK	Bank of England Decision
29-30	JN	Bank of Japan Decision				17	EZ	ECB Decision
						18	JN	Bank of Japan Decision



Recent editions

- [A broadening tightening cycle](#), 23 Sep 2026
- [1999 meets 1990](#), 9 Jun 2026
- [The Impact of the Middle East](#), 10 Mar 2026

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Sea change: the market implications of the threat to the maritime commons

September 25, 2026

The sea is overwhelmingly the most important conduit for global commerce. It is used to transport 80% of the world's trade in goods and 90% of the trade in oil. Even in a world in which services trade is likely to grow in importance owing to AI, the world's oceans remain supreme, with submarine fibre optic cables carrying close to all of the world's internet traffic.

Painstaking multilateral negotiations after 1945 established a legal architecture to enable the almost frictionless movement of commerce across the seas and delimit nations rights' to restrict freedom of navigation. This process learned from the hugely damaging experience of two world wars, and preceding centuries of maritime warfare which had devastating effect on international trade and finance.

Recent years have seen a growing threat to this paradigm. The frequency of maritime supply shocks is increasing at an alarming rate. After the COVID pandemic generated significant friction in maritime supply chains, geopolitics has come to the fore, including Russia's Black Sea blockade in 2022, Houthi attacks on commercial shipping in the Red Sea in 2023, a resurgence in piracy off the Horn of Africa and most recently the twin blockades of the Strait of Hormuz and Bab el Mandeb Strait. This has dovetailed with an apparently reduced US willingness to uphold freedom of navigation, perhaps as part of a broader trend away from supporting the 'rules of the game' that underpin an open international economy. In short, the sea is increasingly a theatre of geoeconomic competition, a trend that could accelerate further with climate induced changes to Arctic maritime routes.

We argue that economists and market participants should be more alive to these risks. The lexicon of tariffs has dominated the market's attention over recent years, but tariff induced supply shocks have proved far less damaging than physical supply shocks. Disruption to ocean commerce has generated outsized impacts on both the prices of goods and the volumes of cross border trade.

We distinguish between three forms of maritime shocks of increasing intensity. From localised disruption that can generate short lived effects on prices, to more serious regional disruptions like the one at present in the Gulf, to global events (disruptions with the capacity to generate a global recession). In the latter category, we argue that the Indo Pacific Region, especially the South China Sea, East China Sea and Taiwan Strait, increasingly qualify as a major risk. We end our report with an analysis on the sensitivity of different economies to maritime shocks, noting that East Asian and European nations stand out as by far the most vulnerable.

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Introduction

It is difficult to overstate the importance of the sea when it comes to world trade. Maritime transport moves four fifths of all goods traded world wide.¹ Over 90% of exported oil is seaborne.² After energy, coal, iron ore, grain and manufactured goods make up the remaining 60% of all seaborne cargo. The 'container revolution,' starting from the late 1960s, dramatically improved the handling efficiency of cargo, and has been seen as one of the primary drivers of global trade growth during the second half of the 20th century.³ The sea doesn't just ship goods, its floor carries electricity and data, with fiber optic submarine cables carrying 99% of the world's international internet traffic.⁴

If the world's oceans were the arteries for the world's second great globalisation from 1945 onwards, the third decade of the 21st century have seen their disruption. The last six years have seen a dramatic rise in maritime based supply shocks.

The 2020 pandemic challenged supply chains globally. This included container shipping, which faced public health restrictions causing a large share of the global carrier fleet to be held in port. More recently, geopolitical events have come to the fore. Russia's 2022 invasion of Ukraine was accompanied by a maritime exclusion zone in the Sea of Azov and parts of the Black Sea, as well as illegal Russian attacks on merchant shipping, paralysing trade in the region. In response to Israel's military activities in Gaza following the 7/10 attacks, the Houthis began illegal attacks on commercial shipping transiting the Red Sea, which resulted in a severe fall in trade volumes. After an international task force formed in 2009 successfully curbed Somali pirate attacks around the Horn of Africa, recent years have seen a resurgence of hijacking. And this year has seen the twin shocks of Iran's closure of the strait of Hormuz and the Houthi blockade on Saudi shipping.

As a result, risk based analysis of maritime choke points has proliferated. We argue, however, that disruptions of this kind should be seen not as isolated risk events, but as part of a trend. An understanding of the legal and historical basis of maritime commerce suggests that the rules of the game are changing. Near frictionless maritime passage is being disrupted by simultaneous challenges to its legal architecture and, more crucially, the US and its Western partners' willingness to enforce these rules. The Trump administration's recent toying with tariffs on the strait of Hormuz, a break from established international norms, is but one example.

There has been far more oxygen dedicated by markets to discussing the impact of tariffs in recent years by the economic community, but the impact of tariffs on trade are relatively trivial compared to the potential disruption caused by maritime trade blockages. Indeed, in this piece, we argue that physical supply shocks to trade can be among the most disruptive types of economic event. The global shipping industry is highly sophisticated, but its network can also prove fragile. We identify three forms of supply shock of increasing intensity: shocks that increase freight rates, regional blockades with some scope for rerouting or substitution of maritime routes, and global disruptions to the freedom of the seas.

1 [UNCTAD](#) statistical bulletin, April 2025

2 [Stanford Energy](#), March 2025

3 Bernhofen et al, "Estimating the effects of the container revolution on world trade." [Journal of international Economics](#), January 2016

4 [United Nations](#)



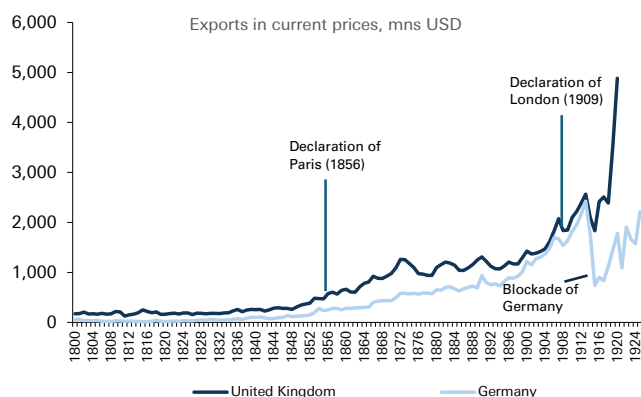
Our work over the last eighteen months has built a framework to understand the implications of shifts in US political economy for the international economy. In particular, we have examined how the US's increasing unwillingness to act as a 'benevolent hegemon' upholding the international order could affect markets, focusing on the dollar, global imbalances and interest rates, and the more frequent use of economic coercion.⁵ Shifts in the maritime arena are yet another example of this shift. Increasingly, the sea is becoming a theatre of geoeconomic competition, while the 'Pax Americana' that has thus far mostly guaranteed free movement across the seas is coming under threat. We end the report with specific case studies on the potential implications of a disruption to commerce around critical maritime routes.

A short history of maritime law and war

"The bare principles of international maritime law are no more than a reasonable compromise between a belligerent's contention, that he should be permitted to stop an enemy's commerce, wherever it is to be found, and the neutral's rejoinder, that he is no-one's enemy, and he has a right to trade and traffic with any country with which he is at peace." **"Official Foreign Office Account of Britain's Economic Blockade of Germany, 1914-1919," published 1937**

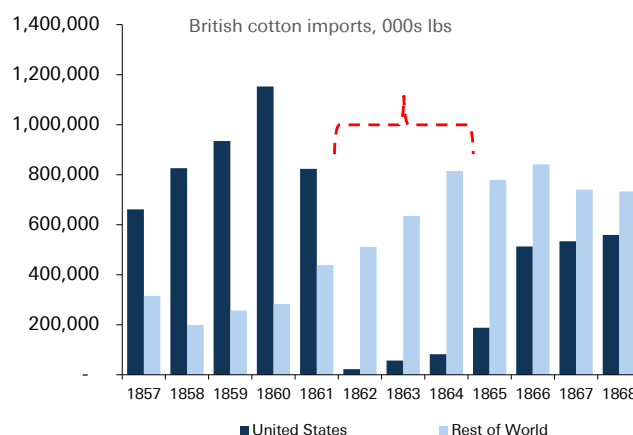
The maritime blockade dates back to the Peloponnesian Wars in 405 BC, but it was not until the early 19th century that a recognisably modern framework of the rights of belligerents and neutrals, and legal basis around maritime commerce, started to be drawn up. The blockade and counter-blockade of Napoleonic France saw precedents established for what was acceptable for 'prizes,' namely the ships and goods that warring parties were allowed to seize and confiscate, the law of 'blockade,' and other important concepts.

Figure 1: Growing trade in the 19th century incentivised the creation of the legal architecture of maritime commerce, which was sharply disrupted by the First World war



Source : Deutsche Bank, Federico, G. and Tena-Junguito A. (2019): World trade, 1800-1938: a new synthesis. Revista de Historia Económica-Journal of Iberian and Latin America Economic History, Vol 37, n.1. <https://doi.org/10.1017/S0212610918000216>

Figure 2: The Union's blockade on confederate shipping devastated the South's cotton exports



Source : Deutsche Bank, data from Powell, J. "Cotton, Liverpool and the American Civil War," PhD thesis October 2018

5 See, [FX Special Report](#), "The Unruly Hegemon: what a decline in US leadership means for markets," 13th March 2025, [FX Special Report](#), "How Defence Spending will reduce global imbalances and weaken USD," 6th May 2026 and [EM Special Publication](#), "Geoeconomics 101: implications for the dollar, emerging markets and beyond," 21st January 2026



These were developed further during the mid-19th century as the rapid rise in global trade led merchants, manufacturers and governments to push for greater protections for mercantile shipping. The 1856 Declaration of Paris provided more stringent legal definitions of blockade and what constituted the contraband of war (trade between neutrals and belligerents that assisted the war effort and could therefore legitimately be seized). This was developed upon by the 1907 Hague Conference. Britain, then the preeminent economic and maritime power of the day, took the lead at these conferences, but also pursued a strategy of Imperial Defence in the late 19th century based on the fortification of its coaling stations and ports across the globe, as well as developing a 'two-power' naval standard,' whereby its naval strength would be at least as powerful as its next two rivals combined.

This underscores that while the establishment of legal norms helped maritime commerce become more predictable and orderly, ultimately their survival depended on the willingness of their signatories to enforce them. Maritime economic warfare during the 19th and 20th centuries often saw these norms suspended. The results could be devastating. Britain's economic blockade of Germany during WW1 debilitated the German war effort as well as causing mass starvation. It saw a ~65% collapse in German exports based on trade data from the Federico Tena database (figure 1). The Union's blockade of Confederate Ports during the American Civil War crippled the confederacy's war effort, with export volumes to Britain falling by close to 100%.

After the disruption caused by the second world war, the growth in global trade over the second half of the 20th century once again motivated the codification of rules of the sea. Successive UN conferences from the mid 1950s onwards culminated in the 1982 UN Convention on the Law of the Sea (UNCLOS), which settled several ambiguities in international law. Among these were geographically defined territorial rights for maritime states, including the twelve mile territorial limit, and the concept of innocent passage, as well as the status of international straits and transit passage. Another increasingly important provision of UNCLOS was the establishment of a legal framework governing the laying and protection of undersea cables.

Although the Convention was ratified by almost every industrial nation, the United States has remained a notable exception. Initially, this was due to concerns in the Reagan presidency that to do so would transfer US claims on seabed ownership. Since then, the US has come close to ratification under Presidents Bush and Obama. Nevertheless, as we shall see below, up until recently the US has largely upheld and enforced the broad provisions of UNCLOS through its Navy and Coast Guard as accepted customary law, with the exception of Part XI of the agreement which regulates the mineral resources of the seabed. Indeed, the US has been described as the 'Convention's best supporter,' despite not ratifying the treaty.⁶

Some important principles in present day maritime law

Territorial waters. The sovereign territory of coastal states. Under the 1982 UN Convention on the Law of the Sea this was defined as up to 12 nautical miles from the low water mark of a state's coast. Some ambiguities exist with respect to defining this limit, including whether states utilise continuous baselines to define their coast, or straight baselines (mapping coastal points from a to b in a straight

⁶ Leah Cole, Lawfare, Not Warfare, [Harvard Kennedy School report](#)



line).⁷

International strait. A passage of water used for international navigation between one part of the high seas or exclusive economic zone and another. Where the strait is narrow enough such that it is entirely made up of certain states' territorial waters, the law of transit passage applies. This law provides freedom of movement for ships and aircraft, provided their journey is for the purpose of continuous and expeditious transit. The concept of transit passage was recognized in the 1982 UN Convention in the Law of the Sea.⁸

Innocent passage. The concept of innocent passage is more restrictive than transit passage. It provides a right of passage through the territorial sea of all states, "so long as it is not prejudicial to the peace, good order, or security of the coastal State." Innocent passage can be suspended or safe routes established if there are risks to pollution, smuggling, security concerns etc. The regime of innocent passage was recognised in the 1958 Convention on the Territorial Sea.⁹

Blockade. A blockade constitutes a maritime or airborne operation to prevent enemy and neutrals from entering an airport, port or coastal area. There is no established body of international law that defines a military blockade, but military custom generally agrees on several principles.¹⁰ A blockade must be declared and notified to all states, including details such as geographical scope. A blockade needs to be 'effective,' in other words appropriately policed by military or other means. Finally, it must be impartial, and not applied on an ad hoc or discriminatory basis to neutral vessels of different states. A 'maritime exclusion zone' or 'maritime interdiction zone' constitute less severe versions of blockades. In the former case, they are announced by states as part of military operations whereby enemy ships in the area are liable to be fired upon. In the latter case, they are usually introduced as part of counter-terrorism, piracy or smuggling operations, whereby ships are liable to be boarded and searched as part of operations.

Freedom of the seas and the Pax Americana

Despite certain flaws, UNCLOS is widely regarded by practitioners, governments and scholars as one of the most successful multilateral treaties ever created. Among its achievements have been the regulation of international maritime trade, the de-limitation of coastal territorial and economic claims and ultimately the preservation of the right to free navigation on the high seas.

The United States, along with other Western partners, have been active enforcers of UNCLOS. In the late 1980s, the Reagan administration launched Operation Earnest Will to protect commercial shipping from mines and missiles fired by Iran in the Gulf of Hormuz.¹¹ More recently, the United States was the largest contributor to the multinational anti piracy Operation Ocean Shield in the Horn of Africa between 2009-2016. Outside of military operations, one of the main ways through which the United States has freedom of the seas is through Freedom of Navigation Operations (FONOPS), which involve the transiting of naval vessels through certain stretches of water such as the Taiwan strait in order to push back

⁷ More details on legal [definition](#).

⁸ More details on legal [definition](#).

⁹ More details on legal [definition](#).

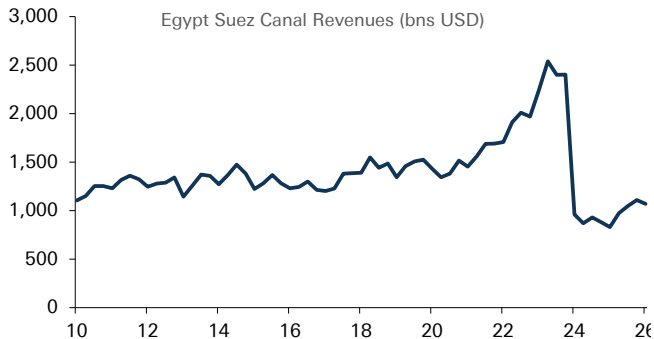
¹⁰ See [Naval College Review of Theories of Blockade](#), Winter, 2021

¹¹ [Revisiting the Tanker War](#), WOTR, March 2024



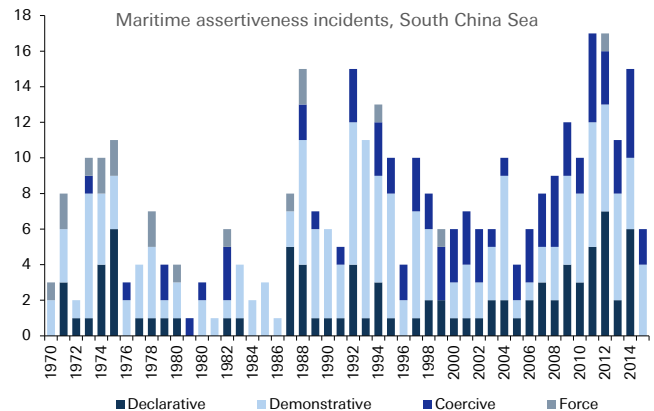
against territorial claims or other navigational restrictions seen as out of line with international law.¹²

Figure 3: Egypt's Suez Canal revenues show little recovery from their collapse in late 2023 due to Houthi attacks in the Red Sea



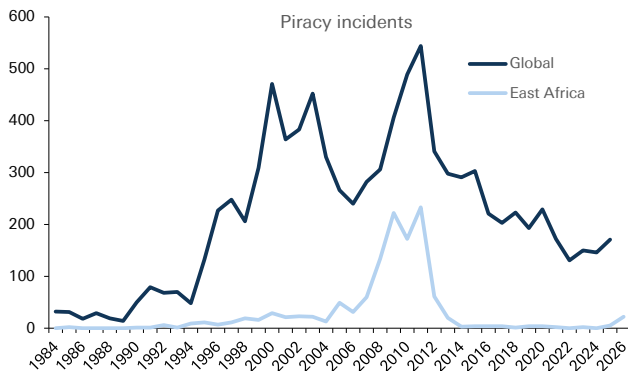
Source : Deutsche Bank, Haver Analytics

Figure 4: Territorial incidents in the South China Sea have picked up markedly in recent years



Source : Deutsche Bank, Maritime Awareness Project

Figure 5: After a decade of calm, piracy incidents in East Africa are picking up again



Source : Deutsche Bank, Piracy Reports compiled by IMO

Figure 6: Damage to the C-Lion 1 submarine cable has been ascribed to sabotage



Source : TeleGeography Submarine Cable resource

Prior to the Iran war, the most visible challenge to freedom of navigation in the Middle East region was the start of Houthi attacks on commercial shipping in the Red Sea. Although initially focused on ships with unspecified 'links' to Israel, these attacks broadened out to incorporate a wide range of ships, dramatically reducing traffic flows through Suez since 2023. A useful visual manifestation of this are Egyptian Suez revenue numbers which have still to meaningfully recover from the drop of traffic (figure 3). Operation Prosperity Guardian, the taskforce launched by the United States was largely ineffective in restoring commercial traffic. This

¹² for an example of FONOPS, see US Navy [website](#). The US, and other international navies, also enforce UNCLOS through the conduct of counter-terrorism, counter-drug smuggling and environmental operations



blockade was subsequently expanded by the Houthis in July this year to incorporate attacks on Saudi vessels and energy infrastructure.

Another important development has been the growing assertiveness of actors in the South China Sea. Data from the Center for Strategic and International Studies suggests that occurrences of maritime assertiveness have dramatically increased over the last decade (figure 4). The CSIS notes that even if incidents have moved away from the outright use of force towards more coercive and demonstrative actions, China, the Philippines and Vietnam have each sought to both claim and counter each other's military/security, fisheries and natural resource rights.¹³ Tension has been especially pronounced in the Taiwan strait, with China objecting to FONOPs from the US, UK, New Zealand, Australia and Germany in recent years, while both Taiwan and China have conducted military exercises in the strait.

Still further threats to maritime security have come from non-state actors or as a result of asymmetric activities designed to undermine or challenge commercial networks. One of these has been the resurgence of pirate attacks off the Horn of Africa, with 15 reported incidents so far this year after joint-coalition operations had all but quashed Somali pirate attacks over a decade ago (figure 5).¹⁴ Another extremely important area of maritime connectivity that is seemingly under threat concerns undersea cables and pipelines. After incidents such as the sabotage of the Nord Stream 2 pipeline in 2022 and more recently telecommunications cables such as the C-Lion in the Baltic (figure 6), there is increasing focus among policymakers as to the vulnerability and economic cost of attacks on sub-sea infrastructure.

Why is Hormuz so important? The blockade of the straits of Hormuz began in early March this year, shortly after the instigation of US and Israeli military attacks on Iran. Iran's blockade, the US counter-blockade and the subsequent MoU adopted in June this year (in practice suspended later that month), represent major departures from the rules based maritime framework enforced by the US and other Western allies.

First, by deliberately targeting neutral vessels, Iran's 'blockade' was a clear violation of both UNCLOS and the established military practice of blockade. This is simply the latest of numerous examples of Iran's flouting of international law and transit rights through the strait, albeit probably the most severe recent example.

Second, while the US's initial blockade on Iranian ships headed to or departing from Iranian ports had a solid legal basis, the subsequent expansion of this blockade at a global level, including ships of any flag state deemed to be carrying contraband of war, has been widely considered to be legal overreach.¹⁵

Third, Iran's plans to impose a transit toll on vessels through the strait of up to 7% of cargo value is a violation of transit rights established under UNCLOS. Iran is not a signatory to the convention, and it has argued that its actions are permissible under the narrower definition of the rights of innocent passage. Nevertheless, even under the narrower 1958 Convention on the Territorial Sea to which Iran is a signatory, this action is considered by maritime law experts to be illegal.¹⁶

¹³ [NBR Special Report 99](#), May 2022

¹⁴ Al Jazeera, 31st August 2026

¹⁵ See the following [legal analysis](#).

¹⁶ For a detailed analysis, see July 2026 [Al Jazeera report](#)



Most surprising, however, has been President Trump's apparent endorsement of a toll based system, to apparently enable the US to recoup the cost of military operations on Iran. Although some analysts have pointed to navigational or tug fees as offering a potential precedent for these actions, this is misleading. Navigational assistance or tugs are supplied for the safe passage of commercial vessels (e.g. in busy shipping lanes with physical obstacles) and consistent with UNCLOS.¹⁷ By contrast, the US's willingness to countenance transit tolls as an acceptable resolution mechanism to the crisis in turn calls into question the White House's support for the post 1945 architecture. This commitment has also been challenged by other actions taken by the Trump administration, including the recent push for deep sea mining on the US continental shelf, in violation of UNCLOS's provisions for the deep sea bed being a common global good.¹⁸ Still a further concern with respect to the US's posture towards maritime security has been the dramatic reduction in US Navy FONOPS in the strait of Taiwan, which have fallen more than 50% from the previous Biden and Trump administrations.¹⁹

Looking ahead, it seems increasingly likely that the sea becomes another theatre of geoeconomic competition, which may in turn be motivating US policy actions. When it comes to Greenland and the Arctic, for example, climate induced changes to maritime routes, especially the possible emergence of viable routes through the Northwest Passage and Transpolar Sea Route, could have informed recent US policy shifts.²⁰ Similarly, there have been reports that Russia is considering withdrawing from UNCLOS on account of territorial claims over the Arctic. UNCLOS governs the Arctic, although a number of states have recently sought to extend the limit of their claims on the continental shelf in that region.²¹

Further examples of increased US attention on strategic maritime considerations in its hemisphere were the respective export blockade on Venezuela earlier this year, whose legality was a subject of discussion although ultimately fell under the umbrella of sanctions enforcement, fuel blockade in Cuba and attention directed towards the Panama Canal.

From global commons to economic warfare?

Markets should care about these developments for the simple reason that supply shocks emanating from major disruptions to maritime trade and digital routes can have severe global economic ramifications. In previous research we have discussed how shifts in US policy over recent years have called into question the maintenance of 'global public goods,' such an open international economic environment. Much in the same way as the Trump administration's increased use of economic coercion using tariffs has undermined confidence in multilateral trade institutions and the 'rules of the game' when it comes to settling trade disputes, should the US cease to enforce international maritime law, the risk could be that a 'race to the bottom' ensues in which state actors can leverage maritime coercion for political or economic purposes.

17 FT Alphaville's coverage of this issue in an August 3rd report appears to disregard widely accepted international customary maritime law as well as UNCLOS

18 See, [Lowy Institute Report](#), 24th October 2025

19 See "[Breaking Defence](#)" August 2026

20 Greenland will be strategically important for both of these routes, which may have motivated US territorial claims. See [Belfer Centre](#), January 2025

21 See [Arctic Institute](#) report April 2024

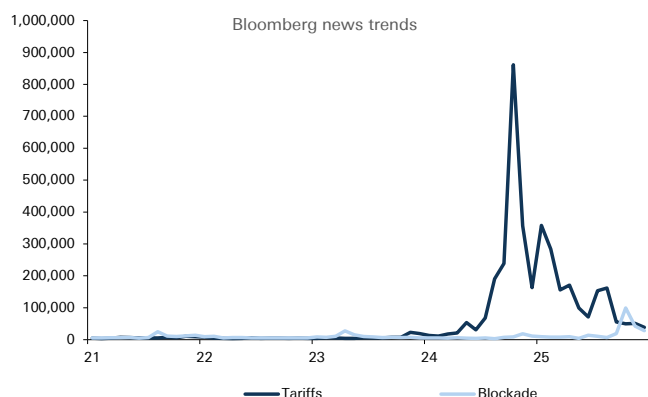


Surprisingly, maritime trade disruptions has still meaningfully lagged behind tariffs in the market lexicon (see Bloomberg news trend for 'tariffs' and 'blockade' figure 7). Recent years has shown that physical supply shocks have larger economic effects than tariffs, however. It is difficult to detect a protracted impact on US import prices, for example, from US tariffs imposed either in the first or second Trump administration, but the impact from the pandemic, Russian invasion of Ukraine and more recently the Hormuz crisis has been much more pronounced (figure 8). More formally, the IMF recently modelled the impact of port delays on US inflation at port and product level, finding that a one hundred hour delay in ports lifts US inflation by 50bp over the following five months.²²

The economic impact extends well beyond price effects. For countries that are especially exposed to maritime imports, with limited substitutability in the near term for affected products, import volumes will also be hit. This was evident during the recent disruption to energy markets in Asia, which saw Bangladesh, South Korea, Thailand and Vietnam imposing energy rationing, and work from home requirements ordered in Indonesia and Malaysia. Figures 9 and 10 track import volumes in Korea and China and clearly demonstrate the impact both on energy imports (first chart) and overall import volumes (second chart) of recent maritime disruptions. In Pakistan, the force majeure declared by Qatar on LNG deliveries has forced Pakistan to instigate widespread load shedding on its electrical grid.

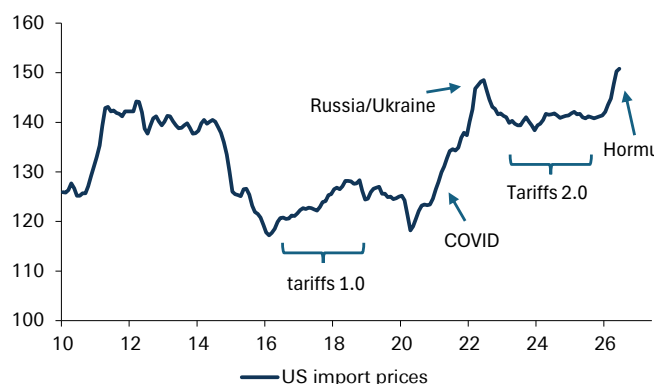
At a global level, the IMF's downgrades to global growth on account of tariffs last year amounted to 0.2% over two years, but its downgrade in the aftermath of the Ukraine war was a cumulative 1%.²³ In short, then, in a world in which supply shocks have become the dominant concern for economic policymakers, the risks to maritime commerce should occupy the first rank of worry.

Figure 7: Maritime disruption has featured a lot less than tariffs in the market's mind



Source : Deutsche Bank, Bloomberg Finance LP

Figure 8: Even as physical supply disruptions have had larger economic effects than tariffs



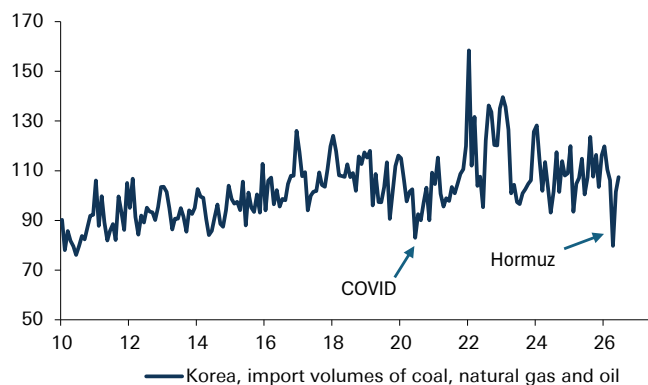
Source : Deutsche Bank, Haver Analytics

22 IMF [Working Paper](#), "From Ports to Prices," February 2026

23 IMF WEOs April 2022 and October 2025.

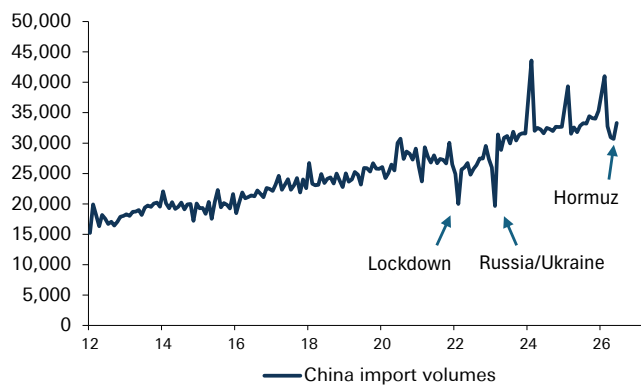


Figure 9: Physical supply shocks have also had material effects on import volumes, especially in Asian economies such as Korea...



Source : Deutsche Bank, Haver Analytics

Figure 10: ...or China



Source : Deutsche Bank, Haver Analytics

Assessing the importance of maritime shocks

With this in mind, the first step for market participants to be better prepared for maritime shocks is to distinguish between their severity. We distinguish between three broad kinds of shocks, and identify some salient characteristics of each.

1. Localised maritime disruption, factors generating higher insurance costs, country specific sanctions.

Examples of the least severe, but still relevant, maritime supply shock would include the sanctioning of Russian seaborne oil by the United States and EU after 2022, Houthi attacks on commercial shipping in the Red Sea in 2023 or pirate attacks off the Horn of Africa.

More limited maritime shocks of this nature can be disruptive, but their economic effects are generally short lived owing to the capacity of the international shipping industry to substitute routes or exporters or otherwise manage risks. They are more likely to manifest themselves in terms of **price** rather than **volume** effects.

Important transmission mechanisms for prices are increased freight rates, which in turn owe to higher insurance costs or demand bottlenecks. This was evident, for example, during the surge in Somali piracy attacks in 2011 which resulted in increased costs for the shipping industry due to ransom and kidnap insurance, but in many cases did not stop ships transiting close to the horn of Africa.²⁴ In these more localised disruptions, shipping companies can choose to re-route. This was the case following the Houthi attacks in 2023 which saw numerous vessels choose to take the lengthier Cape of Good Hope route from Europe to Asia and is reflected in the recent rise in ship ton miles relative to ship tons transported internationally according to UN data (figure 12). Similarly, in response to Russia's blockade of the Black Sea, Ukraine was able to successively reroute a proportion of its grain exports via the Danube or overland routes.²⁵ Still a third option available to ships in these more limited disruptions include re-flagging vessels. Russia has employed re-flagging of oil carriers as part of its 'shadow fleet' following Western sanctions.

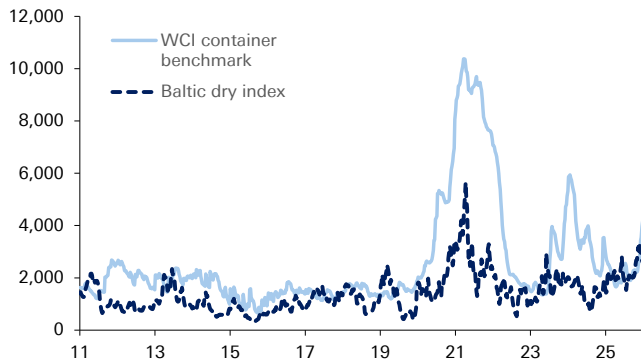
²⁴ See 'From the Oceans' report [here](#)

²⁵ [Council for Foreign Relations Report](#), February 2024



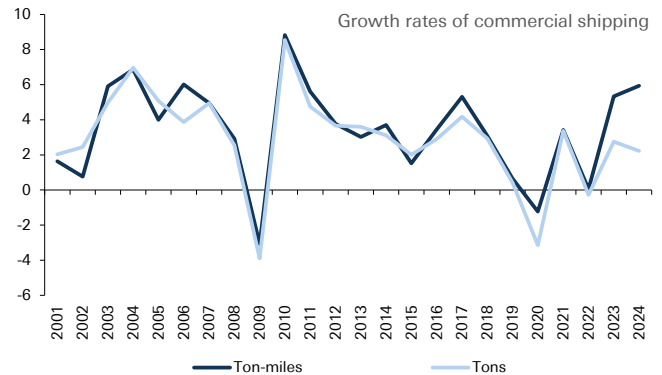
Indeed, 'flags of convenience' are often employed to avoid both the threat of interception by belligerent states, regulatory requirements and in certain cases tax and registration costs.²⁶

Figure 11: Container freight rates tend to be more sensitive to geopolitical shocks, but both container and bulk shipping have been affected



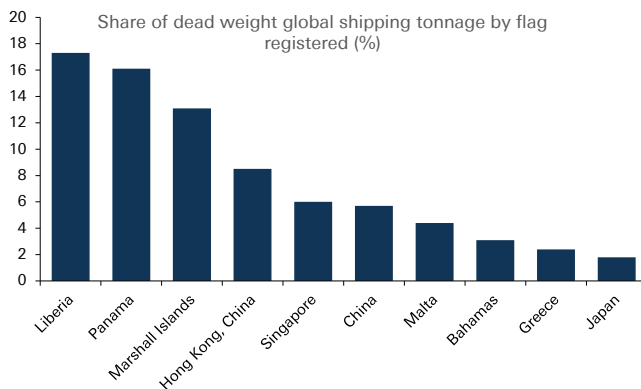
Source : Deutsche Bank, Bloomberg Finance LP

Figure 12: Distances travelled are outstripping tons transported in terms of growth rates of maritime trade, owing to recent disruptions



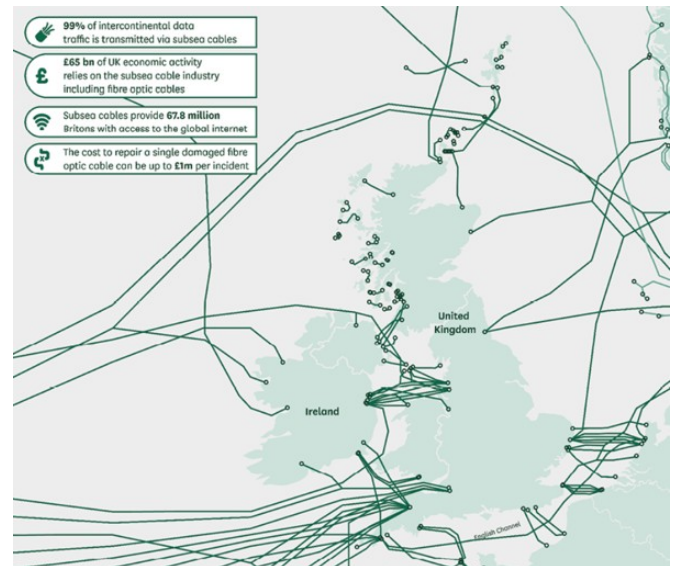
Source : Deutsche Bank, UNCTAD Review of Maritime Transport

Figure 13: 'Flags of convenience' dominate global commercial shipping



Source : Deutsche Bank, UN Maritime factsheet

Figure 14: The UK is one of the most vulnerable countries globally to disruption to sub sea cables



Source : House of Lords

Another example of more limited maritime disruption would be to undersea cables. A recent UK House of Lords Joint Committee addresses the risks associated with the UK's extensive network of over 60 undersea cables. It concluded that the severing of a number of cables was manageable from a communication continuity perspective due to redundancy and rerouting, but that a severe disruption (large

26 [PBS report](#)



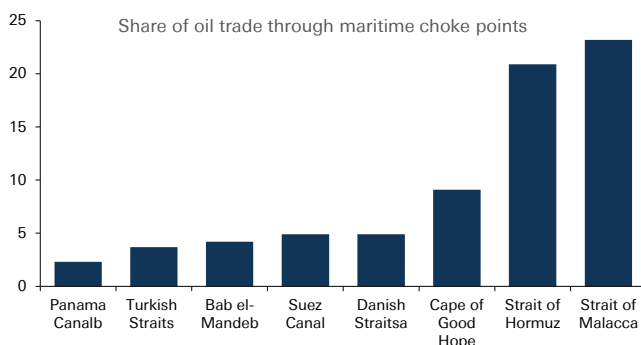
scale attacks on cable infrastructure) could have more severe economic and financial consequences.²⁷

2. Regional disruption or blockade with limited capacity for trade substitution.

An obvious example of this more severe form of maritime supply shock is the current crisis in the Gulf. Owing to the share of crude oil transported via the Strait of Hormuz (21%) and Bab el Mandeb (4.2%), as well as the share of LNG (20%) transported through the region, and its importance for other commodities such as nitrates, the effect of the twin blockades of the Straits of Hormuz and Bab el Mandeb has proved much more difficult for suppliers and the international shipping industry to work around. Moreover, war risk premiums from the insurance market have become prohibitively high, or in certain cases cancelled altogether for high risk vessels.²⁸ Re-flagging of vessels can be of limited use in the event of maritime blockade, where interdiction extends to all commercial vessels regardless of flag.

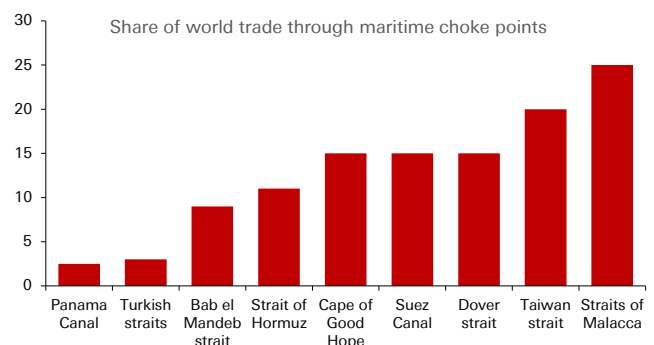
Figures 15 and 16 illustrate the relative importance of different maritime choke points for oil and world trade respectively. It should be noted, however, that the share of trade transiting through these locations does not, on its own, determine the severity of maritime shocks. Geography matters. There are alternatives to transit through the strait of Malacca for commercial vessels transiting to East Asia, even if these involve higher costs and lengthier journeys. By contrast, the 'twin blockade' of Hormuz and Bab el Mandeb has proved problematic due to the fact that it effectively has disrupted both of the main routes of supply for Saudi oil even if, over time, land based alternatives will become available. Correspondingly, the economic impact of these developments has been more severe, and manifested themselves in physical supply shortages, in other words **volume** as well as **price** effects.

Figure 15: The straits of Malacca and Hormuz are overwhelmingly the world's most important transit points for oil



Source : Deutsche Bank, IEA

Figure 16: But there are multiple choke points for world trade, including the Taiwan strait, Dover strait and Suez Canal



Source : Deutsche Bank, various sources including the UN, IEA, think tank reports

²⁷ [House of Lords Report](#) September 2025

²⁸ [Lloyds List](#) March 2026



Figure 17: The 'twin blockades' of Hormuz and Bab El Mandeb have meaningfully hindered global oil exports



Source : Deutsche Bank, map generated by Copilot

3. Global disruption or an extremely severe regional disruption.

The third category of severity has not occurred in market participants' living memory with the only analogies being the shocks to international commerce occurring during the First and Second World Wars. Maritime blockades imposed during these conflicts, and reciprocal attacks on commercial shipping both of belligerents and neutrals across the world, devastated international trade and financial flows.

Outside of a world war, however, another form of severe maritime shocks would involve disruption in the Indo Pacific maritime region, more specifically the South China Sea, East China Sea and Taiwan Strait. First, the region is immensely important from a global trade perspective. The South China Sea carries over a third of global maritime trade and hosts the world's six busiest container ports, while the wider Indo Pacific region host two thirds of global commerce. The straits of Taiwan and Malacca are the two single most important global maritime choke points. Second, economies in the region are highly dependent on maritime commerce for export and energy needs, with economies such as Japan, Korea and Singapore almost entirely dependent on maritime trade of their energy needs. Third, the region carries a complex and dangerous set of competing territorial, economic and strategic claims.



Figure 18: The two most important choke points for global trade are in the South China Sea and Taiwan Strait



Source : Deutsche Bank, map generated by Co-Pilot

Fully elucidating the complexities of this region are beyond the scope of this paper, but briefly these revolve around a series of bilateral and multilateral territorial claims. First, in the East China Sea between China, Japan and Taiwan, principally focusing on the Senkaku/Diaoyu islands and to a lesser extent the contested atoll of Okinotori. Second in the South China Sea between China, Malaysia, Vietnam and the Philippines predominantly focusing on the Spratly Islands, Scarborough Shoal and Paracel Islands. Third, with respect to contested sovereignty over Taiwan. These are in turn related to economic claims concerning natural resources (especially the exploitation of potential oil and gas reserves) and fisheries.²⁹

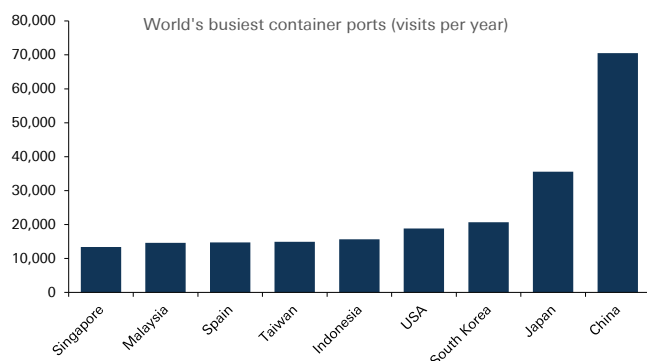
The legal architecture under UNCLOS has been used as an arbitration mechanism over many of these claims (for example recently with respect to both Okintori and in a dispute between the Philippines and China over the nine dash line which the latter claims provides wide ranging sovereign and economic rights over the South China Sea). Even more importantly, UNCLOS has provided the legal basis for the United States and other allies to assert freedom of navigation rights in the region. Among these has been the Taiwan Strait, where UNCLOS's definition of an international strait has underpinned the US and other nations' rights to both freedom of navigation and overflight, as well as military exercises.³⁰ This illustrates once again the importance of the existing legal architecture governing maritime rights, and the potential negative spillovers should the architecture or states' willingness to enforce it, unravel.

29 For a useful discussion of these claims, see Triki, ".[Understanding the South China Sea Crisis: State Claims, International Interventions and Implications](#)," Asia and the Pacific Policy Studies, January 2026. See Also, [War on the Rocks](#), One Sea, Near Sea, Whose Seas? May 2024

30 See, for example, [CIMSEC](#), Naval Operations and the Right to Operate Freely in the Taiwan Strait, 2022

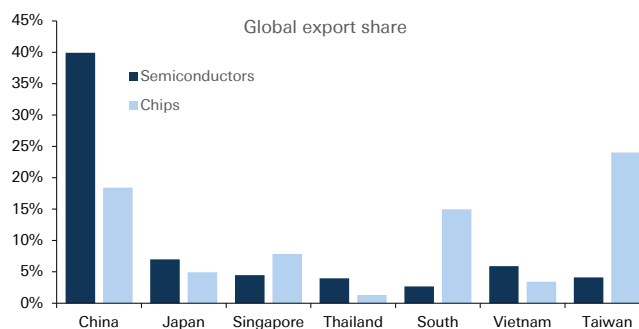


Figure 19: Almost all of the world's busiest container ports operate in the Eastern Pacific



Source : Deutsche Bank, Atlantic Council

Figure 20: The region also dominates market share in electronics



Source : Deutsche Bank, Observatory of Economic Complexity

In this context it is disquieting that both military planners and policy analysts have been dedicating increasing attention to the possibility of China adopting an economic approach, largely through maritime channels, to attempt to settle the long standing dispute over Taiwanese sovereignty.³¹ On the one hand, a blockade or quarantine of Taiwan might be seen as a positive relative to the alternative of a more kinetic military approach, by carrying less human costs, as well as reducing the scope for intervention by other powers.³² On the other hand, such an action is designed to maximise the economic costs for Taiwan, with major consequences for the global economy.

Of even greater concern would be a Taiwan crisis that spills over into a broader regional maritime crisis. Recently, rivals to China, especially Japan and the Philippines, have sought to enhance security cooperation with Taiwan, which risks creating what Chinese military planners have called a 'Three Seas Problem' in which the entire eastern seaboard of China becomes one theatre of operations.³³ Another concern among analysts relates to a potential counter-blockade by the United States of the Strait of Malacca. Still a further potential axis of escalation concerns the Sunda Strait. Indonesia has been increasingly assertive in its sovereign claims over its archipelagic waters, leading some US analysts to worry a non-aligned Indonesia could close the strait as part of its non-alignment policy.³⁴

The importance of Taiwan in the semi-conductor supply chain is well understood, as illustrated by the proliferation of reports on the subject, and even a crisis that was limited to simply a blockade of that country, in the absence of any wider escalation, could lead to significant losses of global output.

The knock on consequences of wider regional maritime disruption, even if this were confined to the purely economic sphere, would be far more disruptive. Consider a scenario in which a Chinese maritime blockade or quarantine on Taiwan was met

31 [Atlantic Council Strategy Paper](#), "A Maritime Blockade of Taiwan By the People's Republic of China, December 2023

32 The difference between a blockade and quarantine largely boils down to whether it is executed via military actors like naval forces or non-military forces, in China's case the People's Armed Forces Maritime Militia. See [here](#)

33 Luo, Shuxian "A Rock and China's Three Seas Problem," [War on the Rocks](#), August 2026

34 [CSIS Report](#), "Why the US Navy Must Resume FONOPS in Indonesia," December 2025



with a counter-blockade from the United States and its allies Japan and Korea extending from the Taiwanese Strait to the Strait of Malacca. Such an approach may be an appealing alternative to direct military confrontation with China. This would have a chilling effect on maritime trade throughout the region, leading to sharp rises in insurance premiums, port closures and the practical cessation of commercial shipping. With the region dominating global container trade (figure 19), electronics exports (figure 20), as well as myriad other consumer and intermediate goods, the effect on international trade would only be comparable to twentieth century world wars, and would almost certainly be consistent with a global recession.

Assessing country vulnerability

The final port of call for this report is to assess how threats to the freedom of the seas impact financial variables.

First, the increased frequency of maritime disruption is a negative supply shock, and as such likely to put upward pressure on inflation and downward pressure on demand. The former is likely to dominate when it comes to interest rates. Under a severe maritime disruption, this may be exacerbated by the pressure placed on exchange rates, especially if central banks seek to curb volatility by defending currencies. For example, in the event of a severe disruption to Taiwanese trade, it is possible to envisage the central bank stepping in to prevent financial markets from exacerbating the shock, utilising foreign exchange reserves and therefore principally treasuries in doing so.

Figure 21 provides an illustrative scorecard of relative vulnerability to maritime shocks. Economies are more vulnerable if they have a larger share of maritime trade or energy imports relative to GDP, or if they are an island state. They are less vulnerable if they have a stronger navy (measured by WDMMW database). Strikingly, the most vulnerable economies are all located in Asia or Europe, with Singapore, the Philippines, Vietnam, the Netherlands and Korea all highly open economies with a heavy reliance on maritime trade, but with limited naval resources. In contrast, larger economies with multiple land borders like the US, Russia, China and Brazil score as much less vulnerable. Translating this into actionable trade or investment ideas is not the point of the table below, but it should serve as a guide for investors and corporates as to the potential vulnerability by country to severe maritime disruption, and the increasing use of the sea as a theatre for geoeconomic competition.



Figure 21: Maritime vulnerability scorecard

	Maritime imports as share of GDP	Seaborne imports of oil and gas % GDP	Naval power index	Island or limited land alternatives?
Singapore	17%	8%	43.2	Yes
Philippines	11%	4%	37	Yes
Vietnam	20%	3%	2	-
Netherlands	14%	3%	37.3	-
Korea	11%	3%	122.9	Yes
Australia	10%	0%	48.9	Yes
Belgium	17%	0%	7.6	-
Spain	7%	2%	56	-
Japan	6%	1%	121.3	Yes
Germany	12%	0%	54.7	-
Indonesia	5%	2%	137.3	Yes
UK	4%	1%	88.3	Yes
India	4%	4%	100.5	-
Italy	6%	2%	80.7	-
France	7%	1%	92.9	-
Türkiye	8%	0%	80.5	-
Brazil	7%	0%	39.9	-
China	3%	2%	319.8	-
Russia	4%	0%	242.3	-
USA	4%	0%	323.9	-

Source : Deutsche Bank, Bloomberg Finance LP, WDMWW database



Appendix 1

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UNCERTAINTY



Deutsche Bank
Research

M&A Spotlight:

When rates rise ...

17 September 2026

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Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

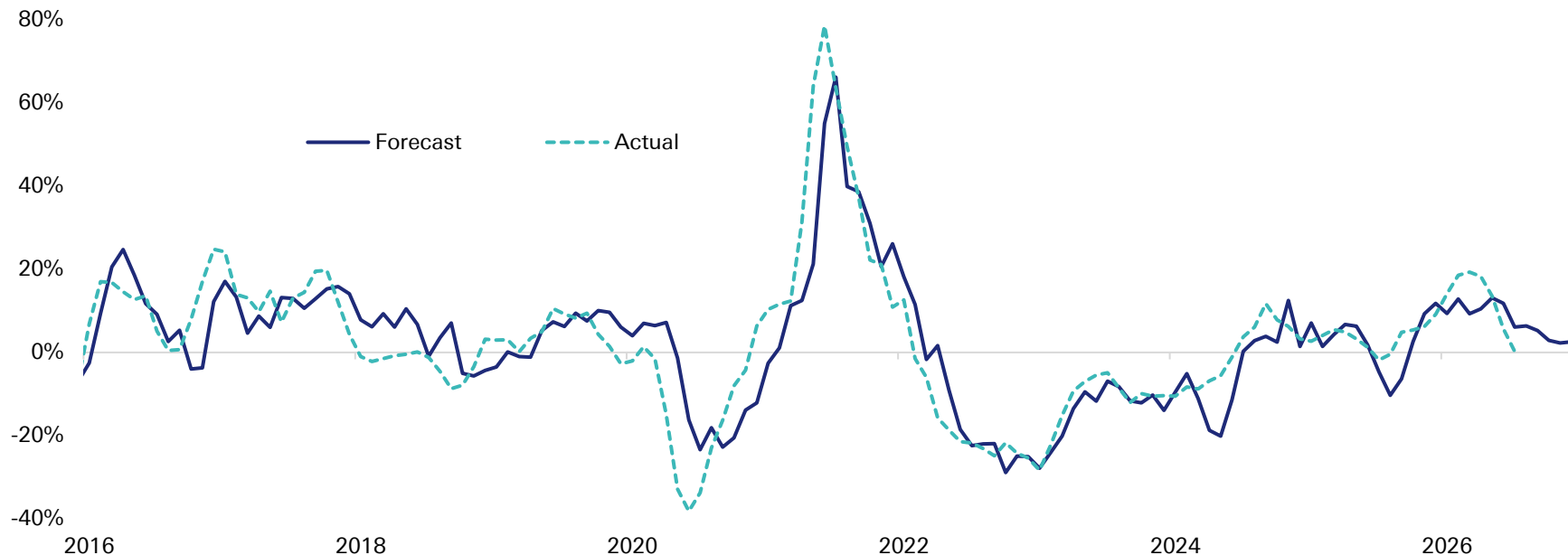
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GEOPOLITICS

Our M&A forecast model: In the US we see low but positive growth over the coming 3 months
Mid-term elections are incentivising some CEOs to wait for greater legislative certainty. That is combining with the reemergence of greater geopolitical uncertainty and the second round of rising oil prices to dampen near-term forecasts



Our 3m-forward forecast for growth in US M&A deal numbers

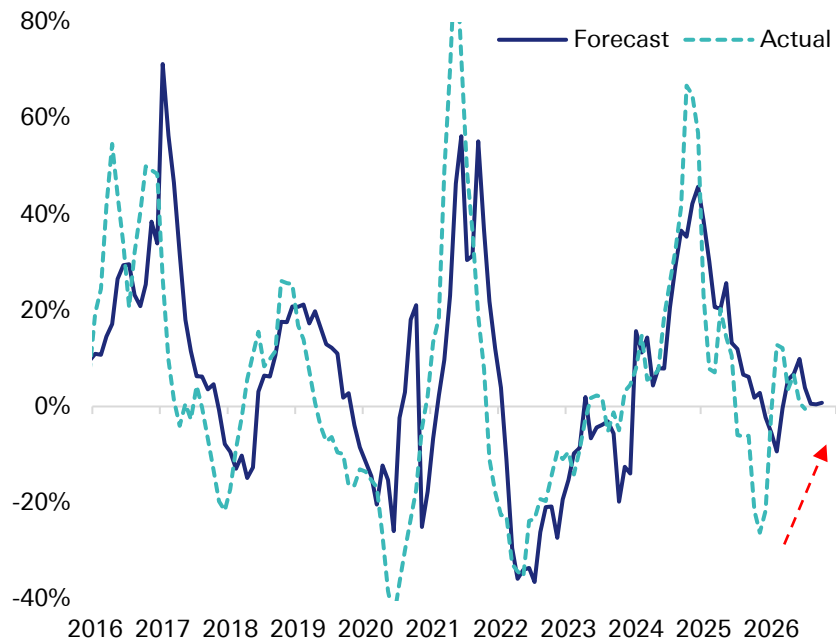


Source: Bloomberg Finance L.P., Dealogic, Deutsche Bank

Our M&A forecast model: In the UK and Europe, our model shows similar low but positive growth
In the UK, this will be encouraging given the market is recovering from a significant drop over the last 24 months

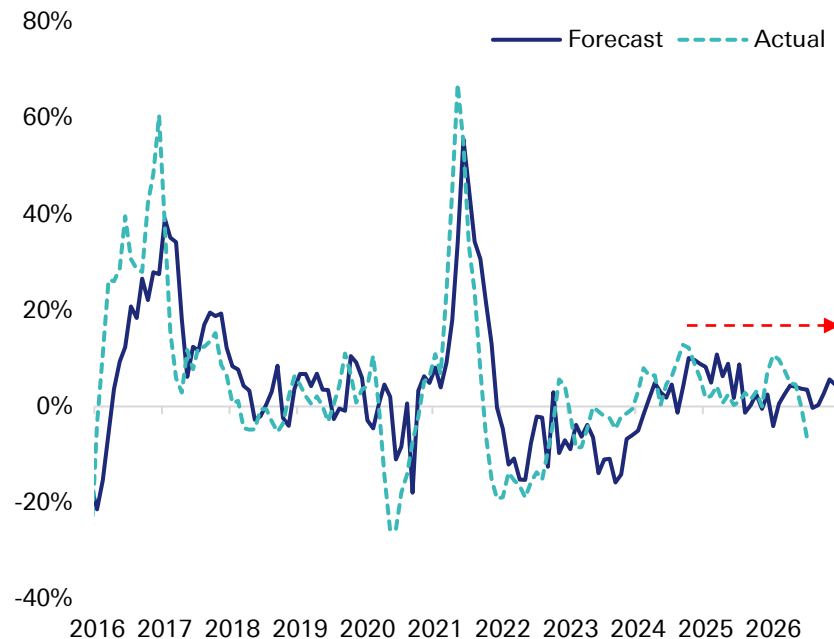


Our 3m-forward forecast for growth in UK M&A deal numbers



Source: Bloomberg Finance L.P., Dealogic, Deutsche Bank

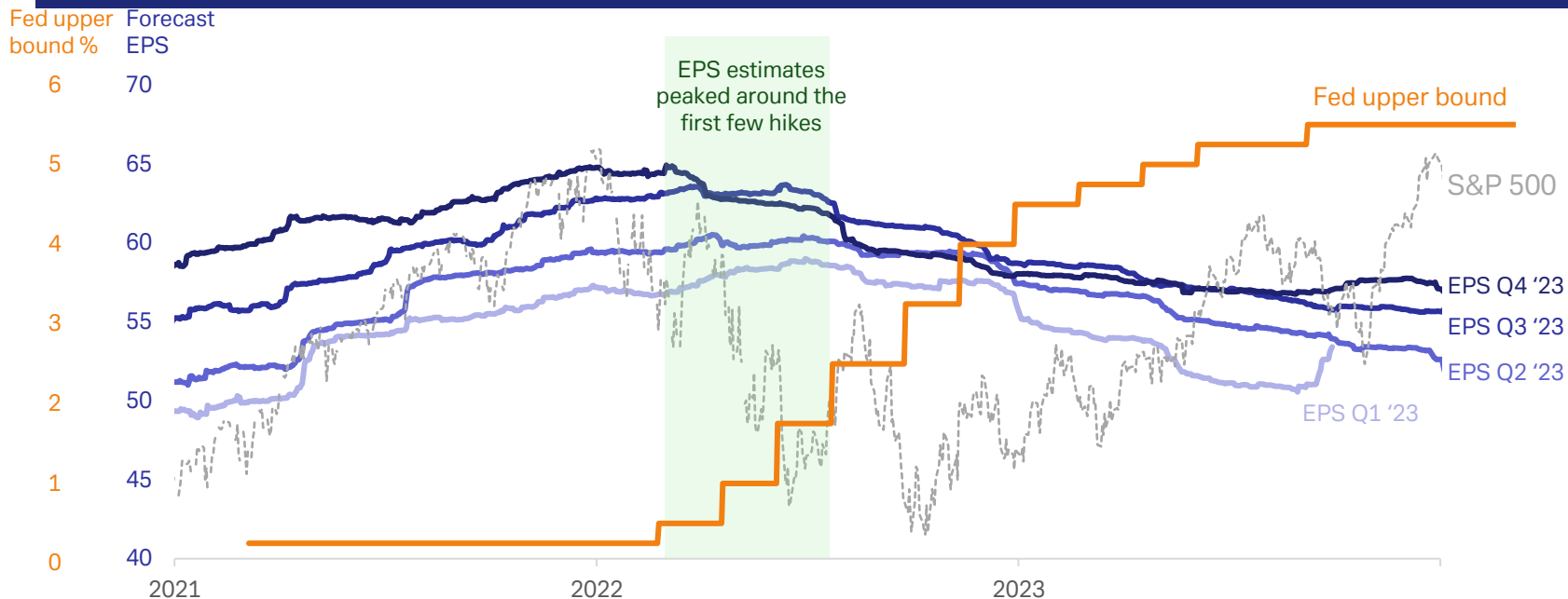
Our 3m-forward forecast for growth in Eurozone M&A deal numbers



Last time around ... Earnings estimates peaked just after the 2022-23 hiking cycle began
It took a few rate rises for investors to materially cut their EPS estimates. However, the extent of the hikes in that cycle were significantly higher than they are likely to be this time around.



EPS estimates for the S&P 500 during the 2022-23 hiking cycle



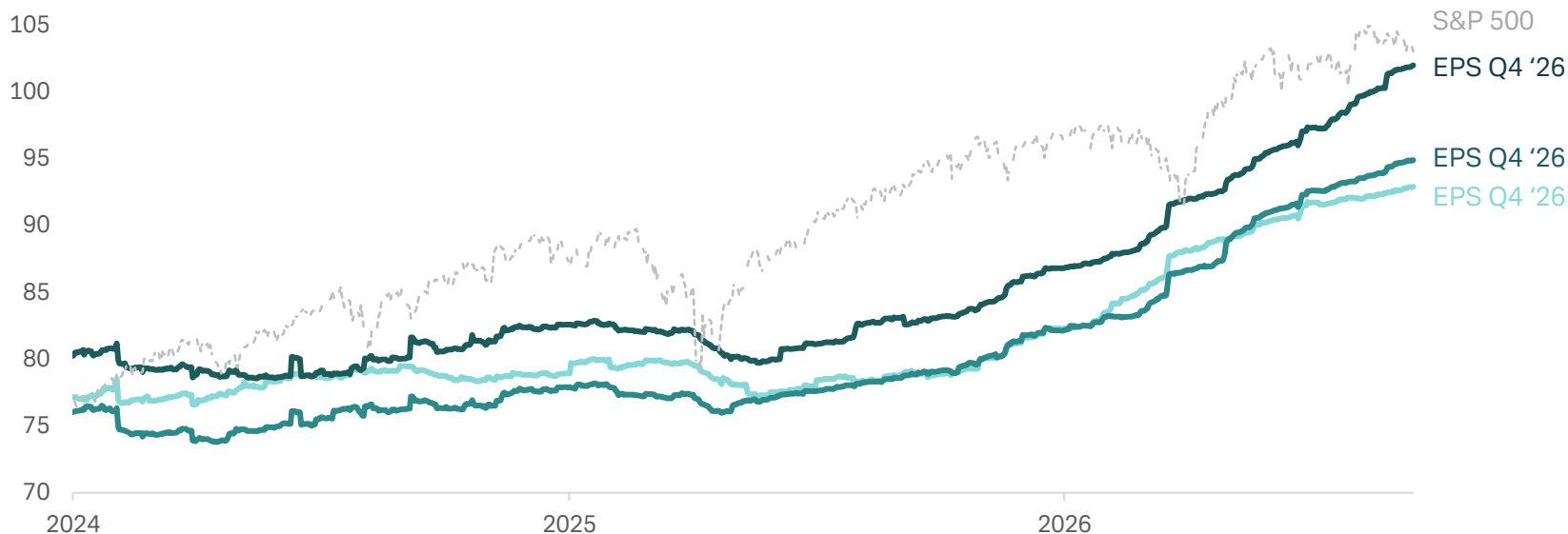
Source: Bloomberg Finance L.P., Deutsche Bank

This time around ... Earnings estimates have accelerated into the rate hike

So far, expectations for rate rises have not caused a substantial rerating of EPS estimates, including in tech companies. The effect may become more pronounced if bond yields breach the 5.00% barrier for a longer period of time.



EPS estimates for the S&P 500 for selected future quarters



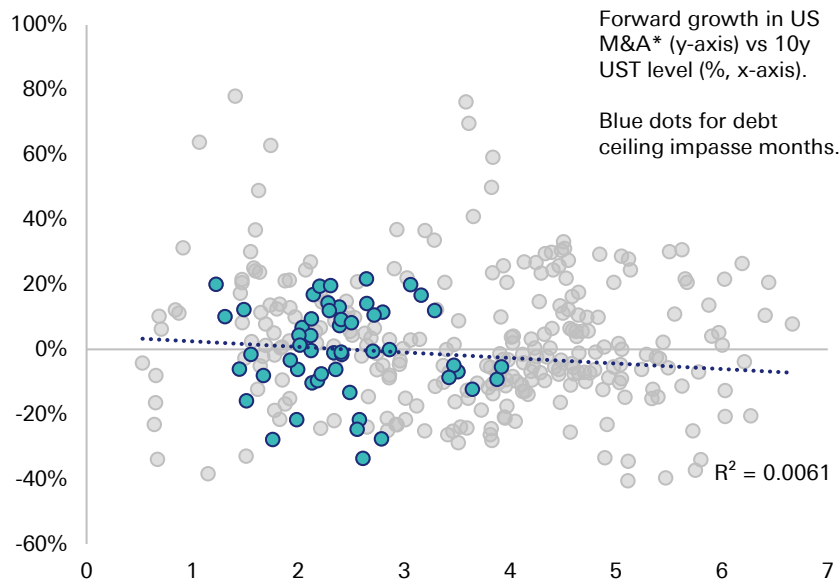
Source: Bloomberg Finance L.P., Deutsche Bank

Positive news: Throughout recent history, M&A is relatively unaffected by treasury yields

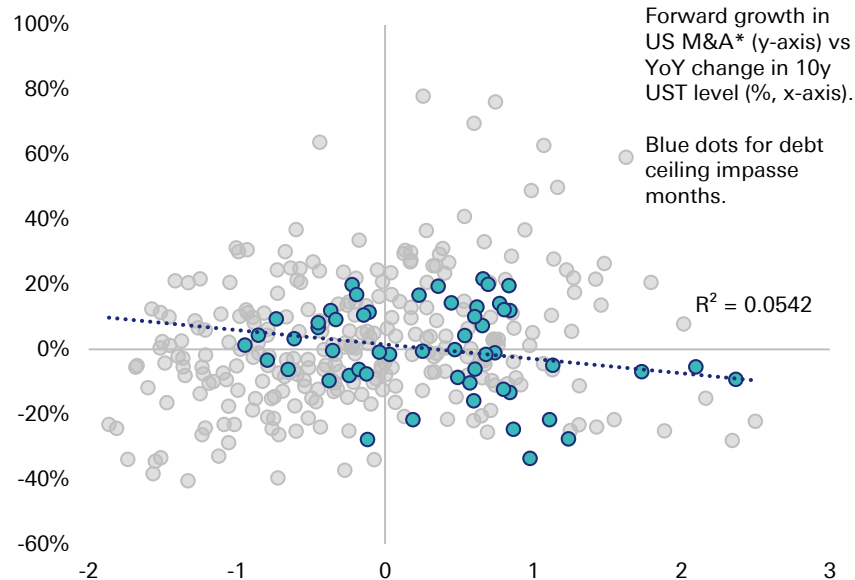
The only material exception to this is when there is an extreme change in the steepness of the 2-10 yield curve – ie. When the yield curve steepens >150 basis points in 12 months.



M&A v the level of the 10yr yield



M&A v the change in the 10yr yield



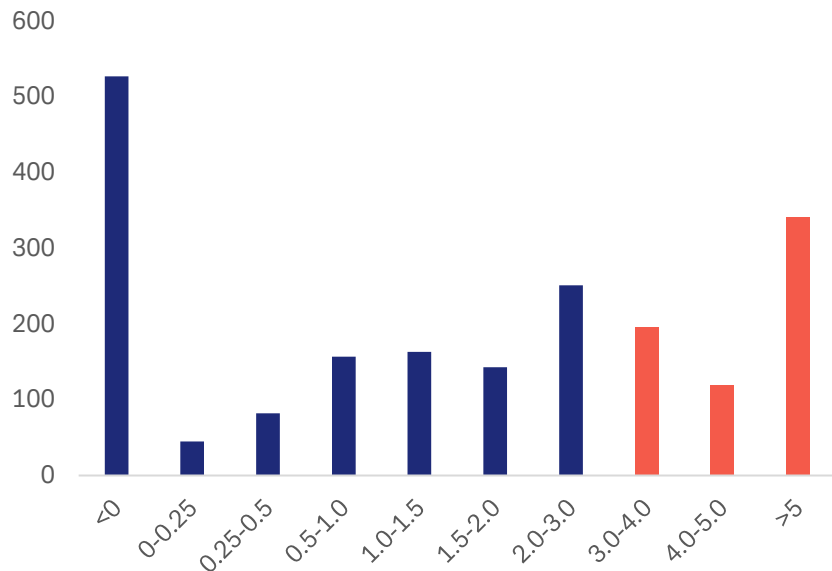
Source: Dealogic, Bloomberg Finance L.P., Deutsche Bank

Make no mistake, higher nominal rates will have a material impact on some firms

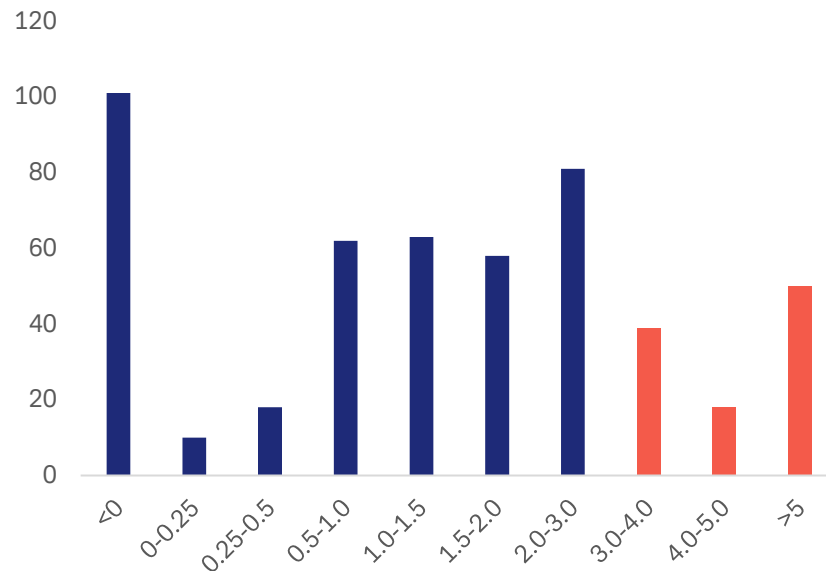
Firms with high debt loads may be more open to M&A opportunities if expectations move towards a more drawn out rate hike cycle



Number of companies with net debt/ebitda in each bucket (Russell 3000)



Number of companies with net debt/ebitda in each bucket (Stoxx 600)

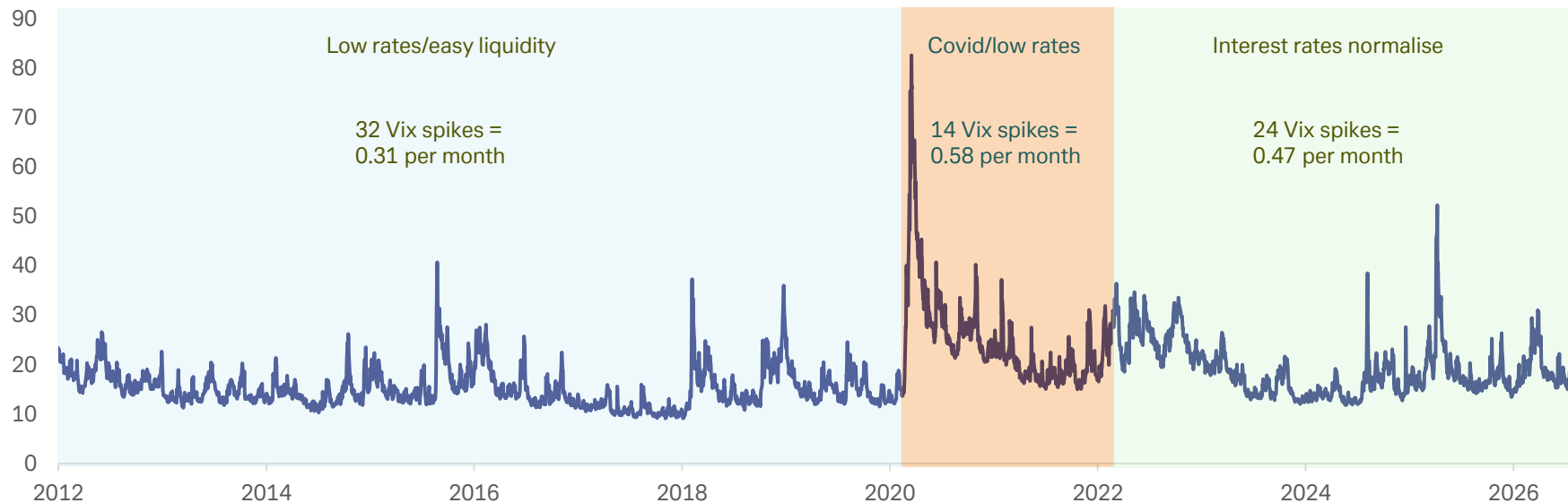


Source: Bloomberg Finance L.P., Factset, Deutsche Bank

The impact of higher rates will be amplified by a market that is showing more volatility
Higher rates tend to bring pre-existing problems to the surface. Given there has been a higher frequency of “volatility events” since rates normalised in 2022, higher rates today could catalyse even more volatility events, even if they are contained.



Vix volatility gauge



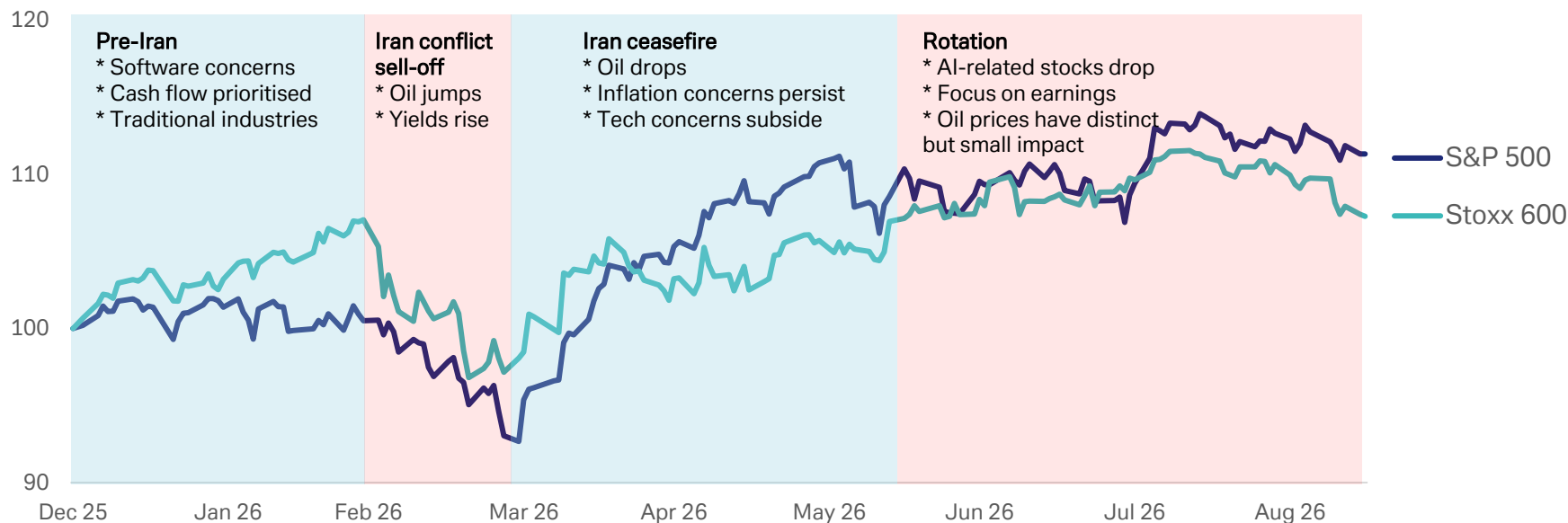
Source: Factset, Deutsche Bank

Already this year, equity markets have lurched between different narratives

Volatility in 2026 has been less dramatic than it was in 2025, however, the changing narratives indicate investors remain skittish



Four stages in equity markets in 2026



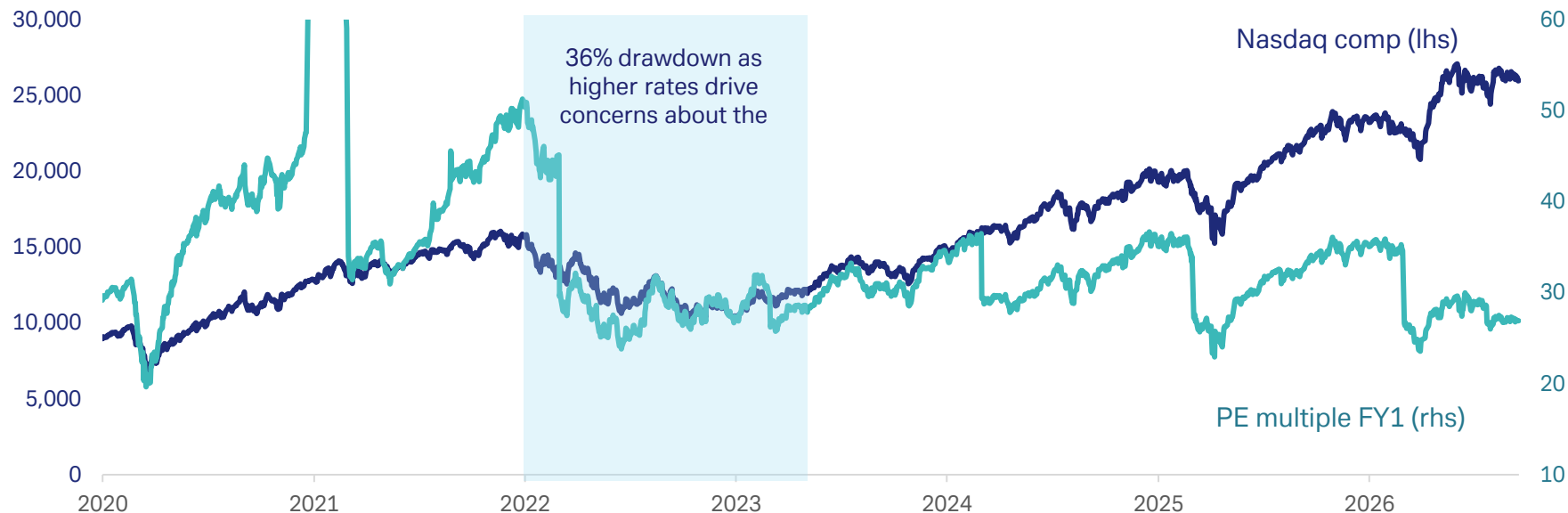
Source: Factset, Deutsche Bank

Broader markets are benefiting from the lack of a tech sell-off, despite higher bond yields

That is a stark contrast to the 2022-23 rate-rise period and is partly due to the strong earning growth at Big-Tech firms. If current earnings forecasts remain intact, current multiples are undemanding compared with recent history. The key thing is expectations about future rate rises and how they affect terminal rates.



Nasdaq composite (price and valuation)



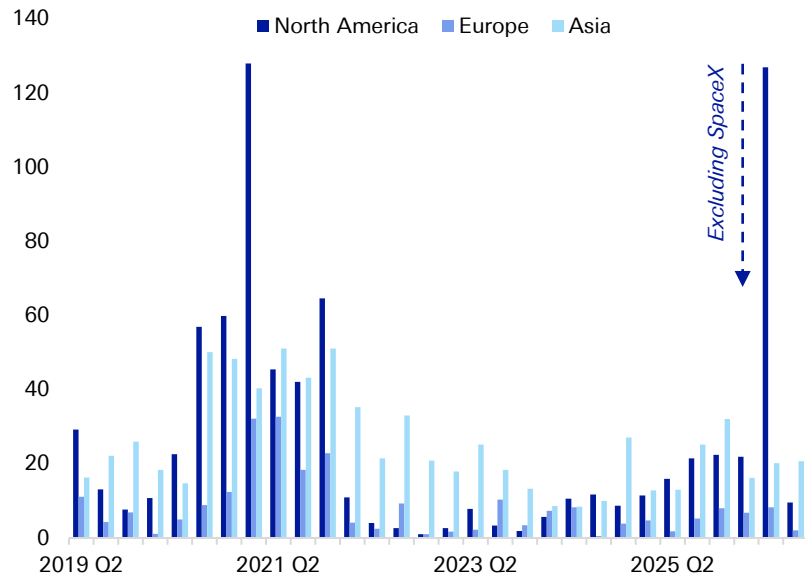
Source: Factset, Deutsche Bank

IPOs are recovering but are still well below 2021 levels ... excluding the very largest

More AI-related businesses are likely to come to the market even after larger firms go public. Tech IPOs have been relatively muted despite the boom in financing this decade. Investors' appetite for AI and big capital markets issuances may push more managers to exit tech holdings while the mood lasts. There have been just c\$7bn in pure-play AI IPOs in the US since 2025.

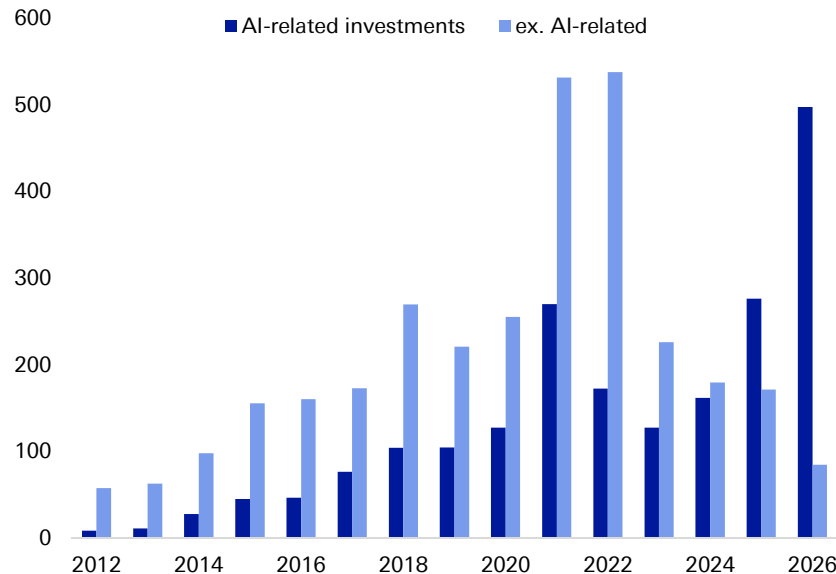


IPO deal values by quarter \$bn



Source: Dealogic, Deutsche Bank

Worldwide VC investment in AI (\$bn)

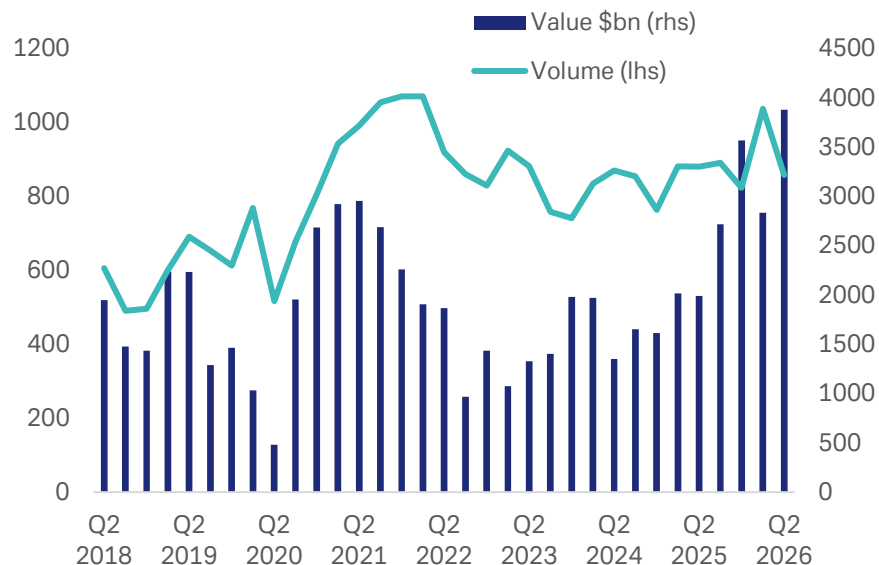


M&A continues to be dominated by big deals

Deal values continue to climb even though deal numbers remain in a tight range. We believe a period of background economic calm is needed for the CEOs of smaller firms to gain the confidence needed to pursue deals.

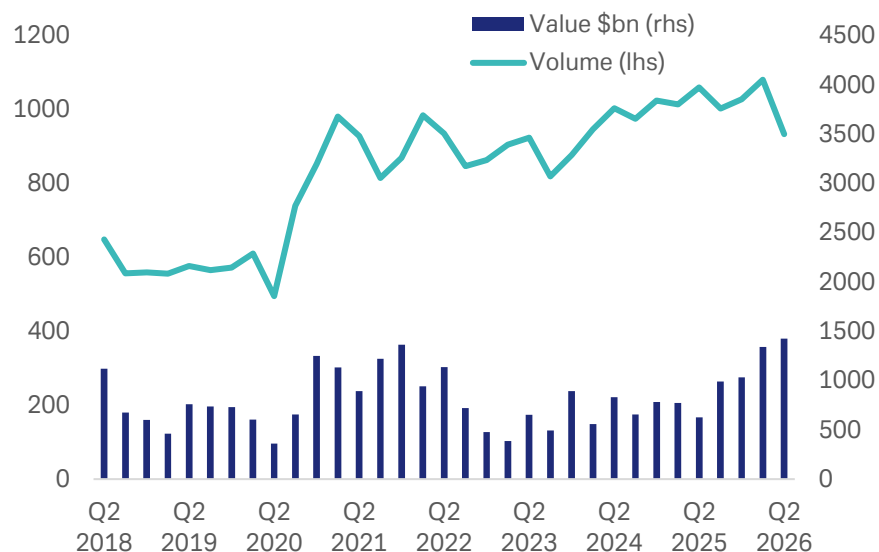


North American M&A



Source: White & Case, Deutsche Bank

Western European M&A



CEO confidence: Leaders expect the economy to be resilient to the current Iran conflict

Still, surveys show that although CEOs are confident about their own company, they are less confident in their industry, and less still in the economy more broadly. That is keeping a lid on some M&A for now.



CEO confidence in the economy 1yr from now



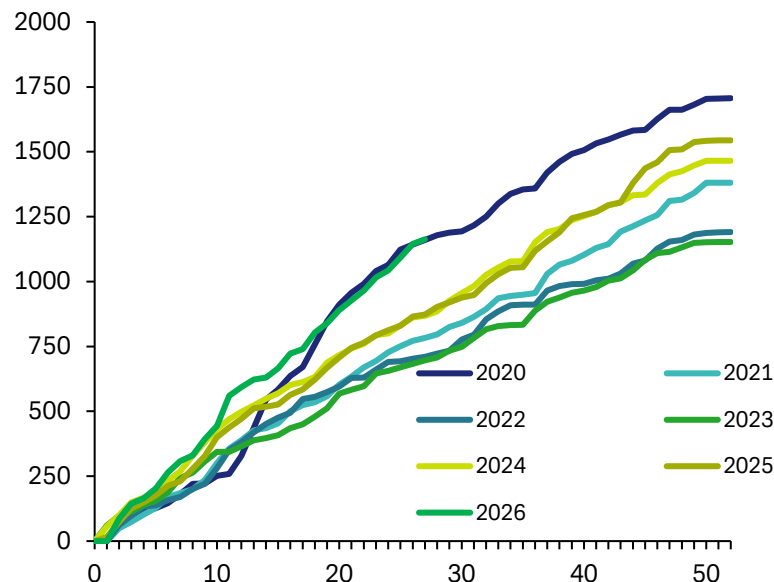
Source: Bloomberg Finance L.P., Deutsche Bank

Financing is abundant ... but can be selective

Debt issuance is still strong and tracking alongside the strongest level this decade. Private capital dry powder continues to push to record levels

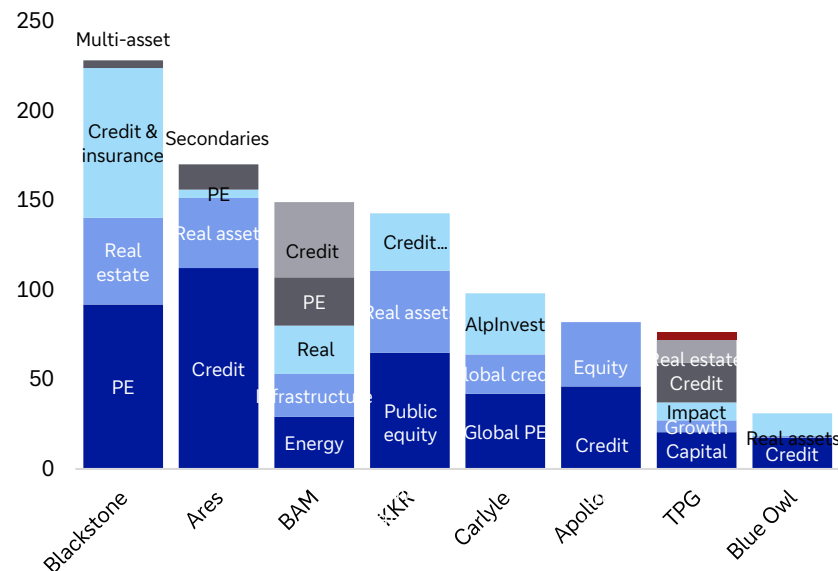


\$IG gross supply (\$bn) by week



Source: Bloomberg Finance L.P., Deutsche Bank

Dry powder by manager and segment, Q2 2026 (\$bn)

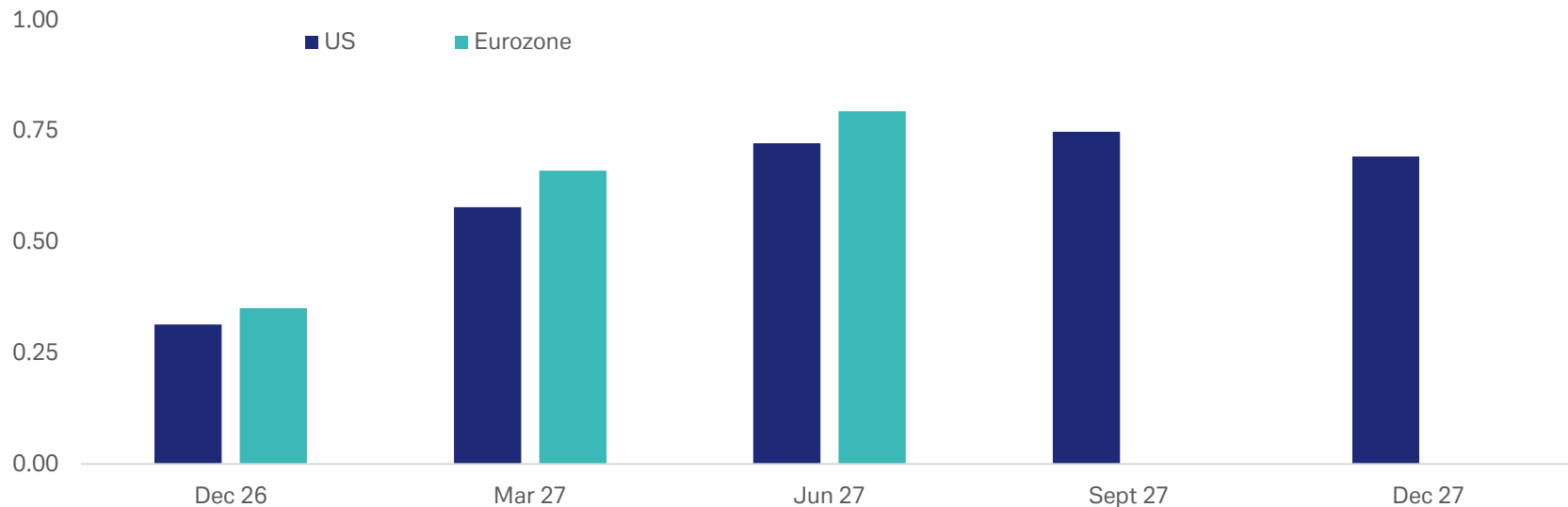


Markets are forecasting an easing of rate-rise (and thus inflation) expectations in 2027

The risks in the current environment may skew towards higher yields as expectations for the conflict in the Middle East lengthen and its impact becomes more severe



Implied interest rate rises by the end of each period (pp)



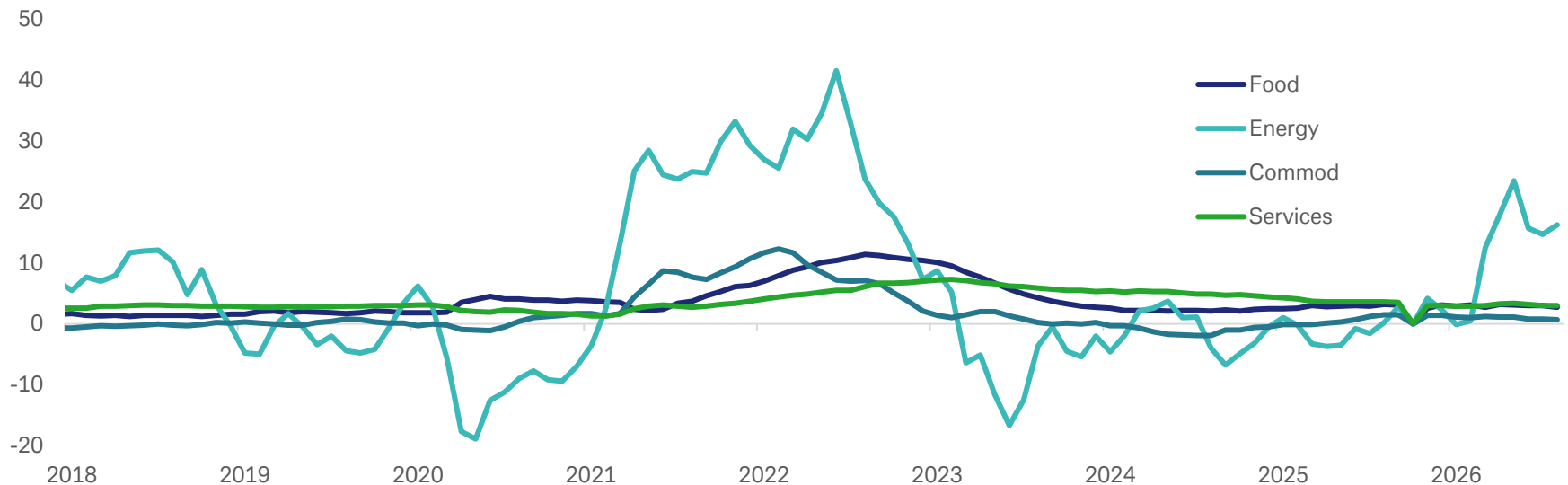
Source: Bloomberg Finance L.P., Deutsche Bank

The last energy shock lasted about 2 years

Iran's assertive push to control energy flows around the Middle East makes it unfeasible to project the length of energy shock



US CPI components (yoy % change)



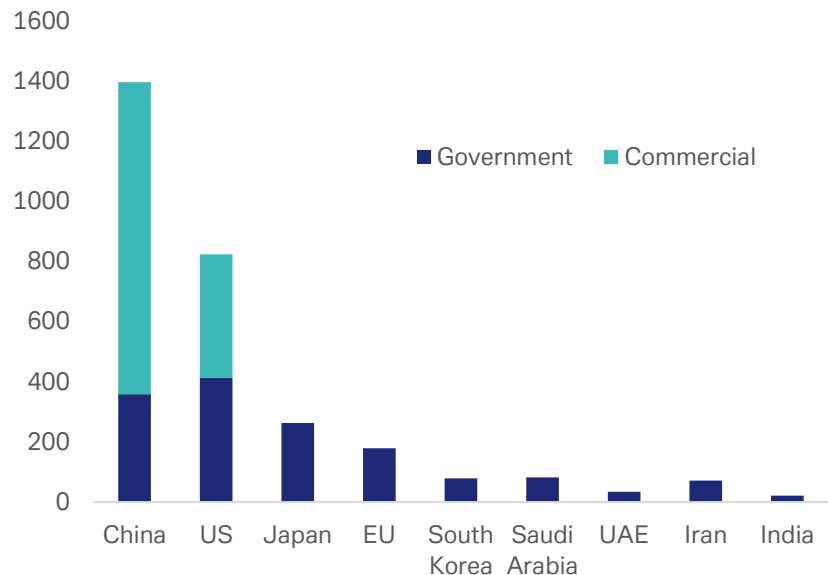
Source: Haver Analytics, Deutsche Bank

The trajectory of the energy shock will depend, in part, on oil inventory buffers

Industry executives have warned over the last two weeks that energy inventories have buffered the world economy for the last 6 months but that those buffers are approaching a cliff edge



Crude oil inventories pre-Iran conflict (m barrels)



Source: RAND, Deutsche Bank

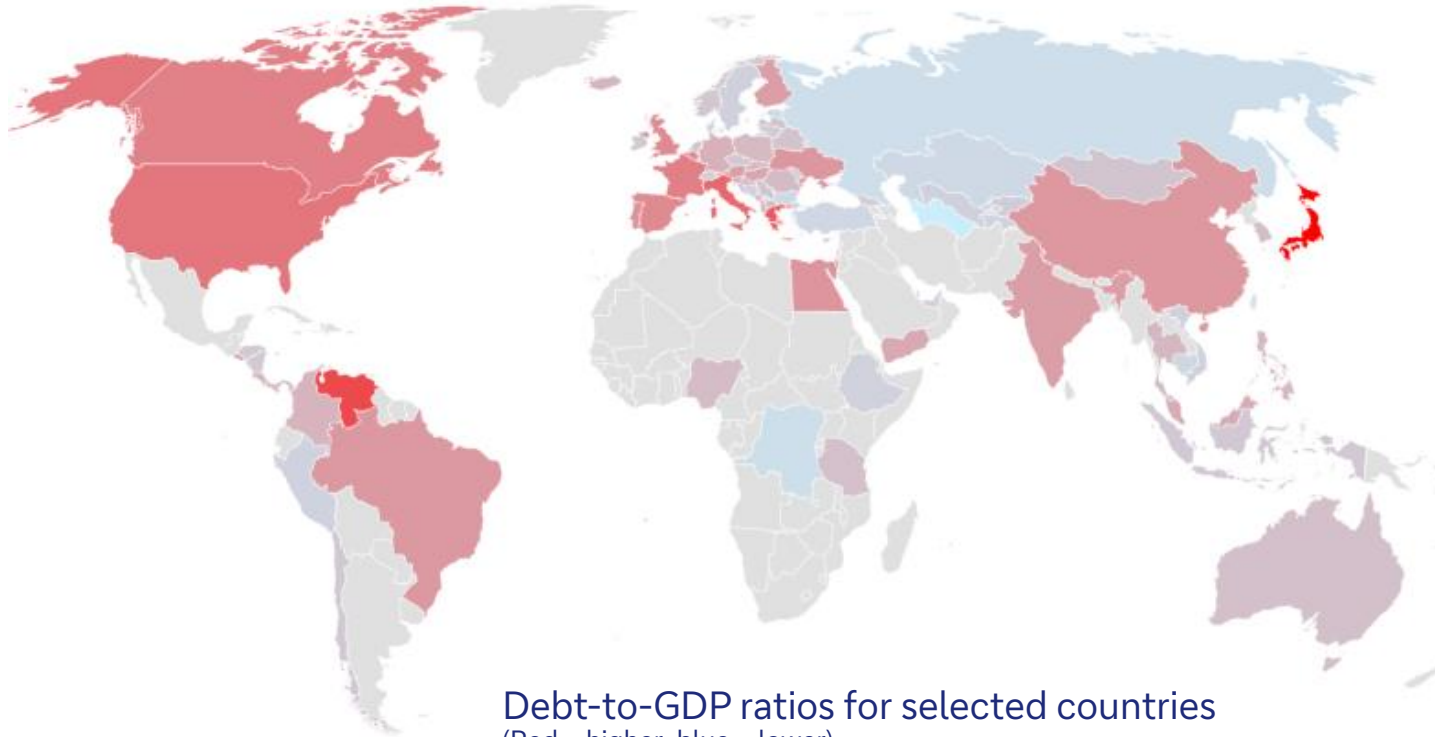
The way China uses its Strategic Petroleum Reserve is critical for world energy prices

RAND estimates China held between 1.1bn and 1.4bn barrels as of May this year – equivalent to 110-140 days of net imports.

That compares with the 400m barrel release coordinated by the IEA in March which had only a limited impact on markets.

Sovereign debt around the world

Confidence in indebted sovereigns tends to ebb and flow and markets usually experience at least one major 'risk-off' moment regarding sovereign debt each year. Upcoming elections in Europe will be a major focus point for investors over the coming months.



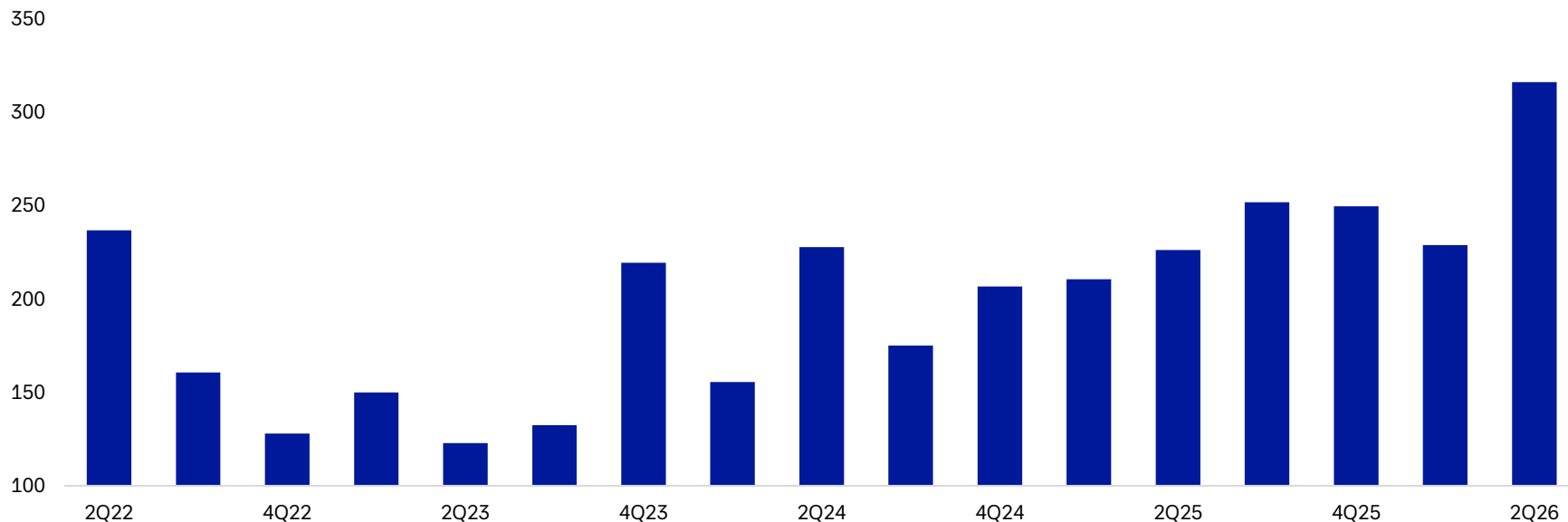
Source: IMF, Deutsche Bank

Private capital is searching for deals 1/2

Aggregate fundraising growth has shot up (in part due to one-off factors) and managers are confident in continued strength



Fundraising by selected large, listed private capital managers (\$bn)



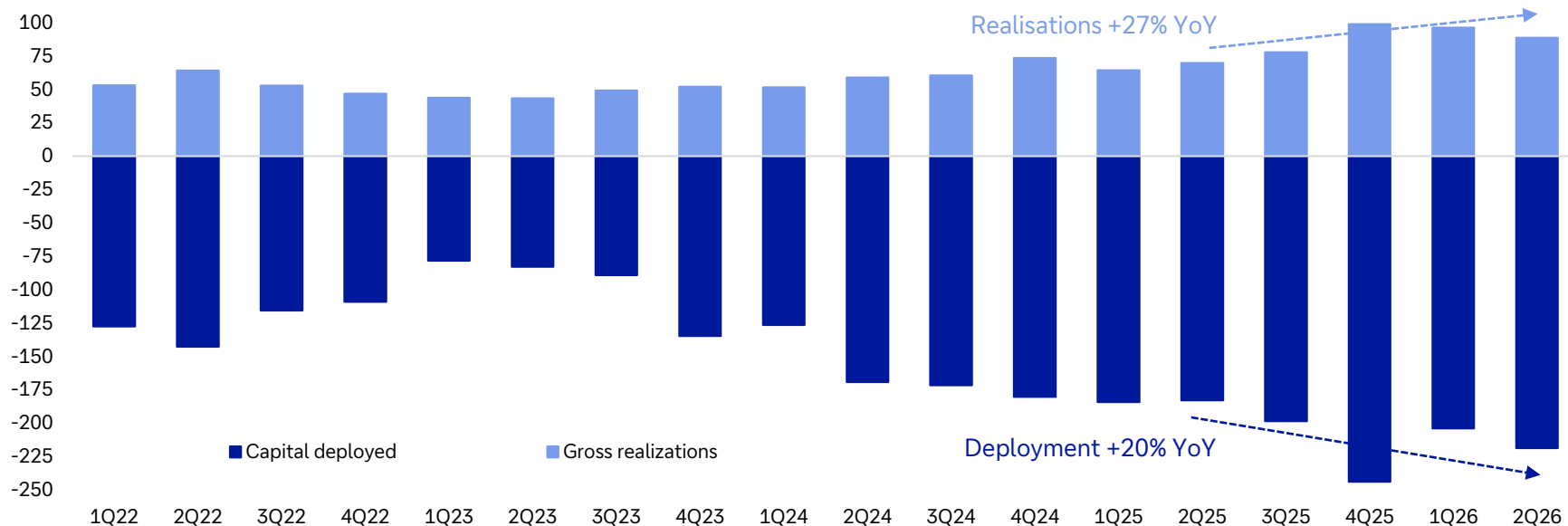
Source: Deutsche Bank company research - [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank

Private capital is searching for deals 2/2

Manager confidence has been helped by the growing rate of deployment and realisations. Through to the end of the year, realisations growth will be in focus given the growing macro uncertainty



Selected private managers, deployment v realisations (\$bn)



Source: Deutsche Bank company research - [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank

Appendix 1



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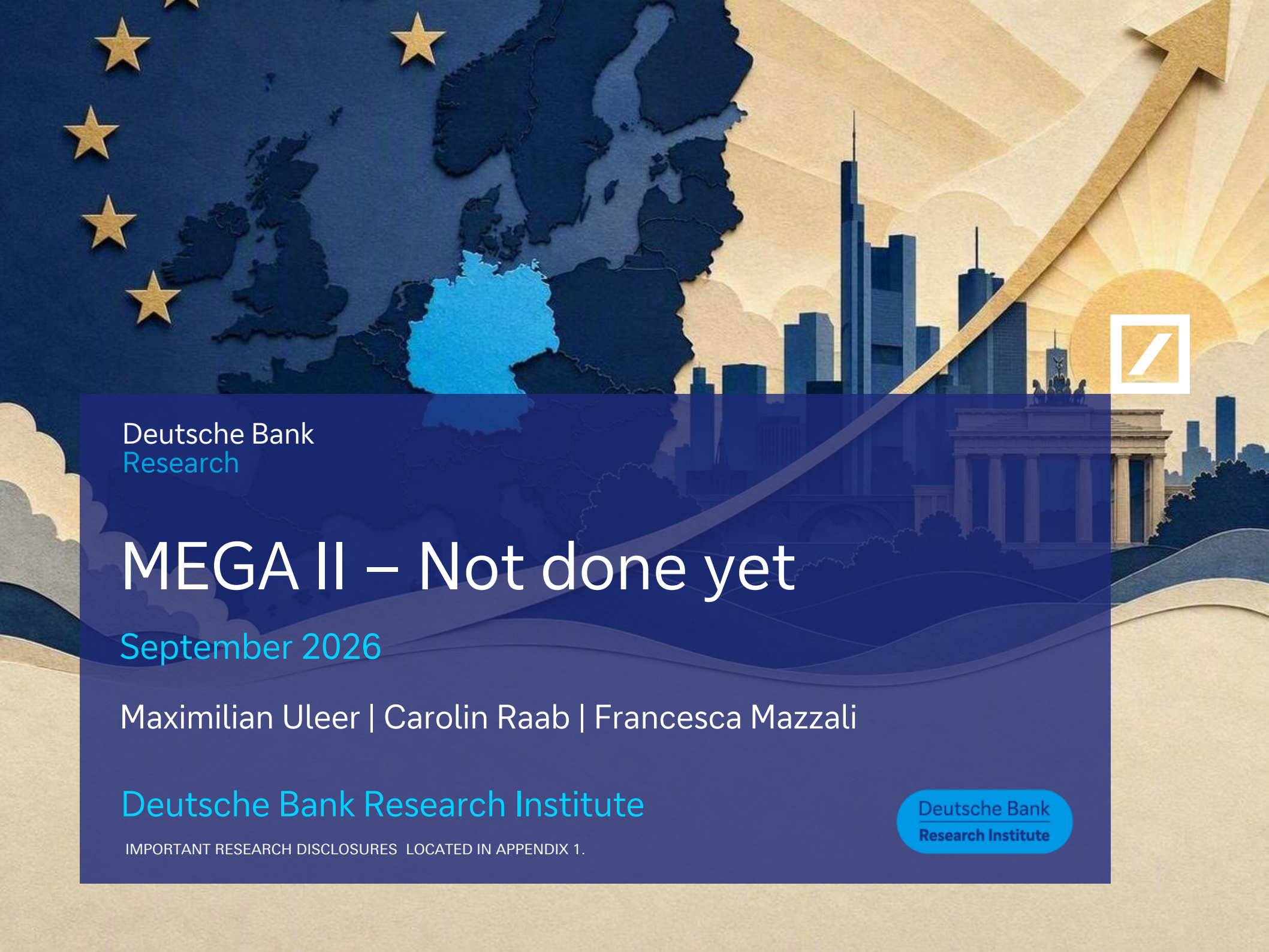
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Deutsche Bank
Research

MEGA II – Not done yet

September 2026

Maximilian Uleer | Carolin Raab | Francesca Mazzali

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Europe is back and Germany's potential seems not yet fully appreciated by markets



In early 2025, we made a bold call: we argued that Europe's 15-year history of underperformance has come to an end (MEGA and [here](#)). We projected the P/E valuation gap between US and European equities to narrow from a record 9.0x at the end of 2024 to 4.3x. Since then, the gap has indeed tightened to 4.7x. With Germany's potential still being underestimated by markets, we expect more to come for Europe.

Looking ahead, we see several structural factors supporting Europe's relative attractiveness:

- **Valuations:** After the tightening, we expect the gap to stabilize. Higher US growth justifies a premium but most of the below arguments argue against a renewed widening.
- **Earnings:** We expect the European earnings recovery to continue, delivering double-digit growth in 2026. While US earnings are still expected to grow faster, uncertainty around the sustainability of future growth, particularly in the Tech sector, remains elevated.
- **Germany:** The US and China are likely to continue growing faster than Europe, but much of that growth has been accompanied by widening deficits and rising debt. US fiscal deficit and debt metrics now exceed European levels. At the same time, Germany's fiscal pivot could support a relative recovery in European growth, an opportunity that equity markets have yet to fully embrace. A recovery in global manufacturing activity provides an additional boost to Europe's economy. Germany could become the powertrain of Europe's economic recovery. Having reduced its debt-to-GDP ratio over the past decade despite macro headwinds, it now has the fiscal capacity to support growth through a substantial spending plan including EUR 1tr in new debt. The economy is showing greater resilience than many expected, manufacturing activity is improving, international investor interest is beginning to re-emerge and reforms are advancing. We estimate the pension reform alone could unlock €10-15bn of annual inflows into European equities.
- **Diversification:** Europe's equity market is well diversified across sectors, providing broader return drivers than more concentrated markets such as the US and Emerging Markets. This helps cushion performance during sharp market downturns.
- **AI:** While Europe is unlikely to catch up with the leading AI developers, it is also less exposed to risks surrounding the monetisation of heavy AI investment. Instead, it should benefit indirectly from AI spending through its advanced manufacturing capabilities, its exposure to AI infrastructure and through growing AI adoption itself.
- **Political stability:** While Europe's fragmented political landscape is a structural disadvantage, external pressures from US and Chinese trade policies and ongoing geopolitical challenges have strengthened coordination across the EU on trade, defence and fiscal matters.

Closing valuation differential



In March 2025, we called for a decline in the valuation gap between US and EU equities. We have now moved significantly closer to our target.

P/E difference S&P 500 and STOXX 600 (12m fwd)



*DB forecast based on the P/E difference before the start of the Russia-Ukraine war (02/24/2022). Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

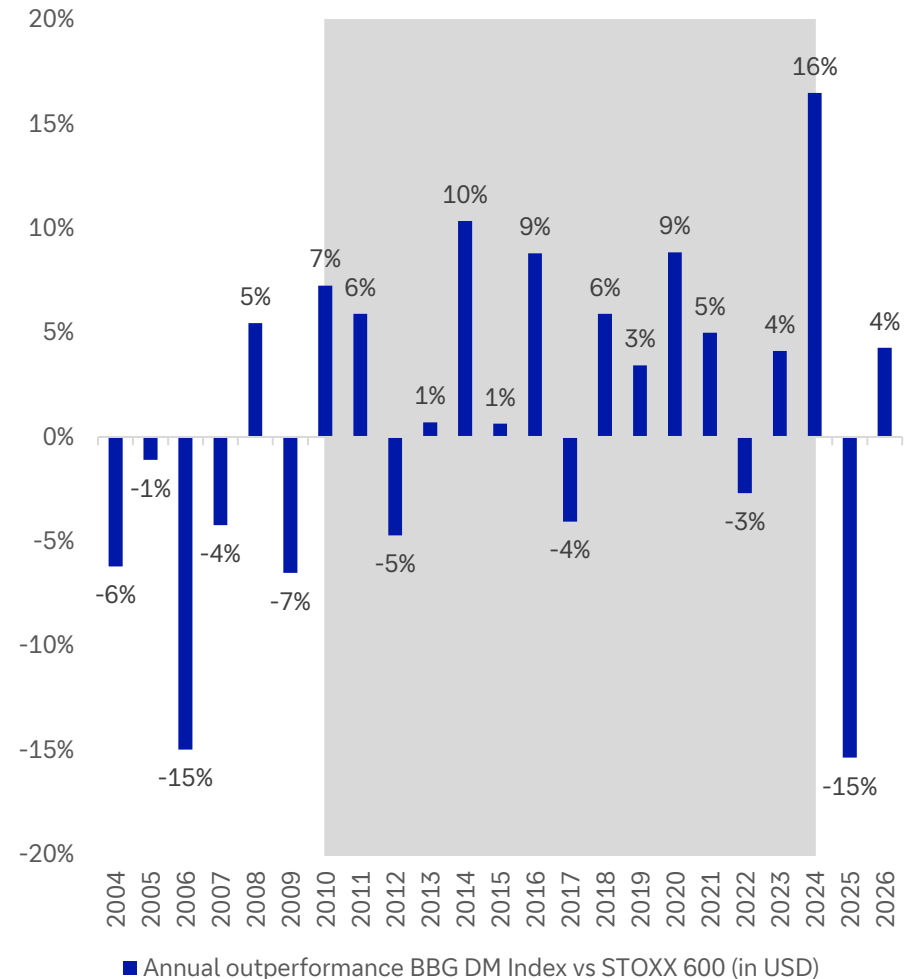
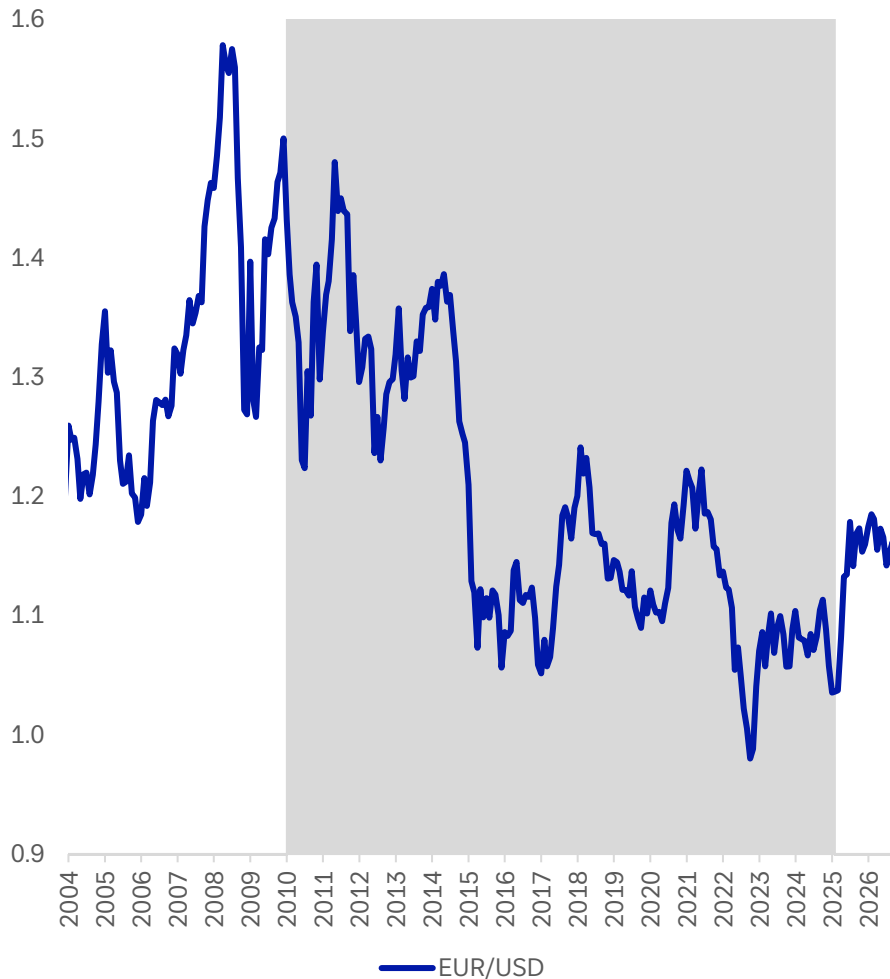
US Dominance... from 2010 to 2024?



European underperformance coincided with a 15-year USD rally. Will the USD continue to grow in its importance over the next 10 years?

In 2010-2024, the USD strengthened...

...supporting RoW outperformance



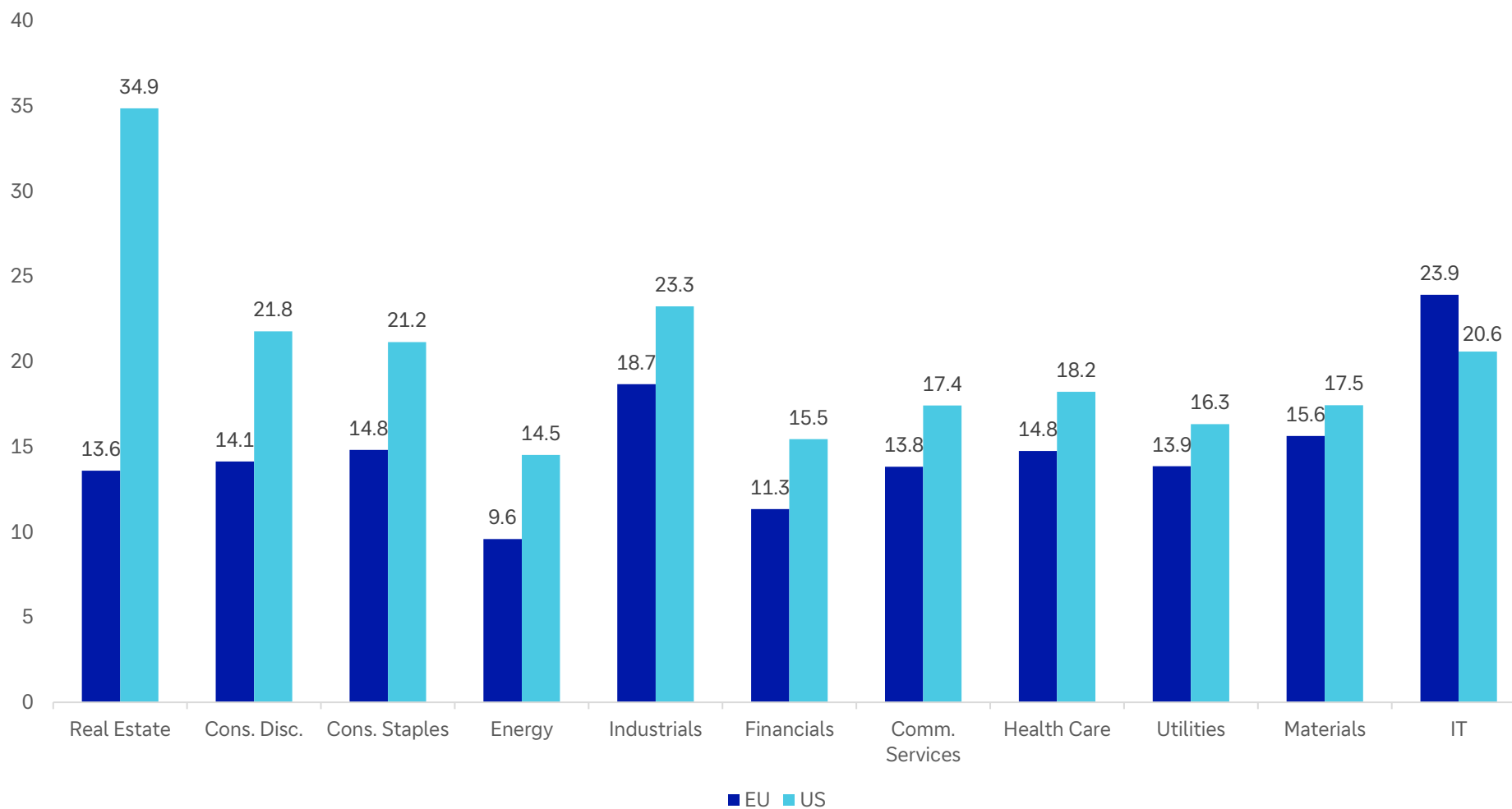
Note: Right chart is total return, 2026 is year-to-date. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Valuation Gap: applies across sectors



The valuation premium no longer stems from the IT sector. Still, every US sector, ex IT, trades at a premium versus the European sector.

Current sector P/E (12m fwd), sorted P/E difference US vs EU sectors



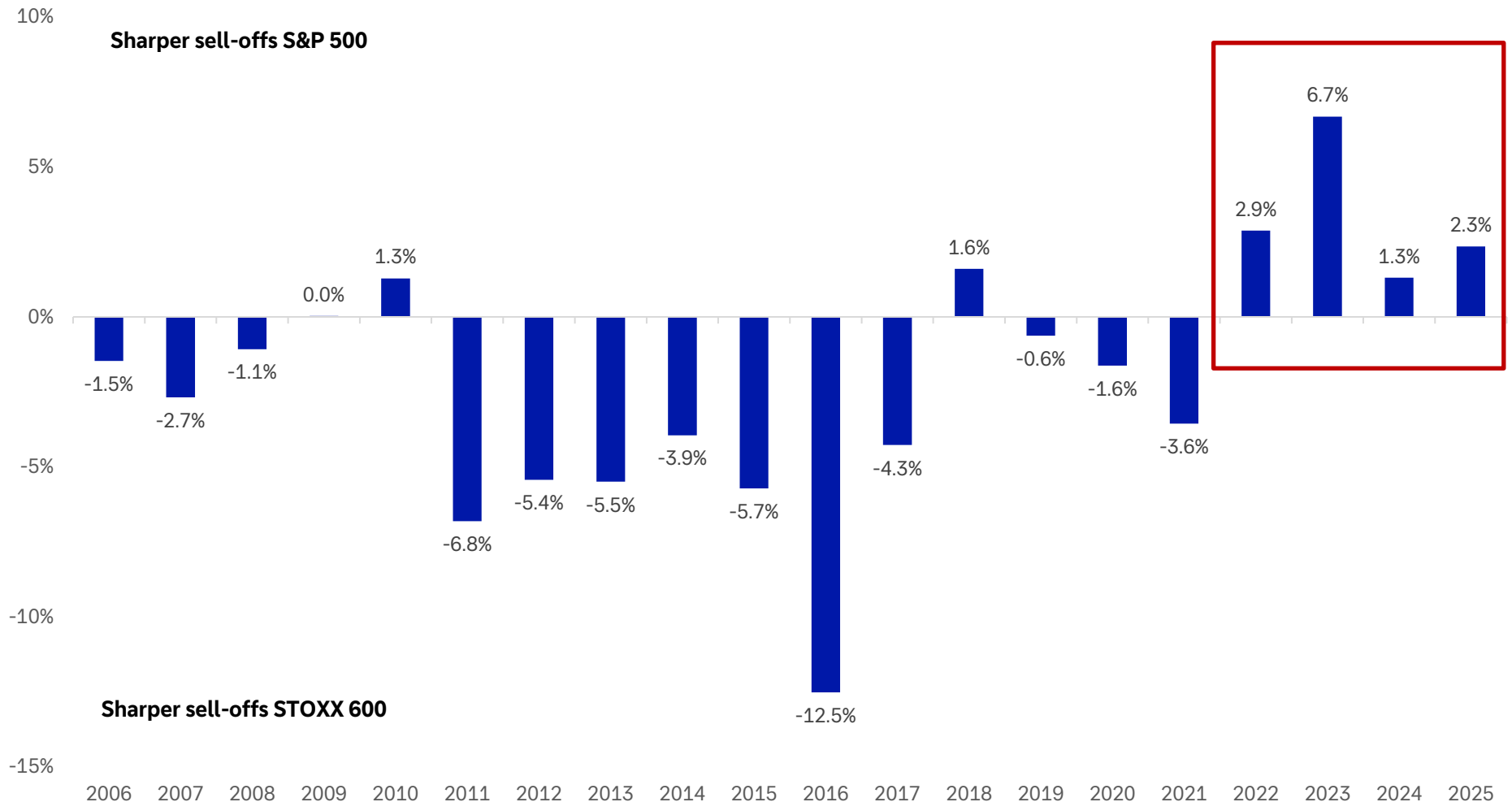
Reference indices: S&P 500 and STOXX 600, GICS sectors. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Concentration: leads to sharper sell-offs



„When the US sneezes, Europe catches a cold” no longer holds true. Drawdowns are significantly sharper in US equities. “When the US sneezes, Europe says Gesundheit”.

Difference in maximum drawdown of S&P 500 vs STOXX 600 (annual, local currency)



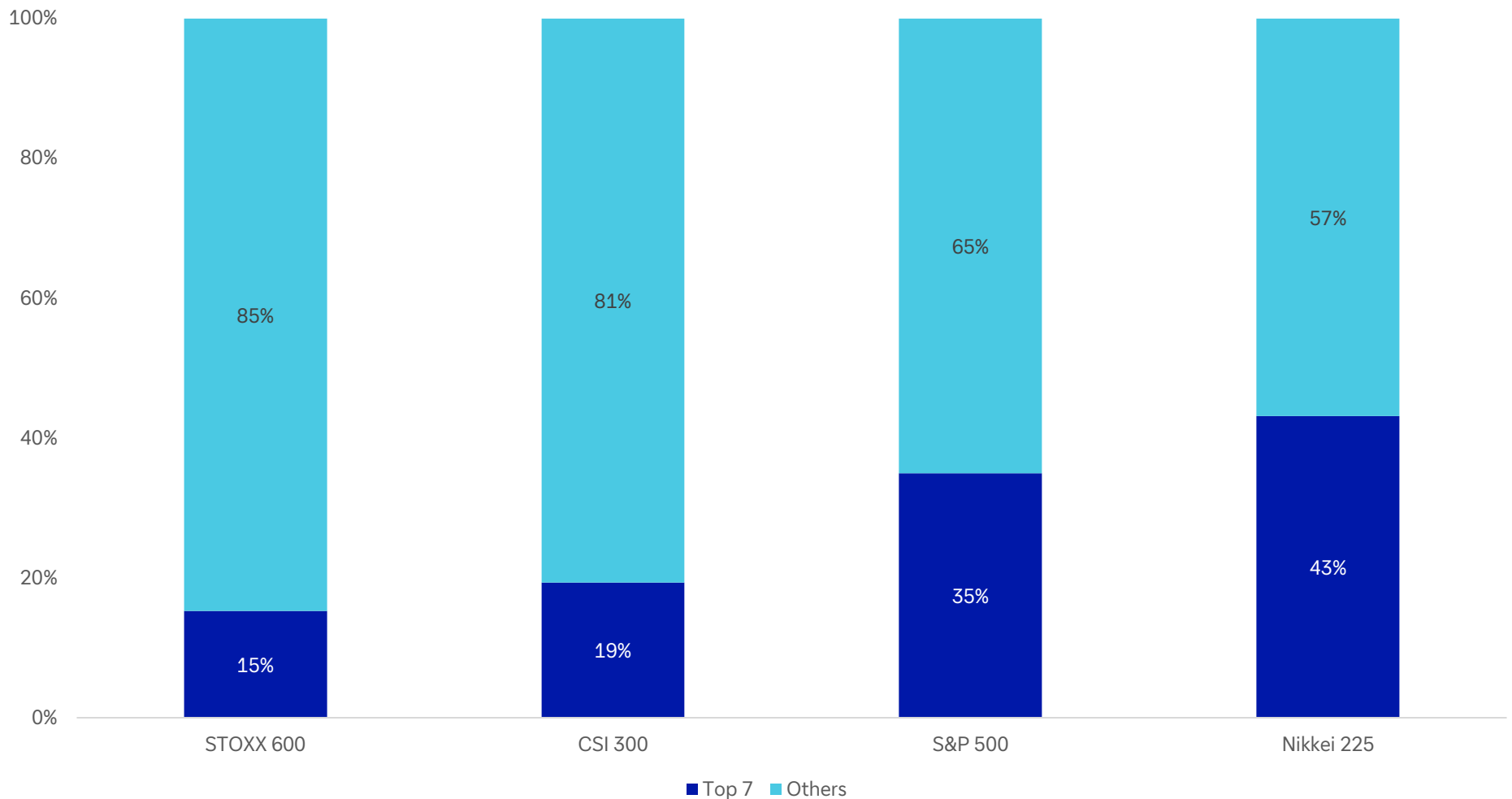
Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Concentration: in index weights



The European STOXX 600 is well diversified. The 7 largest corporates make up only 15% of the index weight vs 43% in the Nikkei and 35% in the S&P 500.

Current index weights



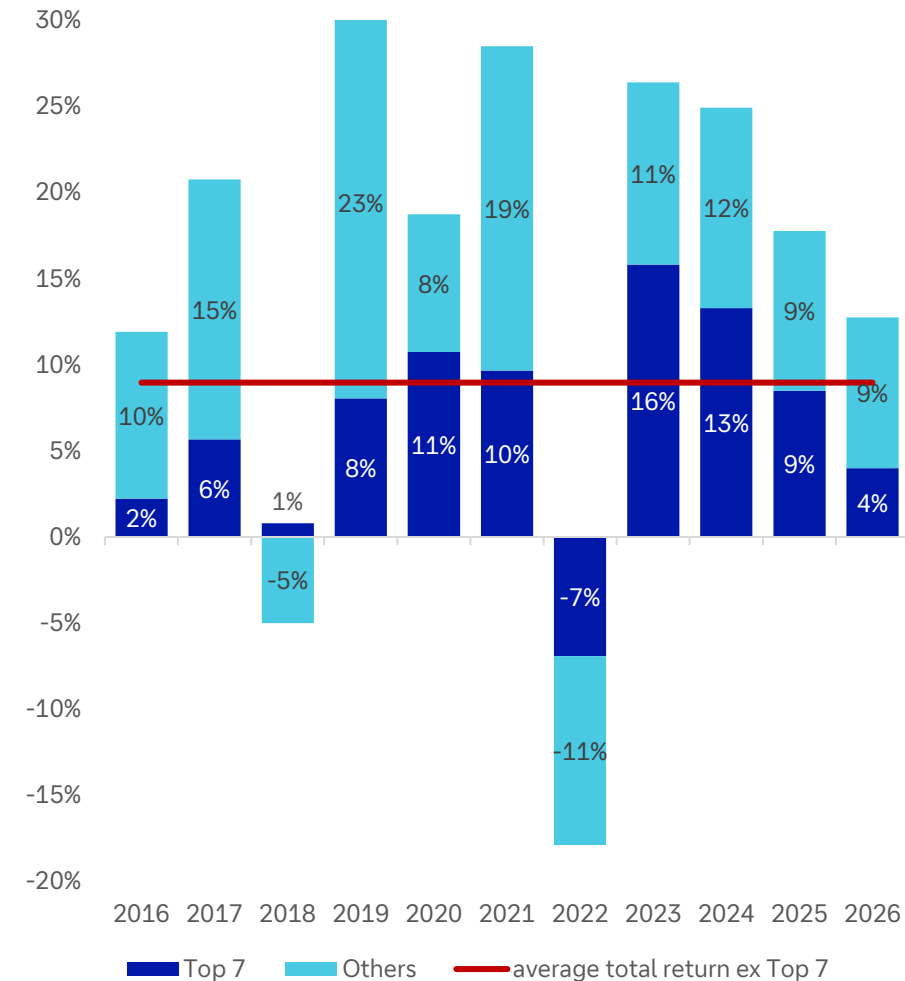
Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Concentration: in performance

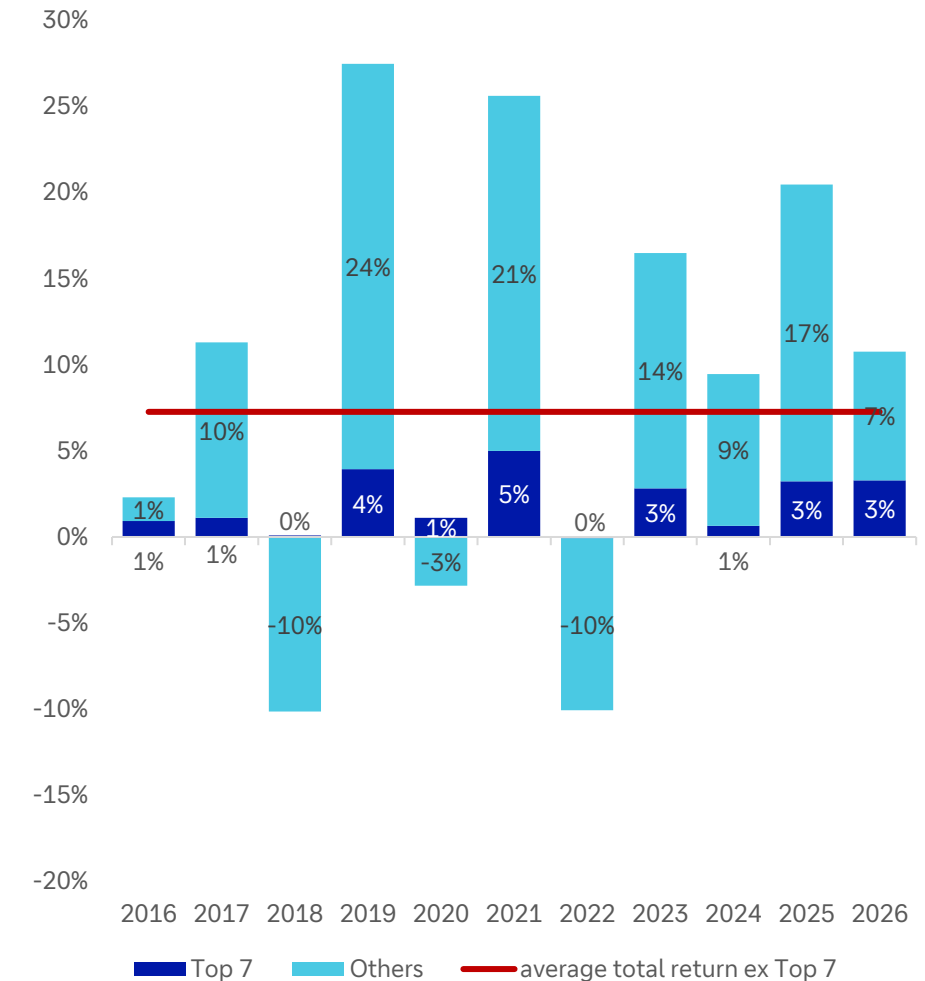


On average over the past 5 years, the S&P returned 16% per year vs 12% in the STOXX 600. Without the top 7 companies, the S&P would have delivered an average return of 8% vs 10% for the STOXX ex top 7.

S&P 500 – total return contributions



STOXX 600 – total return contributions



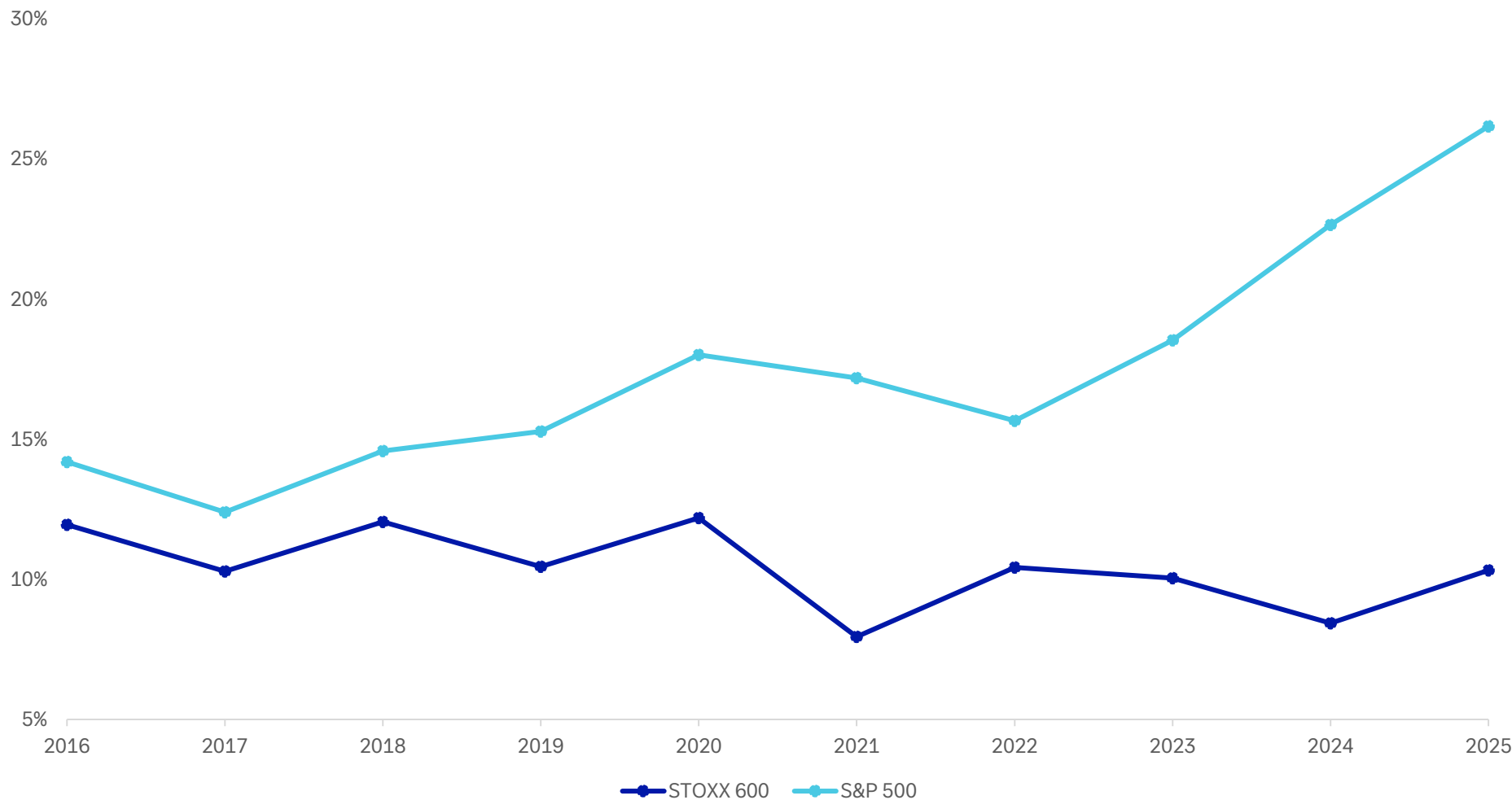
Note: 2026 is YTD. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Concentration: in earnings



The top 7 US companies contributed 53% to earnings growth since 2021. In the STOXX 600, the top 7 only accounted for 33% of profit growth over the same period.

Top 7 companies: share of aggregate index earnings



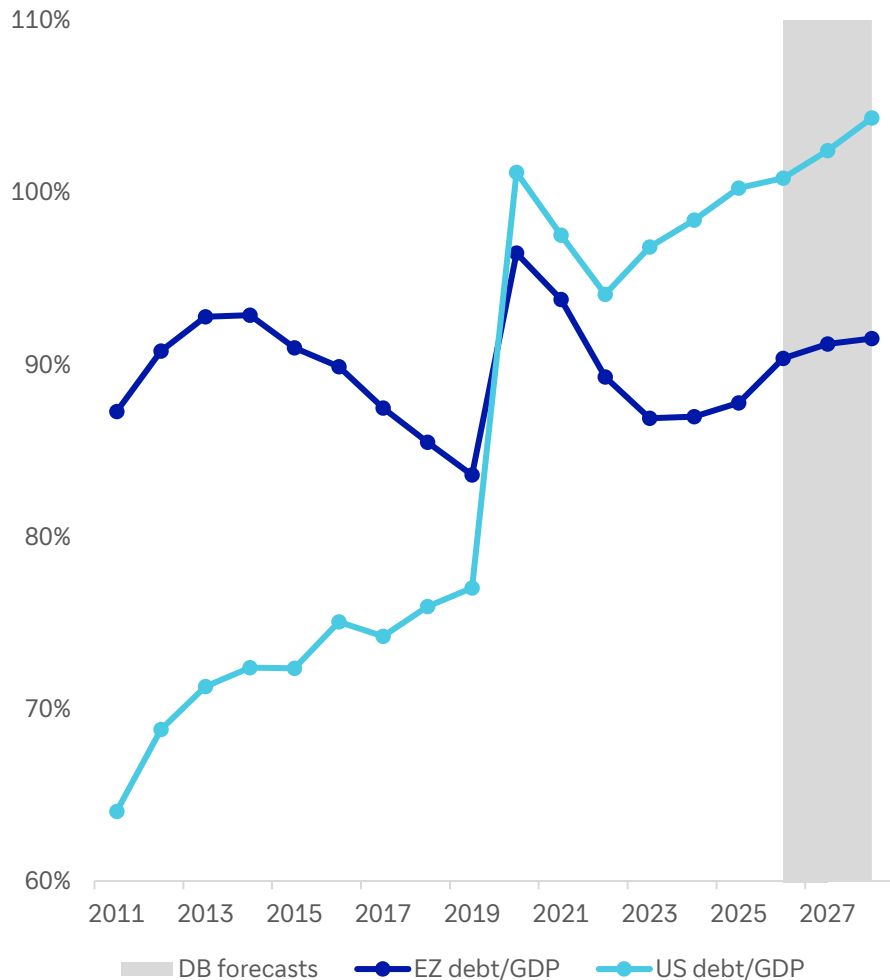
Note: based on current index members and current Top 7. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Fiscal Risks – look increasingly similar

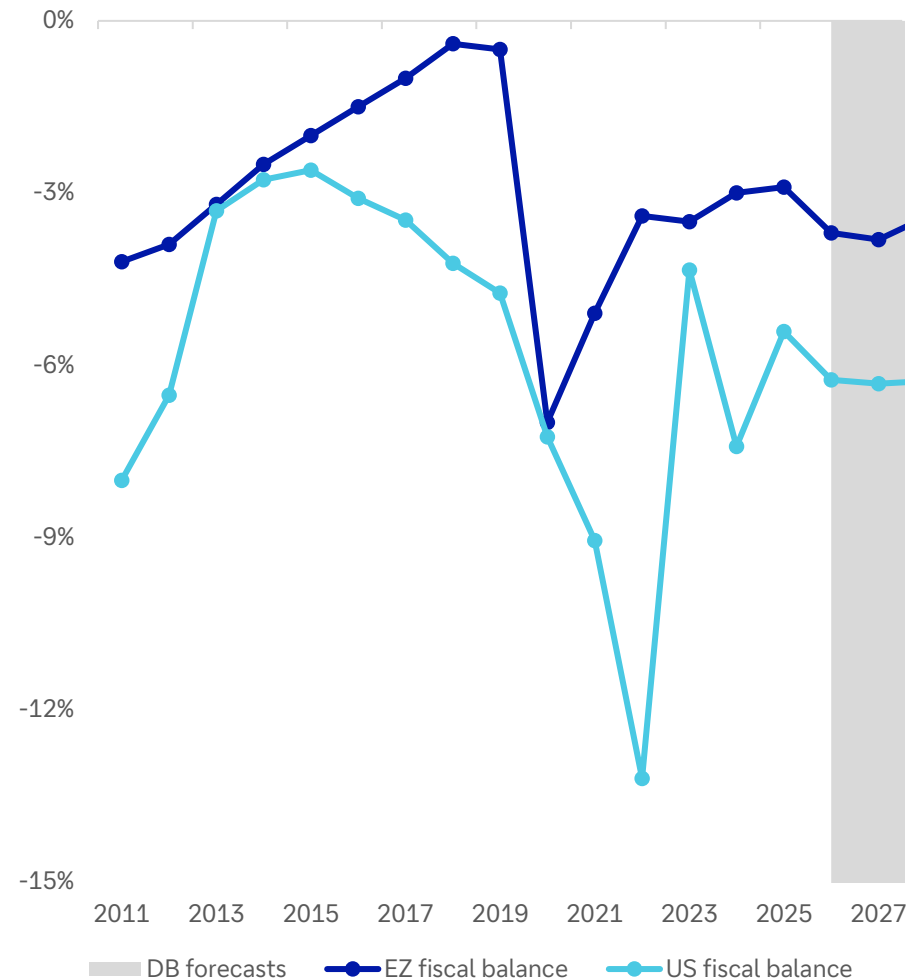


Despite the German fiscal expansion, debt ratios are expected to worsen more in the US than in the Eurozone.

Debt-to-GDP: Eurozone vs US



Fiscal balance (% GDP): Eurozone vs US



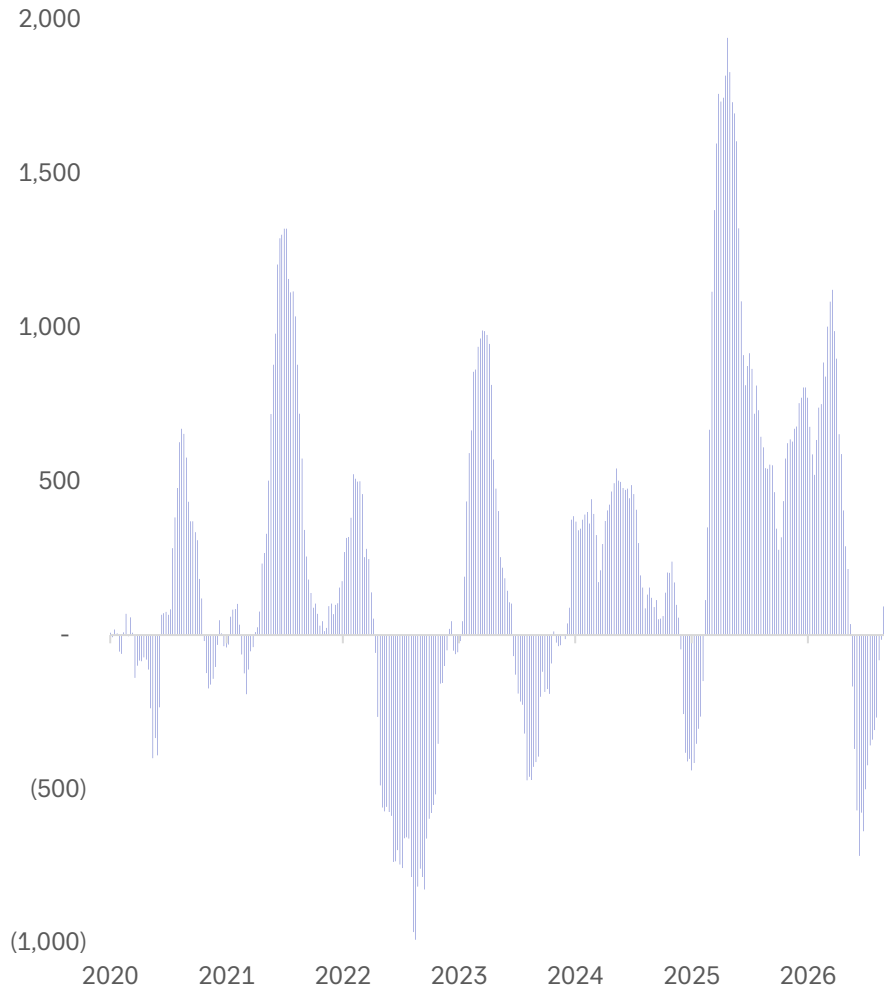
Note: Debt-to-GDP is net for US and gross for EZ to align with DB forecasts. Source: Deutsche Bank Research 09/15/2026

Flows – US dominates, but less than it did

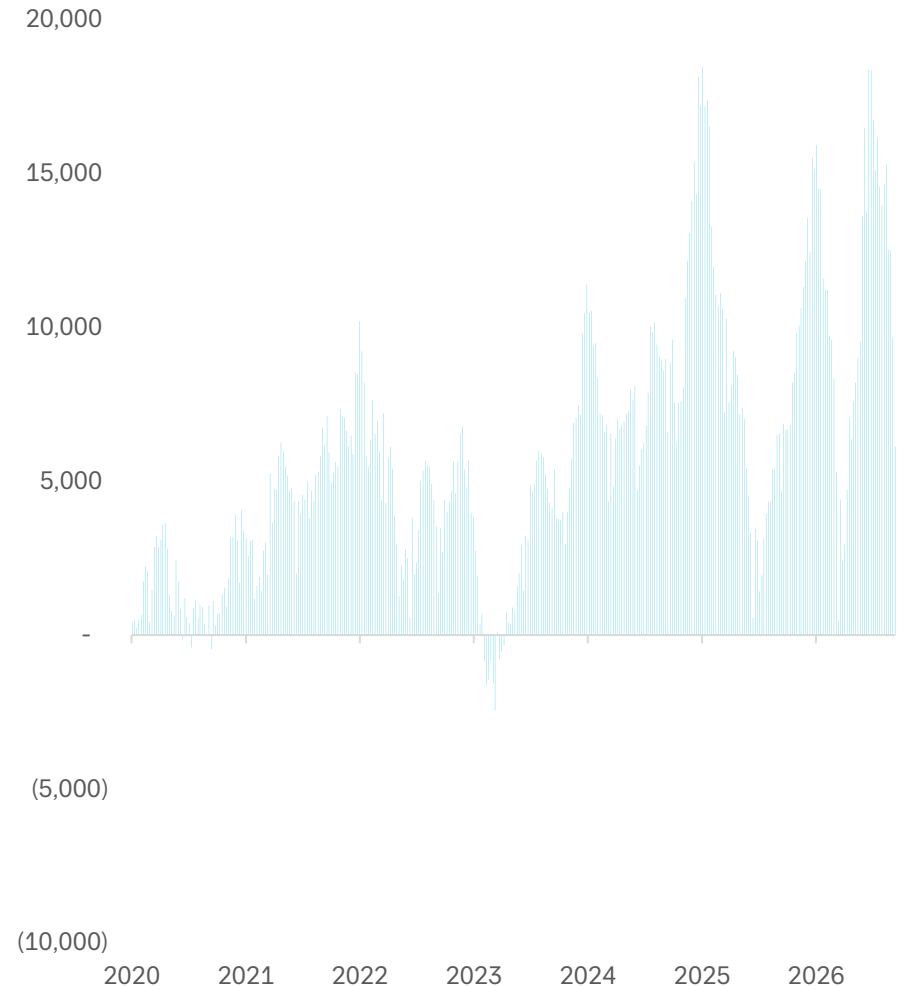


Inflows into European-focused ETFs have picked up significantly over the past two years but remain much smaller and more volatile than US flows.

Europe



USA



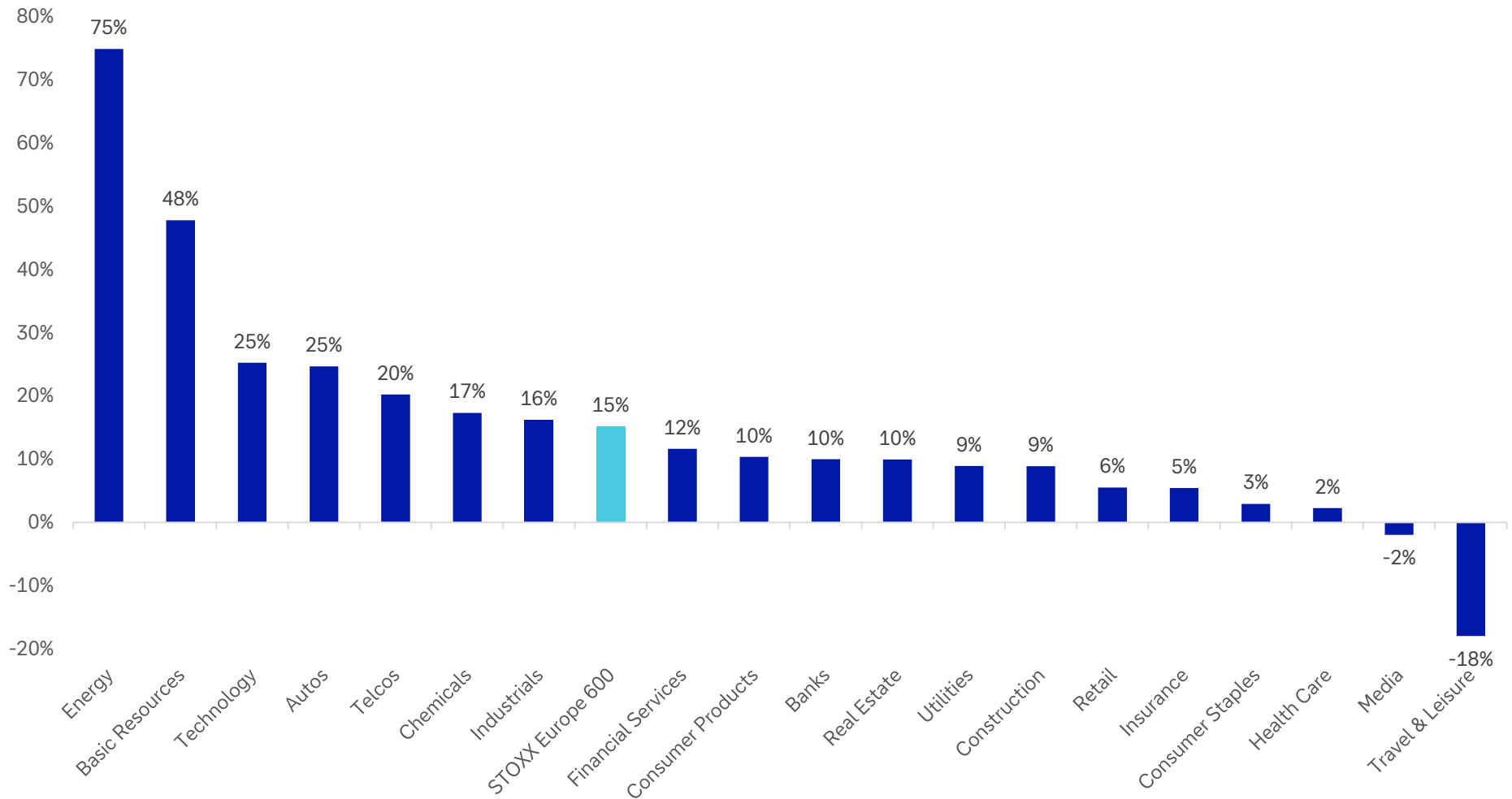
Note: Aggregate fund flow, 12w moving average, based on 50 largest ETFs for the region, in mn USD. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Earnings – Broad-based growth



The outlook for European earnings is encouraging. Almost all sectors should see positive earnings growth this year.

STOXX 600: Consensus expected earnings growth, FY26



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026



Deep dive: Germany

Don't underestimate Germany's capacity to change



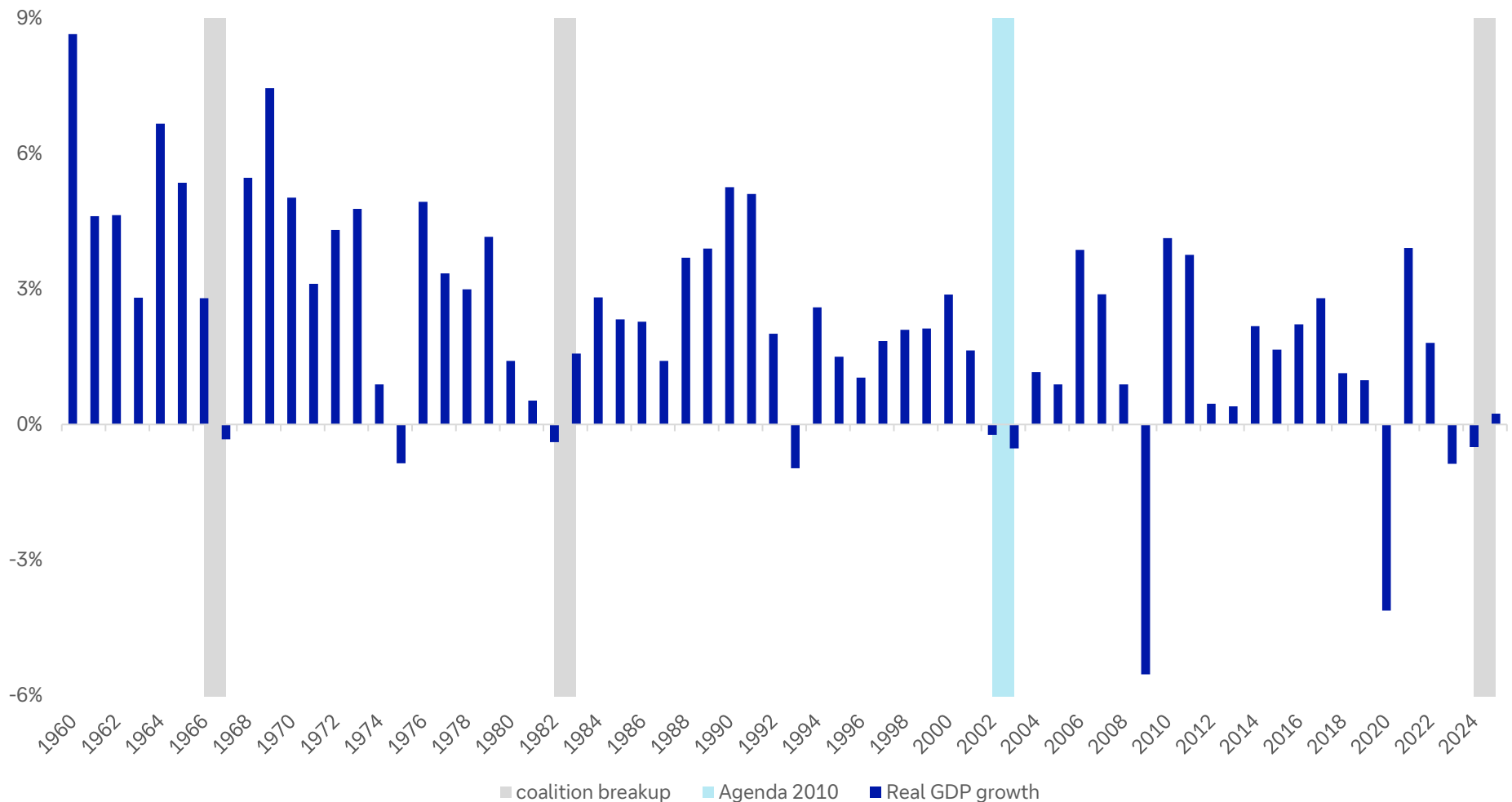
Germany has the potential to become the driving force behind Europe's economic revival. Despite years of macro headwinds, it reduced its debt ratio over the past decade, creating the fiscal space for a major spending package. Economic activity is stabilising, manufacturing conditions are improving, reforms are progressing, and global investor interest is returning. Germany has repeatedly demonstrated an ability to adapt and reinvent itself, with major reform waves following the coalition collapses of 1966 and 1982 and the implementation of Agenda 2010 in the early 2000s.

- **Fiscal:** Few countries have faced as many challenges as Germany in recent years: rising Chinese competition in key industries, the energy shock following Russia's invasion of Ukraine, large-scale immigration, US tariffs, and a three-year manufacturing downturn. Despite these headwinds, Germany is the only G7 economy to have reduced its debt-to-GDP ratio over the past decade. The government plans to raise around €1tn of additional debt by 2030, on top of the current debt stock of €2.8tn. Net credit borrowing is expected to increase from €180bn in 2026 to more than €200bn in 2027, while spending on defence and investment is set to reach roughly €120bn each this year, up more than 30% year-on-year.
- **Macro:** Germany's economy is showing greater resilience towards the ongoing geopolitical headwinds than expected. The manufacturing PMI climbed above 54 in August, its strongest reading in over four years, and both the Ifo and ZEW surveys surprised positively. Factory orders are gradually improving and GDP growth in H1 exceeded expectations, prompting consensus GDP forecasts to be revised up. Although higher energy prices may create short-term headwinds, the burden remains manageable, with energy costs representing just slightly more than 1% of sales for listed German companies.
- **Politics:** Germany's reform momentum is strengthening. The government is combining fiscal stimulus with measures to lower energy costs, boost corporate investment, accelerate infrastructure projects, reform the labour market, and modernise the pension system. We estimate the pension reform alone could unlock €60-100bn of annual capital market inflows. At the same time, political risks appear manageable. Regional elections are unlikely to affect federal policymaking, and we see little risk to the government's reform agenda. If anything, weaker results for the coalition parties could increase pressure to accelerate reforms.
- **Equities:** Markets have yet to price in Germany's improving outlook. While economic indicators continue to strengthen, investor flows into German equities remain weak and the MDAX has lagged the DAX by 7% since late June. As the index with the greatest exposure to the domestic economy, the MDAX is the purest listed play on Germany's recovery and has historically shown a much stronger relationship with German GDP growth. A combination of fiscal stimulus, structural reforms, and more industry-friendly policies creates substantial catch-up potential following years of underperformance. Importantly, valuations remain attractive, with the MDAX trading at its cheapest relative level in history.

BIG PICTURE: Recession → Reforms → Growth

Don't underestimate Germany's capacity to change. After the coalition break-ups of '66 and '82, Germany introduced reforms and re-emerged stronger. Reforms were also implemented when growth turned in 2003 under the Agenda 2010.

Germany was in recession in '66 & '82, but recovered quickly after the political crisis was resolved



Source: Bloomberg Finance LP, Bundesbank, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion

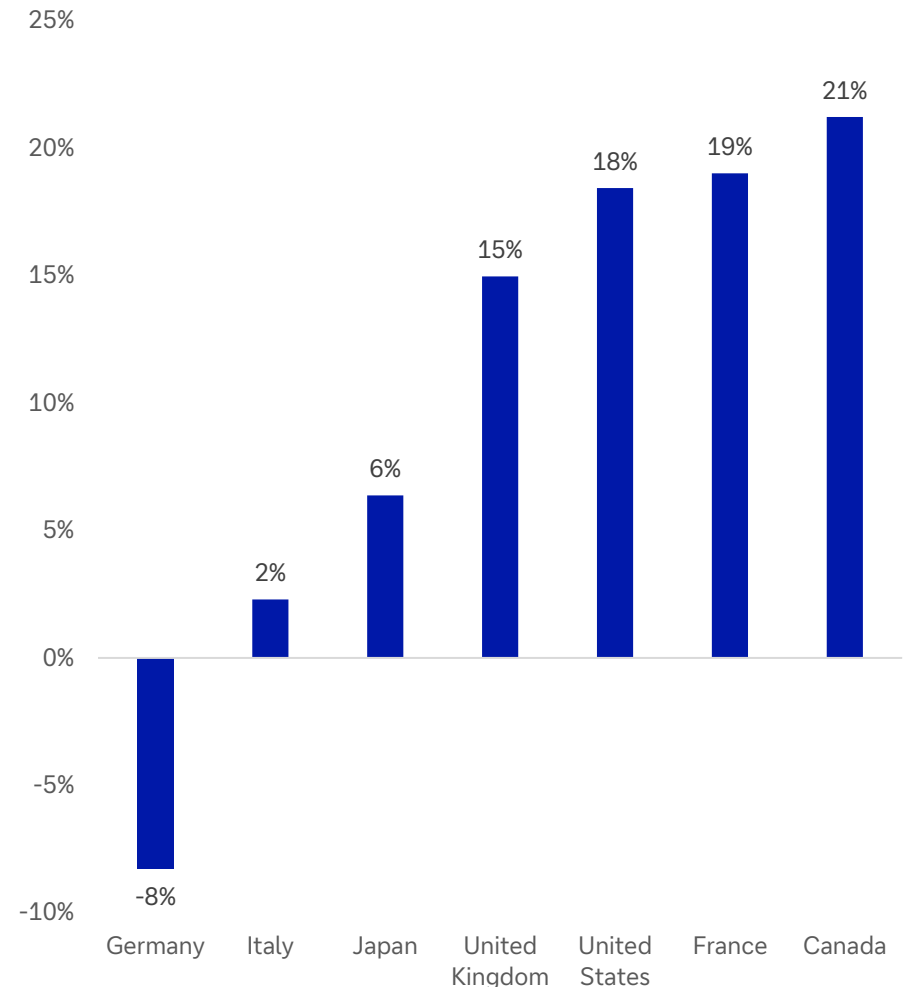


Germany is the only G7 country to have reduced its debt relative to GDP over the past decade.

Key takeaways

- Over the past decade, few countries were facing as many headwinds as Germany: growing competition from China in its key industries, the energy transformation following the war in Ukraine, mass immigration, US tariffs and a 3-year manufacturing crisis.
- Despite all of this, Germany managed to be the only G7 country to have reduced its debt relative to GDP in the last 10 years.
- German debt-to-GDP dropped by 8 percentage points while e.g. US growth was largely debt-financed, with the US' debt-to-GDP ratio increasing by 18 percentage points over the same period.

Change in debt-to-GDP over the past 10y (pp)



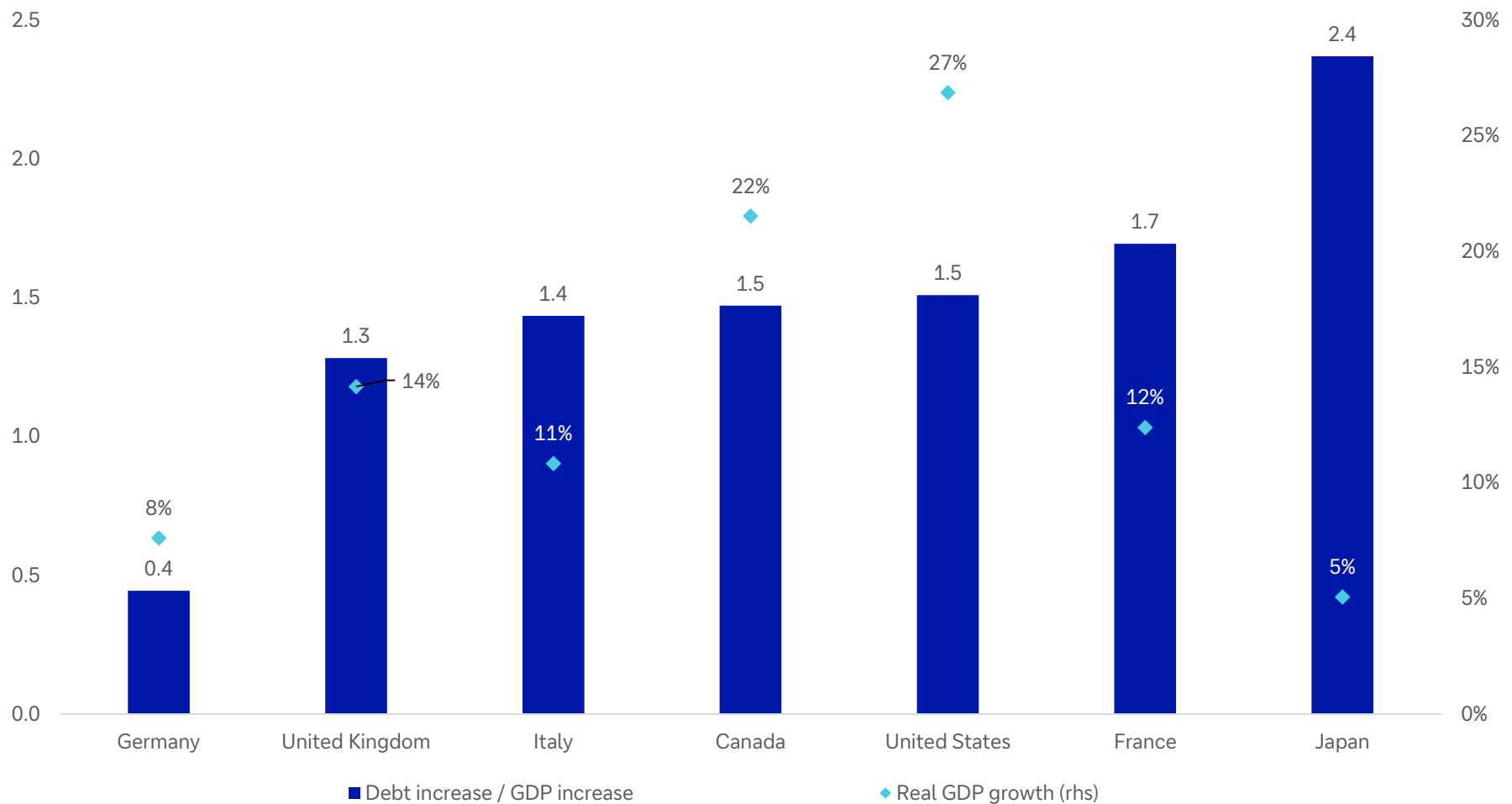
Note: All statistics in local currency, 2025 vs 2015, debt-to-GDP based on nominal GDP and gross general government debt. Source: IMF, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion



Germany is the only G7 country that increased debt by significantly less than 1 Euro for every Euro of GDP growth.

Bang for the buck: Germany financed its growth at a lower cost (10y)



Note: All statistics in local currency, 2025 vs 2015, debt increase / GDP increase in nominal terms. Source: IMF, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion



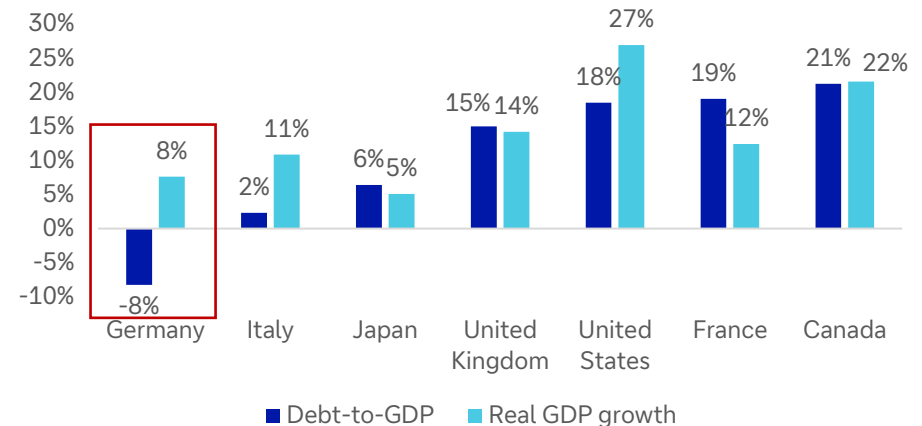
Germany managed to grow its GDP despite fiscal consolidation.

An additional €1 trillion in debt by 2030 could unlock significant growth.

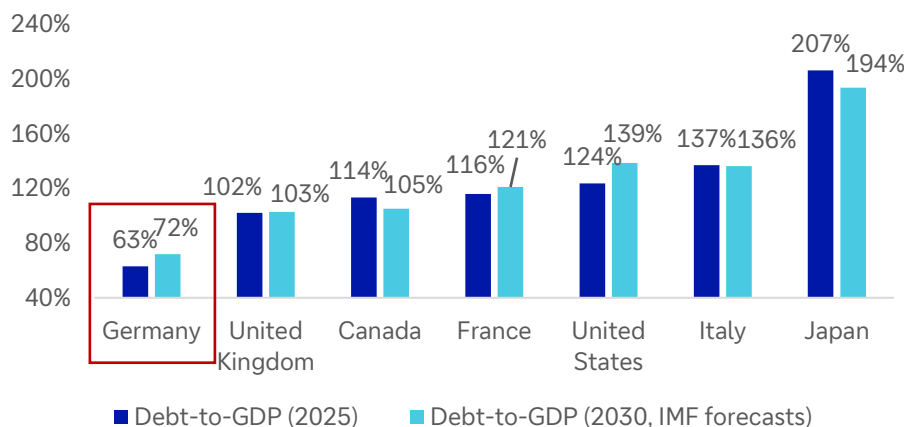
Key takeaways

- The German government plans to increase debt by more than €1 trillion by 2030, coming from €2.8 trillion in 2025 (gross general government debt). Almost 70% of the debt increase will be related to the core budget, vs 30% in off-budget funds.
- A large share of the additional debt (>€600bn) will be allocated to defence and security-related spending, almost €300bn to the special infrastructure fund (especially transportation and digitalization), and the remaining amount to other core-budget borrowing.
- For more information, see [here](#).

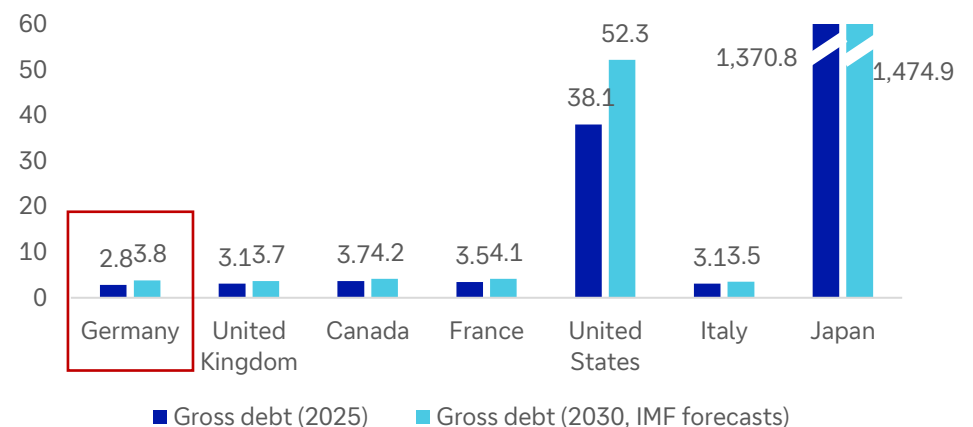
10y changes in debt and GDP across the G7



IMF projections of debt-to-GDP



Gross debt (trillions, local currency)



Note: All statistics in local currency, 10y changes are 2025 vs 2015, debt-to-GDP in nominal terms based on gross general government debt.
Source: IMF, Bundestag, Deutsche Bank Research 09/14/2026



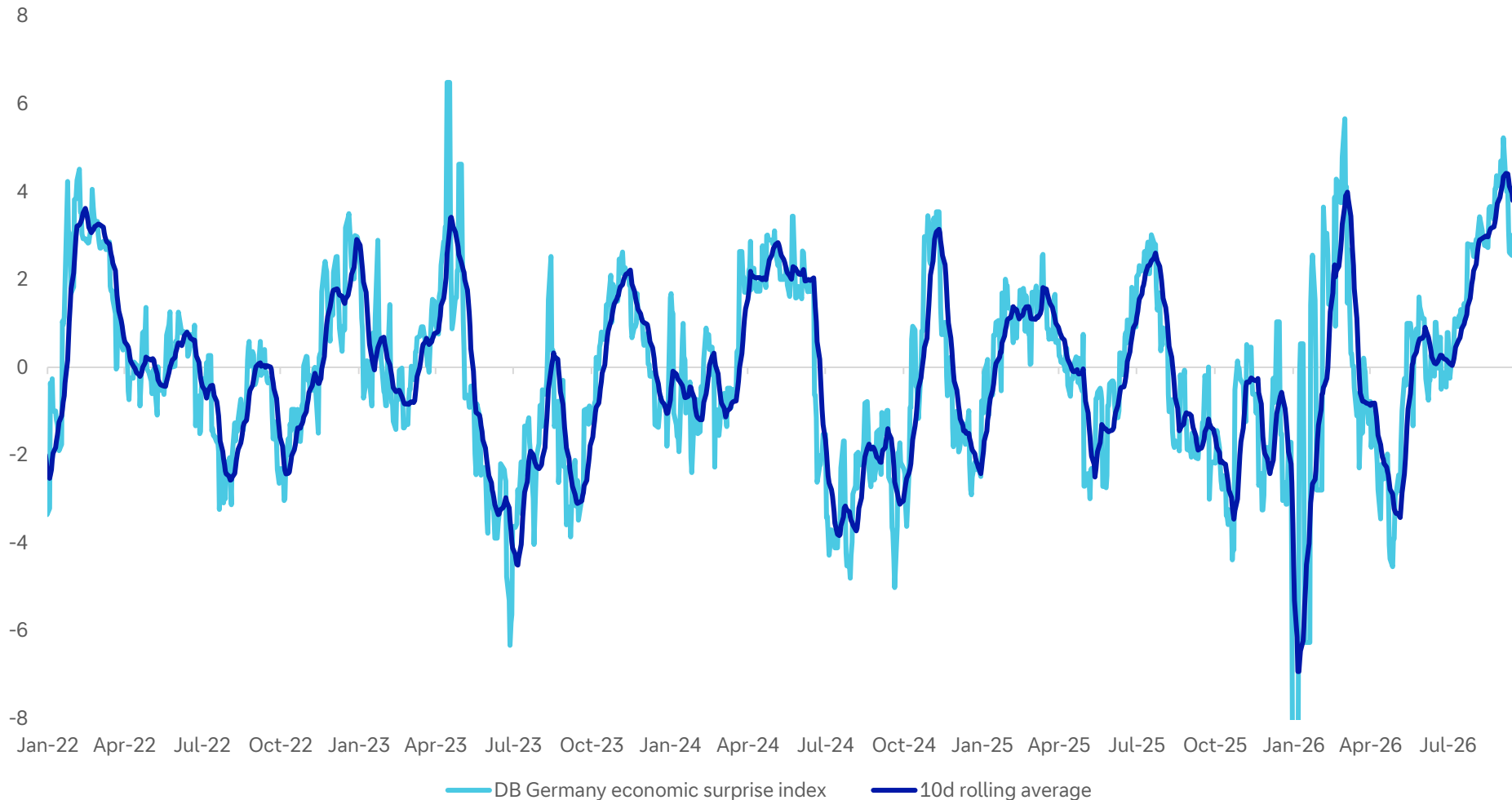
Macro & political backdrop

Macro backdrop – back on track



Most recent macro data came in much better than expected in Germany.

DB Germany economic surprise index has reached its highest level since 2021



Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

Macro backdrop – back on track

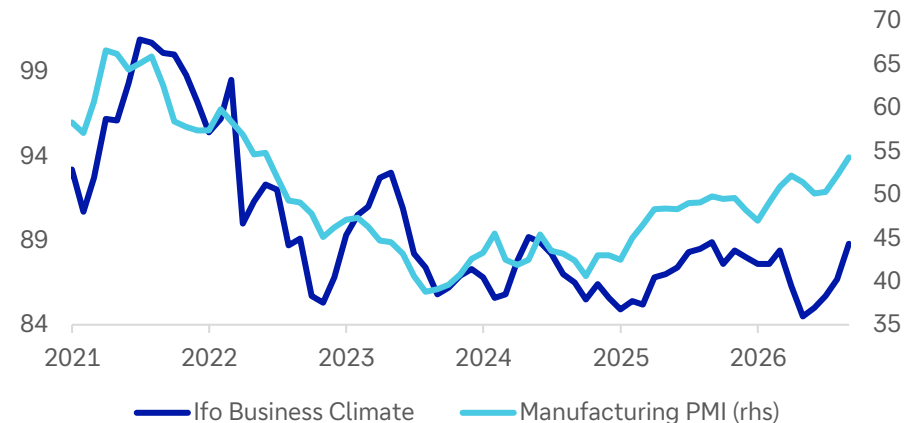


Green shoots have emerged in the German economy over recent months.

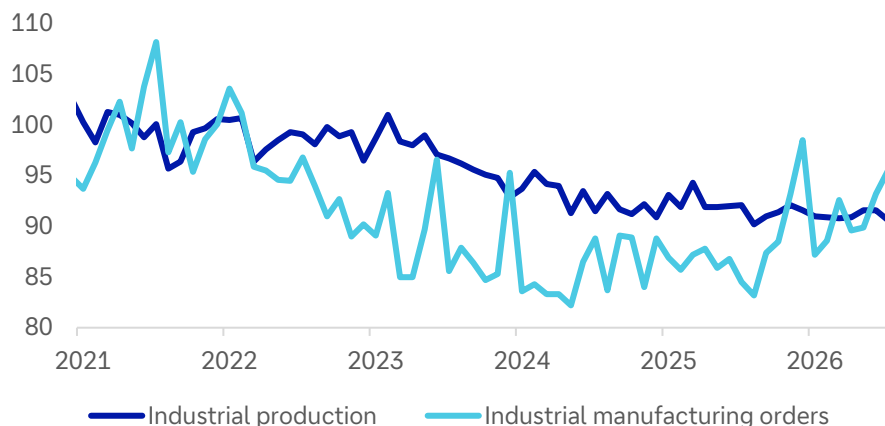
Key takeaways

- The German economy is more resilient than expected. The manufacturing PMI rose above 54 in August, its highest level in more than four years. Ifo and ZEW surveys also both surprised to the upside in August.
- Factory orders continue to edge higher, albeit still driven mainly by large orders.
- After GDP proved stronger than expected in H1, we recently upgraded our '26 GDP forecast to 1%.

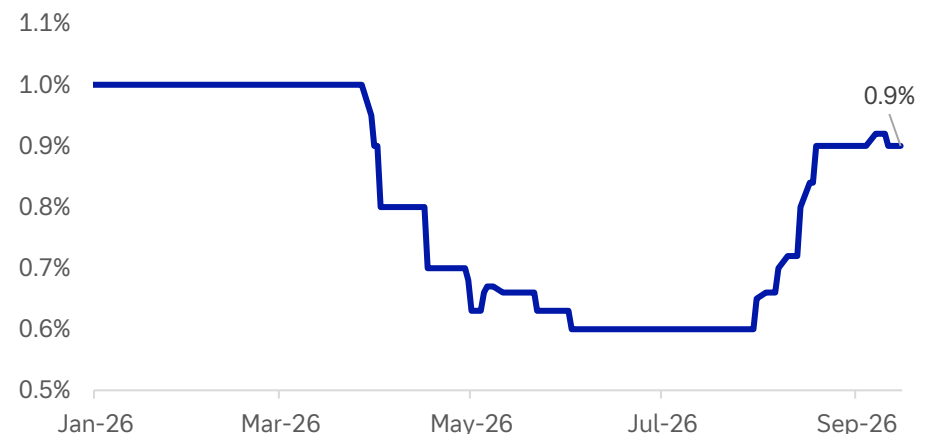
Corporate sentiment recovering



Factory orders accelerated in July



2026 GDP forecasts revised upwards*



*Median of Bloomberg consensus forecasts. Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

Regional elections

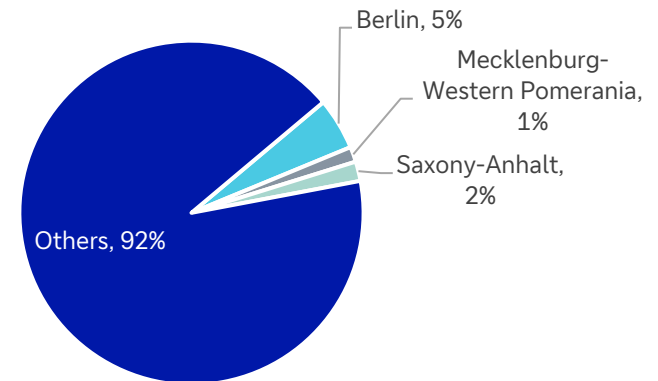


Higher uncertainty from regional elections. But relevance limited and pressure on government to increase pace of reforms is growing.

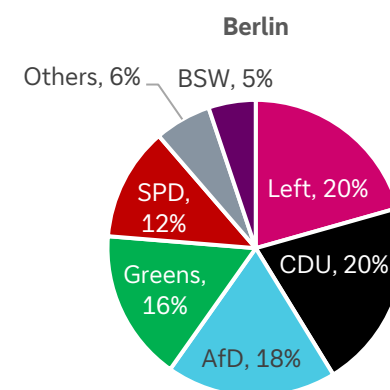
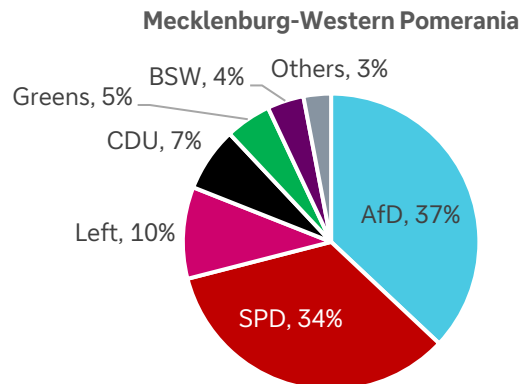
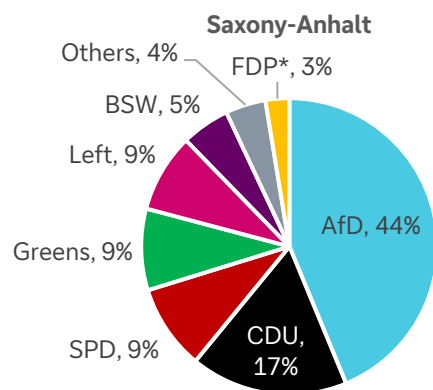
Key takeaways

- Contrary to many concerns, the Saxony-Anhalt election triggered little to no market reaction. While the AfD will likely enter government, the economic relevance of Saxony-Anhalt within Germany is limited. In Meck-Pom and Berlin, AfD support is weaker, though fragmented parliaments may complicate coalition building.
- Regional elections have limited implications for federal politics. More importantly, we don't see the elections derailing the government's reform agenda. If anything, weaker results for coalition parties could increase pressure to accelerate reforms.

2025 German GDP: breakdown by State



Regional elections: latest results and polls (MP and Berlin elections on 20th September)



*below 5% hurdle, not in parliament

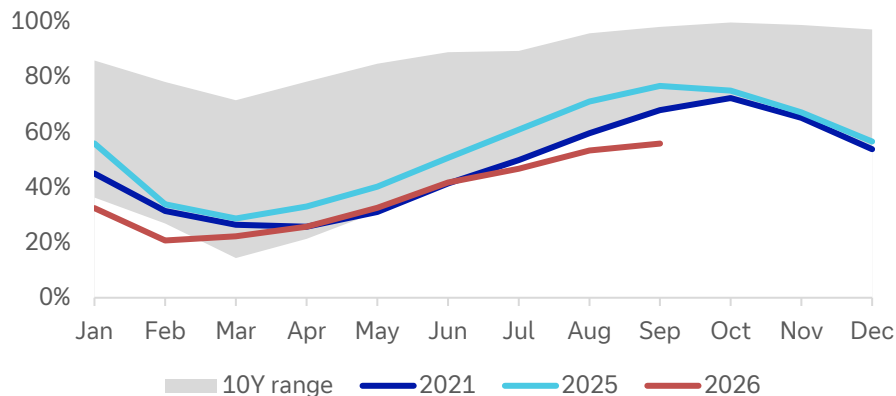
Source: Wahlrecht.de, Statistikportal des Bundes und der Länder, Deutsche Bank Research 09/15/2026

Macro headwinds – manageable?

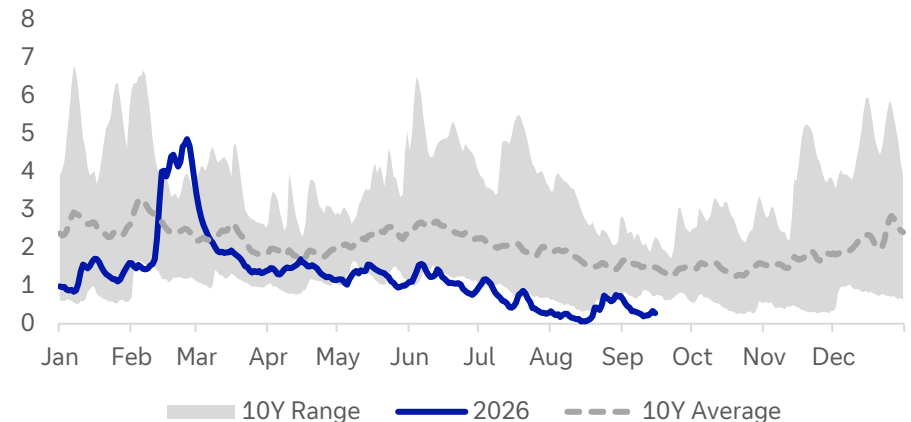


Energy costs to weigh on short-term sentiment. However, current levels are still manageable. We estimate energy costs to be 1.2% of sales for listed German companies.

Gas storage level at only 56%



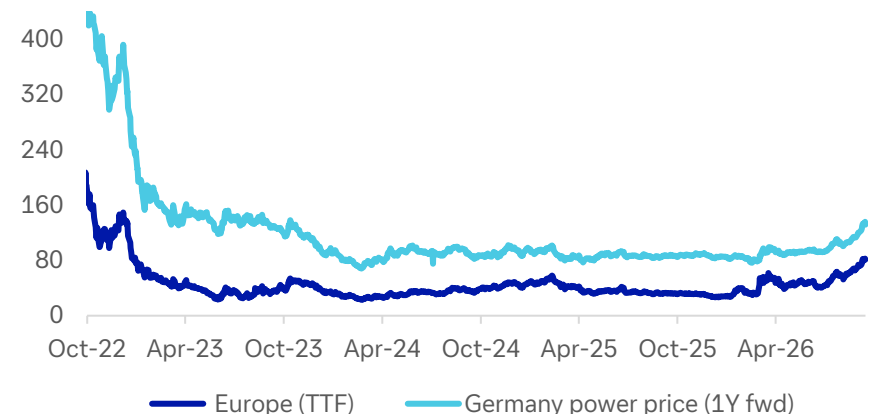
Rhine level at Kaub remains too low



Bund yield at highest since 2009



Gas and power prices up, but far below '22



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

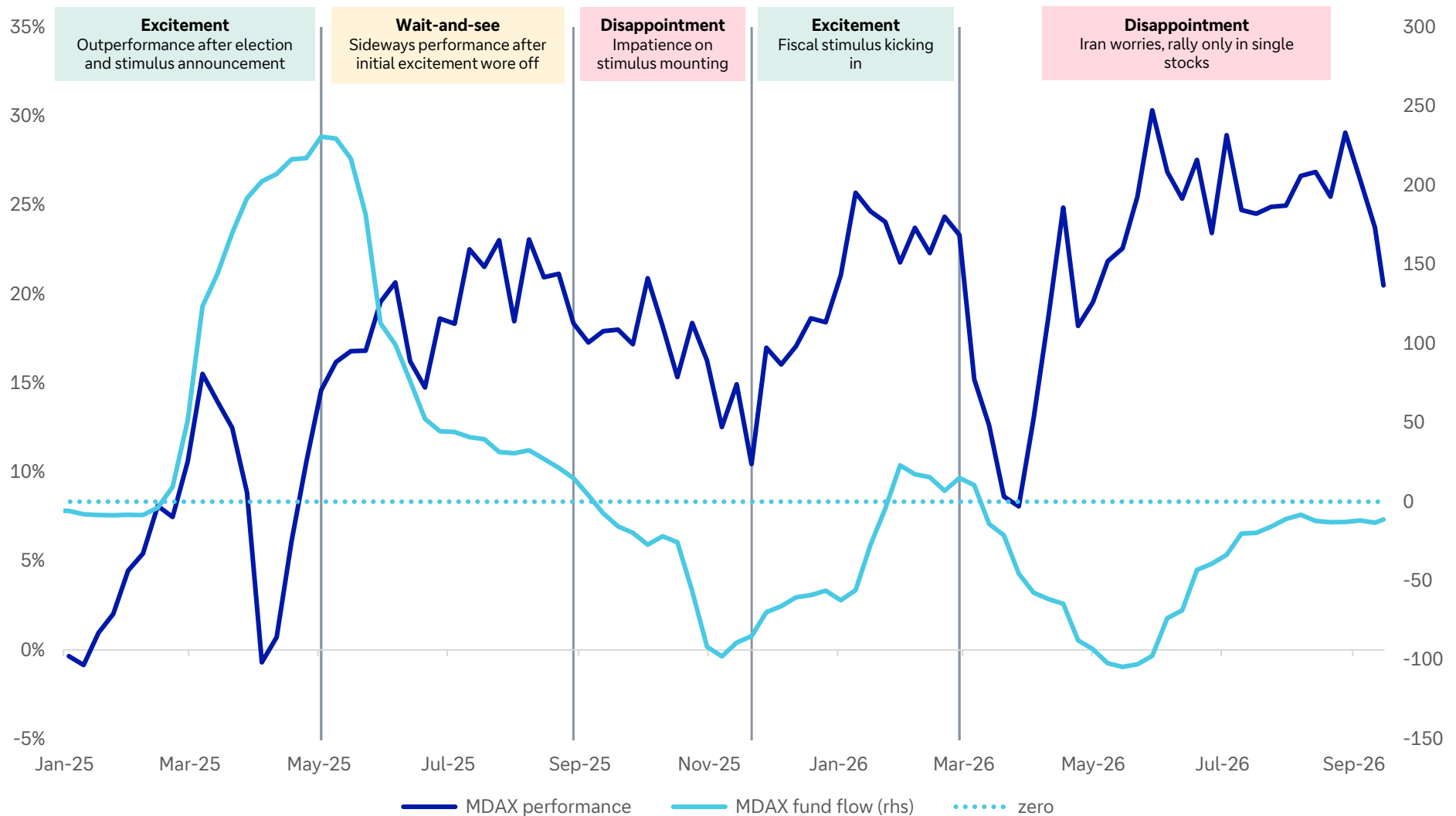


Equity markets

Germany: Rollercoaster of emotions



We have been going through many emotions. Bring patience to benefit from the recovery.



Aggregate fund flows (12-week mov. avg., mn EUR) based on up to 10 largest ETFs for the index. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

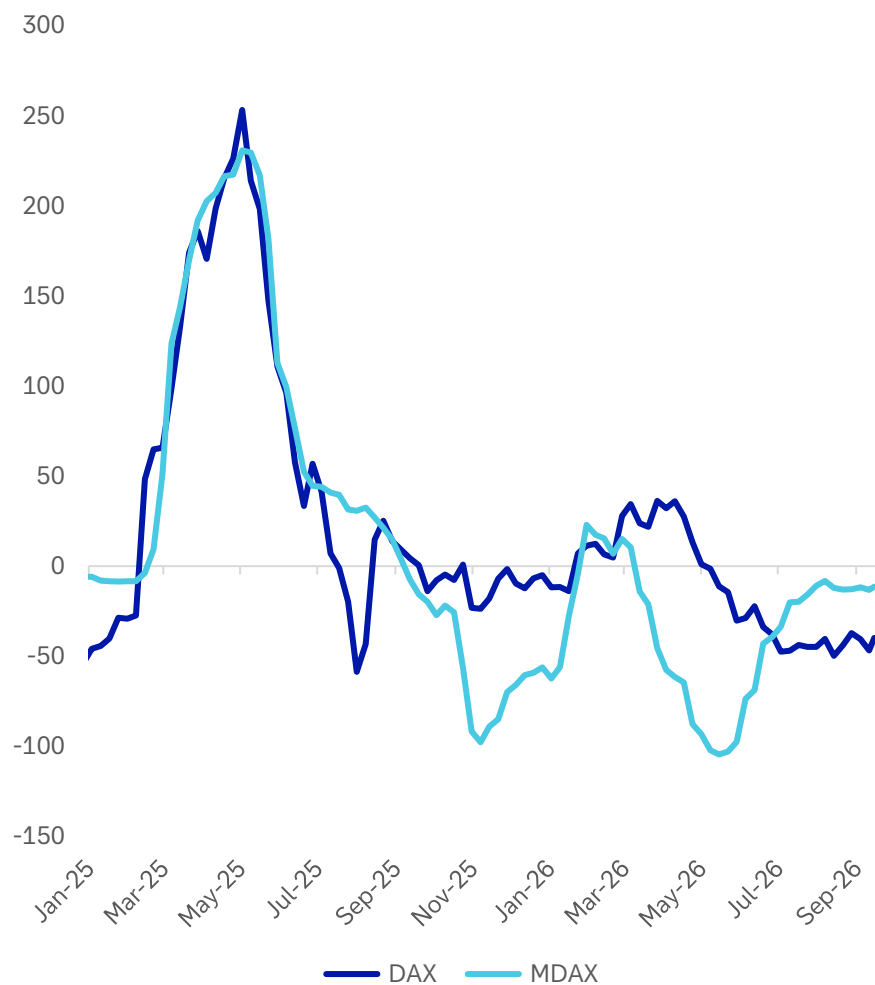
German ETF flows slowly recovering



Despite the improvement in macro indicators, flows into German equities are still depressed. The MDAX underperformed the STOXX 600 by 2% since end of June.

DAX & MDAX ETF aggregate fund flow

YTD performance (total returns, local CY)



Note: Fund flows proxied by flows into and out of largest (up to) 10 ETFs for the index, 12-week moving average, mn EUR. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Equity implications: DAX vs MDAX

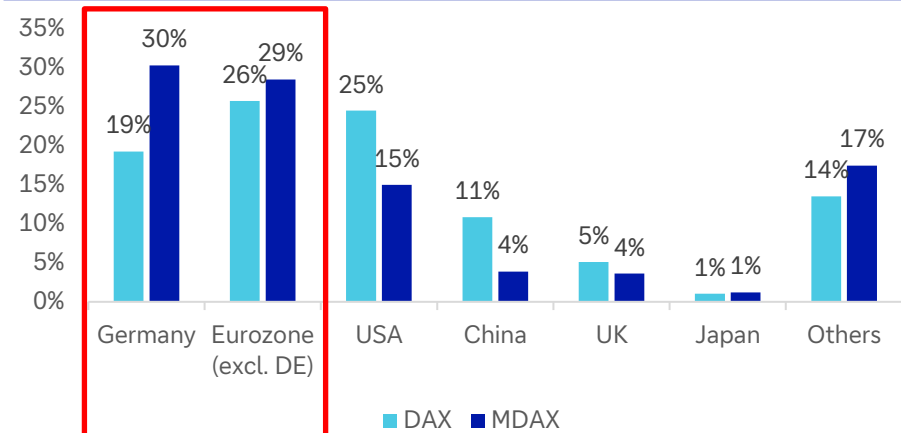


We think the MDAX is a better proxy for German growth and should benefit most from economic improvements and new reforms.

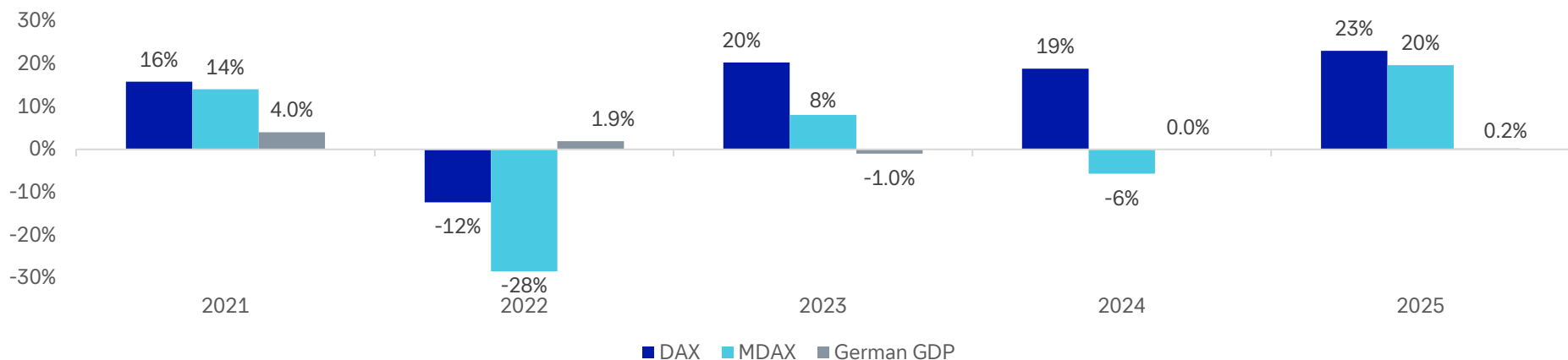
Key takeaways

- The DAX's exposure to Germany is limited; a better proxy for Germany-specific news flow is the MDAX.
- The MDAX has a higher revenue exposure to Germany than the DAX (see right chart) and its performance is linked much more closely to German GDP growth (see below chart). The new reform agenda and a more industry-friendly policy approach in Germany bear significant catch-up potential for the MDAX which has underperformed the DAX significantly over the past years (see [slide 30](#)).

Regional revenue exposure (FY24)



MDAX is the better proxy for German growth



All exposures are best estimates by our DB company analysts and are not necessarily based on published company data.
Source: Bloomberg Finance LP, Company Data, Deutsche Bank Research 09/15/2026

MDAX still offers high catch-up potential



German mid caps underperformed large caps massively over the past years. The improving macro backdrop and new reforms should support the MDAX.

Total return since 2022



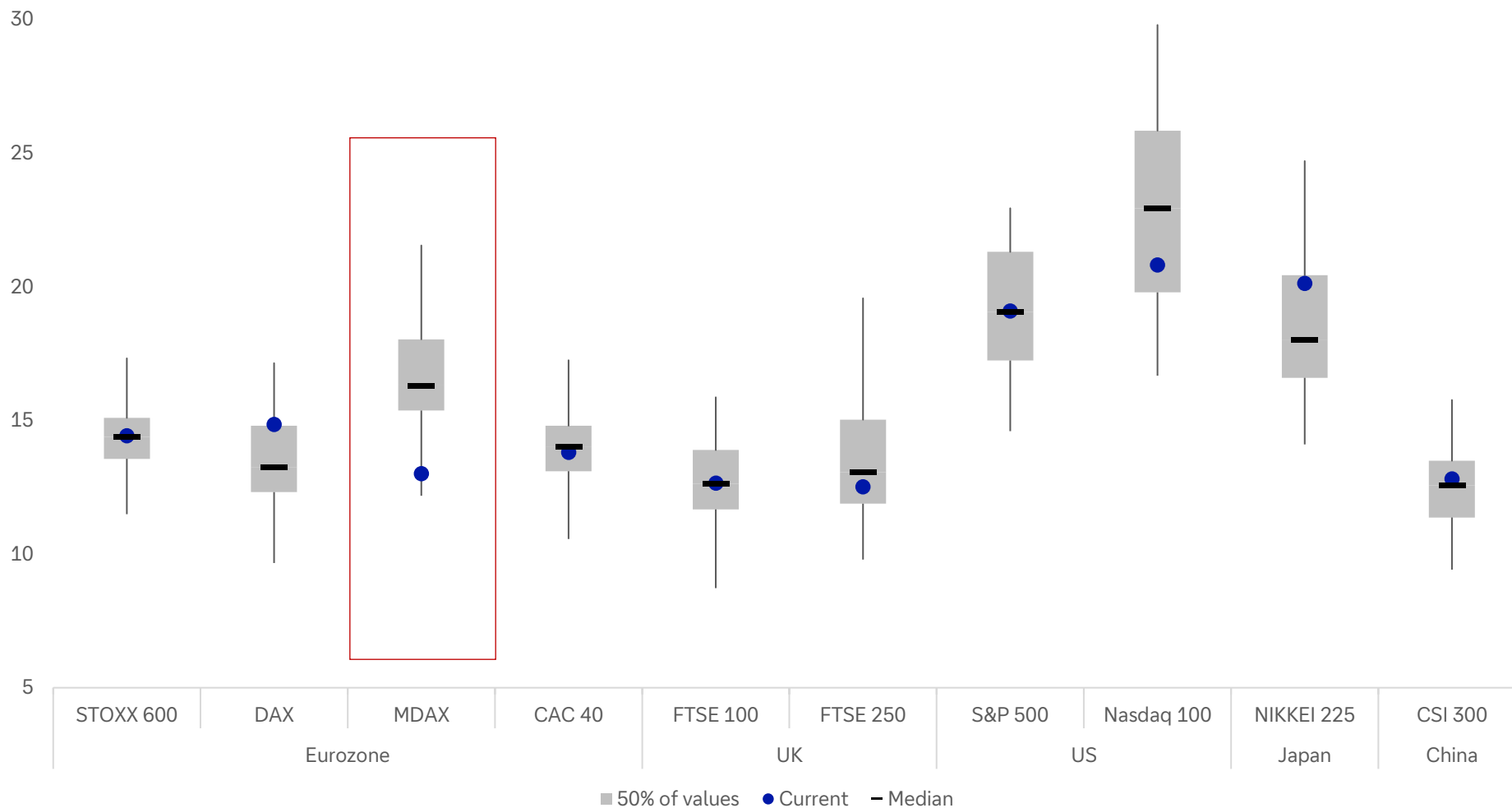
Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Valuations



In our view, European valuations are not elevated. Within Europe, the MDAX trades at lowest value relative to history

Valuation vs 10y median (12m fwd P/E)



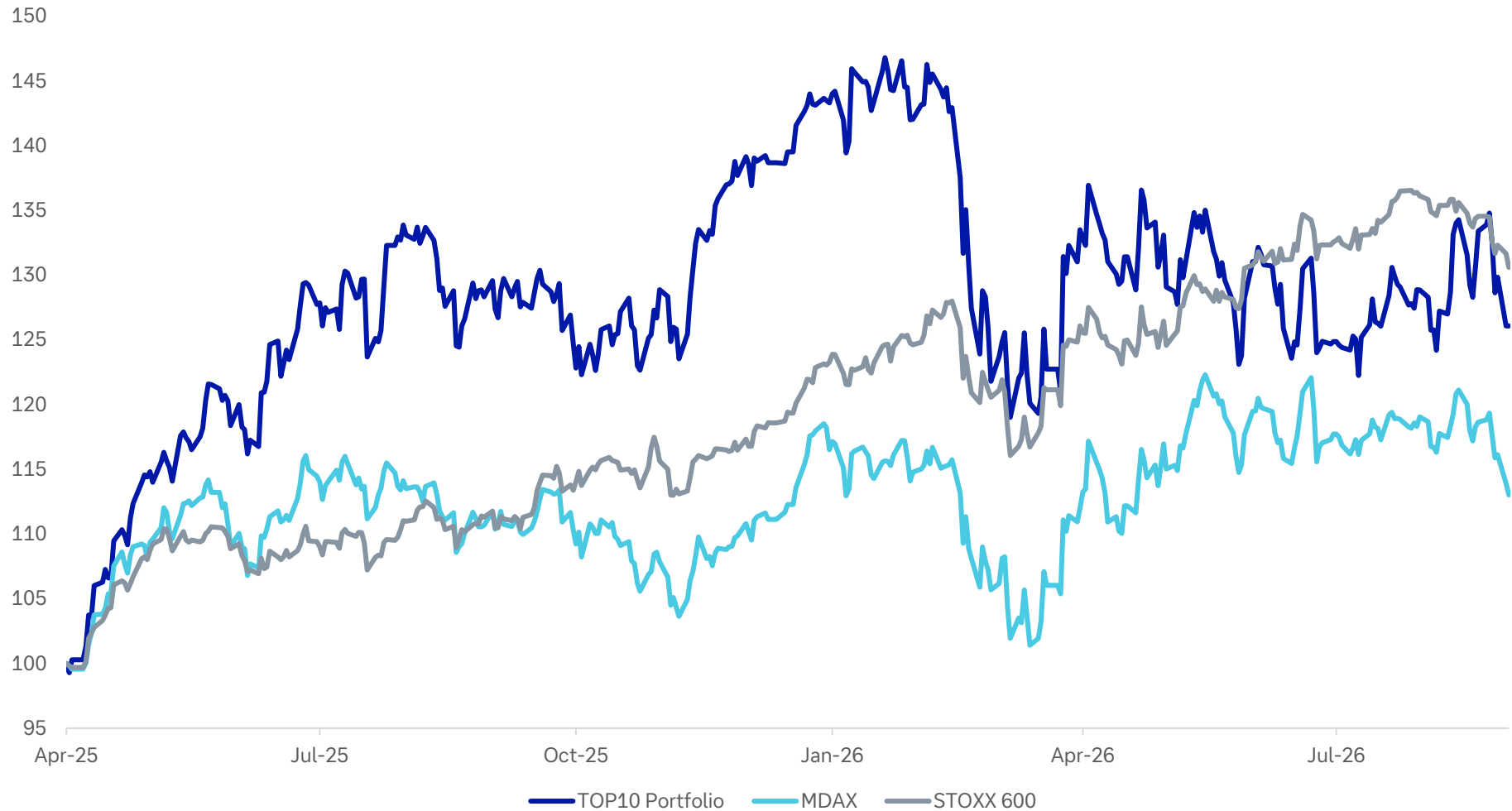
Time period: Aug 2016 – today.

Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

TOP10 Germany stock basket



Total return since inception



Note: German top picks portfolio inception was on 04/15/2025. . Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026



Reform agenda

Germany: Reform hammer in full swing



Area	Detailed measures	Status	Area	Detailed measures	Status
Economic Growth	– "Growth Booster": – 30% accelerated depreciation on machinery and equipment, 75% on EVs – Gradual corporation tax cut from 15% to 10% and R&D tax credit expansion	– In place – Jul '25 - Dec '27 for depreciation – 2028-2032 for corporate tax cut	Health Care System	– Restricting free family insurance – Higher contributions by higher earners – More co-payments by patients and less benefits (braces, dentures, homeopathy) – Price caps for hospitals/pharma	– Approved in parliament in July, – Effective in 2027/2028
Energy Prices	– Industrial power price: electricity price for corporates lowered to 5 c/kWh – Earlier abolition of gas storage levy – Lower grid fees – Electricity tax cuts to EU minimum level is extended indefinitely – Reduction of energy tax on petrol and diesel by 17 ct/l for two months	– In place – Industrial power price until Dec '28 – Reduced energy tax only during May + June '26	Private Pension System	– "Frühstart-Rente" (early start pension): monthly public grant of €10/child (ages 6-18), to be invested into capital market products, tax-free until the payout phase. Withdrawals generally allowed from age 65.	– Approved by the Cabined in August, Parliamentary approval pending
Private Pension System	– "Altersvorsorgedepot" (old-age provision account): retirement savings account with tax benefits, allowing investment into products without guarantees (e.g., ETFs)	– Approved, to become effective on Jan 1, 2027	Statutory Pension System	– Gradual increase in retirement age to 67.5 – Limiting early retirement programs – Age for old-age part-time work up to 58 – Raising number of contributors, e.g. self-employed, MPs, managers – Introduction of mandatory funded component: 2% additional contribution into centralized pension fund.	– Approved by coalition committee, Parliamentary approval pending
Sector-specific	– "Construction turbo" law to accelerate planning and approval procedures – Air ticket tax cut by up to ~€11/passenger – Autos: EV subsidies of up to €6,000 for buyers, extension of car tax exemption for EVs (up to 10y)	– Construction turbo until 2030 – Air ticket tax: since 1st July – EV subsidy 'til '29, EV tax exemption 'til 35	Labour Market	– Easier extension of fixed-term contracts – relaxed dismissal protection for top earners – Preferential tax treatment if a new job is taken up quickly – Medical certificate from first day of illness	– Approved by coalition committee, Parliamentary approval pending
Public Procurement	– Law to faster and simpler public procurement processes – No need to start a contract award process for federal public contracts up to 50k EUR	– In place since 1st July	Income Tax	– Increase of basic tax-free allowance – Higher deductions for families and employees – Flattening of the tax rate for low income – Increase of "rich-tax" to 45% on taxable income >€250k (47% for income >€280k)	– Approved by the Cabinet in September, Parliamentary approval pending
Labour Market	– Retirees can earn up to €2,000/m tax-free when working while on retirement	– In place since Jan '26	Bureau-cracy	– Removal of some reporting obligations – Reduction of need for company representatives – Applications (e.g for construction projects) to be automatically approved after 3m – Aim to cut federal ministries staff by 8%	– Approved by coalition committee. – Bureaucracy Reduction Act passed in May '26
Infra-structure Accelerat or Act	– Prioritizes approval process for critical infrastructure (e.g., railways, highways) – Digitalization of approval processes – Easing of environmental checks	– Passed by Bundestag on June 26			

Source: Federal Ministry of Finance, Federal Government, Bundesregierung, Deutsche Bank Research 09/15/2026

Germany: pension reform

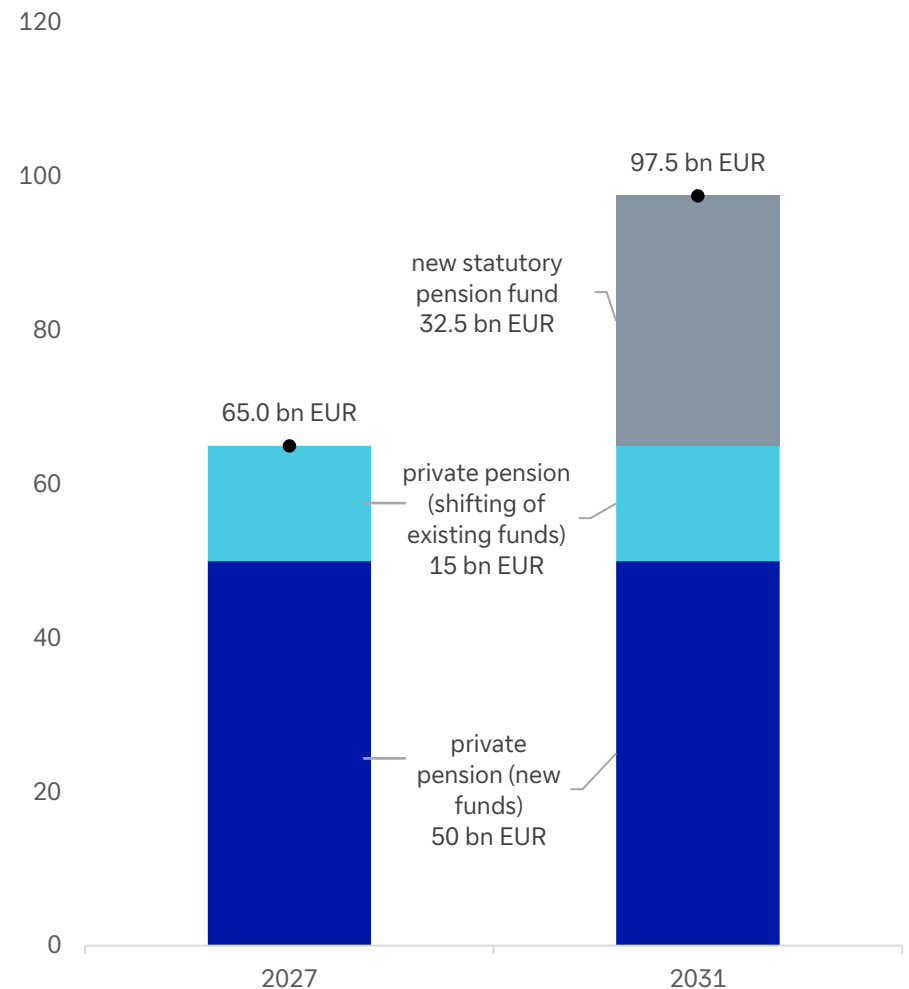


Pension reform could lead to additional annual inflows of 10-15 bn EUR into European equities over the next years.

Key takeaways

- The planned reform of Germany's pension system could generate **significant inflows into European equities** by shifting retirement savings towards capital markets.
- On the private pension side, the new “**old-age provision account**” encourages savers to move funds from low-yield deposits and insurance products into higher-return investments. If half of the eligible population invests the annual amount required to receive the full tax benefit (€2,300), **inflows could reach €50bn per year**.
- We also expect assets to **migrate from the existing “Riester” system**. With total estimated Riester assets of €150bn, a 10% annual reallocation would generate a further **€15bn of inflows**.
- In parallel, the government plans to introduce a funded component to the statutory pension system, equivalent to 2 pp of gross salaries, to be invested through a central **public pension fund**. Contributions are planned to rise from 0.5pp in 2028 to 2pp by 2031, implying annual **inflows of €8bn initially and €30-35bn by 2031**, according to our economists.
- Overall, we estimate these reforms could generate **€60-100bn of additional annual capital market inflows**, of which around **€10-15bn (15%) may find its way into European equities**. To put this into context: the top 50 largest European ETFs saw inflows of c. €40bn in 2025.

Projected inflows into capital markets



Source: BMAS, Deutsche Bank Research 09/15/2026

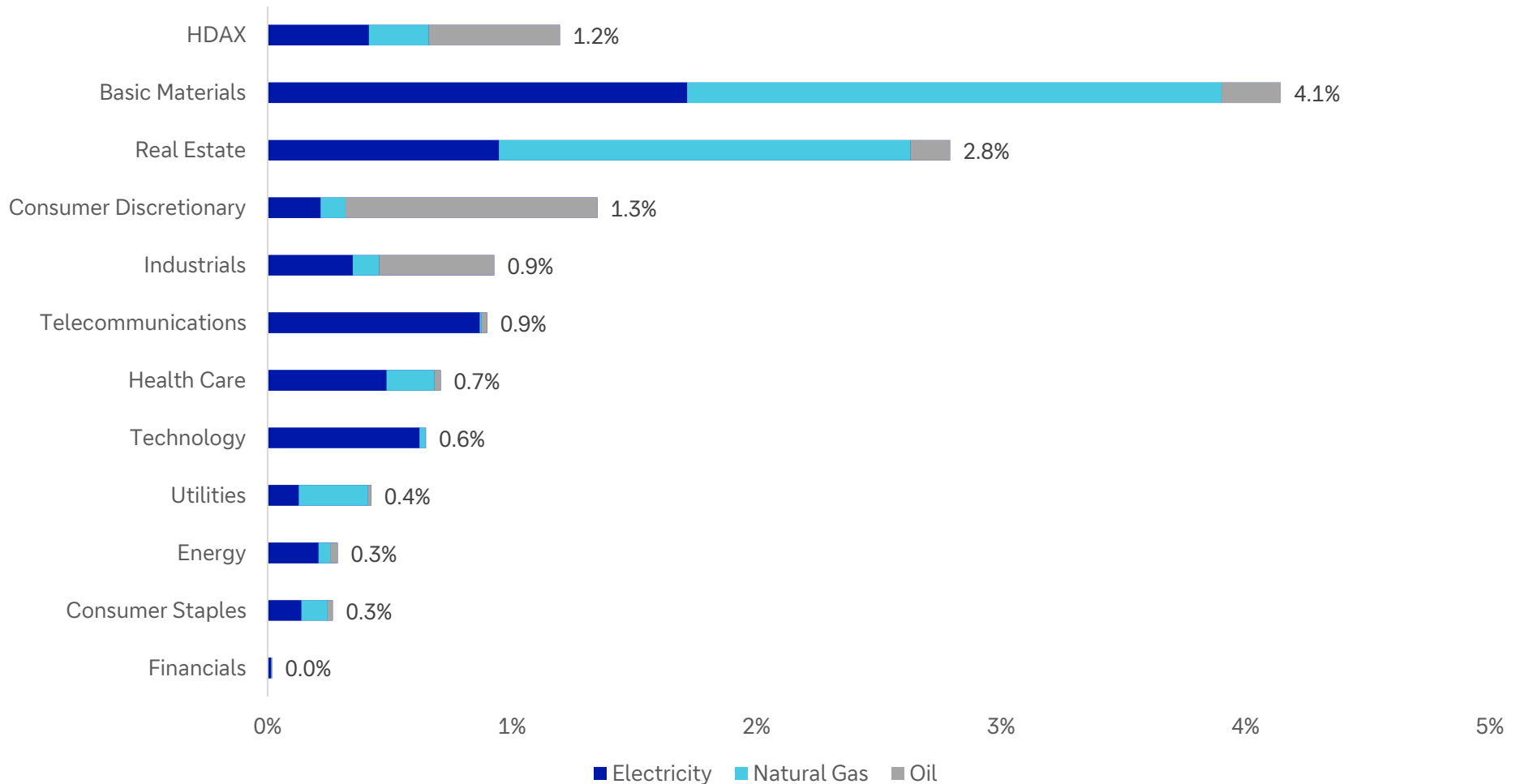


Appendix

Energy costs comprise only 1.2% of HDAX revenues



HDAX energy cost as % of sales*



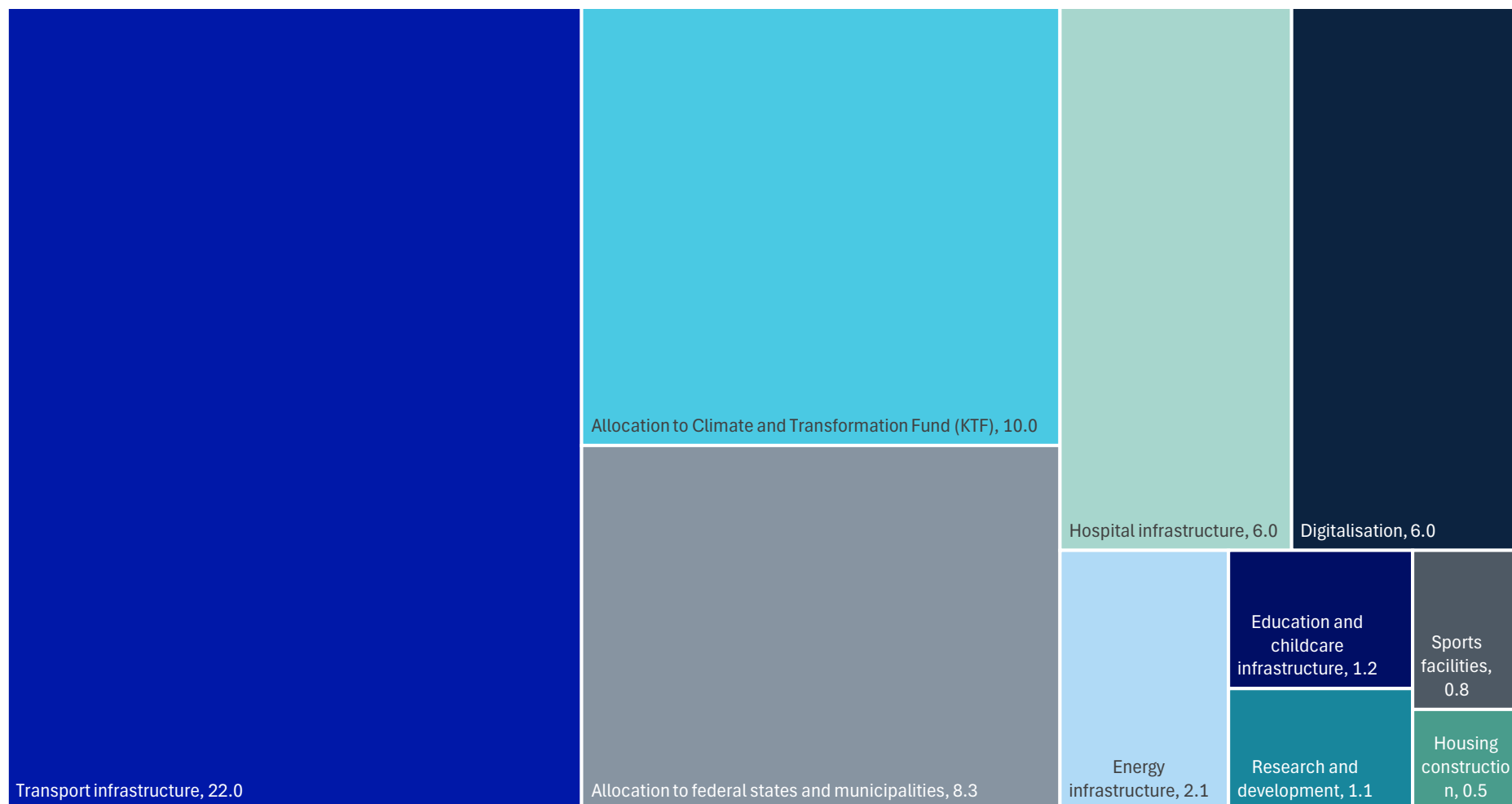
*N=50, based on 3y average German power prices, TTF natural gas and Brent oil. Source: Bloomberg Finance LP, Deutsche Bank Research 09/11/2026

Where to invest?



Most of the higher spending is going into Defence and Infrastructure. Defence has already rallied. We expect sectors with order intake from Infra spend to benefit most.

Planned investments from the special fund for Infrastructure & Climate Neutrality, 2026, mn EUR

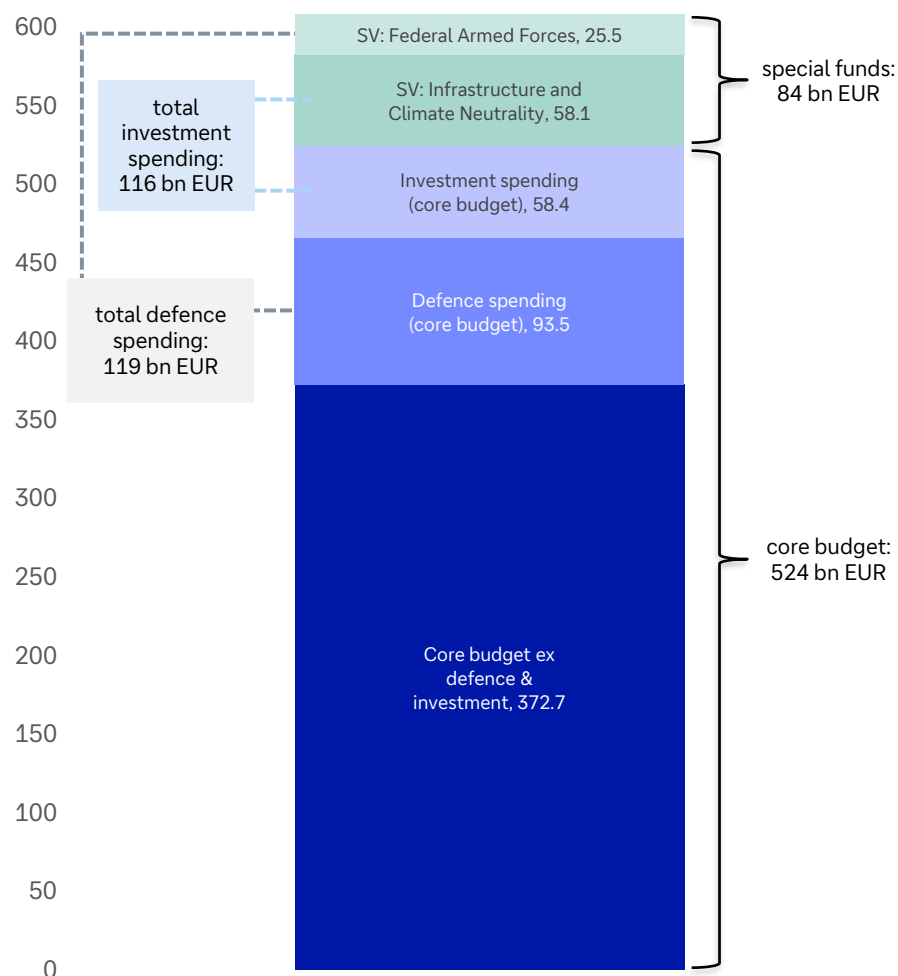


Source: Bundesfinanzministerium, Deutsche Bank Research 09/15/2026

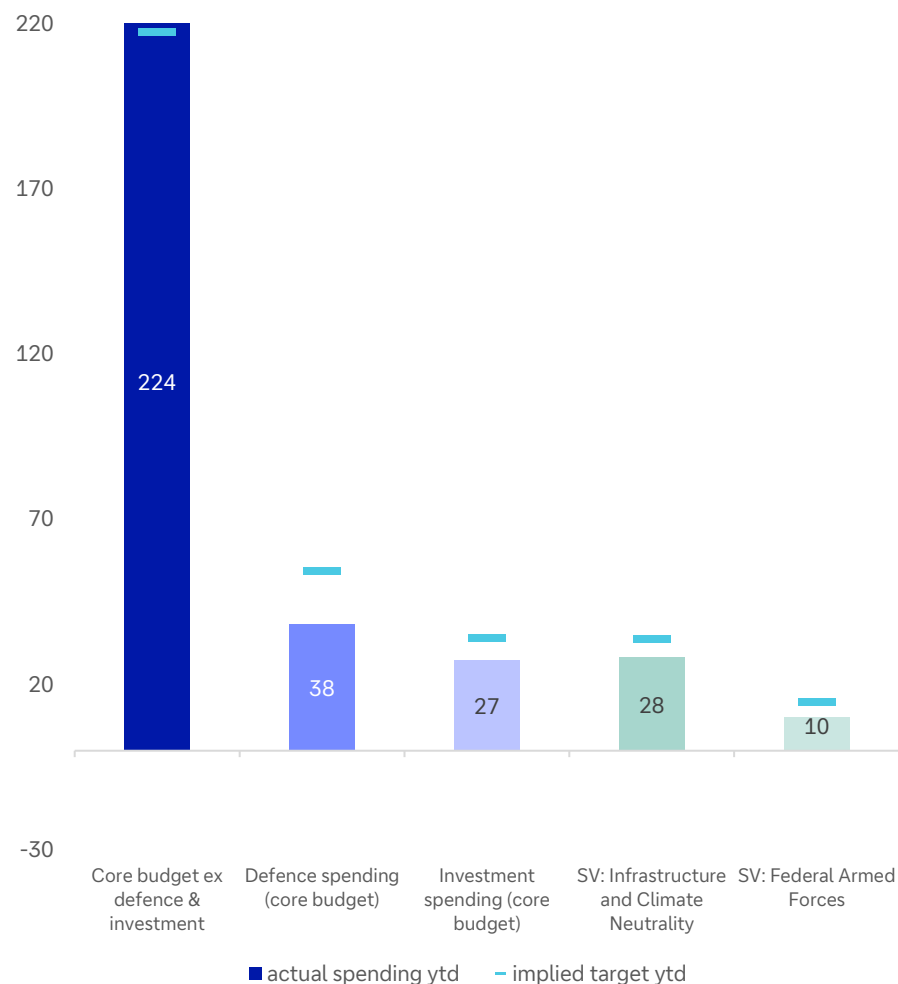
Germany: fiscal expansion underway, with room for spending to accelerate



2026 planned fiscal spending



Spending still below target year-to-date

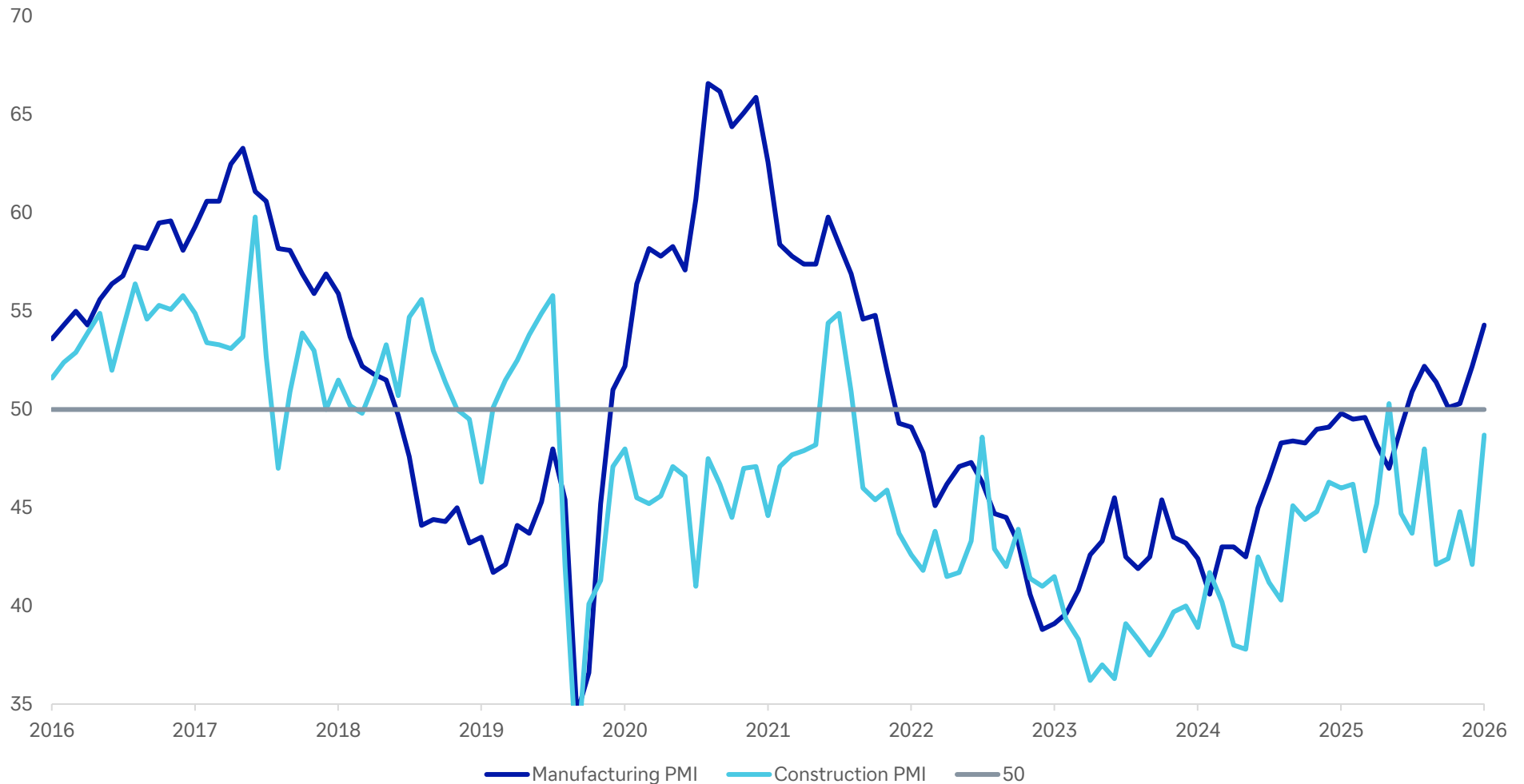


All figures in bn EUR. Actual spending data as of end of July. Source: Bundesfinanzministerium, Deutsche Bank Research 09/07/2026

Manufacturing PMI highest in over 4 years, Construction PMI catching up



Germany Manufacturing PMI



Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

German reforms: example from the past

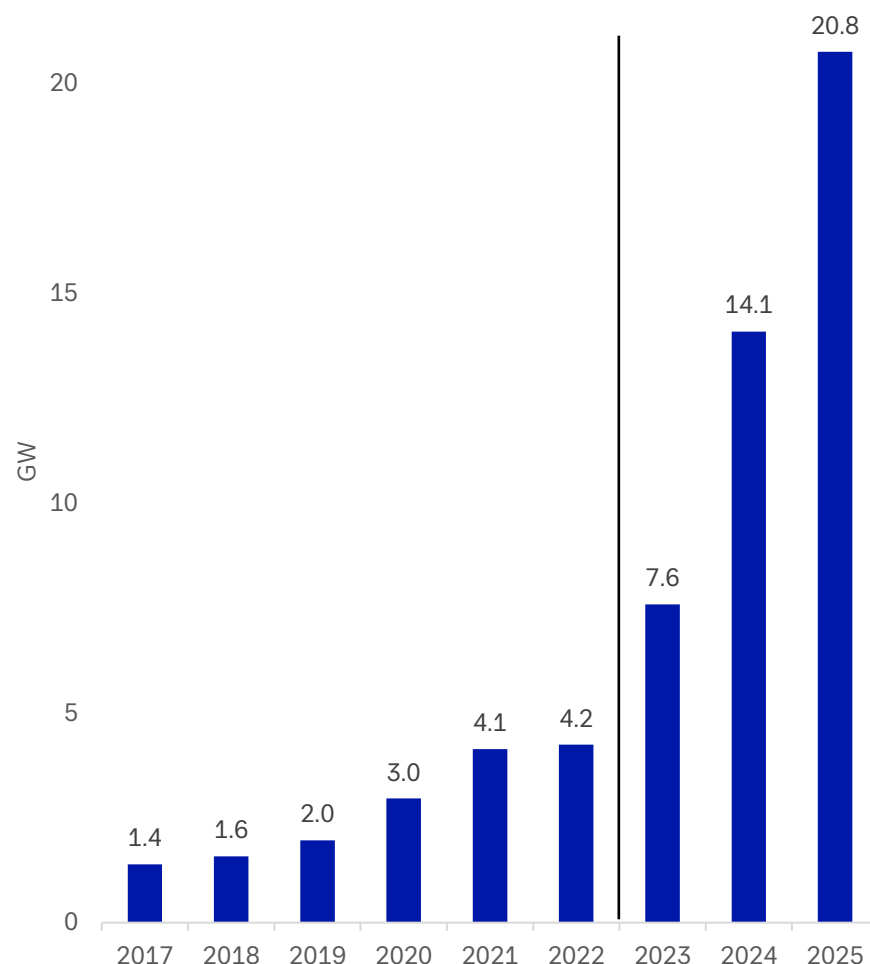


Onshore & Offshore Wind Act

Key takeaways

- In 2023, Germany introduced the Onshore Wind Act and revised the Offshore Wind Energy Act to accelerate the energy transition and streamline renewable energy development.
- For the first time, federal states face legally binding requirements to allocate land for wind power. The share of Germany's land designated for onshore wind is set to increase to 1.4% by 2027 and 2.0% by 2032.
- Policy reforms are already delivering results. In H1 2025, Germany approved 20.8 GW of new onshore wind capacity, nearly four times the level seen in 2022, marking a record pace of expansion.

Newly approved onshore wind energy capacity



Source: SMARD, Statista, Deutsche Bank Research 09/15/2026

German reforms: example from the past

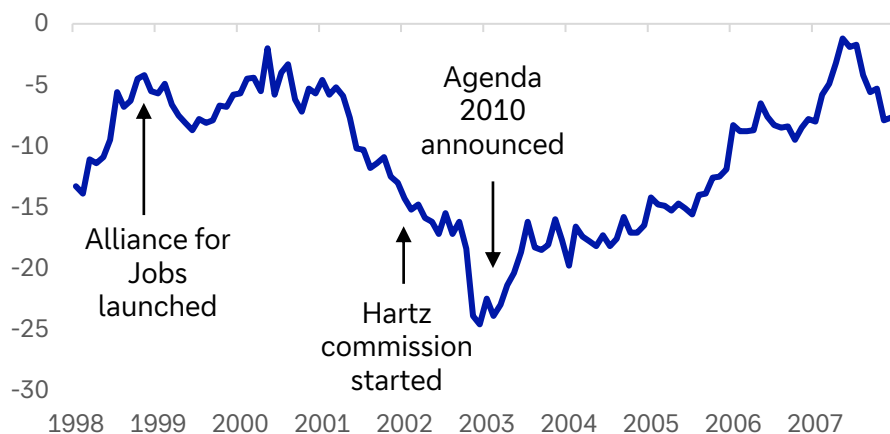


Agenda 2010

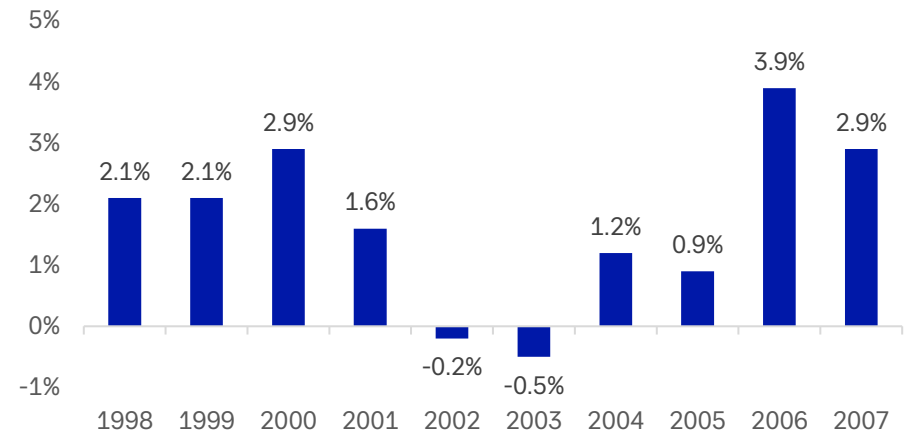
Key takeaways

- Facing stagnant growth and rising unemployment, Germany launched a series of labour-market reforms between 2002 and 2005, beginning with the Hartz Commission and culminating in Agenda 2010.
- The reform package tightened unemployment benefits, expanded labour-market flexibility, improved job placement, and reduced income and corporate taxes to strengthen incentives for work and investment.
- While initially controversial, the reforms helped restore confidence: GDP growth turned positive in 2004, consumer sentiment improved, and equity markets rebounded.

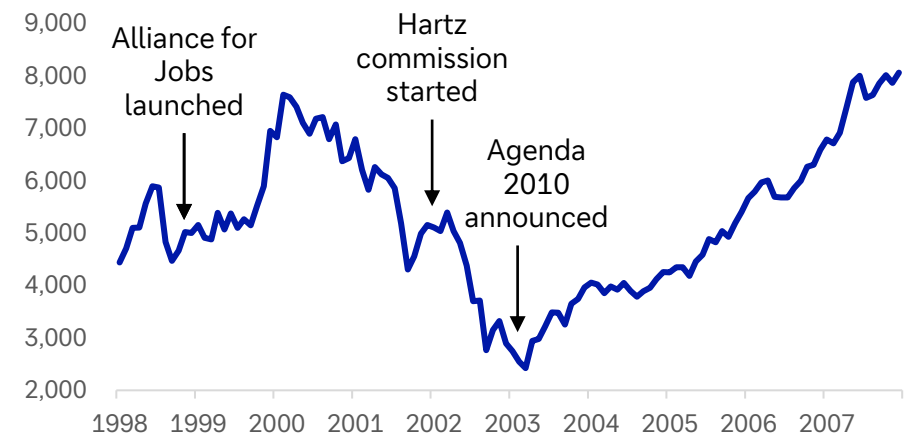
Consumer confidence



Germany annual GDP growth



DAX



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Appendix 1



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Deutsche Bank
Research

ASSET TOKENIZATION 101

The Next Phase of Financial Infrastructure

September 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.



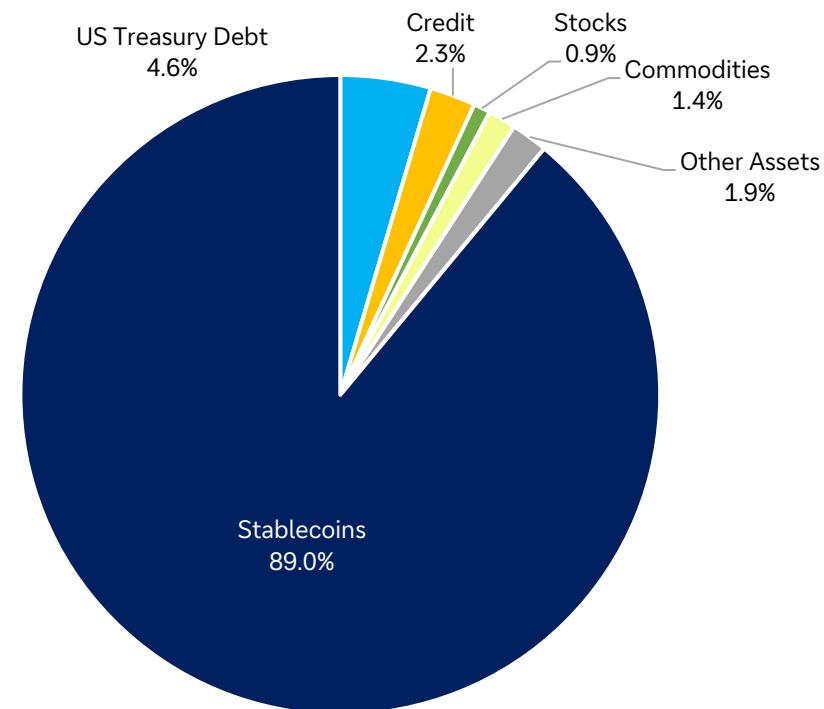
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EXECUTIVE SUMMARY: Tokenization expands into market infrastructure



- The global tokenized assets market, excluding stablecoins, grew from ~\$10bn in Jan 2025 to ~\$39bn in Sep 2026. US Treasury debt remains the largest segment outside stablecoins, with money market funds leading near-term adoption. Despite the US Digital Asset Market Clarity Act failing to advance through to the Senate yesterday, we remain positive about the momentum to tokenize both assets and capital markets in the coming years.
- Institutions are increasingly applying tokenization to settlement, funding and collateral management. Notable examples include DTCC's tokenization service, DLT-based intraday repo and collateral initiatives, and the Eurosystem's Pontes solution, which connects DLT platforms to TARGET Services and is expected to launch later this month.
- A broad tokenized capital market requires progress across three areas: assets, settlement money and market infrastructure. The tokenized RWA market remains small relative to traditional markets, and token issuance does not automatically create secondary-market liquidity.
- In this chartbook, we broaden our focus from tokenized assets, such as bonds and stocks, to the infrastructure through which they are issued; and review (i) market size; (ii) adoption trends; (iii) the regulatory drivers of tokenization; and (iv) barriers that will determine the pace of growth.

Market share of tokenized assets, Sep 14 2026



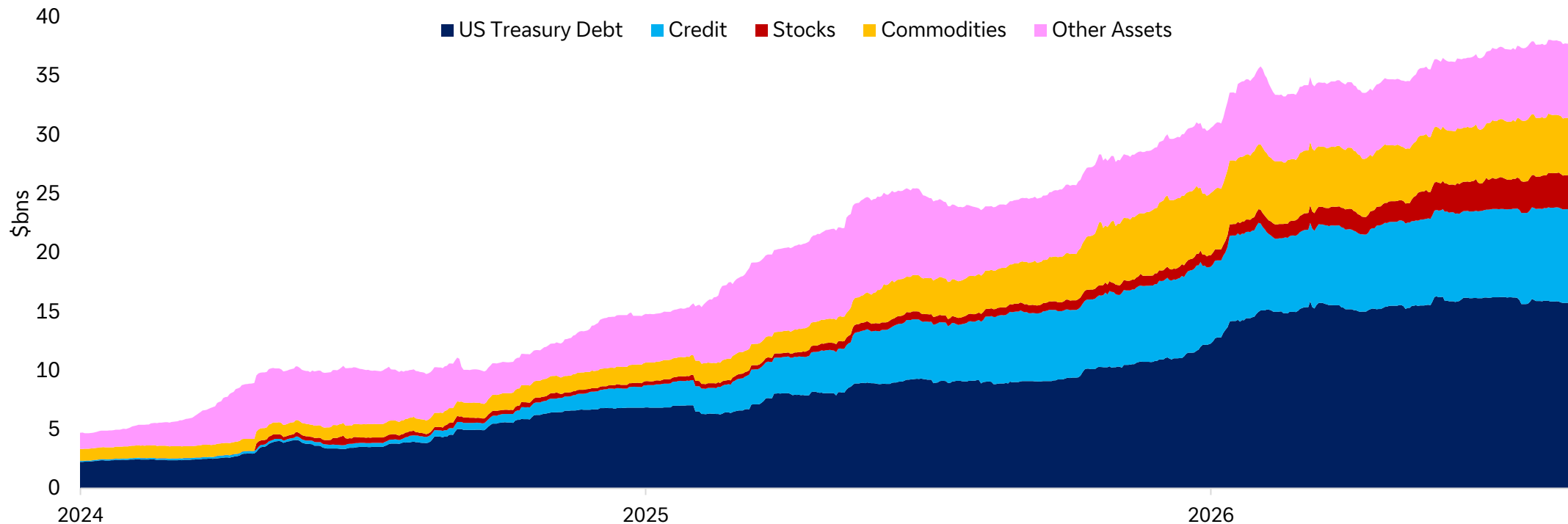
Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" denotes tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Note: Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, private equity, venture capital and real estate.



MARKET SIZE: US Treasuries continue to dominate

- US Treasury debt remains the largest on-chain tokenized asset segment excluding stablecoins, reaching \$15bn in September 2026. Much of that is driven by tokenized money market funds (MMF).
- Demand reflects appetite for highly liquid, short-term risk-free collateral assets, with the ability to generate yield.

Distributed tokenized asset value, excluding stablecoins



Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" refers to tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, stocks, private equity, venture capital and real estate.

DEFINITION : How asset tokenization works



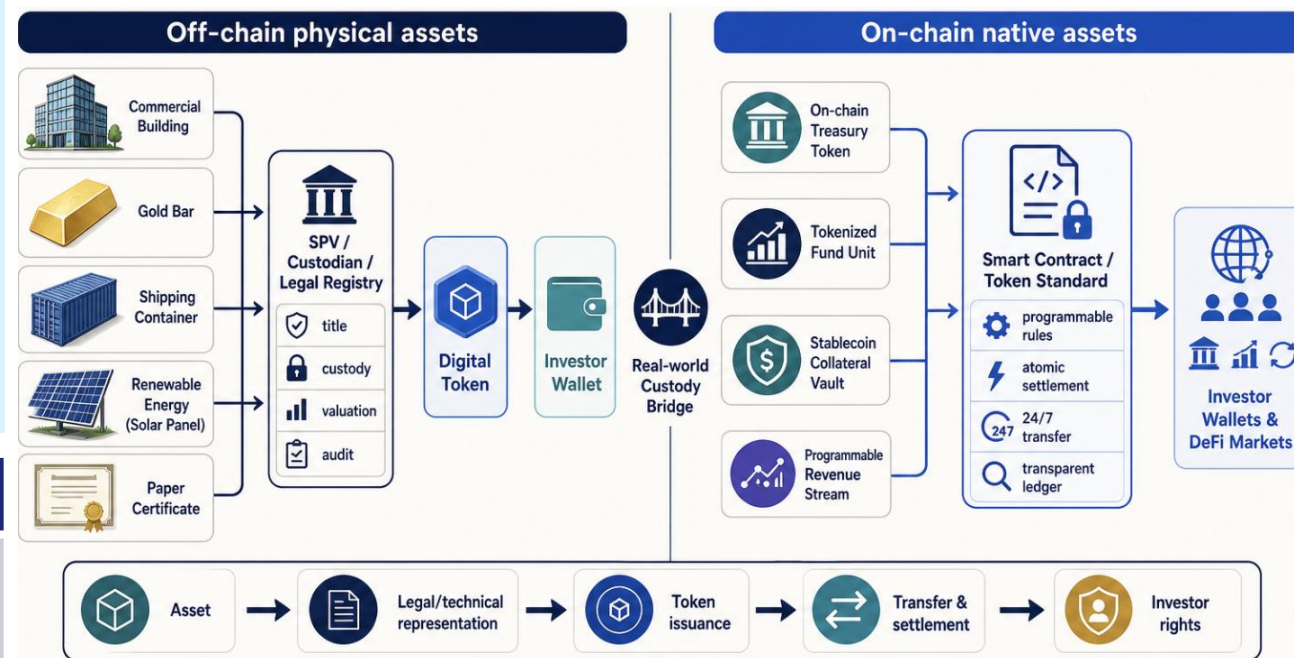
- Tokenization records assets or claims as digital tokens on a distributed ledger, using public or permissioned networks.
- Potential benefits include faster transfers, automated servicing and smaller investment units.
- Shared records can reduce reconciliation, but trading hours, redemption terms and investor access remain product-specific.

Two issuance models:

- **On-chain (or native):** The security or fund share is issued directly on a distributed ledger.
- **Off-chain (or tokenized representation):** The underlying asset remains outside the blockchain, but ownership claims and economic rights are represented digitally on-chain.

Component	Traditional form	Tokenized arrangement
Settlement assets	Bank deposits, central bank reserves, cash	Stablecoins, tokenized deposits, CBDCs, central bank money available through DLT platforms
Issuance assets	Treasury bills, government bonds, money market funds, real estate, commodities etc	Tokens representing securities, fund shares or claims on underlying assets
Market infrastructure	Custodians, exchanges, clearing houses and transfer agents	DLT-based issuance, settlement and servicing, often operated by existing institutions

Tokenization can enable automated transfers and settlement



Native digital securities are issued on-chain but may still require custodians, registrars and other intermediaries, according to the [SEC](#).

Sources: Deutsche Bank Research. Image generated by AI.



STABLECOIN SPOTLIGHT: The settlement side of tokenization

- Stablecoins account for ~89% of the combined value of stablecoin and distributed tokenized assets. We assess them separately because their principal role is payment and settlement.
- They support near-instant, round-the-clock transfers for remittances, business payments and treasury operations. Cross-border savings depend on transaction fees, FX costs, compliance and conversion into bank money.
- [Chainalysis](#) projects adjusted annual transfer volume could rise from \$28tn in 2025 to \$719tn by 2035; higher scenario approaching \$1.5 quadrillion.

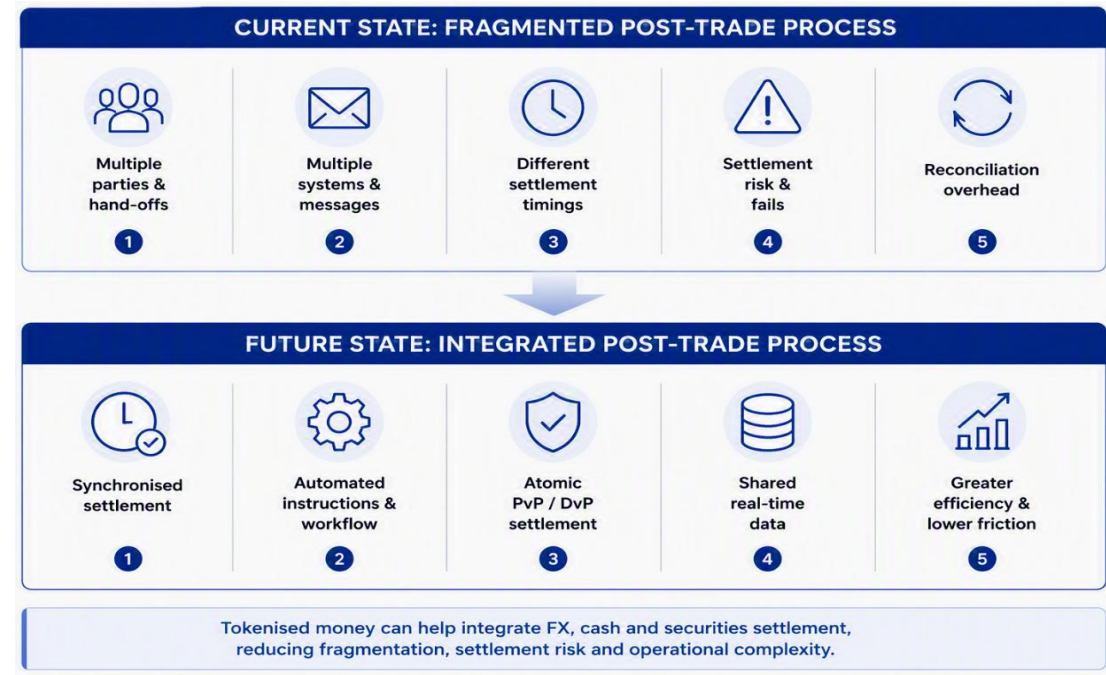
Regulation and competition

- The US Treasury has [proposed](#) implementing rules for payment stablecoins. The GENIUS Act takes effect on the earlier of 18 January, 2027, or 120 days after final implementing regulations are issued.
- Stablecoins could increase competition for bank deposits and payments. Exchange rewards are distinct from issuer-paid interest, which GENIUS prohibits for permitted payment stablecoins.

Risks

- Trust depends on reserve quality, legal redemption rights and operational resilience. Stablecoins remain dependent on their issuers and supporting institutions.

A more integrated post-trade process



Sources: Deutsche Bank Research, taken from DB's [FX Special Report: Tokenised money and the future of FX - Deutsche Bank Research](#)

Tokenized deposits and central bank money provide alternative settlement options. Connecting the cash and securities legs of a transaction can improve settlement efficiency and reduce fragmentation.

For more on tokenized money and settlement, particularly for FX, read [here](#).

INSTITUTIONAL ADOPTION: Tokenization expands into funding, settlement and collateral



- Early efforts focused on tokenized claims on traditional securities (i.e. BlackRock's BUIDL, Franklin Templeton FOBXX), while the underlying assets remained off-chain.
- Financial institutions are now extending tokenization to the infrastructure through which assets are distributed, traded, settled and financed.

DLT adoption: small, but intraday repo offers a practical use case

- Global DLT fixed-income issuance [reached €4.8bn](#) in 2025, up 48% from 2024.
- DLT intraday repo (i.e. JPM Kinexys and Broadridge's DLR) [can accelerate](#) collateral transfers and settlement, helping institutions manage short-term funding needs.
- [A Finadium study](#) estimated that using intraday DLR for 15% of activity could reduce intraday liquidity buffer needs by 8–17%.

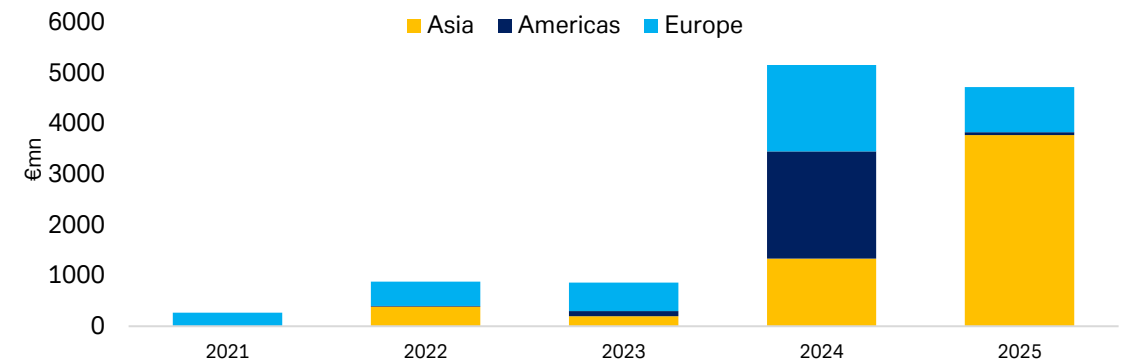
Tokenized securities on blockchain rails

- DTCC to launch its tokenization service in Oct 2026, covering Russell 1000 constituents, ETFs tracking major indices, and US Treasury bills, notes and bonds.
- Initial production trades took place in July with >30 participating firms. Tokens are recorded on DTCC's Hyperledger Besu network or on Canton, settling on either. The SEC granted a three-year authorization.

Official-sector initiatives and interoperability

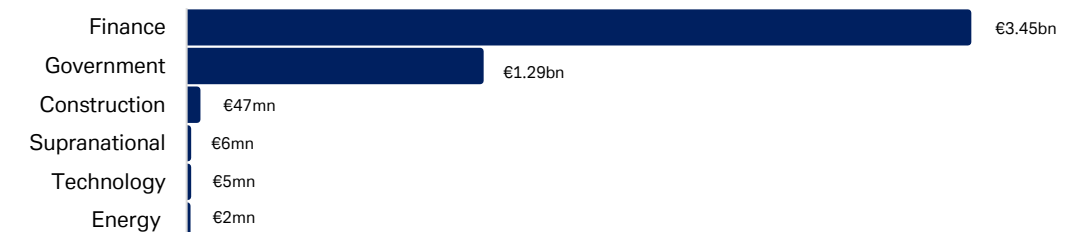
- Project Agorá, led by the BIS and IIF with central banks and financial institutions, demonstrated a prototype for cross-border settlement using tokenized deposits and central bank reserves. Real-value testing is the next phase.
- Clearstream, DTCC and Euroclear published a joint interoperability framework with BCG in March 2026 to address fragmentation across digital asset markets.
- Eurosystem's Pontes solution connects DLT platforms to TARGET Services for settlement in central bank money.

Digital bond issuance by region



Source: Deutsche Bank Research, Association of Financial Markets Europe.

Global digital bond issuance by issuer type in 2025



Source: Deutsche Bank Research, Association of Financial Markets Europe.

SHORT-TERM ADOPTION: Tokenized Treasuries and money market funds lead



- Tokenized cash-management funds, including BUIDL and USYC, combine short-term yields with more flexible transfers and collateral use.
- Potential benefits include faster settlement and lower transaction costs, making Treasury exposure more accessible across eligible digital platforms.
- Direct Treasury bond tokenization [accounted for](#) 27% of overall US Treasury tokenization in 2025, indicating adoption beyond fund structures.
- These assets can also serve as collateral in repo and margin arrangement, supporting demand beyond investment returns.

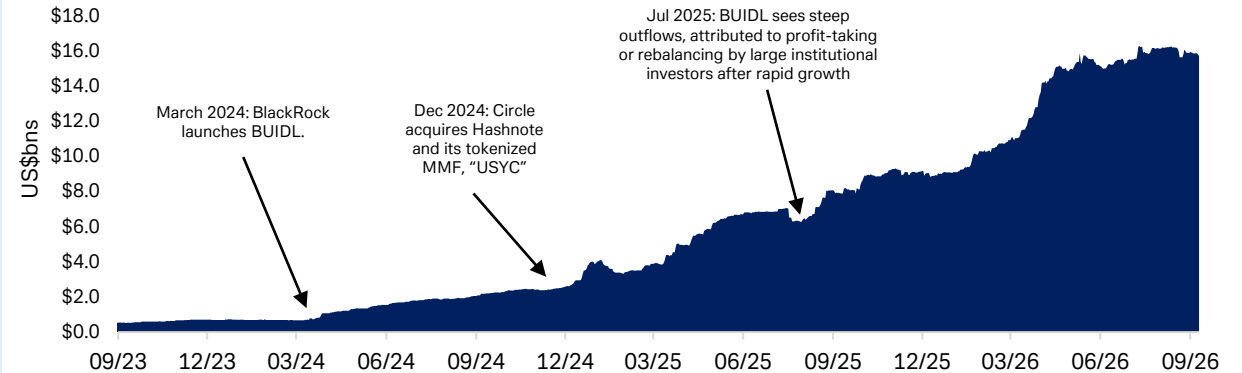
Secondary-market constraints

- Tokenized bonds are often held to maturity. Limited investor participation and weak interoperability between platforms constrain secondary-market liquidity.
- Some exchanges are attempting to improve liquidity by connecting digital and conventional markets through dual listings, including SIX digital exchange and Swiss Stock exchange).

Risks

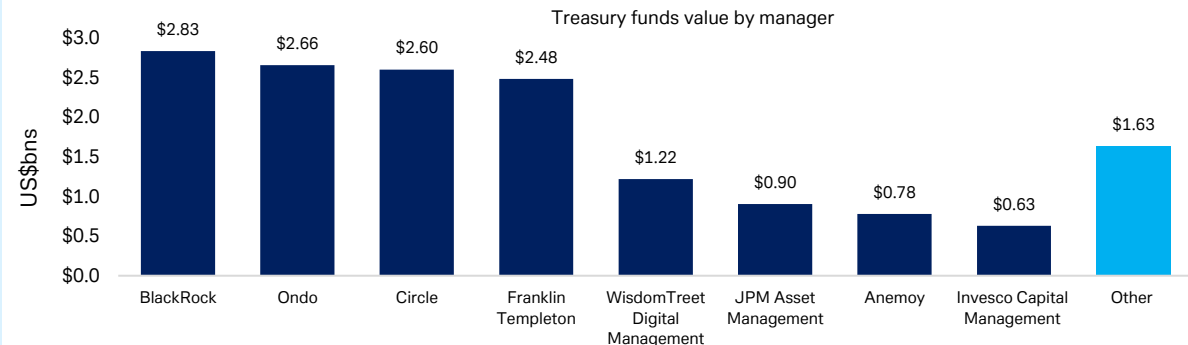
- Redemption terms can differ from the settlement hours and liquidity of underlying assets, creating potential liquidity mismatches.
- Blockchain transparency can improve visibility but may also accelerate withdrawals during stress.
- Using tokenized fund shares across multiple products and as collateral can create additional channels of contagion.

Tokenized US Treasury market value reaches \$15bn



Source: Deutsche Bank Research, RWA Database. Updated Sep 10, 2026.

Largest providers of tokenized Treasury products



Source: Deutsche Bank Research, RWA.xyz. Data from September 10, 2026. TMMF asset class includes tokenized US treasuries, corporate bonds, and non-US government debt.

MEDIUM-TERM ADOPTION: Tokenized equities remain small; DTCC's launch could broaden adoption



- Tokenized equities remain small (<1%) relative to traditional stock markets.
- [68% of equity tokenization](#) in 2025 consisted of direct tokenization of individual stocks or baskets of stocks, vs. tokenized equity funds.
- Suggests that blockchain adoption is increasingly focused on the securities themselves (i.e. DTCC method) rather than through third-party platforms (i.e. Ondo and xStocks).

Developments

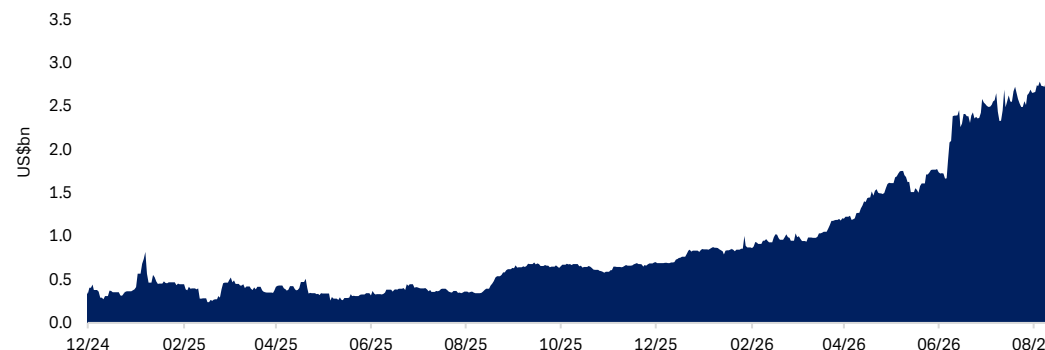
- DTCC's tokenization service, scheduled to [launch in Oct](#), will allow eligible existing securities to move across blockchain infrastructure within its regulated framework.
- These tokenized securities will retain the entitlements, investor protections and ownership rights associated with their traditional form.
- NYSE has announced plans for a tokenized securities platform designed to support 24/7 trading of US-listed equities and ETFs, with near-instant settlement and stablecoin funding.

Benefits: Reduce barriers for foreign investors; automated post-trade processes and longer trading hours; reduce information asymmetry (e.g., embedding listing or disclosure rules directly into the tokens).

Risks & limitations

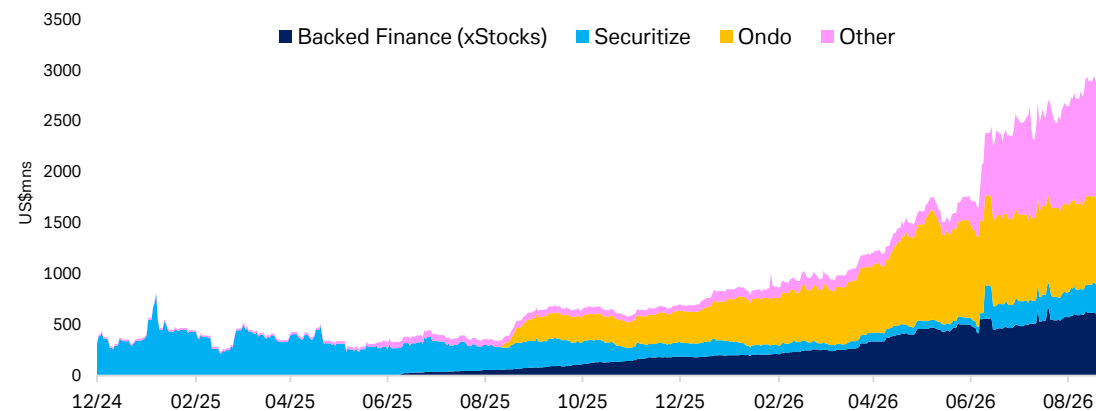
- Tokenized equities require clear legal ownership, enforceable investor rights and alignment with existing securities rules.
- Under some third-party wrapper structures, the token provides exposure to a share without making its holder a shareholder in the underlying company.
- Investors may therefore face issuer or custodian risk and may not receive the same voting, dividend or other rights as direct shareholders.

Market value of tokenized equities



Source: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

Tokenized equities by platform



Sources: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

LONG-TERM ADOPTION: Private credit expands through record-keeping; real estate is growing fast but from a negligible base



Private credit

- Private credit stands ~\$8bn in total, including asset-backed credit (\$2.6bn), specialty finance (\$2.5bn), corporate credit (\$1.9bn), and diversified credit (\$798mn).
- Activity includes lending linked to digital assets and selected corporate credit products.
- A larger use case is blockchain-based record-keeping: represented credit stands at ~\$35bn, over 4x the distributed market.
- Companies like Figure Technologies and Tradable use blockchains to support loan administration, ownership records, payments tracking, and servicing.

Challenges

- Tokenization remains concentrated among a small number of issuers and lacks a deep, and liquid secondary market.
- Legal fragmentation, credit quality and servicing arrangements remain important risks. Blockchain records do not remove the underlying borrower's default risk.

Real estate

- The market expanded from ~\$35mn in 2025 to \$226mn in Sep 2026, although it remains small.

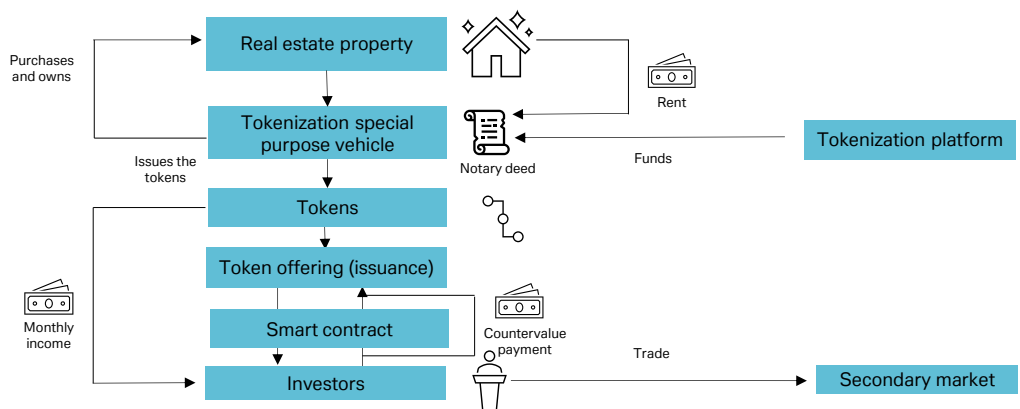
Benefits

- Fractional ownership can lower minimum investment sizes (typically for \$100-\$1,000 per unit) and widen access, particularly where traditional capital markets and mortgage markets are limited.

Challenges

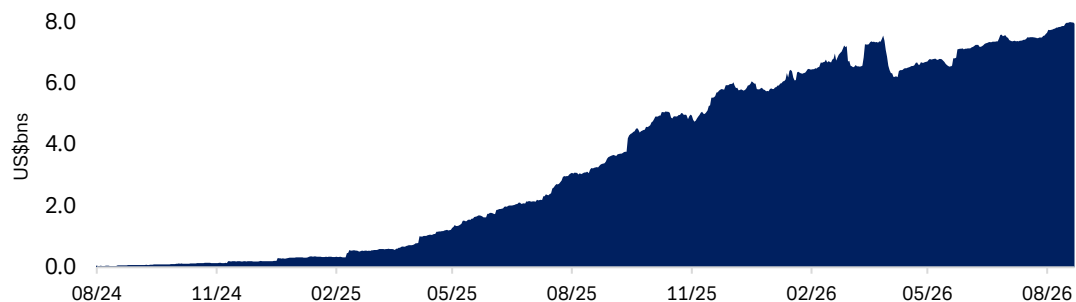
- A [BIS study](#) found that liquidity improvements can depend on platforms offering to buy tokens back. These arrangements can help investors exit but expose platforms to liquidity and solvency risks.

Real estate tokenization process



Source: Deutsche Bank Research, BIS, "When bricks meets bytes: does tokenization fill gaps in traditional real estate markets?"

Tokenized credit market stands at ~\$8bn



Sources: Deutsche Bank Research, RWA.xyz. Updated September 10. Excludes platforms that issues tokens using blockchain as a recordkeeping layer (i.e. Represented Layer)

BARRIERS TO ADOPTION: Weak secondary market liquidity and lack of regulations continue to challenge adoption



Main barriers to adoption

- **Secondary-market liquidity remains low.** Issuance growth has not consistently translated into active trading or deep markets.
- **Regulatory uncertainty:** Fragmented or incomplete frameworks slow institutional adoption and complicate integration with traditional financial infrastructure.
 - Legal uncertainty persists over whether token ownership conveys ownership of the underlying asset or a claim on it.
 - In segments like money market funds, rapid cross-border growth can increase regulatory arbitrage, capital-flow volatility and AML/CFT compliance challenges. Policymakers must adapt frameworks as the technology evolves.
- **Fragmented infrastructure:** Public blockchains, permissioned institutional ledgers and hybrid platforms often lack interoperability, fragmenting markets. Clearstream, DTCC and Euroclear published a joint framework to address this in March 2026.
- **Limited integration** with existing custody, settlement, and record-keeping arrangements prevents institutions from scaling tokenized products.
 - Proprietary infrastructure and centralized controls also raise questions about ownership and governance.

Sources: Deutsche Bank Research.

Jurisdiction	Regulatory status	Main implication
	EU MiCA + DLT pilot regime	Comprehensive crypto-assets regulation. Tokenized financial instruments and tokenized deposits remain subject to securities and deposits regulation respectively (e.g. MiFID). National pilots (e.g., EU's DLT Pilot Regime) are testing tokenized capital-markets applications; focus is on capital market infrastructure modernization.
	UK Cryptoassets Regime / BoE Systemic Stablecoins Regime Live - Digital Securities Sandbox, Digital Gilt pilot Program (DIGIT) and Great British Tokenised Deposits (GBTD) Project	UK Cryptoassets and Stablecoins Regime finalised in 2026. FCA and BoE launched the Digital Securities Sandbox for firms to test DLT-based blockchains; HSBC first bank to go live. The FCA has also identified fund tokenization as a strategic priority through 2025-2030. FCA/BoE launched joint vision for wholesale tokenised markets in 2026 – work ongoing. DIGIT / GBTD development ongoing.
	GENIUS Act to begin in Jan 2027 with final rule out by Nov 26; CLARITY Act stalling	GENIUS establishes a federal framework for stablecoins. Key controversy is whether yield or reward mechanisms should be permitted in stablecoins and other digital assets; oversight is fragmented across the SEC, CFTC, banking regulators and state authorities. CLARITY Act, market structure bill, addressing yield proposal and related items failed its procedural vote in Sep. Attention will refocus on US federal regulators for digital asset rules in absence of market structure.
	China opposes permissionless crypto, not tokenization itself	Strict rules on cryptocurrencies remain; China's focus remains on e-CNY and state-approved blockchain networks. Recent reforms suggest the e-CNY is moving away from CBDC towards a tokenised deposits model, reflecting limited retail CBDC uptake alongside existing mobile payment platforms solutions.
	Project Ensemble Sandbox (Aug 2024); new regulatory roadmap for digital assets (Feb 2025); progress ongoing.	HKMA has supported tokenized green bonds; tokenized deposits; tokenized money-market and investment products. Project Ensemble has progressed to real-value transaction testing using tokenised deposits. HK to announce a new digital bond platform in H22026 to support issuance and settlement of tokenized bonds.
	Project Guardian (c. 2022)	Institutional tokenization sandbox; money market funds and bonds are tokenized and settled on-chain. Focus is increasingly on commercialisation and scaling successful pilot projects across funds, bonds, deposits and FX.
	Financial Instruments and Exchange Act, overseen by the Financial Services Agency	Legally recognizes security tokens through FIEA since 2020. Subjects security tokens (STOs) to the same regulations as traditional securities, including registration and disclosure, custody and investor protection obligations.
	Abu Dhabi's ADGM regulatory framework treats security tokens as regulated financial instruments.	Tokenized securities classified as "digital securities" under ADGM. Actively supports blockchain-based projects that allow businesses to test tokenization ideas via its sandbox program.

OUTLOOK: Tokenized capital markets could become a core part of financial by the 2030s



- Tokenization will **modernize financial infrastructure** rather than eliminate intermediaries. We expect a **gradual transition onto digital rails**, with banks retaining key roles across payments, FX, treasury, financing, collateral and trading settlement.

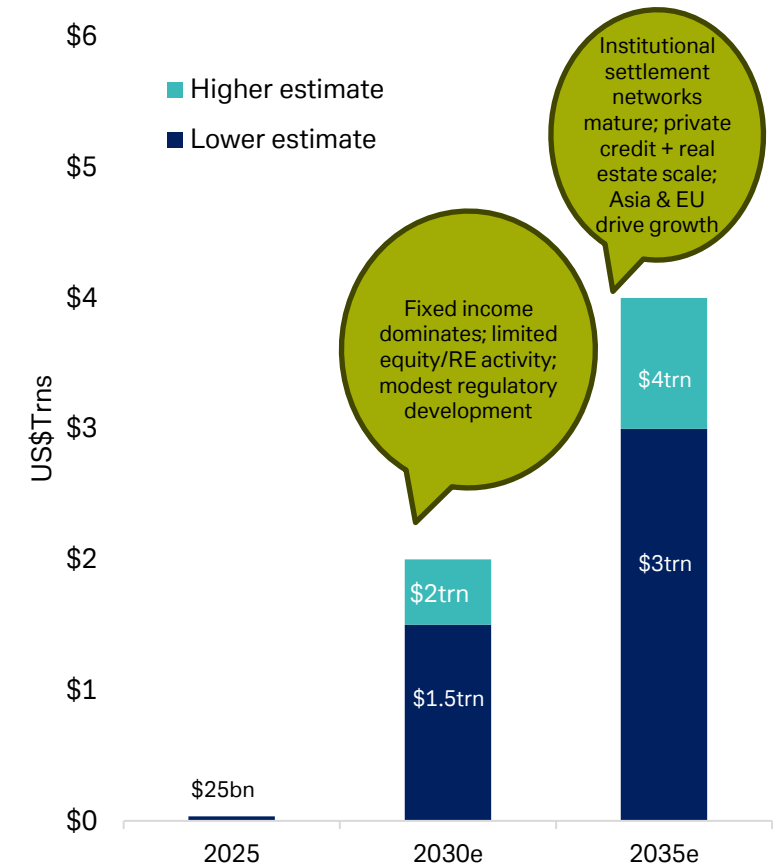
Assumption, on the positive side reflects...

- So far, tokenization has achieved its greatest scale in stablecoins, with activity concentrated in crypto trading and payments gaining ground.
- Short-term: Adoption will be driven by tokenized US Treasuries and money market funds, expanding as yield-bearing collateral in repo and margin arrangements.
- Long-term: The opportunity extends beyond assets to the infrastructure supporting issuance, settlement, custody, collateral management and repo markets.
- We are seeing **3 growth catalysts**: (i) **infrastructure**: major financial market infrastructures (DTCC, NYSE, Nasdaq) are integrating tokenization into core workflows; (ii) **liquidity**: Growth of regulated on-chain money (stablecoins and tokenized deposits) providing a trusted settlement foundation; (iii) and **regulation**: US 2025 GENIUS ACT continues to encourage global digital asset adoption, although the failure of the CLARITY act to advance yesterday likely rules out progress in this congressional session until Jan 2027. Looking ahead, focus will turn back to the SEC and CFTC for crypto rules.

...Risks to adoption

- **Accelerated liquidity runs**: Tokenization may create a new form of systemic liquidity risk that spreads faster than traditional finance.
- **Money market funds yield-chasing**: Declining interest rates or rising competition may push MMF issuers into riskier assets, less-liquid assets to boost yield, undermining their safety.
- **Crypto contagion**: Tight integration with crypto markets via stablecoin reserves and collateral risks transmitting digital asset volatility to conventional financial markets.
- **Operational risks**: Smart-contract vulnerabilities, cyberattacks, network outages could halt trading and prevent access to the underlying assets.
- **Financial Crime**: Traditional institutions must scale up compliance and AML controls to manage illicit activities on decentralized networks.

Conservative forecasts for tokenized real-world assets (ex stablecoins)



Source: Deutsche Bank Research.

AUTHORS



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She is the author of [*Democratizing Finance*](#) published by Harvard University Press.



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She holds a dual MA/MSc in International History from Columbia University and the London School of Economics, and a BA in History from Barnard College, Columbia University.

Appendix 1



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